



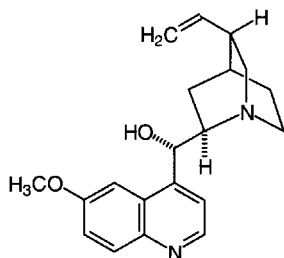
quinhydrone

quinhydrone an addition compound of one mole each of hydroquinone and *p*-benzoquinone. It is a reddish-brown crystalline substance with a dark-green lustre, useful as an anti-oxidant, in photography, and as a redox system in the **quinhydrone electrode**.

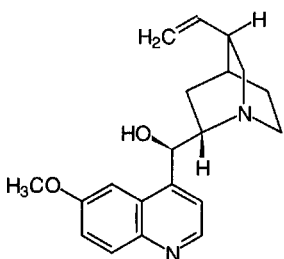
quinhydrone electrode a half-cell consisting of a bright platinum (or sometimes gold) electrode immersed in a test solution to which has been added a little **quinhydrone**. It has an electric potential related to the pH value of the solution and thus is useful as a secondary hydrogen electrode for the electrometric measurement of pH.

quinic acid the D enantiomer ($1\alpha,3\alpha,4\alpha,5\beta$)-1,3,4,5-tetrahydroxycyclohexanecarboxylic acid occurs commonly in plants, either free or as esters (e.g. **chlorogenic acid**), and may make up as much as 2–10% of the dry weight of leaves. It may be formed by the **shikimate pathway**. Its catabolism to protocatechuic acid has been extensively studied in *Aspergillus nidulans* and *Neurospora crassa*.

quinidine β -quinine; 6'-methoxycinchonan-9-ol; a stereoisomer of **quinine** noted for its cardiac anti-arrhythmic effects.



quinine 6'-methoxycinchonan-9-ol; a bitter-tasting alkaloid obtained especially from the bark of cinchona, any of a number of trees or shrubs of the tropical genus *Cinchona* (especially the Javan species *C. lidgeriana* and various South



American species). It was formerly much used (and sometimes is still needed) for the prevention and treatment of malaria; it

binds to the DNA of the malarial parasite, *Plasmodium* spp., and is thought thereby to inhibit biosynthesis of the parasite's nucleic acids. Quinine and drinks containing it (e.g. tonic water) help to prevent cramp. *Compare* **quinidine**.

quinoid or **quinonoid** of, derived from, or resembling a **quinone** (def. 2).

quinol an alternative name for **hydroquinone**.

4-quinolone antibiotic a generic term for any of a wide range of synthetic antibacterial compounds whose molecular structures contain a 4-oxo-1,4-dihydroquinoline or a 4-oxo-1,4-dihydrocinnoline nucleus; e.g. **cinoxacin**, **ciprofloxacin**, **nalidixic acid**, **oxolinic acid**. Their antibiotic activity derives from interaction with the A subunit of bacterial DNA gyrase (see type II **DNA topoisomerase**); this results in trapping of bacterial DNA in the complex it forms with the A subunit. Rejoining of the DNA strands broken by the enzyme is thereby arrested.

quinone 1 a common name for **p-benzoquinone**. **2** any member of a class of diketones derivable from aromatic compounds by conversion of two CH groups into CO groups with any necessary rearrangement of double bonds. Simple quinones are usually *p*-quinones, i.e. derivatives of *p*-benzoquinone (*compare* **hydroquinone** (def. 2)), but they may also be *o*-quinones, i.e. derivatives of *o*-benzoquinone (*compare* **catechol** (def. 2)). Naturally occurring quinones form a large, varied, and widespread group of compounds; they include numerous pigments, various electron carriers (e.g. **plastoquinone**, **ubiquinone**), and vitamins K. *See also* **semiquinone**.

+quinone *comb. form* denoting a **quinone** (def. 2); e.g. benzoquinone, anthroquinone, naphthoquinone.

quinonoid an alternative term for **quinoid**.

quinoprotein a proposed name for any of a number of bacterial oxidoreductase enzymes within the sub-subclass EC 1.1.99 that have pyrroloquinoline-quinone (see **methoxatin**) as co-enzyme or prosthetic group. The category includes alcohol dehydrogenase (acceptor) (EC 1.1.99.8) of methylotrophic bacteria, glucose dehydrogenase (pyrroloquinoline-quinone) (EC 1.1.99.17), and a number of dehydrogenases acting on other specified (types of) alcohols or polyols.

quinovosamine 2-amino-2,6-dideoxy-D-glucose; an amino derivative of **quinovose** present in bacterial polysaccharides.

quinovose *symbol*: Qui; a trivial name for 6-deoxy-D-glucose. α -Quinovose occurs, e.g., as its 6-sulfo derivative (6-deoxy- α -D-glucopyranosyl 6-C-sulfate) in the sulfoquinovosyl diacylglycerols, which form one of the two major groups of the glycerolipids of chloroplasts. The sulfoquinovosyl derivative also occurs in some **cardiac glycosides**.

quisqualate receptor *see* α -amino-3-hydroxy-5-methyl-isoxazole-propionic acid receptor.



Rr

r 1 symbol for **ribo+** (def. 1); it is used as a prefix in (one- or three-letter) symbols for ribonucleosides for emphasis or clarity, especially to distinguish from 2'-deoxyribonucleosides. 2 symbol for reaction. 3 *abbr.* for rat (as a prefix to an abbreviation for the name of a peptide hormone); denoting derived (or as if derived) from a rat or rats; e.g. rGH.

r symbol for 1 concentration ratio. 2 rate of concentration change. 3 radius. 4 interatomic distance.

R symbol for 1 a residue of the α -amino acid L-arginine (alternative to Arg). 2 a residue of an incompletely specified base in a nucleic-acid sequence that may be either adenine or guanine. 3 a residue of an unspecified purine nucleoside (alternative to Puo). 4 an unspecified univalent group in a formula of an organic compound. The group must be attached by means of carbon and derived from an aliphatic, carbocyclic, or heterocyclic compound. Up to three such groups, when different, may be designated R, R', and R'', or R¹, R², and R³; for more than three such groups, superscript numerals only should be used. 5 röntgen (a non-SI unit of radiation exposure). 6 resonance effect (in electron displacement).

R symbol for 1 electric resistance. 2 thermal resistance. 3 gas constant. 4 molar refraction.

R stereochemical descriptor denoting *rectus*; i.e. the absolute configuration of a chiral compound in which the priority (obtained using the Cahn-Ingold-Prelog sequence rules) of the substituent groups on the chiral centre decreases in a clockwise (right-handed) direction. For use as a prefix in chemical nomenclature, the descriptor is placed in parentheses and connected with a hyphen, e.g. (R)-alanine. For chiral compounds in which the relative but not the absolute configuration is known, *rectus* configuration is denoted by the prefix (R*)-. See **sequence rule**. Compare **S**.

R_B symbol for R_B value.

R_f or **R_F** symbol for R_f value.

R_p symbol for phosphorylation state ratio.

R_s symbol for saturation ratio.

RAB a gene encoding a Rab protein – any of a family of GTP-binding proteins similar to Ras proteins (see **RAS**) and probably involved in membrane traffic, etc. Example, Rab-2 from human and dog (identical sequences); database code RAB2_HUMAN, 212 amino acids (23.52 kDa).

rabbit aorta-contracting substance *abbr.*: RCS; a name given (in 1969) to an unstable factor released from guinea-pig lung that caused contraction of isolated strips of rabbit aorta. The activity was probably due to the presence of thromboxane A₂ (see **thromboxane**).

rabbit reticulocyte system a cell-free system, from lysed rabbit reticulocytes, that translates added eukaryote mRNAs from a wide variety of heterologous sources. Endogenous mRNA is destroyed using a Ca²⁺-dependent ribonuclease, micrococcal endonuclease; this enzyme is inactivated by adding EGTA. Heterologous mRNA is then added and is translated by the system.

rac- *abbr.* for racemo-.

RACE *abbr.* for rapid amplification of cDNA ends; a variation on the **polymerase chain reaction** designed for amplification of cDNA corresponding to low-abundance mRNA.

racemase any enzyme that interconverts the two enantiomers of a chiral compound; such enzymes are categorized in subclass EC 5.1 (which also includes **epimerases**). If more than one chiral centre is present, a racemase must invert configuration of all the chiral centres. Thus, the name methylmalonyl-CoA racemase, sometimes used for methylmalonyl-CoA epimerase (EC 5.1.99.1), is incorrect – the enzyme does not change configurations of the chiral centres of the CoA moiety.

racemate any homogeneous phase containing equimolar amounts of enantiomeric molecules.

racemic *symbol*: (±)- or (formerly) dl-; denoting the presence of equimolar amounts of the dextrorotatory and levorotatory enantiomers of a compound, regardless of whether the (optically inactive) product formed is crystalline, liquid, or gaseous. A homogeneous solid phase composed of equimolar amounts of enantiomeric molecules is termed a **racemic compound**, and a mixture of equimolar amounts of enantiomeric molecules present as separate solid phases is termed a **racemic mixture**; see also **racemate**. A (±)-product may be resolved into its (+)- and (–)-components. The term is derived from **racemic acid** [from Latin *racemus*, bunch of grapes], the optically inactive mixture of (+)- and (–)-tartaric acids sometimes found during the manufacture of wine. By extension, the term is applied to DL- and (RS)-mixtures. See **D/L convention**, **optical isomerism**, **sequence rule**. See also **racemo-**.

racemize or **racemise** to induce the formation of a mixture of the two possible enantiomers from one of the pure isomers of a chiral compound. —**racemization** or **racemisation** *n*.

racemo- *abbr.*: rac-; prefix (in chemical nomenclature) designating a racemate. It may be used as an alternative to the symbol (±)- or (for certain classes of racemic compounds) the symbol DL-. Compare **meso-**.

rachitic of, pertaining to, or afflicted with **rickets**.

rachitis an alternative name for **rickets**.

rachitogenic tending to cause **rickets**.

rad 1 symbol for radian. 2 *symbol*: rad or (to avoid possible confusion with **rad** (def. 1)) rd; the cgs unit of absorbed dose of ionizing radiation, equivalent to an absorption of 0.01 joule of energy per kilogram (= 100 ergs per gram) of irradiated material. Whereas the rad represents, in practice, about the same amount of energy as the **röntgen** (the exact equivalence depending on the material irradiated), it applies to any type of ionizing radiation in any medium. 1 rad = 0.01 Gy. 3 *abbr.* for radiation.

radial 1 of, resembling, or relating to a radius or ray; in rays. 2 emanating from the centre of a circle or a common central point. 3 spreading out uniformly in all directions on a surface. —**radially** *adv.*

radian *symbol*: rad; the SI supplementary unit of plane angle. One radian is equal to the angle subtended at the centre of a circle by an arc whose length is equal to the radius. It is dimensionless. There are 2 π radians in a circle; 1 rad $\approx$ 57.295 78°.

radiant energy *symbol*: *Q* or *W*; the energy of any form of electromagnetic radiation. The SI derived unit is the **joule**.

radiant intensity *symbol*: *I*; the **radiant power** of radiation from a point source per unit of solid angle; i.e. $I = d\Phi/d\Omega$, where Φ is the radiant power and Ω is the solid angle measured in steradians. The SI unit is the watt per steradian (W sr⁻¹).

radiant power or **radiant flux** *symbol*: *P* or Φ ; the rate of flow of energy of radiation; i.e. $\Phi = dQ/dt$, where *Q* is the radiant energy. The SI derived unit is the watt.

radiation 1 the emission of energy from a source and its transmission as particles, rays, or waves, especially as electromagnetic rays or waves, sound waves, or streams of subatomic particles. 2 energy so transmitted, especially the corpuscular and electromagnetic rays emitted in the decay of radionuclides. 3 **ionizing radiation**. 4 divergence from a common point, especially radially from a central point or source.

radiation absorbed dose see **absorbed dose**, **rad** (def. 2).

radiation biology an alternative name for **radiobiology**.

radiation-chemical reaction any chemical reaction that is initiated by the absorption of radiation but is distinguishable from a **photochemical reaction** by its lack of specificity (i.e. by

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giving rise to a variety of reaction products) and by occurring at higher radiation energies (i.e. by absorption either of electromagnetic radiation from the mid-ultraviolet region onwards or of high-energy particulate radiation) and thus by resulting always in the formation of detectable ions.

radiation chemistry the branch of chemistry concerned with radiation-chemical reactions. *Compare radiochemistry.*

radiation damage the totality of the harmful changes induced in a cell or tissue by radiation. The harmful effects may include formation of **thymine dimers** in DNA or of radicals from polyunsaturated acyl groups in lipids.

radiation dose equivalent *see* **dose equivalent**.

radiation inactivation method a technique, based on **target theory**, that enables the size of the functional unit of a biologically active macromolecule (or of its complexes with other molecules) to be estimated; it can be used on unpurified (e.g. membrane-bound) material. Determinations are made of the biological activity remaining in a sample after exposure to various large doses of ionizing radiation (electrons or X- or gamma radiation of at least 1 MeV), and the value, D_{37} , of the radiation dose, expressed in Mrad, required to reduce the activity to 37% of (i.e. e^{-1} times) its original value is obtained; the relative molecular mass, M_r , of such a macromolecule may then be found from the empirical relationship

$$M_r = 6.4 \times 10^5 / D_{37}, \text{ where } D_{37} \text{ is determined at } 30^\circ \text{C.}$$

radiation source a quantity of radioactive material used as a source of ionizing radiation.

radical (*in chemistry*) 1 any molecular entity, charged or uncharged, that possesses an unpaired electron (but normally excluding any paramagnetic metal ion); often formed by homolysis of a covalent bond. Radical character is indicated in a formula by a centred dot symbolizing the unpaired electron and placed (if possible) beside the symbol for the atom of highest spin density; e.g. $\text{HOO}\cdot$, $\cdot\text{CH}_3$. The term **free radical** is now preferably restricted to any radical that does not form part of a **radical pair**. 2 a former name for **group** (def. 2).

radical anion *see* **radical ion**.

radical cation *see* **radical ion**.

radical centre the atom in any polyatomic **radical** (def. 1) on which the unpaired electron is largely localized.

radical ion any **radical** (def. 1) that carries an electric charge. One carrying a negative charge is termed a **radical anion** (e.g. the benzene radical anion, $\text{C}_6\text{H}_6^{\cdot-}$); one carrying a positive charge is termed a **radical cation** (e.g. the benzene radical cation, $\text{C}_6\text{H}_6^{\cdot+}$).

radical pair any two **radicals** (def. 1) in close proximity in liquid solution, with a solvent cage. They may have been formed simultaneously, e.g. by homolysis, or have come together by diffusion. Whilst together, correlation of their unpaired electron spins occurs and manifests itself as a distortion of nuclear magnetic resonance spectra.

radical scavenger any substance that can react readily with, and thereby eliminate, radicals. In biological systems or materials, radical scavengers may function as **antioxidants** or may protect from damage by ionizing radiation. Among the important radical scavengers that function in animal tissues at physiological concentrations are the water-soluble substances ascorbate, glutathione, and the purine bases, and the fat-soluble substances tocopherols, the retinols, the carotenes, and the ubiquinones.

radioactivate to convert a stable nucleus into one undergoing radioactive decay by bombardment with high speed particles, including protons, alpha particles, neutrons, and deuterons. These are formed in a cyclotron or synchrotron from a radioactive source, and accelerated through a potential difference of a few thousand volts by a magnetic field before being focused on the substance to be converted.

radioactivation analysis or **activation analysis** a method for the qualitative and/or quantitative estimation of the chemical elements in a sample. It depends on the identification and de-

termination of the radionuclides formed when the sample is bombarded with neutrons or other particles.

radioactive having the properties associated with a **radioactive isotope**.

radioactive concentration the **activity** (def. 2) per unit quantity of any material in which a **radionuclide** occurs. *See also specific activity.*

radioactive decay *see* **decay** (def. 1b).

radioactive isotope or **radioisotope** any isotope of a chemical element whose nucleus is unstable and emits alpha, beta, or gamma rays. The product is another element, which may be stable or unstable; if the latter, it decays further. Thus, the principal isotope of uranium, ^{238}U , emits an alpha particle and decays to ^{234}Th , which emits a beta particle forming ^{234}Pa ; this chain continues to the formation of ^{206}Pb , which is stable.

radioactive label or **radiolabel** a **radioactive isotope** incorporated into a molecule to confer on it the property of radioactivity.

radioactive tracer or **radiotracer** a radioactive substance that is added to a metabolic system in quantities (mass) too small to perturb the system, in order to follow, by isolating the radioactive intermediates formed, the sequence of transformations undergone normally by the same (nonradioactive) molecule under similar conditions.

radioactivity 1 the emission of radiation from certain **nuclides** by the spontaneous transformation of their **nuclei**. 2 (*loosely*) the radioactive material itself or its emitted radiation. *See activity* (def. 2).

radioallergosorbent test *abbr.*: RAST; a test used primarily for quantifying the levels of antigen-specific immunoglobulin E (IgE) in serum. A specific allergen, which has been covalently coupled to an insoluble **Sephadex** carrier, is incubated with serum containing the unknown amount of IgE. The carriers are then washed and further incubated with radiolabelled antibodies to IgE, the amount of radioactivity bound being a measure of the amount of IgE. The advantage of this method over the **radioimmunosorbent test** is that IgE specific to a particular antigen can be selectively quantified.

radioassay any type of **assay** in which a radioactive isotope is used in the measurement. The term embraces, e.g., **immuno-radiometric assay**, **radioimmunoassay**, and competitive protein-binding assay.

radioautogram or **radioautograph** or **autoradiogram** the pattern formed after development of an X-ray film that has been exposed to a paper, gel, or thin-layer chromatogram containing radioactive compounds.

radioautograph 1 an alternative term for **radioautogram**. 2 to prepare a radioautogram. —**radioautography** *n*.

radiobiology or **radiation biology** the study of the interaction between radiation and biological material.

radiocarbon any radioactive isotope of carbon, usually carbon-11 or carbon-14.

radiochemical purity *abbr.*: RCP; the proportion of the total **activity** (def. 2) that is present in the stated chemical form in a radioactive material.

radiochemistry the branch of chemistry concerned with the compounds of radioactive nuclides. *Compare radiation chemistry.*

radiochromatogram a chromatogram of substances some or all of which contain a radioactive isotope.

radiofrequency radiation electromagnetic radiation encompassing wavelengths in the region 1 mm (300 GHz) to 30 km (10 kHz) (frequencies in parentheses). This range includes microwave radiation and lies above that of infrared radiation.

radiography the use of X-rays to examine the internal structures of a body. An X-ray film is placed behind the body, through which X-rays are passed. After exposure, there are areas of the film in which development has not occurred due to absorption of the X-rays by structures of the body, thus yielding a pattern of these structures. —**radiographer** *n*.; **radio-graphic** *adj.*; **radiographically** *adv.*

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radioimmunoassay

radioimmunoassay *abbr.*: RIA; a highly sensitive method of assay of nonradioactive material. A known amount of antibody (Ab) directed against the substance (antigen, Ag) to be assayed is saturated with a mixture of Ag and radioactive Ag (Ag*), so that total Ag (Ag + Ag*) is in excess. When non-radioactive Ag is added to Ab along with Ag*, Ag and Ag* compete for binding to Ab, so that less Ag* will be found in the antibody-antigen complex as the ratio Ag:Ag* increases. If the antibody-antigen complex is then separated from free antigen, the total amount of antigen (the substance being assayed) can be calculated from the ratio Ab-Ag*:Ag* and the known **titre** of antibody. It is often convenient to attach the antigen to plates so that the amount of ligand bound to the plate is proportional to the amount of test antibody.

radioimmunosorbent test *abbr.*: RIST; a test used primarily for quantifying total serum immunoglobulin E (IgE) levels. Antibodies specific for IgE, covalently linked to an insoluble dextran, are incubated first with a known amount of radio-labelled IgE and then with the serum containing the unknown amount of IgE. From the decrease in the amount of radiolabel bound to the immobilized antibodies, due to competition by the unlabelled IgE in the serum, the total amount of IgE in the serum can be estimated.

radioiodinate to incorporate one or more atoms of **radioiodine** (usually iodine-125 or iodine-131) into a molecule.

radioiodine any radioactive isotope of **iodine**, notably iodine-125 (half-life 60 days) and iodine-131 (half-life 8 days).

radioisotope a radioactive isotope.

radiolabel 1 a radioactive label. 2 to incorporate a radiolabel.

radioligand any ligand containing a radioactive isotope; *see also* **Scatchard plot**.

radiology the study and use of radiation and radioactive substances, especially in the diagnosis and treatment of disease. It includes the use of diagnostic X-rays in **radiography**, the therapeutic use of ionizing radiation to treat disease (**radiotherapy**), and the experimental and diagnostic use of radioisotopes introduced into the body. —**radiological** *adj.*; **radiologically** *adv.*; **radiologist** *n.*

radiolysis the cleavage of chemical bonds by high-energy radiation. —**radiolytic** *adj.*; **radiolyse** *vb.*

radiometer 1 any instrument used to measure radiant energy. 2 **Crookes radiometer** a specific type of radiometer in which the different behaviour of a polished surface, which reflects energy, and a blackened surface, which absorbs it, is utilized to cause a rotor to turn at a rate related to the intensity of radiation. [After William Crookes (1832–1919), British chemist and physicist.] —**radiometric** *adj.*; **radiometry** *n.*

radiomimetic 1 describing any drug or other chemical substance, typically an alkylating agent, whose effects in living systems (e.g. carcinogenic, immunosuppressive, mutagenic) resemble those produced by **ionizing radiation**; more loosely, describing any such agent able to cause gene and chromosome mutations. 2 a radiomimetic (def. 1) substance or agent.

radionuclide a radioactive nuclide. —**radionuclidic** *adj.*

radionuclidic impurity any radionuclide, other than the nuclide that should be present, in a radioactive material.

radionuclidic purity or **radiopurity** (of a radioactive material) the proportion of the total **activity** (def. 2) that is in the form of the stated **radionuclide**.

radiotherapy *see* **radiology**.

radiotracer an alternative name for **radioactive tracer**.

radixin a protein of the erythrocyte band 4.1 family present in a variety of eukaryotic cells; it shares about 75% homology with human **ezrin**. It is an actin barbed-end capping protein, highly concentrated in the undercoat of the cell-to-cell **adherens junction** and the cleavage furrow in the interphase and mitotic phase. Example from human: database code RAD1_HUMAN, 583 amino acids (68.49 kDa).

radyl (*in chemical nomenclature*) a term used to indicate any acyl, alkyl, or alkenyl group where this cannot be specified in detail.

RAF any of a family of genes encoding Raf proteins; there are at least three Raf proteins, Raf-1, A-Raf, and B-Raf. These are protooncogene cytoplasmic serine/threonine protein kinases (EC 2.7.1.-) of the Ras/Mos family; they show similarity to members of the protein kinase C family in both their conserved C-terminal catalytic and regulatory N-terminal domains. Example, Raf-1 (human), which has binding domains for ATP, phorbol ester, and diacyl glycerol: database code KRAF_HUMAN, 648 amino acids (72.97 kDa). This protein is involved in transduction of mitogenic signals from the cell membrane to the nucleus. In the mouse, *raf* is the transforming gene of the mouse sarcoma virus, 3611, v-Raf having a truncated regulatory domain that causes constitutive activation of kinase activity. The c-Raf-1 protein kinase, a 74 kDa protein, has been found in all mammalian cells tested. Raf-1 is an important mediator of signals involving cell growth, transformation, and differentiation and is activated by a wide variety of extracellular stimuli. It has **MAP kinase** kinase activity. In many cell types, Raf functions immediately downstream of the G-protein Ras (*see* **RAS**) in the activation pathway; although Ras does not directly activate Raf, it mediates translocation of Raf-1 from cytoplasm to plasma membrane where it is activated. In some cells Raf may be phosphorylated and activated by members of the protein kinase C family. The first physiological substrate to be discovered for Raf kinase was a protein kinase (MEK) of yeast that is required for meiotic recombination and which on phosphorylation can activate **MAP kinase**.

raffinose or **gossypose** or **melitose** or **melitriose** the trisaccharide *O*- α -D-galactopyranosyl-(1 $\rightarrow$ 6)-*O*- α -D-glucopyranosyl-(1 $\rightarrow$ 2)- β -D-fructofuranoside; it occurs in plants almost as commonly as sucrose, being present, e.g., in cereal grains, cotton seeds, and many legumes. It is the first member of a series in which galactosyl residues are attached to sucrose, others being the tetrasaccharide **stachyose** and the pentasaccharide **verbascose**; these are all synthesized from sucrose by transfer reactions in which an α -D-galactopyranoside of *myo*-inositol (i.e. galactinol) is the donor of the galactosyl residues.

r.a.m. *abbr.* for relative atomic mass.

Ramachandran plot or **conformational map** a plot that shows the ranges of bond angles that are permissible and the main types of structure of an L-polypeptide chain (e.g. α helix, β -pleated sheet). As the peptide bond, $\text{—C}^{\alpha}\text{—NH—}$, is planar, a polypeptide chain has only two degrees of freedom per residue, the twist about the $\text{C}^{\alpha}\text{—N}$ bond axis, ϕ , and that about the $\text{C}^{\alpha}\text{—C}^{\beta}$ axis, ψ . Some foldings will be easily achieved; others will be impossible because they would bring neighbouring unbonded atoms or groups within van der Waals contact distances. The Ramachandran plot shows the permitted range within a plot of ϕ against ψ . [After G. N. Ramachandran (1922–).]

Raman, (Sir) Chandrasekhara Venkata (1888–1970), Indian physicist; Nobel Laureate in Physics (1930) 'for his work on the scattering of light and for the discovery of the effect named after him'.

Raman scattering the inelastic scattering of a photon by a molecule, producing in the scattered light weak radiation of frequencies not present in the incident light. The sample does not have an absorption band at the wavelength of the incident light (*compare* **fluorescence**) and the frequency differences between the weak Raman lines in the spectrum and the exciting line are characteristic of the scattering substance and independent of the frequency of the exciting line. [After C. V. Raman.]

Raman spectrum a plot of **Raman scattering** over a range of frequencies.

Ramón y Cajal, Santiago (1852–1934), Spanish anatomist and histologist; Nobel Laureate in Physiology or Medicine (1906) jointly with C. Golgi 'in recognition of their work on the structure of the nervous system'.

Ramsay, (Sir) William (1852–1916), British physical and inorganic chemist; Nobel Laureate in Chemistry (1904) 'in recognition of his discovery of the noble gases'.

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nitron of his services in the discovery of the inert gaseous elements in air, and his determination of their place in the periodic system'.

RAN a gene encoding a small nuclear G-protein, first identified as a well-conserved gene distantly related to *H-RAS*. Its product, Ran, interacts with **RCC1**, and the complex is involved in regulating cell-cycle progression and messenger RNA transport. In the absence of RCC1, GDP-Ran predominates and this results in activation of **M-phase-promoting factor** and chromatin condensation. The presence of RCC1 activates nucleotide exchange on Ran, increasing the amount of GTP-Ran and preventing premature chromatin condensation. Example from *Mus musculus*: database code RAN_MOUSE, 216 amino acids (24.42 kDa). See also **nuclear pore**.

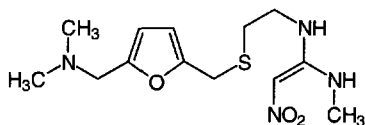
random coil the condition in solution of a linear polymer that exhibits little side group interaction, or resistance to rotation about the bonds in its chain. There are no preferred conformations. Quantitative expressions have been developed for the average radius of gyration. The term has formerly been incorrectly applied to various states of protein chains.

random copolymer a polymer of more than one species of monomer, in which the sequence of species within the polymer is not predetermined by controlled steps during synthesis, but depends only on random collisions between the reactant molecules. An example is the sequence of bases in RNA synthesized from nucleoside diphosphates by **polynucleotide phosphorylase**.

random mechanism or **random pathway** (for an enzymic reaction involving two substrates, $A + B \rightleftharpoons C$) a mechanism in which the enzyme, E, may first react with either substrate and the resulting enzyme-substrate complex then reacts with the second substrate, e.g. $A + E \rightleftharpoons EA$ then $EA + B \rightleftharpoons EAB$, or, $B + E \rightleftharpoons EB$ then $EB + A \rightleftharpoons EAB$, subsequently $EAB \rightleftharpoons E + C$. Compare **ordered mechanism**.

random sample a selection of part of a population made in such a way that no individual is favoured over any other individual. See also **sample**, **statistical sample**.

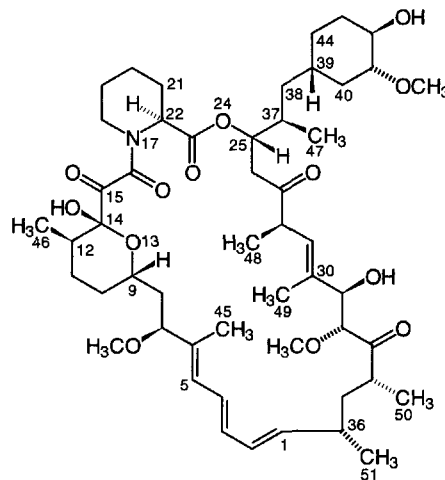
ranitidine a competitive antagonist at histamine H_2 receptors, being the major and pharmacologically more selective analogue of the imidazole derivative, **cimetidine**. It inhibits histamine- and pentagastrin-evoked gastric acid secretion and cardiac stimulation and also inhibits gastric secretion evoked by physiological stimuli, agonists at muscarinic cholinergic receptors, and gastrin. It is used in the treatment of gastric and duodenal ulcers. *One proprietary name*: Zantac (hydrochloride).



RANTES *abbr. for* regulated upon activation, normal T-cell expressed and secreted chemokine; *other names*: small inducible cytokine A5; a chemokine secreted by T cells and macrophages on stimulation by mitogens, that acts as a chemo-attractant, and stimulates basophils and eosinophils. Example from human (precursor): database code SISD_HUMAN, 91 amino acids (10.07 kDa); residues 1–23 are the signal, 24–91 RANTES.

ranunculeic acid another name for **columbinic acid**.

rapamycin an immunosuppressant (and antifungal agent) from *Streptomyces hygroscopicus*; it is used to reduce rejection of organ transplants. It is a lipid-soluble macrocyclic lactone, with the unusual feature of a cyclohexane ring with hydroxyl and methoxyl substituents. It competes with **FK506** for a common binding site on **FKBP**. See also **cyclosporin**.



rapamycin

RAPD PCR *abbr. for* random polymorphic DNA PCR; a variation of the **polymerase chain reaction** in which many short (≈ 10 nucleotide) primers are used. The products of the reaction from different sources can lead to recognition of polymorphisms and physical maps.

rape or **colza** a herbaceous plant, *Brassica napus*, of the mustard family, Cruciferae; it resembles a turnip, and the seeds yield valuable **rapeseed oil**.

rapeseed oil oil derived from the seed of **rape**; it is widely used as a lubricant and cooking fuel, and for the manufacture of animal foodstuffs, soap, and synthetic rubber. It contains **erucic acid**, which has cardiotoxic effects varying with species; rape varieties of low erucic acid content have recently been developed.

rapid-reaction kinetics the kinetic analysis of an enzyme reaction before the steady state is reached. The **stopped-flow method** is commonly used, in which two independent sources of substrate and enzyme are mixed immediately before flow through a spectrophotometer cell. An opposing syringe stops the flow immediately before activating the detection and recording system, such that reaction times of a few milliseconds can be achieved. Characteristically, reaction rates may be appreciably faster than during the steady state. In the **quenched flow** modification of the method, a further reagent is added to stop the reaction immediately before detection and recording. See also **relaxation kinetics**.

Rapoport-Lübering shunt a unique feature of erythrocyte glycolysis in which the glycolytic intermediate 3-phosphoglyceroyl phosphate is converted by bisphosphoglycerate mutase to 2,3-bisphosphoglycerate, which is in turn converted to 3-phosphoglycerate and inorganic phosphate by bisphosphoglycerate phosphatase, both reactions being catalysed by a single enzyme molecule. The shunt bypasses the near equilibrium phosphoglycerate kinase reaction, one of the two ATP-generating reactions of glycolysis. Since 2,3-bisphosphoglycerate is an important intracellular modulator of hemoglobin function, regulation of the Rapoport-Lübering shunt balances the energy needs and oxygen-transporting function of the red cell. [After Samuel Mitja Rapoport (1912–) and J. Lübering.]

rare earth (*strictly*) an oxide of a **rare earth metal**, or (*more loosely*), a rare earth metal itself.

rare earth metal or **rare earth element** scandium, yttrium, or a lanthanoid.

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RAS an oncogene (*v-ras*) first discovered in the retroviruses Harvey and Kirsten rat sarcoma viruses; three closely related (85% homology) oncogenes are now known, *Ha-ras*, *Ki-ras*, and *N-ras*. Their cellular counterparts encode 21 kDa guanine nucleotide-binding proteins that play a pivotal role in regulating growth and differentiation in nearly every eukaryotic cell studied. There are two subfamilies, the Rho and Rac proteins (involved in relaying signals from the cell-surface receptors to the actin cytoskeleton) and the Rab family (involved in regulating the traffic of intracellular transport vesicles). Ras proteins undergo a number of modifications of the N and C termini that result in their localization to the plasma membrane. Ras binds both GTP and GDP with high affinity. It is considered active in the GTP-bound form and inactive in the GDP-bound form. The intrinsic rate of nucleotide exchange and GTPase activity is very slow. Ras-GDP predominates in resting cells, but when Ras is activated by occupied receptors, specific guanine nucleotide exchange factors (GEFs), e.g. **Sos**, enhance nucleotide exchange leading to Ras-GTP, GTP exceeding GDP in cytoplasm. Ras-GTP then activates downstream effector proteins such as Raf-1 (see **RAF**). Activated GTPase activating protein (GAP) causes the formation of inactive Ras-GDP. The retroviral oncogene, *v-ras*, encodes a protein that differs from the *c-ras* product as a result of a point mutation. If this affects amino acids 12, 13, 59, 61, or 63 the protein fails to hydrolyse its bound GTP and so persists abnormally in its active state, constitutively transmitting an intracellular signal for cell proliferation. Alternatively, some transforming mutations increase the rate at which Ras can exchange guanine nucleotides, e.g. when amino acids 116, 117, 119, or 146 are affected, again increasing the amount of Ras-GTP. Activated *ras* oncogenes have been found in up to 30% of all human tumours analysed, with the most common point mutations being at codons for amino acids 12 and 13. In some types of malignancy, the figure is nearly 100%. Two genes corresponding to *H-ras* occur in yeast; they encode **G-proteins** that inhibit **adenylate cyclase**. 3-D structure known (including GDP bound to guanine nucleotide-binding domain) for human protein H-Ras-1: database code NRL_4Q21, 168 amino acids (19.07 kDa).

RAST *abbr.* for radioallergosorbent test.

rate coefficient *see* **rate constant**.

rate constant or **rate coefficient** *symbol:* k ; a constant relating the concentration of a reactive species to the velocity of a reaction; thus for a first order reaction (see **order of reaction**) such as: $EA \rightarrow E + A$, the rate of reaction, v , is proportional to the concentration of the reactant EA, so $v = k[EA]$, where k is the first-order rate constant. For a second order reaction, e.g. $E + A \rightarrow EA$, the rate is proportional to the product of the reactant concentrations, i.e. $v = k[E][A]$, where k is the **second-order rate constant**.

rate-determining step *see* **rate-limiting step**.

rate-limiting step the slowest step in a reaction sequence; i.e. the step with the smallest rate constant.

rate of consumption *see* **rate of reaction**.

rate of conversion *symbol:* ξ ; for a specified chemical reaction, the rate of increase of the extent of reaction at any moment in time, regardless of the conditions under which the reaction is carried out: $\xi = d\xi/dt$, where ξ is the extent of reaction and t is time. In the case of a reaction with one-to-one stoichiometry, such as a typical simple enzyme-catalysed reaction, the extent of reaction is equal to the amount of a reactant consumed or the amount of a product formed. Rate of conversion is an extensive quantity (see **extensive property**), its dimensions are (amount of substance) divided by time, and it is expressed in mol s^{-1} (compare **rate of reaction**). It is therefore independent of the volume of the system. Where this volume remains constant, as during a typical enzyme-catalysed reaction, the rate of conversion is equal to the rate of reaction multiplied by the volume: $\xi = vV$. The term was formerly known in physical chemistry as **rate of reaction**. *See also* **catalytic activity**.

rate of formation *see* **rate of reaction**.

rate of reaction or **velocity of reaction** or (sometimes) **reaction rate** *1 symbol:* v ; for a specified enzyme-catalysed reaction of a typical kind (i.e. one involving a single substrate, S, a single product, P, one-to-one stoichiometry between them, and constant-volume conditions), the **rate of consumption** of substrate or the **rate of formation** of product at any moment in time is given by:

$$v = -d[S]/dt = d[P]/dt,$$

where $[S]$ and $[P]$ are respectively the concentrations of substrate and product, and t is time. A similar set of relationships applies where there are several reactants in one-to-one stoichiometry, but if the overall stoichiometry is not one-to-one – a situation rarely arising in enzyme kinetics – the relationships are more complicated. Where intermediates are formed in amounts comparable to those of the reactants, there is time-dependent stoichiometry and a rate of reaction for the complete reaction cannot be specified. Rate of reaction is an intensive quantity (see **intensive property**), its dimensions are (amount-of-substance concentration) divided by time, and it is expressed in $\text{mol L}^{-1} \text{s}^{-1}$ (compare **rate of conversion**). *See also* **initial velocity**, **limiting rate**. *2 (formerly, in physical chemistry)* the extensive quantity now termed **rate of conversion**.

rate theory of agonist action a theory of agonist action stating that the response to an agonist is proportional to the number of receptor-agonist associations per unit time, i.e. to the rate of receptor occupation, the binding of agonist to receptor being reversible. The response is therefore maximal at time zero and falls exponentially to an equilibrium value.

Rayleigh *see* **Strutt**.

Rayleigh ratio *symbol:* R ; the ratio of the intensity of light scattered by particles (molecules) in solution at an angle θ from the incident beam, I_θ , multiplied by the square of the distance, r , from the particle to the detector, divided by the intensity of the incident light, I_0 , i.e. $R = I_\theta r^2 / I_0$. The **reduced Rayleigh ratio**, $R_\theta = I_\theta r^2 / I_0 (1 + \cos^2 \theta)$. Many literature references do not distinguish between these two. *See also* **Rayleigh scattering**. [After J. W. Strutt, Lord Rayleigh.]

Rayleigh scattering the elastic scattering of photons of light by molecules or atoms of the substance through which the light is passing. When light passes through a liquid or gas, atoms or molecules are polarized by the electric vector of the light, thus inducing a rapidly fluctuating dipole in the atoms or molecules in the light path. The fluctuating dipole leads to the emission of electromagnetic waves in various directions of the same frequency as the incident radiation, and this emission is seen as scattered light. *See also* **Rayleigh ratio**. [After J. W. Strutt, Lord Rayleigh.]

Rb *symbol* for rubidium.

RB a human **tumour suppressor gene** that encodes a nuclear phosphoprotein, **retinoblastoma protein**. Inactivation or loss of **RB** on both copies of the chromosome (13q14) leads to **retinoblastoma**.

R band *see* **Giemsa's stain**.

RBC or **rbc** or **r.b.c.** *abbr.* for red blood cell. *See* **erythrocyte**.

Rbe *abbr.* for relative biological effectiveness (i.e. **quality factor**).

RBS *abbr.* for René Borghgraef solution; the *proprietary name* for any of a range of general-purpose surfactants and decontaminating agents, especially for laboratory use, that have combined detergent and chelating properties. They are identified with a number or other designation, e.g. RBS 25, RBS 35, RBS solid.

RCC1 a chromatin-bound guanine nucleotide release (or exchanging) factor. Its function is required for the normal coupling of the completion of DNA synthesis and the initiation of mitosis, and in its absence mitosis initiates before DNA replication is complete. RCC1 therefore appears to be involved in the regulation of this process. It is also necessary for RNA export from the nucleus. It binds to and acts on the nuclear G-protein, **Ran**. Example from the yeast *Candida albicans*:



database code RCC_CANAL, 464 amino acids (51.27 kDa). [From regulator of chromosome condensation.]

RCF *abbr.* for relative centrifugal force.

R chromophore a **chromophore** that gives rise to low-intensity absorption bands at long wavelengths. *Compare* **K chromophore**. [From German *Radikal*.]

RCP *abbr.* for radiochemical purity.

RCS *abbr.* for rabbit aorta-contracting substance.

rd *symbol* for rad (def. 2) (alternative to rad).

Rd *abbr.* for rubredoxin.

rDNA **1** *abbr.* for ribosomal DNA. **2** (*sometimes*) *abbr.* for recombinant DNA; use not recommended because of possible confusion with ribosomal DNA.

Re- *prefix* (in chemical nomenclature) denoting *rectus* (i.e. right, or clockwise) configuration of a face of a trigonal atom. *See* **Re/Si convention**.

Re *symbol* for rhenium.

Re *symbol* for Reynolds number.

RE *abbr.* for (eukaryotic) response element.

reactancy of enzymes the number of reactants involved in an enzyme-catalysed reaction in a given direction. The numbers of kinetically important substrates or products in an enzyme mechanism are designated by Uni (1), Bi (2), Ter (3), or Quad (4). Thus, e.g., Uni Uni indicates one substrate and one product, and Ter Bi indicates three substrates and two products.

reactant any material participating in a chemical reaction in a given direction.

reaction (in chemistry) any process in which molecules interact with one another, leading to chemical or physical change.

reaction centre *see* **photosystem**. *Compare* **photosynthetic reaction centre**.

reaction intermediate a transient chemical species, with a lifetime appreciably longer than a molecular vibration time, formed directly or indirectly from the reactants of a chemical (often enzymic) reaction, and that further reacts (directly or indirectly) to give the reaction products.

reaction rate (*sometimes*) an alternative term for **rate of reaction**.

reaction time an alternative name for **latent period** (def. 1).

reactivate to restore activity to any enzymic or cellular system.

reactive describing any molecular entity or centre that is capable of reacting readily. Thus, a **reactive centre** is the point in the molecule at which electron configurations favour a reaction.

reactive oxygen a highly active oxygen species, such as **superoxide ion**.

reactivity the degree to which a molecule or centre is reactive.

reading (in molecular biology) the process of one-way linear decoding of a nucleotide sequence during, e.g., **transcription** or **translation**.

reading frame any sequence of bases in DNA that codes for the synthesis of either a protein or a component polypeptide. The point of initiation of reading determines the frame, i.e. the way in which the bases will be grouped in triplets as required by the genetic code. For example, the sequence

–ATGGTCA–

could be grouped

–A,TGG,TCA–

or

–AT,GGT,CA–

or

–ATG,GTC,A–

depending on the point at which the reading frame begins. *See also* **open reading frame**.

readthrough the continuation of DNA transcription beyond a normal termination sequence, due to a failure of RNA polymerase to recognize the termination signal. Readthrough may also occur in translation, where a mutation has caused a ter-

mination codon to be converted to a codon for an amino acid (a sense codon); this is the underlying etiology of several **hemoglobinopathies**. *See also* **readthrough protein**.

readthrough protein a product formed by translation of a messenger RNA when one of its termination codons is misread or is recognized by a **suppressor tRNA**. Polypeptide-chain synthesis then continues until the next termination codon is reached, resulting in the production of a variant protein with a C-terminal extension.

reagent **1** any chemical substance that reacts or participates in or is necessary for a reaction. **2** a substance used for the detection or determination of another substance by chemical or microscopical means, usually in solution; e.g. the **Folin–Ciocalteu reagent**.

reagent grade a level of purity of a chemical; chemicals of this quality conform to the specifications established by the Committee on Analytical Reagents of the American Chemical Society and may often be sufficiently pure to use as primary standards in an analytical laboratory, with information concerning the actual percentage of impurities or at least the maximum limit of impurities being given on the label.

reagin **1** a term originally used to describe a heat-labile antibody-like substance in the serum of many people suffering from atopic allergies and that has now been identified as immunoglobulin E (IgE). The **immediate hypersensitivity** (type I) generated by the binding of IgE to its specific antigen on the surface of mast cells involves the release of cell mediators via the reaginic or anaphylactic mechanism. **2** a special kind of antibody response produced by syphilitics that has specificity for **cardiolipin** antigens.

real-time (in computing) indicating that a computer monitors a process continuously as it occurs, in distinction to the analysis of recorded data.

rearrangase or **S–S rearrangase** an alternative name for **protein disulfide isomerase**.

rearrangement reaction a reaction in which molecular bonds are rearranged without the loss of any atom from the molecule, e.g. as in the conversion of glucose 6-phosphate to fructose 6-phosphate.

recA a gene in *Escherichia coli* encoding a protein, RecA, involved in **genetic recombination** and in the SOS response for DNA repair (*see* **SOS repair**). The protein has two activities: (1) a DNA-dependent ATPase in the presence of single-stranded DNA that catalyses the ATP-dependent uptake of single-stranded DNA to form the heteroduplex required in recombination. (2) In the presence of DNA fragments it becomes a protease (RecA*; EC 3.4.99.37) that cleaves a specific Ala–Gly bond in certain repressor molecules, including its own repressor (*see* **lexA**). This allows all the SOS genes to be activated. Example from *E. coli* and *Shigella flexneri* (identical sequences): database code RECA_ECOLI, 352 amino acids (37.80 kDa); 3-D structure known; eight motifs. *See also* **exodeoxyribonuclease V**, **recombinational repair**, **recombinase**.

recalcitrance or **persistence** the ability of a substance to remain in a particular environment in an unchanged form.

recBC a gene in *Escherichia coli* for **exodeoxynuclease V**, an ATP-dependent nuclease specific for double-stranded DNA; it mediates **genetic recombination**.

recDNA (*sometimes*) *abbr.* for recombinant DNA.

receptor any cellular macromolecule that undergoes combination with a hormone, neurotransmitter, drug, or intracellular messenger to initiate a change in cell function. Receptors are concerned directly and specifically in chemical signalling between and within cells. Cell-surface receptors, e.g. the **acetylcholine receptor** and the **insulin receptor**, are located in the plasma membrane, with their ligand-binding site exposed to the external medium. Intracellular receptors, e.g. steroid-hormone receptors, bind ligands that enter the cell across the plasma membrane.

receptor assay **1** the determination of receptor numbers and the dissociation constant of receptor–ligand binding. The

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receptor cross-talk

binding at various concentrations of ligand is measured, often by use of a radioactive ligand, and is analysed by means of a **Scatchard plot**. **2** the determination of the concentration of a ligand by utilizing a purified receptor in a manner similar to that used in **radioimmunoassay**; in this case the receptor, rather than an antibody, performs the role of specific binding agent.

receptor cross-talk interaction between two receptors such that one receptor, when activated, modulates the action of the other. This type of interaction often involves the production by the first receptor of a **second messenger**, which then modulates the action of the second receptor.

receptor-destroying enzyme an enzyme that brings about chemical change in a (usually membrane-bound) receptor, leading to inactivation of the receptor. Knowledge of the specificity of the enzyme provides information about the nature of the groups or linkages exposed to enzyme attack, and thus to receptor structure or orientation in the membrane. Examples are **sialidase** and **trypsin**.

receptor-effector complex the complex that results when an occupied receptor binds to another protein (the **effector**), the activation of which by the occupied receptor leads to signal transduction.

receptor-mediated describing a response that requires the intermediary action of a receptor, especially one located in a membrane.

receptor-mediated endocytosis a form of **endocytosis** involving the internalization (i.e. passage into the cell interior) of stretches of membrane into which receptors have clustered after binding of ligand. The cytoplasmic surfaces of these stretches of membrane are coated with **clathrin** and are termed **coated pits**. See also **endosome**.

receptor transformation see **transformation** (def. 3).

recessive describing an **allele** that lacks effect when accompanied in the same diploid by a **dominant** form of the same gene. Most (not all) defective mutant genes are recessive to their normal functional counterparts. —**recessivity** *n*.

recipient bacterium see **bacterial conjugation**.

reciprocal **1** expressed in the reverse direction, in response to an initial application of any force or stimulus. **2** (in *mathematics*) the inverse value, obtained when the values of numerator and denominator are reversed; the reciprocal of *x* is obtained by evaluating $1/x$.

reciprocal plot a plot in which the units of both the ordinate and the abscissa are reciprocals of the values that have been determined. Compare **Lineweaver-Burk plot**.

recircularization or **recircularisation** the restoration of circularity to a plasmid vector following the insertion of recombinant DNA. The plasmid used for gene cloning is a small circular molecule. This is cut with a **restriction endonuclease** to create linear DNA molecules. The foreign DNA fragment with the same cohesive ends is then annealed to the cut plasmid to form a recombinant DNA circle. The DNA molecule containing the foreign DNA insert is then covalently sealed with **DNA ligase**. Recircularization also occurs during the replication of some DNA phages by a **rolling circle** replication mechanism.

recognition any specific binding interaction between molecules. Such interaction between two molecules is based on the existence of **recognition sites**, one (or more) on each of the interacting molecules, that are tailored to fit in such a way that highly specific binding occurs; i.e. the two molecules uniquely 'recognize' each other. See also **chiral recognition**, **molecular recognition**.

recognition sequence the part of the sequence of a macromolecule that is able to recognize and bind to a particular sequence in another macromolecule.

recognition site **1** (in *proteins*) a site for **recognition** of another protein, involving a specific sequence of amino acids; the latter may be in a single chain (e.g. target sites for protein phosphorylation), or may involve amino acids in separate chains brought into juxtaposition by quaternary structure. **2** (in *nucleic acids*) a site for recognition of a protein, involving

ing a specific sequences of bases; see **restriction endonuclease**. **3** (in *pharmacology*) a region of a receptor macromolecule to which an endogenous agonist binds.

recombinant clone a population of bacteria or other host derived from a single cell into which a vector containing **recombinant DNA** has been introduced.

recombinant DNA a fragment of deoxyribonucleic acid (DNA) that has been inserted into a cloning vector, thereby leading to its use in the isolation of a clone of cells characterized by the presence of the fragment. The term is derived from the concept that insertion into the vector is a form of **genetic recombination**. [Note: The abbreviation rDNA has sometimes been applied to recombinant DNA, but this use is discouraged (rDNA having been pre-empted for ribosomal DNA); while alternatives such as recDNA or rtDNA have been suggested, IUBMB considers a standard abbreviation unnecessary.] See also **recombinant DNA technology**.

recombinant DNA technology an area of biotechnology concerned with the manipulation of **recombinant DNA**. It has many important applications, including (1) DNA sequencing, which may lead to the ability to predict the primary structure of a protein where this is the product of a cloned gene, (2) the synthesis of recombinant protein by a suitable expression system, and (3) the production of **DNA probes** (def. 3) for use in hybridization techniques. The technology has led, e.g., to the production of human insulin for the treatment of diabetics, human growth hormone for administration to children with growth defects, and a vaccine for hepatitis B. DNA probes are essential for the detection of aberrant genes which are the cause of numerous diseases and the technique has the potential for gene therapy. See also **genetic engineering**.

recombinant protein a protein coded for by a gene – **recombinant DNA** – that has been cloned in a system that supports expression of the gene and translation of messenger RNA (see **expression system**). Modification of the gene by recombinant DNA technology can lead to expression of a mutant protein. Proteins coexpressed in bacteria will not possess post-translational modification, e.g. phosphorylation or glycosylation. Eukaryotic expression systems are needed for this; see, e.g., **baculovirus**.

recombinase **1** an enzyme that recognizes specific sites involved in the joining of V-region gene segments in L- and H-chain gene segment pools in the generation of antibody. **2** RecA protein (see **recA**) is also a recombinase, the activity of which is increased in *Escherichia coli* following damage to DNA. RecA binds to a region of single-stranded DNA and catalyses the formation of duplex DNA; hydrolysis of ATP is involved. A recombinase recognizes and binds to each of two recombinant sites on different DNA molecules or within the same DNA. One DNA strand in each site is cleaved at a specific point within the site, and the recombinase becomes covalently linked to the DNA at the cleavage site through a phosphotyrosine (sometimes a phosphoserine) bond. The transient protein-DNA linkage preserves the phosphodiester bond lost in cleaving the DNA; cofactors such as ATP are unnecessary in subsequent steps. The cleaved DNA strands are rejoined to new partners, with new phosphodiester bonds created at the expense of the protein-DNA linkage. The result of the initial breakage and rejoining process is the **Holliday junction**. To complete the reaction, the process must be repeated at a second point within each of the two recombination sites. In summary the recombinase can be viewed as a site-specific endonuclease and ligase in one package.

recombination see **genetic recombination**, **recombinant DNA**, **immunoglobulin synthesis**.

recombinational heteroduplex see **heteroduplex**.

recombinational repair a mechanism to restore the integrity of DNA when replication of DNA stops at a **thymine dimer** but resumes after the dimer is passed, creating a gap. The undamaged parental strand recombines into the gap, a process promoted by RecA protein (see **recA**). The new gap in the parental



recoverin

strand is filled by the action of **DNA polymerase** and **DNA ligase**. The thymine dimer itself is not repaired in this process but time is allowed for the excision system to repair this damage later.

recoverin an EF-hand Ca^{2+} -binding protein, apparent molecular mass 26 kDa; *also known as cancer-associated retinopathy protein*. It selectively localizes to the retina and pineal gland and is implicated in the activation of retinal rod guanylate cyclase by rhodopsin, possibly being involved in the blocking of the phosphorylation of rhodopsin, by forming an inhibitory complex with rhodopsin kinase. It has been identified as an autoantigen in cancer-associated retinopathy, a putative autoimmune process. Example from human: database code RECO_HUMAN, 199 amino acids (22.97 kDa).

recovery the portion of material or activity that remains after any extraction or purification, usually expressed as a percentage of the amount that was present as starting material.

recruitment the action by any cell, cell constituent, or molecule that brings about the cooperation of any other component in performance of any function.

recrystallize or **recrystallise** to dissolve a crystalline substance and subsequently crystallize it from the solution, usually to increase the purity of the substance. —**recrystallization** or **recrystallisation** *n*.

rectangular hyperbola a **hyperbola** whose asymptotes are at right angles. It is sometimes said that, for enzymes that follow so-called **Michaelis kinetics**, the plot of initial rate against substrate concentration has the shape of the curve of a rectangular hyperbola. It is true that the Michaelis–Menten equation does yield a curve that is part of a rectangular hyperbola, but the axes of the hyperbola must be rotated through 45° , and displaced, if they are to coincide with the axes of the Michaelis–Menten plot.

rectification **1** (*in physics*) the process of obtaining a direct current from an alternating electrical supply. **2** (*in physiology*) the regulation of membrane conductance by the voltage across the membrane; strong rectification implies that ionic channels carrying current are open at some membrane potentials and shut at others. Conductance is low at very negative membrane potentials, but increases as the potential becomes less negative. **3** (*in chemistry*) the process of purifying a liquid by distillation.

rectus *Latin* right; used to describe the right, or clockwise, configuration of one of the two faces of a compound containing a trigonal carbon atom. It is designated by the prefix *Re-*. Compare *sinister*. See also *Re/Si* convention.

recycle to pass again through a system one or more times after return to a starting condition or state.

red drop a phenomenon in photosynthesis, in which the quantum yield is relatively constant in light of wavelengths between 500 and 600 nm, but drops dramatically at longer wavelengths.

redistil or (*US*) **redistill** to distil again that which has already been distilled, usually as an aid to purification of liquids.

red marrow see **bone marrow**.

red muscle dark, skeletal, voluntary muscle that is relatively rich in myoglobin and cytochromes, and derives energy for contraction from a high oxidative capacity, in contrast to **white muscle**, which has a high glycolytic capacity. The distinction between muscle types should more strictly be based on muscle-fibre types. Type I fibres have a high oxidative capacity based on a high activity of tricarboxylic-acid cycle and respiratory chain enzymes. Type IIA fibres have high oxidative and high glycolytic capacity. Type IIB fibres have low oxidative and high glycolytic capacity. The proportions of these fibres vary in different muscle types; accordingly the muscle may have the ability to engage in very rapid but quickly exhausted 'fast twitch' contractions characteristic of white muscle, or less powerful but more sustained contractions typical of red muscle.

redox *abbr.* for reduction–oxidation; see **oxidation–reduction**.

redox balance the ratio of the number of reductions to the number of oxidations in a given metabolic pathway. Compare **redox state**.

redox couple the oxidized (electron acceptor) and the reduced (electron donor) forms of the species of a given **redox reaction**; e.g. NAD^+ and $(\text{NADH} + \text{H}^+)$. The potential of any redox couple in solution can be measured against a reference electrode, and a table of relative potentials constructed. See **redox potential**.

redox hypothesis a now discounted model for the **Pasteur effect** in glycolysing muscle extracts. Addition of redox indicators that changed the redox state in the direction of oxidation decreased the rate of lactate formation; it was therefore suggested that the effect of oxygen might be connected with a change of the redox state in the same direction.

redox indicator see **indicator** (def. 3).

redox potential or **oxidation–reduction potential** *symbol:* E_0 ; for a particular **redox couple** at a given pH and temperature, the potential of an electrode (usually bright platinum) immersed in a solution containing 1 M oxidant and 1 M reductant, relative to a standard **hydrogen electrode**. The more 'reducing' the couple, the more negative is E_0 ; the convention in biochemistry is that the more 'oxidizing', the more positive is E_0 . Some standard potentials, measured against the hydrogen electrode, are: $\text{NAD}^+/\text{NADH} + \text{H}^+$, -0.32 V; fumarate/succinate, $+0.032$ V; cytochrome *b*(Fe^{3+})/cytochrome *b*(Fe^{2+}), $+0.22$ V; cytochrome *c*(Fe^{3+})/cytochrome *c*(Fe^{2+}), $+0.25$ V; $\text{O}_2 + 2\text{H}^+/\text{H}_2$, $+0.82$ V.

redox reaction a chemical reaction in which the oxidation of one reactant is coupled to the reduction of a second reactant, electrons being transferred from the reactant being oxidized to that being reduced.

redox state the ratio of the concentration of the oxidized member of a **redox couple** to the concentration of the reduced member; e.g. $[\text{NAD}^+]/[\text{NADH}]$ or $[\text{Fe}^{3+}]/[\text{Fe}^{2+}]$. The term should only be applied to a particular redox system and not to a heterogeneous system.

red shift any shift of the peaks of a spectrum to longer wavelengths (i.e. towards the red end of the spectrum), including a shift from the red to infrared. Compare **blue shift**.

red tide a bloom of the dinoflagellate *Ptychodiscus brevis* (*Gymnodinium breve*), an organism that produces **brevetoxins** and causes epidemics of food poisoning.

reduce to cause or undergo **reduction**.

reduced Rayleigh ratio see **Rayleigh ratio**.

reducing agent any substance acting as a **reductant**. Compare **oxidizing agent**.

reducing end the end of an oligo- or polysaccharide molecule that carries a potential hemiacetal or hemiketal (reducing) group.

reducing sugar any **sugar** that will reduce heated alkaline solutions of cupric salts, such as **Fehling's** or **Benedict's solutions**, due to the presence of a hemiacetal or hemiketal group.

reductant the chemical species that donates an electron in an **oxidation–reduction reaction**; i.e. the species that undergoes **oxidation** in such a reaction. Compare **oxidant**.

reductase a donor:acceptor oxidoreductase enzyme (EC class 1) when named after the more oxidized alternative substrate and when O_2 is not a substrate. See **oxidoreductase**.

reduction the chemical process by which oxygen is withdrawn from, hydrogen is added to, or (more generally) an electron is added to (with or without addition of a proton) a molecular entity. Compare **oxidation**. —**reductive** *adj.*; **reduce** *vb*.

reduction division the first division of **meiosis**.

reduction potential the electrode potential that measures the tendency of one half of a **redox reaction** to proceed so as to gain electrons. See **oxidation potential**, **redox potential**.

reductive alkylation the reaction of amino groups with an alkyl carbonyl group, followed by reduction of the resulting **Schiff base** with a reducing agent such as sodium borohydride. See **reductive amination**.

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reductive amination

reductive amination the process whereby ammonia or a primary or secondary amine reacts with ketones and aldehydes in the presence of reducing agents such as H_2 and a metal catalyst, or more conveniently, $NaBH_4$ or its derivative $NaBH_3CN$, to produce primary, secondary, or tertiary amines. In a formal sense, the amine is alkylated in this reaction while the carbonyl group is reduced, hence the term reductive amination, or alternatively **reductive alkylation**. Amino acids have been made from oxo acids by this process. A variation of reductive amination, termed the **Eschweiler-Clarke reaction**, involves the methylation of an amine to the tertiary amine stage using formic acid as the reducing agent and formaldehyde as the methylating reagent.

reductive carboxylic-acid cycle or **reductive tricarboxylic-acid cycle** a metabolic cycle used by members of the Chlorobiaceae for CO_2 fixation. The usual operation of the tricarboxylic-acid cycle is reversed, with three enzyme replacements, and other supplementary reactions are used. Initially one CO_2 reacts with succinyl-CoA to form oxoglutarate; oxoglutarate synthase (EC 1.2.7.3) replaces the usual **oxoglutarate dehydrogenase complex**. A second CO_2 converts oxoglutarate to isocitrate with use of **isocitrate dehydrogenase**, and isocitrate proceeds to citrate. Citrate is then cleaved by ATP citrate (*pro-S*) lyase to acetyl-CoA and oxaloacetate; the latter recycles to succinyl-CoA with fumarate reductase (NADH) replacing succinate dehydrogenase. These operations reverse the usual tricarboxylic-acid cycle and convert two CO_2 to acetyl-CoA. A net gain of oxaloacetate is possible by conversion of acetyl CoA and one CO_2 to pyruvate (pyruvate synthase), pyruvate to phosphoenolpyruvate (phosphoenolpyruvate synthase) and **phosphoenolpyruvate** and one CO_2 to oxaloacetate (**phosphoenolpyruvate carboxylase**). The overall result is that four CO_2 yield a net gain of the four-carbon oxaloacetate.

reductive methylation a specific form of **reductive alkylation**, in which the alkyl group is methyl. Formaldehyde is often used as a reagent in this case.

reductive pentose-phosphate cycle or **Calvin cycle** other names: carbon reduction cycle; C_3 cycle/pathway; more appropriately, Calvin-Bassham cycle; the cyclic set of reactions, occurring in chloroplasts in the majority of higher plants (**C_3 plants**), that results in the fixation of CO_2 as glucose using the ATP and the NADPH formed in the light reactions of photosynthesis. Briefly, CO_2 reacts with ribulose 1,5-bisphosphate under the action of **ribulose-bisphosphate carboxylase** (Rubisco), EC 4.1.1.39, to form two molecules of 3-phosphoglycerate; these are phosphorylated (by ATP) to 1,3-bisphosphoglycerate, which in turn is reduced (by NADPH) to glyceraldehyde 3-phosphate. This is then converted by aldolase, transketolase, and other enzymes to fructose 6-phosphate and ribulose 5-phosphate; the latter being phosphorylated (ATP) to ribulose 1,5-bisphosphate, completing the cycle. See also **Hatch-Slack pathway**.

reductone any reducing substance with a chemical structure containing the stabilized enediol $-C(OH)=C(OH)-$. Reductones are generated when monosaccharides are heated for prolonged periods with strong alkali, whereby the enols originally produced are broken down to strongly reducing fragments. Another notable example is ascorbic acid, and in some contexts reductones are described as '**apparent ascorbate**', i.e. the reductive acids that are present in certain processed foods and react similarly to ascorbate, interfering with its determination.

Re-face (of an atom or molecule) see **Re/Si convention**.

reference something that acts as a standard in relation to quality, quantity, or type against which other values or characteristics can be compared; e.g. reference spectrum.

reference electrode any electrode, selected by convention, against which the electronegativity of compounds is measured. The standard biochemical reference is the **hydrogen electrode**, normally a platinum electrode in hydrochloric acid through which hydrogen gas is passed. A cell is constructed such that another electrode is immersed in a test solution and the poten-

tial difference between these two half-cells is measured. For convenience, however, the reference electrode is often a **calomel electrode**, and after measurements have been made against this, the electronegativity against the hydrogen electrode can be calculated from the known difference between this and the hydrogen electrode.

reflect to change the path of electromagnetic radiation, when it meets a surface, such that it returns partially or completely in a direction on the same side of the surface as the source.

reflectometer see **densitometer** (def. 1).

reflex (in physiology) an automatic or involuntary response of the nervous system to a stimulus. A reflex may be a monosynaptic spinal reflex (e.g. a simple motor response to a sensory stimulus), involving transmission through the spinal cord only, or polysynaptic and involving synaptic transmission in the brain. A **conditioned reflex** is one that is learned through repeated association of a stimulus with either rewarding (e.g. food) or aversive (e.g. punishment) events.

refolding (for proteins) the restoration of three-dimensional structure in a manner that may or may not achieve a full return to the native state following **denaturation**.

refraction the change in direction undergone by a wave of electromagnetic radiation on passing through a boundary between two media in which its velocity differs. **Snell's law** states that $n_1 \sin \theta_1 = n_2 \sin \theta_2$, where n_1 and n_2 are the **refractive indices** of the two media, and θ_1 and θ_2 are the angles between the radiation and the normal to the boundary in the corresponding media. [After Willebrord Snell (1591–1696), Dutch astronomer and mathematician.]

refractive index abbr.: RI; symbol: n ; for a given medium, a measure of the degree of **refraction** induced by the medium when light or other electromagnetic radiation passes through it. The refractive index of a medium is the ratio of the speed of light in a vacuum, c_0 , to the speed of light in the medium, c ; i.e. $n = c_0/c$. For many practical purposes the speed of light in air may be used instead of that in a vacuum.

refractivity a measure of the extent to which a medium will cause the deviation of a ray of light or other electromagnetic radiation entering its surface. If the **refractive index** of a medium is n , its refractivity is $(n - 1)$. The **specific refractivity** is given by $(n - 1)/\rho$, where ρ is the density of the medium. The **molecular refractivity** is the specific refractivity multiplied by the relative molecular mass.

refractometer an instrument for measuring **refractive index**.

refractory 1 (of a clinical condition) resistant to treatment. 2 (of a cell or tissue) resistant to stimulation, as by a drug, hormone, or nerve impulse, especially during a period following previous stimulation. See also **tachyphylaxis**. 3 (of a substance or material) resistant to heat, corrosion, mechanical deformation, or other aggressive agents causing alteration. 4 a substance or material having such resistance.

Refsum's disease a rare autosomal recessive inherited disease with neurological symptoms, including retinitis pigmentosa, peripheral neuropathy, cerebellar ataxia, and nerve deafness. It is caused by a defect in **alpha oxidation** of fatty acids, which is important for the oxidation of **phytanic acid**. The defect results in the accumulation of this acid in tissues and serum of patients. The symptoms are reduced by restriction of dietary dairy products and meat from ruminants, which contain phytanic acid.

regeneration 1 (in chemistry) the process of restoring to a state of preparedness for use, as in the case of an ion-exchange resin washed with acid or alkali to restore it to the H^+ or OH^- form. 2 (in biology) the growth of new tissues or organs to replace ones lost or damaged, e.g. through injury or surgery. In some tissues, such as liver, epithelial tissue, or pancreas, regeneration occurs by simple duplication of cells, with the **regeneration time** varying from one week to a year. In other systems, such as epidermis, renewal is from specialized **stem cells**. Regenerating liver can be used as a model in which systems involved in proliferation are more highly activated than



regeneration time

in normal liver. It is a unique property of liver that it is possible to remove two thirds of the organ yet it regenerates to form an organ of similar total weight to the original, albeit with somewhat different organization of the lobes. Many plants can regenerate a complete plant from a shoot segment or a single leaf or from tissue culture cells.

regeneration time the time taken for a tissue or organ to regain its former size or weight following damage or removal. For example, if two-thirds of the liver of a rat is removed surgically the liver regains its former weight in about two weeks. The proliferation of cells can be measured by the use of [^3H]thymidine to label the cells synthesizing DNA.

regio+ *comb. form* denoting, or relating to, an area or site.

regioselective pertaining to a chemical change that occurs with greater frequency at one site than at a number of other possible sites of similar type, usually involving a structural or positional isomer.

regiospecific describing a chemical reaction that occurs specifically in only one of two possible ways, but not including stereospecificity (*see* **stereospecific**).

registration peptide one of the postulated extensions of the α chains of **tropocollagen**, when synthesis has just been completed; two $\alpha 1$ extensions and one $\alpha 2$ extension interact to form a stable assembly with the correct chains brought together into register.

regression coefficient *see* **regression line**.

regression line a line fitted to a series of two variable quantities, x and y , so as to minimize the sum of the squares of the distances parallel to the y axis of the observations from the line. The regression line of y on x has the equation (**regression equation**): $y = a + bx$ where $a = \Sigma y - b \Sigma x/n$ and b , the **regression coefficient**, is the slope of the line and is given by:

$$b = (n \Sigma xy - \Sigma x \Sigma y) / (n \Sigma x^2 - (\Sigma x)^2)$$

where n is the number of pairs of observations. In an analogous manner a regression equation, line, and coefficient may be obtained for x on y .

regucalcin a hepatic Ca^{2+} -binding protein, and also a GTP-binding protein, that modulates the hormonal regulation of plasma membrane Ca^{2+} , Mg^{2+} -ATPase.

regulation (*in metabolism*) control of the rate of a metabolic system, especially in intact organs and cells as a result of interaction from components of related systems. —**regulatory adj.**

regulator gene or **regulatory gene** any gene that does not contribute structural information to the proteins whose synthesis it controls, but that determines the production of a specific cytoplasmic substance, a **repressor**, that inhibits information transfer from a **structural gene** (or genes) to protein. It may control the synthesis of several different proteins, which are determined by a coordinated group of structural genes (*see* **operon**). Although, when it was put forward, the concept of the regulatory gene was helpful, it is now realized that the genome codes for many different proteins that control the transcription of the structural genes (*see*, e.g., **attenuators**, **transcription factors**), and that can be classed as gene-activator proteins; these will possess two domains, one to recognize a specific regulatory DNA sequence, the other to contact the transcription machinery.

regulator-site hypothesis a hypothesis to explain the control of insulin release from pancreatic islet B cells by glucose. It postulates that the glucoreceptor is an enzyme that is activated (or inhibited) by glucose and catalyses the production from a precursor of an activator, which controls the release process. *Compare* **substrate-site hypothesis**.

regulatory enzyme an enzyme that regulates the rate of a metabolic pathway; under regulating conditions, it is the enzyme of lowest activity in the pathway. Commonly, the rate is altered either by variation in the amount of regulatory enzyme (as a result of increased or decreased synthesis or degradation), or by variation in its activity by covalent modification (often by phosphorylation or dephosphorylation by kinases or

phosphatases) or by allosteric modifiers. Classic examples are **phosphofructokinase** and glycogen phosphorylase (*see* **phosphorylase**).

regulatory gene *see* **regulator gene**.

regulatory sequence a sequence in a DNA molecule concerned with regulating the expression of a gene; e.g. a **promoter** or **operator**.

regulatory site a specific site on an enzyme that is distinct from the substrate-binding site and that binds a feedback inhibitor or other regulatory molecule. *Compare* **allosteric site**.

regulatory subunit any of the subunits of a multimeric enzyme (composed of nonidentical subunits) that has a regulatory rather than a catalytic function.

regulon a group of genes, whether linked in clusters or unlinked, that respond to a common regulatory signal.

rehydrate to restore water to material that has been lyophilized (*see* **freeze-dry**) or otherwise dehydrated.

Reichstein, Tadeus (1897–1996), Polish-born Swiss chemical engineer, organic chemist, and botanist renowned for his isolation and study of 29 different steroids from the cortex of the adrenal gland and (in 1933) for devising a method of synthesizing ascorbic acid, vitamin C, from glucose that is still in use industrially; Nobel Laureate in Physiology or Medicine (1950) jointly with E. C. Kendall and P. S. Hench 'for their discoveries relating to the hormones of the adrenal cortex, their structure and biological effects'.

reiterated describing nucleotide sequences that occur many times within a genome. These occur especially in higher organisms. *See also* **repetitive sequences**.

REL a gene family encoding transcription factors of the κB family; *v-rel* is the oncogene of avian reticuloendothelial virus strain T (REV-T), encoding p69^{v-rel}. Rel and v-Rel form homo- or heterodimers that bind to **NF- κB** motifs. v-Rel has lost two N-terminal amino acids and 118 C-terminal residues compared with Rel. Example of Rel product from mouse: database code REL_MOUSE, 587 amino acids (64.96 kDa). Example of p69^{v-rel}: database code TREL_AVIRE, 503 amino acids (55.91 kDa).

relA a gene in *Escherichia coli*, mutations in which can lead to a 'relaxed' phenotype in which amino-acid starvation fails to switch off ribosomal RNA biosynthesis. The gene product is an enzyme active in the presence of ribosomes and uncharged tRNA, and called GTP pyrophosphokinase, EC 2.7.6.5; *systematic name*: ATP:GTP 3'-pyrophosphotransferase; *other names*: pppGpp synthetase I; **stringent factor**; guanosine 3',5'-polyphosphate synthase; it catalyses a reaction between ATP and GTP to form guanosine 3'-diphosphate 5'-triphosphate and AMP. The product is referred to as pppGpp and is converted to ppGpp (guanosine 3'-diphosphate 5'-diphosphate), which is the active nucleotide in evoking the **stringent response**. Database code RELA_ECOLI, 744 amino acids (83.76 kDa).

relative atomic mass or (*formerly*) **atomic weight** *abbr.*: *r.a.m.*; *symbol*: A_r . 1 (of an atom) the ratio of the mass, m_a , of the atom in question, to the atomic mass constant, m_u , which is one-twelfth of the mass of the nuclide ^{12}C ; i.e. $A_r = m_a/m_u$. A_r is a mass ratio, a pure number with no units. 2 (of an element) the ratio of the average mass, $\bar{m}_a$, of the atoms in a specified sample of the element in question, E , to the atomic mass constant, m_u ; i.e. $A_r(E) = \bar{m}_a(E)/m_u$. Since, in general, elements are mixtures of isotopes of differing nucleon numbers (mass numbers), an average mass of the atoms is required in this definition. Furthermore, the isotopic composition of many elements can vary with the origin of the sample and its treatments. Hence there are limits to the precision with which the values of the relative atomic masses of many elements can be given.

relative band rate or **relative band speed** *see* R_b **value**.

relative biological effectiveness *abbr.*: RBE; *an alternative term for quality factor*.

relative centrifugal force *symbol*: G ; the force acting on a particle in a centrifugal field expressed in multiples of the acceleration due to gravity.

relative density

celeration due to gravity, g . $G = \omega^2 r$, where ω is the angular velocity in rad s^{-1} and r the radial distance from the axis of rotation in m. ($G = 1440 \times \pi^2 \omega^2 r$ where ω is in revolutions per minute and r in cm.)

relative density *symbol:* d ; the ratio of the mass density of a substance, ρ , at a given temperature, to the mass density of a reference substance (usually water), ρ^0 at a reference temperature (usually 4 °C); i.e. $d = \rho/\rho^0$.

relative formula mass the relative molecular mass as calculated from the formula of a compound.

relative molecular mass or **relative molar mass** or **molecular weight** *abbr.:* r.m.m.; *symbol:* M_r ; the ratio of the mass of a molecule, m_r , to the atomic mass constant, m_u , which is one-twelfth of the mass of the nuclide ^{12}C ; i.e. $M_r = m_r/m_u$. M_r is a mass ratio, a pure number with no units. The convention for indicating the size of a protein is thus ' M_r ' followed by a number without units, e.g. M_r 10 000; the equivalent **molecular mass**, i.e. its mass in daltons, would be 10 kDa. *See also* dalton, **relative atomic mass**, **unified atomic mass unit**.

relative permittivity *symbol:* ϵ_r ; formerly called **dielectric constant**, the ratio of the permittivity of a substance, ϵ , to the permittivity of vacuum, ϵ_0 , i.e. $\epsilon_r = \epsilon/\epsilon_0$.

relaxation (*in chemistry*) the process of self-adjustment of a molecular system to a new state of minimum free energy following a disturbance.

relaxation kinetics a technique used to study rapid reactions in which a mixture at equilibrium is rapidly adjusted to a new set of conditions. The system is then observed as it approaches its new equilibrium, a process known as **relaxation**. The temperature-jump method is often used, whereby an electric discharge is used to increase the temperature by several degrees in a period of about 1 μs . *See also* rapid-reaction kinetics.

relaxation spectrometry the use of spectrometry to follow the kinetics of a reaction undergoing relaxation from a perturbation. *See* relaxation kinetics.

relaxed control a mechanism of control of plasmid replication in which replication of plasmids greatly exceeds that of the chromosome. *Compare* stringent control.

relaxed DNA circular DNA that is not supercoiled (*see* supercoil), and instead has an unwound loop. The supercoiled form is energetically favoured over the relaxed form. Since the supercoiled DNA is more compact it moves faster when centrifuged or upon electrophoresis, and so the two forms of DNA can be separated.

relaxed form *abbr.:* R form; one of two alternative quaternary structures of a protein, the other being the **tense form** (or taut form, *abbr.:* T form). These forms derive from the concept of two conformational states of a protein, which differ in their affinities for a ligand. Factors affecting the conformational equilibrium also influence the affinity of ligand-binding and vice versa. Such factors may include the ligand itself, which could enhance or reduce the affinity of the protein, giving rise to **positive cooperativity** or **negative cooperativity** of binding respectively. The T form has a lower affinity for the substrate. For example, oxyhemoglobin is R, and deoxyhemoglobin is T; the binding of O_2 to the heme group causes the iron atom to move with respect to the heme plane, and this favours the move from T to R. *See also* allosteric constant, Monod-Wyman-Changeux model (for allosteric compounds).

relaxed mutant a bacterial mutant that lacks the *rel* (relaxed) gene, which exerts a critical control over the **stringent response**. The *rel⁻* mutant is unresponsive to amino-acid starvation, so that rates of RNA synthesis, for example, continue as before. *See also* *relA*.

relaxin a protein hormone ($M_r \approx 5600$) that is responsible in most mammals (including humans) for dilation of the symphysis pubis, softening of the cervix, and various other physiological changes prior to parturition. It is synthesized and stored in the corpus luteum. Like **insulin**, the hormone consists of two polypeptide chains linked by two disulfide bonds, with

an intrachain disulfide bridge also present. Although otherwise dissimilar in primary structure, the two proteins have very similar tertiary structures. As with insulin, synthesis is thought to occur via an inactive, single-chain precursor, in this case **prorelaxin**, and storage is as granules in vesicles. Example from human (precursor): database code REL1_HUMAN, 185 amino acids (21.12 kDa). *See also* **insulin-related growth factors**.

release factor *abbr.:* RF or (formerly) R factor; proteins that participate in the release of a nascent polypeptide chain from a ribosome. There are two classes of release factors, one codon-specific and the other non-codon-specific. In bacteria two codon-specific factors have been identified, RF-1, specific for UAG/UAA, and RF-2, specific for UGA/UAA. In eukaryotes one such factor has been found, eRF-1. These factors act at the ribosomal A-site and require polypeptidyl-tRNA at the P-site. In bacteria the action of RF-1 and RF-2 are aided by the non-codon-specific RF-3 which binds to guanine nucleotides. In eukaryotes a similar factor, eRF-3, has been identified. The genes encoding RF-1 (*prfA*) and RF-2 (*prfB*) have been identified. Mutations in these genes often cause misreading of termination signals, increased frameshifting, and temperature-sensitive growth of the cells. Examples from *Escherichia coli*: RF-1, database code RF1_ECOLI, 360 amino acids (40.41 kDa); RF-2, database code RF2_ECOLI, 365 amino acids (41.19 kDa). *See also* **ribosome recycling factor**.

release-inhibiting factor or **release-inhibiting hormone** any polypeptide that inhibits the release of a hormone or hormones. An example is **somatostatin**.

releasing hormone or **releasing factor** any of a group of peptide neurohormones that are synthesized in various distinct regions of the hypothalamus. They are released in nanogram amounts into the capillaries of the portal vessels of the median eminence of the hypothalamus, then carried to the anterior pituitary where they regulate the production and secretion in microgram amounts of the tropic hormones, e.g. thyrotropin, somatotropin, and the gonadotropins. In the anterior pituitary they act via the **adenylate cyclase** system and are then degraded by proteolysis.

rem *symbol:* rem; a non-SI unit of (radiation) dose equivalent, equal to 0.01 J kg^{-1} of irradiated human (or other mammalian) tissue. It is being superseded by the **sievert**; 1 rem = 10^{-2} sievert. [Acronym for röntgen equivalent man (or mammal).]

REMI *abbr.:* for restriction enzyme-mediated integration (*see* insertional mutagenesis).

remnant a fragment of a molecule or other entity remaining after partial degradation especially **chylomicron** remnant.

renal pertaining to, or of, the kidney.

renal rickets a condition occurring in older children and characterized by the skeletal changes of **rickets**, due to impairment of renal function allowing excessive loss of calcium and phosphate into the urine.

renaturation the return of denatured macromolecules (protein, DNA, etc.) to the conformation they maintained before denaturation. In the case of proteins, reversible denaturation is brought about by disulfide reducing agents and urea, for nucleic acids by heat or salts. *See also* native state.

renewal (of tissue) *see* regeneration (def. 2).

renin or **angiotensin-forming enzyme** or **angiotensinogenase** EC 3.4.23.15; a labile proteolytic enzyme, $M_r \approx 40\,000$, formed from prorenin in plasma and kidney, and secreted into the blood by the kidney. It is also produced in relatively large amounts by the submaxillary gland of certain strains of mice. A glycoprotein aspartic endopeptidase, it is highly specific for the Leu-|-Leu bond in **angiotensinogen**, which it cleaves to generate angiotensin I. Despite its high specificity it resembles other eukaryotic aspartyl proteinases. Example, precursor from human: database code RENI_HUMAN, 406 amino acids (45.01 kDa). It is a **preproprotein**; residues 1–23 are the peptide signal and 24–66 are the propeptide. 3-D structure is known. *Compare* rennin.

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renin-angiotensin system

renin-angiotensin system a system involved in the control of blood pressure. **Renin**, secreted into the blood by the kidney in response to osmotic changes, anoxia, or kidney trauma, acts on angiotensinogen to release angiotensin I (AI) (see **angiotensin**). **Angiotensin-converting enzyme** forms angiotensin II (AII) from AI by release of His-Leu from the C-terminal end. AII stimulates release of the mineralocorticoid **aldosterone** from the glomerulosa cells of the adrenal cortex. Various nonapeptide analogues of AI inhibit converting enzyme and are used in the treatment of renin-dependent hypertension. See also **captopril**.

rennet an extract obtained from the stomachs of young mammals living on milk. It contains rennin (i.e. **chymosin**), an enzyme that clots milk and is used in cheese-making.

rennin an alternative name for **chymosin** (which is preferred because of possible confusion with **renin**). See also **rennet**.

Reoviridae a family of double-stranded RNA viruses (class III in the **Baltimore classification**) with a segmented genome, infecting vertebrates, arthropods, and plants. The virion consists of a nonenveloped icosahedral particle, 60–80 nm across, and containing 10–12 segments of RNA together with **RNA-directed RNA polymerase**; the latter serves to transcribe the negative strand of each RNA molecule into mRNA. This family comprises six groups: the orthoreoviruses, orbiviruses, rotaviruses, cytoviruses, phytoreoviruses, and Fijiviruses. No pathology is associated with the orthoreoviruses infecting humans, but rotaviruses cause gastroenteritis, and orbiviruses cause the disease blue tongue in sheep.

reovirus see **Reoviridae**.

rep an obsolete unit of absorbed dose of ionizing radiation of any kind, equal to the dose of radiation that, upon absorption by a given mass of living tissue, causes the same energy deposition as 1 röntgen of X- or gamma radiation, i.e. about 96 ergs per gram (0.0096 J kg⁻¹) of tissue. See also **rad**, **rem**. [Acronym for röntgen equivalent physical.]

REP abbr. for repetitive extragenic palindromic sequence.

RepA the protein product of the *Escherichia coli* gene *repA* which has a role in the replication of DNA in *E. coli*. Although the principal DNA helicase in *E. coli* is DnaB helicase, there is also a different 68 kDa monomeric protein, which has some helicase activity during certain life-cycle stages of the replication of single-stranded phages. The contribution of RepA protein to the host cell is less clear, but the rate at which *E. coli* replication forks propagate is reduced in *rep*⁻ mutants.

repairase an enzyme concerned in the repair of an interrupted polynucleotide chain.

repair enzyme any of the enzymes making up the enzyme complex responsible for the excision of **pyrimidine dimers** which are formed on exposure of DNA to ultraviolet light, when adjacent pyrimidine residues on a DNA strand become covalently linked. Three enzymic activities are essential for the repair: (1) an enzyme complex comprising the three proteins encoded by the *uvrABC* genes (see **uvr**) that detects the distortion produced by the pyrimidine dimer; (2) the UvrABC enzyme complex also has another activity, described as a specific excinuclease, which cuts the damaged DNA strand eight nucleotides away from the dimer on the 5' side and four nucleotides away on the 3' side; (3) the 12-residue oligonucleotide excised by this specific excinuclease diffuses away and DNA polymerase I enters the gap to effect repair synthesis. Gaps are filled by **DNA ligase**. See also **DNA repair enzyme**, **excision repair**, **recombinational repair**, **mismatch repair**.

repeated genes see **repetitive sequences**.

repeat-induced point (mutation) see **RIP**.

repeat sequence or **repeat 1 (in DNA)** see **repetitive sequences**.
2 (in proteins) any sequence of the same amino-acid residue or group of amino-acid residues that is repeated in a polypeptide. Thus in some cases there are runs of glutamine or cysteine. In *Drosophila melanogaster* there is a protein with a string of 11 cysteine residues, and human androgen receptor protein has a string of 21 glutamine residues. In fact there are at present in

composite databases 44 proteins with this number of glutamine residues. In other proteins, e.g. tropocollagen, there is a regular sequence, except in this case for the last 20 amino-acid residues at each end of the chain. In tropocollagen every third residue is glycine usually followed by proline or hydroxyproline. The intrachain hydrogen bonds involve glycine. 3 the repeats of sugar residues often found in complex polysaccharides.
repeat unit a major periodicity in the structure of a molecule as deduced from X-ray diffraction patterns.

repetitive extragenic palindromic sequences or **palindromic units** palindromic sequences in repetitive DNA in *Escherichia coli* and *Salmonella typhimurium*. These are never found within a coding sequence. They are topoisomerase substrates and potential transcription terminators.

repetitive sequences repeated sequences of nucleotides found in the DNA of eukaryotes. There are various types. (1) **Satellite DNA** typically consists of tandem (head-to-tail) repeats of a few hundred nucleotides. It is so called because those first discovered had an unusual ratio of bases that made it possible to separate the satellite DNA from the bulk of the cell DNA in a centrifuge. (2) **Minisatellite DNA** consists of shorter tandem repeats. The **restriction fragment length polymorphisms** linked to many genes have short sequences of bases in common. rDNA (see **ribosomal DNA**) probes for these sequences are useful in the preparation of a profile of the genome of an individual (fingerprinting). (3) **Transposons** (transposable elements) are also characterized by repeated sequences, often numbering several hundred copies. Primate DNA has large numbers of two types of transposable elements, L1 transposable elements and **Alu sequences**, which comprise about 300 bases. (4) **Repeated genes** for rRNA and histones are also found. In many cases repetitive sequences seem to be maintained solely by their ability to replicate within the genome (the '**selfish DNA hypothesis**'). Rather than conferring benefits their behaviour can sometimes result in loss of fitness to the host, as in some human genetic diseases; mutations can arise due to insertion of transposable elements, to chromosomal rearrangements induced by recombination between repeated sequences, or to amplification of microsatellite sequences.

replacement vector a **cloning vector** in which part of the normal genome is replaced by foreign DNA. It contrasts with an **insertion vector** where the foreign DNA is merely incorporated into the normal genome. For example, a mutant of lambda phage, called λgt-λβ, contains only two *EcoRI* cleavage sites and so the middle segment of the DNA can be removed leaving 72% of the normal genome. A suitably long DNA replacement between the two ends of lambda DNA enables such a recombinant DNA molecule to be packaged into a lambda particle.

replica (in electron microscopy) a reproduction of the surface of an object. There are various ways by which the surface of an object, e.g. membrane of an organelle, may be examined by transmission electron microscopy by means of a replica. Such a replica may be obtained by evaporating a thin film of a heavy metal such as platinum on to the dried specimen. The metal is sprayed from an oblique angle in order to deposit a coating of uneven thickness because a shadow effect is created that gives the image a three-dimensional appearance. The organic material may be dissolved away to provide a replica of the surface of the specimen. Two other methods, **freeze-etching** and **freeze-fracturing**, employ frozen sections.

replica plating a technique for the production of identical patterns of bacterial colonies on a series of agar plates. The method uses velveteen or other fabric stretched on a pad, which is then pressed onto a plate of bacterial cultures so that some bacteria from each colony are transferred to the pad, preserving the spatial pattern of the colonies (in mirror image). If this pad is then pressed lightly onto a fresh culture plate, it will transfer colonies in the same formation to the new plate. This new plate may be prepared using media that will be selective for certain mutants, fermentation reactions, or anti-



replicase

biotic sensitivity. A similar technique can be used in the screening of **hybridomas** for antibody production.

replicase a *loose term* for certain polymerase enzymes; e.g. certain viral **RNA-directed RNA polymerases** can be referred to as RNA replicases.

replicate 1 to make an exact copy of something, as in the **replication of DNA**. 2 (in *electron microscopy*) to make a **replica**.

replication complex the complex of replication proteins that are linked together as a single large unit, of total mass 10^6 daltons, that moves rapidly along the DNA during **replication of DNA**. It enables DNA to be synthesized on both sides of the replication fork in a coordinated and efficient manner. That part of the complex that contains **DNA primase** and **DNA helicase** is known as the **primosome**.

replication-error mutation a mutation that arises during DNA replication as a result of the incorrect copying of the sequence of nucleotides in the parent DNA. Probably fewer than one mistake occurs in 10^9 nucleotides added.

replication fork or **replicating fork** the Y-shaped region of a replicating DNA molecule or chromosome, resulting from separation of the DNA strands and in which synthesis of new strands takes place. The process requires the participation of **helicases** that open the duplex.

replication of DNA the means whereby new strands of deoxy-polynucleotides are synthesized. The structure of double-stranded DNA, with two strands of complementary base pairs, led Watson and Crick to propose a possible means of copying DNA: the two strands could unwind and each act as the template for a new strand, thus forming two helices, each containing one parental strand and one new one. The elegant **Meselson-Stahl experiment** demonstrated that DNA replication was indeed semiconservative. Other mechanisms of replication, 'conservative' and 'dispersive', have been proposed but there is no evidence that they occur. The helix of the double-stranded parental DNA must be unwound and this requires more than 20 proteins in a complex known as a **primosome**. All DNA-dependent DNA polymerases synthesize the new strands in the direction 5' to 3' using as substrates deoxynucleotide triphosphates with the production of pyrophosphate. The new strands are antiparallel to the parental template strand. At low resolution both strands of a double-stranded DNA appear to be synthesized in the same direction in spite of the fact that the polymerases only synthesize the strands in one direction. The problem is resolved by the fact that one strand, the **leading strand** (or continuous strand), is synthesized in a continuous fashion whereas the other strand, the **lagging strand** (or discontinuous strand), is synthesized discontinuously to give **Okazaki fragments** (1000–2000 nucleotides long in prokaryotes and 100–200 nucleotides long in eukaryotes) which are eventually joined by **DNA ligase**. The DNA ligases cannot start a chain and must rely on a priming device which is provided by RNA **primase** which again synthesizes an RNA primer in the direction 5' to 3' using nucleoside triphosphates as substrate. The continuous strand needs to be primed only once whereas the discontinuous strand must be primed repeatedly. Three principal **DNA polymerases** for *Escherichia coli* were discovered in the laboratory of A. Kornberg. **DNA polymerase I**, in addition to possessing synthetic activity *in vitro*, is also a 3'–5' **exonuclease** which gives it an editing function. It also has 5'–3' nuclease activity. **DNA polymerase II** participates in DNA repair but is not required for replication. **DNA polymerase III** is the predominant polymerase for replication of *E. coli* DNA *in vivo*. Eukaryotic cells have at least four DNA polymerases.

replicative able to be replicated. *See also* **replicative form**.

replicative form *abbr.*: RF; the intracellular form of viral nucleic acid that is active in replication. A single-stranded viral nucleic acid, be it RNA or DNA, is first converted into a double-stranded form – the **replicative intermediate** – to enable the viral strand to be replicated preferentially. In the case of a **retrovirus** a hybrid of RNA and DNA is formed.

replicative intermediate *see* **replicative form**.

replicon 1 a structural gene that controls the initiation and replication of DNA. 2 by association, the whole DNA sequence (such as the bacterial chromosome) that is replicated under the influence of a single replicon system.

replisome a **replication complex** hypothesized to account for the initiation of chromosome replication within a living cell.

replisome a particulate fraction that can be isolated from the nuclei of **S phase** Chinese hamster embryo fibroblasts and containing some of the enzyme activities required for DNA synthesis.

reporter gene a gene used to disclose the function of potential regulatory sequences. The sequences are placed in plasmids, upstream of the reporter gene; these chimeric plasmids are then introduced into cells and the expression of the reporter gene is measured as an index of the function of the regulatory sequences under the conditions being investigated. The *lacZ* (β -galactosidase) gene, the *CAT* gene (*see* **chloramphenicol O-acetyltransferase**), and the gene encoding firefly **luciferase** have been used in this way.

reporter group a chemical group that acts as a **probe**.

repress to check or restrain; the effect may be partial or total. *See* **repression**, **repressor**.

repressible describing a metabolic reaction that is capable of undergoing **repression**.

repression (in *genetics*) the act of inhibiting gene expression. An example is **catabolite repression** of the *lac* operon by glucose, mediated through binding of cyclic AMP to **catabolite (gene) activator protein** (CAP). By extension, the gene product itself (enzyme or other protein) and thus any associated metabolic reaction is said to be repressed. *See also* **repressor**.

repressor (in *genetics*) a molecule that specifically prevents the transcription of regulatory genes under control of an **operator**. When the repressor is bound to the operator, transcription cannot occur. An inducer can induce transcription by binding to the repressor to form a complex that has a low affinity for the operator. The classic example of a repressor is the lactose repressor protein, which controls the expression of the *lac operon* in *Escherichia coli*. The protein binds to a specific sequence of 21 nucleotide pairs (called the operator) that overlaps an adjacent RNA polymerase binding site (the promoter) at which RNA synthesis begins. In the presence of lactose, allolactose is formed, which binds to the repressor. This induces a conformational change that causes the repressor to loosen its hold on the DNA so that the gene is derepressed. In other examples, the binding of a signalling molecule may increase the affinity with which the repressor binds to the DNA; e.g. the *trp* operon, encoding enzymes that produce tryptophan in *E. coli* (*see* **trp gene**). *See also* **repression**.

reproducibility the ability to carry out a series of identical experiments to yield results that fall within a small range of error derived by statistical analysis (*see* **standard deviation**, **standard error of the mean**). The limits of error that are acceptable depend on the limitations of experimental technique and the inherent variability of the system under investigation.

reproduction (in *biology*) the production by an organism of new individuals that are more or less similar to itself. Reproductive strategies can be divided broadly into two categories: sexual reproduction, which requires the participation of male and female **gametes** and gives rise to genetic variation among the offspring; and asexual reproduction, which does not involve gametes and leads to the formation of offspring that are genetically identical to the parent. The latter takes various forms; examples include **budding**, **fragmentation**, **parthenogenesis**, **spore formation**, and **vegetative reproduction**.

reptation a term invoked to rationalize the movement and thermodynamics of linear polymers, dividing the chain conceptually into sections (**reptons**), which form the basis for statistical mechanical calculations of the relaxation time of the polymer and viscosity of polymer solutions.

repton *see* **reptation**.

RER *abbr.* for rough endoplasmic reticulum.

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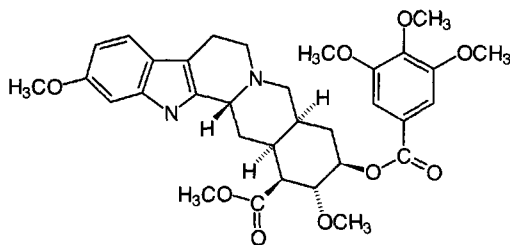
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reseal to join again the ends of ruptured components, as in, e.g., the resealing of plasma membranes after whole cells have been rendered permeable by various techniques, or of fractionated membrane fragments, which may reseal to form closed vesicles. The term can also refer to ligation of the cut ends of DNA after insertion of foreign DNA into a vector.

reserpine (3 β ,16 β ,17 α ,18 β ,20 α)-11,17-dimethoxy-18[(3,4,5-trimethoxybenzoyl)oxy]yohimban-16-carboxylic acid methyl ester; an alkaloid isolated from the roots of certain species of *Rauwolfia* and noted for its antihypertensive and CNS depressant actions. These arise from its inhibition of the intraneuronal Na⁺, Mg²⁺-ATPase pump, which maintains the neuronal vesicular stores of biogenic monoamines (**dopamine**, **norepinephrine**, and **serotonin**).



reserve polysaccharide any polysaccharide that is stored to provide an energy source at times of need. Examples are starch granules in plants, and glycogen granules in animals.

Re/Si convention a nomenclatural system for specifying the **stereoheterotopic** faces of a trigonal atom using the **sequence rule** in two dimensions. This is used to establish a priority sequence for the three ligands, e.g. of *Cabc*. The plane containing C and the first atoms of the ligands is then inspected from an arbitrarily chosen side. Starting with the highest priority ligand, it is determined whether the priority sequence is clockwise, hence *Re* (from Latin *rectus*, right), or counterclockwise, hence *Si* (from Latin *sinister*, left); e.g. for acetaldehyde, CH₃CHO, if the face inspected shows a clockwise ordering, O → CH₃ → H, that face is designated *Re*. There are a number of extensions of this general principle, and the specialist literature should be consulted for details.

residence time the averaged time that a site is occupied by any designated molecule.

resident protein *see* **luminal protein**.

residual body a membrane-lined cytoplasmic inclusion characterized by undigested residues (e.g. membrane fragments or whorls, ferritin-like particles). The term embraces **telolysosome** and (hypothetical) **post-lysosome**.

residue 1 any of the incorporated amino-acid moieties in a peptide or protein. 2 material that remains after any procedure to remove something.

residue mass the relative molecular mass of a residue; i.e. the sum of the **relative atomic masses** of all the atoms composing the residue.

residue weight the molecular weight of a residue; i.e. the sum of the **atomic weights** of all the atoms composing the residue. **Residue mass** is the preferred term.

resin 1 or **rosin** any of a group of naturally occurring amorphous solids or semisolids, typically light yellow to dark brown in colour. Usually brittle and insoluble in water, they are found as plant exudates, often on trees such as pines and firs. 2 the polystyrene base of certain ion-exchange materials used in chromatography. The polystyrene is modified to carry sulfonic acid or carboxyl groups (cation exchangers), or primary or quaternary ammonium groups (anion exchangers).

resistance 1 the ability of a living organism, particularly a bacterium, to resist the effect of a disadvantageous environment or substance, especially an antibiotic. A bacterium may

possess one or more resistance plasmids. 2 *see* **electric resistance**.

resistance donor a strain of an organism, usually a bacterium, that is capable of transferring resistance, usually by conjugation, to a suitable recipient.

resistance factor *abbr.*: R factor; *an alternative name for resistance plasmid*.

resistance plasmid *abbr.*: R plasmid; *a plasmid that carries genetic information for resistance to antibiotics or other antibacterial drugs. Other name: resistance factor (abbr.: R factor)*.

resolution 1 the degree of separation between two compounds being subjected to a transport method such as chromatography or sedimentation equilibrium (i.e. being resolved); complete resolution yields two compounds each uncontaminated by the other. 2 (*in physics and biophysics*) the extent to which closely juxtaposed objects can be distinguished as separate objects using an optical instrument, e.g. a microscope. The degree of resolution is dependent on the **resolving power** of the system. The fineness of detail with which objects can be seen is limited by the wavelength of light used. Only those objects similar in dimensions to the wavelength of the light (or larger) can be resolved. The wavelength of X-rays is from about 10⁻⁸ m to 10⁻¹¹ m and they can thus be used to resolve structures at the atomic level. Thus a structure can be said to have been determined at 3 Å resolution, 7 Å resolution, etc.

resolvase an enzyme that, together with **transposase**, is involved in the replicative transposition of genetic elements in *Escherichia coli*. The 3' ends of the target chromosome, after the first cutting and splicing, serve as replicative primers for copying both the gaps and the two strands of the transposable element (which contains transposase) itself. Ligase action generates a cointegrate, a large circular structure containing both donor and target chromosomes with two freshly replicated copies of the transposable element. Resolvase now catalyses the site-specific recombination between the two elements, resulting in one copy of the transposable element inserted in each of the two chromosomes. Example, *tnpA* gene product from Tn3 which confers resistance for ampicillin; database code TNP3_ECOLI, 185 amino acids (20.59 kDa); it contains a **helix-turn-helix** motif.

resolving power 1 (*in chromatography*) the power to resolve individual compounds (*see* **resolution** (def. 1)). It depends on (1) the efficiency of the system employed; for columns this is given by the **theoretical plate** number; and (2) the relative affinities of different compounds for the stationary and mobile phases. 2 (*in physics*) a the ability of an optical instrument, e.g. a microscope or telescope, to produce separate images of objects placed close together. b the ability of a spectrometer to separate adjacent peaks in a spectrum.

resonance 1 (*in physics*) the induction or amplification of one vibration by another vibration of a periodicity equal to the natural oscillation frequency of the first, which is then said to **resound**. In an electronic circuit, resonance occurs when an applied emf equals the natural frequency of oscillation of the circuit (used, e.g., in tuning a radio receiver). 2 (*in chemistry*) a phenomenon shown by any molecular entity to which more than one **contributing structure** can be assigned, when these structures are of similar energy and differ only in the distribution of the valence electrons; the entity then oscillates between the two structures and adopts, in effect, an intermediate state. Resonance always results in a different distribution of electrons than would be the case if there were no resonance. *See also* **resonance energy**, **resonance hybrid**. 3 *see* **electron spin resonance spectroscopy**, **nuclear magnetic resonance**. —**resonant** *adj.*; **resonate** *vb*.

resonance energy the difference in energy of an actual molecule and the **contributing structure** of lowest energy.

resonance hybrid the structure that results when a chemical entity exhibits **resonance** (def. 2), such as occurs by the sharing of one or more electrons over several bonds. It has a lower en-



ergy than any of the possible formal nonresonance contributing structures, giving additional stability of which the so-called **resonance energy** is a measure. A classic example is carbon dioxide, which may be represented by the **Lewis formula** for each as $\ddot{O}::O::\ddot{O}$, $\ddot{O}::C::\ddot{O}$, and $^+O::C::\ddot{O}^-$. Compare **mesomerism**.

resonance stabilization or **resonance stabilisation** stabilization of molecular structure by resonance; see also **resonance hybrid**.

resonate see **resonance**.

resorb to absorb again. —**resorbent** *adj.*; **resorption** *n.*; **resorptive** *adj.*

resorcinol *m*-dihydroxybenzene; a compound used in the **Seliwanoff test** for ketoses.

resorption see **resorb**.

respiration 1 the process of gaseous exchange between an organism and its environment. In plants, microorganisms, and many small animals, air or water makes direct contact with the organism's cells or tissue fluids, and the processes of diffusion supply the organism with dioxygen and remove carbon dioxide. In larger animals the efficiency of gaseous exchange is improved by specialized respiratory organs, such as lungs and gills, which are ventilated by breathing mechanisms. These organs possess respiratory surfaces, across which gases are exchanged with the blood or other transport medium. This then carries dioxygen to the body tissues, and removes carbon dioxide. 2 or **oxidative metabolism** the various energy-yielding reactions of cells or organisms that require oxygen as the final electron acceptor. See **respiratory chain**. —**respiratory** *adj.*

respiratory burst a phase of elevated metabolic activity, during which oxygen consumption increases, that occurs in neutrophils, monocytes, and macrophages shortly after phagocytosing material. This is associated with increased activity of the hexose monophosphate shunt and production of **NADPH** (see **pentose phosphate pathway**). An oxidase takes electrons from the **NADPH** and transfers them to O_2 creating the reactive superoxide (O_2^-) ion and H_2O_2 in the vacuole. Patients with chronic granulomatous disease never produce a respiratory burst. In some cases this is due to lack of an essential cytochrome *b*. The patients suffer an accumulation of mononuclear cells at sites of chronic inflammation, forming granulomas.

respiratory chain or **electron-transport chain** the sequence of enzymes and other proteins within the mitochondrion and prokaryotic cell membranes by which substrates, including **NADH** and succinate, are oxidized by dioxygen. It is fundamental to tissue **respiration** (def. 2) which has the function of maintaining tissue metabolism through **oxidative phosphorylation**. The chain can be separated into a number of particles (see **respiratory complex**) each of which contains individual elements of the chain. The mechanism involves, in the case of **NADH**, removal by **NADH dehydrogenase** of a proton plus two electrons. In the case of succinate, there is a transfer of two hydride ions (i.e. two protons plus two electrons) by succinate dehydrogenase to the enzyme-bound **FAD**; other flavoproteins can similarly oxidize fatty acyl-CoA or β -hydroxybutyrate or α -glycerophosphate. The protons are initially released to the intermembrane space, and the electrons are transferred by the appropriate flavoprotein to ubiquinone, from which they are transferred singly in sequence to cytochrome *b*, cytochrome c_1 , cytochrome *c*, and cytochrome *ala*₃ at which point they reduce molecular oxygen to promote formation of water with the protons that were formed earlier. Many other components are also involved. In tightly coupled mitochondria, the protons released to the intermembrane space reach the mitochondrial matrix (the space in which they react with molecular oxygen) by transfer through the inner mitochondrial membrane by the F_0F_1 complex of the mitochondrial **ATPase** (see **H^+ -transporting ATP synthase**), with concomitant phosphorylation of **ADP** to **ATP**. See also **chemiosmotic coupling hypothesis**.

respiratory complex any of several groups of physically associated components of the **respiratory chain**. The concept arose

during experiments to purify individual components, when it was found that membrane particles could be isolated each with a consistently characteristic composition. Four main complexes have been identified: complex I, which contains primarily **NADH dehydrogenase**; complex II, which contains succinate dehydrogenase; complex III, which contains cytochromes *b* and c_1 ; and complex IV, which contains cytochrome *ala*₃ together with copper-containing proteins. Complexes I, II, and III all contain additionally a number of proteins rich in iron and sulfur. The F_0 and F_1 subunits of the **ATPase** associated with the respiratory chain have been termed complex V (see **H^+ -transporting ATP synthase**). See also **complex I, II, III, IV**.

respiratory control the regulation, in coupled mitochondria, of the rate of oxidative phosphorylation by the level of **ADP**. As the **ADP** level increases, and the **ATP** level falls, electron flow through the **respiratory chain** and coupled **ATP** synthesis are enhanced. The **respiratory control ratio**, or **respiratory control index**, represents the ratio of the rate of oxygen consumption of mitochondria in the presence of both substrate and **ADP** to the rate in the presence of substrate but the absence of **ADP**; i.e. the ratio of the rate in respiratory state 3 to respiratory state 4 (see **respiratory state**).

respiratory pigment any of a number of pigmented proteins, associated with respiratory processes, that derive their colour from a prosthetic group, usually heme in red proteins such as cytochrome *c*, or flavin nucleotides in flavoprotein dehydrogenases, which are yellow.

respiratory quotient or **respiratory ratio** *abbr.*: **RQ**; a ratio of the rate of carbon dioxide evolution by any cell, tissue, or organism to its rate of dioxygen uptake.

respiratory state or **state of the respiratory chain** any one of the possible steady states of the phosphorylating **respiratory chain** in mitochondria. In state 1 both **ADP** and respiratory substrates are lacking. State 2 is a 'standard state' in which **ADP** has been added to exhaust the endogenous respiratory substrate; in state 2' substrate has been added to exhaust the endogenous **ADP**. In state 3, the 'active' state of rapid respiration, all required components are present and the respiratory chain itself is the rate-limiting factor. State 4, an anaerobic 'resting' state, is characterized by a low respiration rate. In state 5, only oxygen is lacking, the respiration rate then being at zero.

State	[O_2]	[ADP]	Substrate level	Respiration rate	Rate-limiting factor
1	>0	low	low	slow	ADP
2	>0	high	0	slow	substrate
2'	>0	low	high	slow	ADP
3	>0	high	high	fast	respiratory chain
4	>0	low	high	slow	ADP
5	0	high	high	0	O_2

respirometer an instrument for measuring the uptake of oxygen during incubation experiments. The **Barcroft apparatus** and the **Warburg apparatus** were examples of this type of instrument.

response the events brought about by a **stimulus**, especially in a cell, tissue, or organism. See **stimulus-response coupling**, **sensitization**, **desensitization**.

response coefficient *symbol*: **R**; an overall measure of the control exercised by a specific enzyme inhibitor, at its concentration, *I*, on the overall flux, *F*, of a multienzyme system. Hence,

$$R_f^F = (\partial F/F)/(\partial I/I)$$

or

$R = \text{fractional change in flux/fractional change in inhibitor.}$

A response coefficient can also be determined for the effect of *I* on the concentration of a metabolite *S*. Thus

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$$R_I^S = (\partial S/S)/(\partial I/I)$$

It is the product of the **elasticity coefficient** and the **flux control coefficient**.

response element *abbr.*: RE; a DNA sequence (sometimes called a 'module') that is common to promoters/enhancers of genes whose expression is coordinately regulated (*see* **coordinate induction/repression**). Examples: the **estrogen response element** (ERE) has the consensus sequence GGTCANNNTG[A/T]CC; the glucocorticoid response element (GRE) has the consensus sequence TGGTACAAATGTTCT; the **heat-shock response element** (HSE) has the consensus sequence GNGGAANNTCCNNG; the **metal response element** (MRE) has the consensus sequence CGNCCCGGNCNC; the **serum response element** (SRE) has the consensus sequence CATATTAGG; and the **thyroid response element** (TRE), which is common to several enhancers, has the consensus sequence TGACTCA.

response time the interval between initiation of a process and the commencement of the action that it induces.

responsiveness (*in physiology*) the ability of a cell, tissue, or organ to undergo a response. Maximal response normally corresponds to the healthy *in vivo* state, so that the degree of response is an indication of the integrity of the preparation. However, negative regulators may be removed during tissue preparation, thus giving enhanced activity *in vitro*.

restin Reed-Sternberg cell intermediate filament-associated protein; *other name*: cytoplasmic linker protein-170 α -2 (*abbr.*: CLIP-170); a protein that is highly expressed in such cells in Hodgkin's lymphoma and having a large coiled-coil α -helical domain similar to those in myosins. Example from human: database code REST_HUMAN, 1427 amino acids (160.81 kDa).

resting potential the potential difference across the membrane of a cell in the unstimulated state. *See also* **membrane potential**.

resting state an inactive or unstimulated state, especially of a cell. *See also* **cell-division cycle**.

restricted DNA DNA that has been cut by a **restriction endonuclease** within a bacterium; bacteria are protected from the propagation within them of foreign DNA, usually viral DNA, by the possession of restriction endonucleases which degrade the invading DNA. Each nuclease recognizes a specific sequence of 4–8 nucleotides in the DNA. The cleaved sequences, where they occur in the genome of the bacterium itself, are protected from cleavage by methylation at the amino group of an adenine, or either the 5 position of a cytosine or the amino group of a cytosine. The newly replicated strand of the bacterial host DNA, which is protected from degradation by the methylated parent strand with which it forms a duplex, is modified before the next cycle of replication (*see* **restriction-modification system**). Such sequences in the viral DNA are usually not so methylated and hence are degraded first by the restriction enzymes and then by bacterial exonucleases. Invading DNAs are only rarely modified and enabled to reproduce in their new host. Its progeny are, however, no longer modified in the way that permits them to propagate. *See also* **host-controlled modification**.

restricted phage or **restricted bacteriophage** *see* **restricted virus**.

restricted virus a virus whose host range is limited by **host-controlled modification** so that when it infects a related but non-identical host cell its DNA is degraded by the new host cell's **restriction endonucleases**.

restriction (*in molecular biology*) the action or use of **restriction endonucleases**. *See also* **restrictive**.

restriction analysis another term for **restriction mapping**.

restriction endonuclease or **restriction enzyme** or **restriction nuclease** any of a group of enzymes, produced by bacteria, that cleave molecules of DNA internally at specific base sequences. They, therefore, act to 'restrict' the replication of foreign, usually viral, DNA entering the bacterial cell (*see* **restricted DNA**). Some of them also methylate the host bacterial

DNA to protect it from degradation. There are three known types of restriction enzymes. Types I and III each carry activity for both the endonuclease and methylase function whereas type II enzymes do not carry the methylase activity. Type I enzymes cleave the DNA at a random site located at least 1000 bp from the recognition sequence, type III do so 24–26 bp distant from the recognition site. In contrast type II enzymes cleave DNA at specific sites within the recognition sequence. This makes type II enzymes indispensable for DNA manipulation. Around 2500 species of type II restriction enzymes with some 200 differing sequence specificities have been characterized. A restriction endonuclease is named by the first letter of the genus of the bacterium that produced it and the first two letters of its species, followed by its serotype or strain designation, and a roman numeral if the bacterium contains more than one type of restriction enzyme. **Type I** (EC 3.1.21.3) restriction and modification enzymes are complex, multifunctional systems that require ATP, S-adenosyl methionine, and Mg^{2+} as cofactors; in addition to their endonucleolytic and methylase activities, they are potent DNA-dependent ATPases. The subunit composition, three subunits designated R, M, and S, is similar to that of ATPases. They catalyse the endonucleolytic cleavage of DNA to give random double-stranded fragments with terminal 5'-phosphates; ATP is simultaneously hydrolysed. Example, R protein from *Escherichia coli*, enzyme *EcoR*124/3: database code T1R1_ECOLI, 1033 amino acids (119.52 kDa). **Type II** enzymes have activity as site-specific deoxyribonucleases (EC 3.1.21.4), which catalyse the endonucleolytic cleavage of DNA to give specific double-stranded fragments with terminal 5'-phosphates; Mg^{2+} is a cofactor. Example from *E. coli*, enzyme *EcoRV*: database code T2E5_ECOLI, 244 amino acids (28.49 kDa). Such enzymes are the basis of **restriction mapping**. **Type III** enzymes have activity as site-specific deoxyribonucleases (EC 3.1.21.5). The protein has two subunits, products of the *res* (from 'restriction') and *mod* (from 'modification') genes; these are respectively, responsible for binding the system-specific DNA recognition site in the restriction and the methylation of one of the adenosyl residues in the recognition site. A list of restriction endonucleases is given in **Appendix F**. *See also* **restriction-modification system**.

restriction enzyme an alternative name for **restriction endonuclease**.

restriction fragment a fragment of DNA produced by the action of a **restriction endonuclease**. It usually refers to the product of a type II enzyme that recognizes specific base sequences in double-helical DNA and cleaves both strands of the duplex at specific sites.

restriction fragment length polymorphism *abbr.*: RFLP; **polymorphism** within a population of organisms of the size of the fragments of DNA produced as the result of the action of a type II **restriction endonuclease**. It results from a corresponding polymorphism of **restriction sites** in the genome, due to slight differences in base sequence between individual members of the population. RFLP is exploited in constructing physical maps of the genome, and widely used for localizing specific genes and detecting genetic differences between closely related individuals. The analysis of restriction fragments can reveal the presence of a mutation that may itself cause disease or be closely linked to one that does; for example, genetic diseases such as sickle-cell anemia, cystic fibrosis, and Huntington's disease can be detected by RFLP analyses. *See also* **restriction mapping**.

restriction map a pattern obtained as the result of **restriction mapping**.

restriction mapping the use of type II **restriction endonucleases** to cleave genomic or other DNA to produce a characteristic pattern of fragments. The fragments are resolved by gel electrophoresis and visualized. Every DNA molecule produces a unique pattern, which accords with the specificity of the particular restriction enzyme. By comparing the patterns ob-

restriction-modification system

tained from two or more different DNAs, differences between closely similar DNA molecules, such as arise from mutations, may be detected. The use of DNA probes, such as rDNA, to identify particular fragments may often be of value.

restriction-modification system any system whereby host DNA is methylated to render it resistant to the action of the cell's **restriction endonucleases** (see **restricted DNA**); methylation involves the adenine or cytosine residues at the restriction sites. Several enzymes have been implicated in this action: (1) EC 2.1.1.72; *recommended name*: site-specific DNA-methyltransferase (adenine-specific); *other names*: N-6 adenine-specific DNA methylase; modification methylase; it catalyses a reaction between S-adenosyl-L-methionine and DNA adenine to form S-adenosyl-L-homocysteine and DNA 6-methyl-aminopurine. (2) Another type is EC 2.1.1.73; *recommended name*: site-specific DNA-methyltransferase (cytosine-specific); *other names*: C-5 cytosine-specific DNA methylase; modification methylase; it catalyses a reaction between S-adenosyl-L-methionine and DNA cytosine to form S-adenosyl-L-homocysteine and DNA 5-methylcytosine. Examples: (1) *EcoRV* methylase from *Escherichia coli*: database code MTE5_ECOLI, 298 amino acids (34.60 kDa); (2) *EcoP15* *mod* gene product of a type III system (see **restriction endonuclease**); this methylates one of the A residues in 5'-CAGCAC-3'; it is a homotetramer: database code T3MO_ECOLI, 645 amino acids (74.22 kDa).

restriction negative describing an organism that does not contain any **restriction endonuclease**.

restriction site a sequence of bases in a DNA molecule that is recognized by a **restriction endonuclease**.

restrictive 1 describing a strain or line of cells that does not permit the replication of a particular viral DNA, and is thus resistant to infection. 2 describing a set of conditions, such as a temperature range, in which a mutant phenotype is expressed, thus limiting the growth of the organism or causing its death. *Compare* **permissive**. See also **conditional lethal mutation**, **temperature-sensitive mutation**.

restrictive conditions see **conditional lethal mutation**.

restrictive temperature see **temperature-sensitive mutation**.

restrictocin an antitumour polypeptide, M_r 16 000, from *Aspergillus* spp. It cleaves the large RNA of the eukaryotic 60S ribosomal subunit and inhibits protein synthesis in picornavirus-infected cells.

RET a human transforming oncogene derived through fibroblast transfection with T-cell lymphoma DNA. It encodes two forms of receptor tyrosine kinases of unknown ligand; the cytoplasmic oncogenic forms (p76^{RET}, p81^{RET}) are constitutively phosphorylated on tyrosine; they form homo- and heterodimers, and have autophosphorylation activity. A number of variants have been associated with multiple endocrine neoplasia and Hirschprung's disease. Example (precursor): database code RET_HUMAN, 1114 amino acids (124.32 kDa).

retentate or **dialysis residue** or **residue** (in **dialysis**) the moiety of the dialysed material that has not traversed the membrane; the liquid containing this moiety (which may or may not still be entirely in solution). Some authorities prefer the term **non-diffusible material**.

retention coefficient (in **column chromatography**) the ratio of the **void volume** to the **elution volume** for a specific solute. It is a measure of the partition of the solute between the column material and the solvent, and is independent of column size.

retention time the time taken by a compound to traverse the length of a **gas-liquid chromatography** column. The retention time is a characteristic of each compound under defined conditions. It can be used to aid in identification within a series of homologues, e.g. fatty acids, which are characterized by their **carbon number** derived from the retention time.

retention volume a parameter of liquid chromatography, related to the volume at which each compound elutes under defined conditions.

reticulate 1 or **reticular** resembling a net; in the form of a net.

work. 2 to create in the form of a net or network. —**reticulation** *n*.

reticulocalbin a Ca^{2+} -binding luminal protein of the endoplasmic reticulum. It has six **EF-hands**, a 20-residue N-terminal **signal sequence**, and the C-terminal sequence **HDEL**. Example (precursor) from mouse: database code RCA1_MOUSE, 325 amino acids (38.07 kDa).

reticulocyte an immature **erythrocyte** that is capable of protein synthesis, most of the protein synthesized (about 90%) being hemoglobin. It retains intracellular organelles, including mitochondria, and histologically stains for the presence of ribosomes. The name implies misleadingly that it has a **reticulum** (def. 1), which is characteristic of cells that secrete protein, a function not possessed by the reticulocyte.

reticulocyte lysate the preparation obtained by suspending **reticulocytes** in distilled water so that the cells are lysed by osmotic shock. The reticulocytes are usually obtained from rabbits in which **reticulocytosis** has been induced by the administration of phenylhydrazine. The residual RNA is usually removed by the addition of a ribonuclease that is calcium-dependent and is thereby inactivated by the addition of EDTA. Such a lysate is often used for the *in vitro* translation of added mRNA.

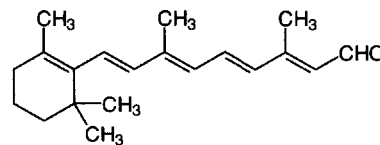
reticulocytosis enhanced levels of reticulocytes in the blood, often due to experimentally-induced increased production. See also **reticulocyte lysate**.

reticuloendothelial system a former name for **mononuclear phagocyte system**.

reticulum 1 the **endoplasmic reticulum**, a netlike structure of membranes found in the cytoplasm. 2 the most anterior chamber of the ruminant stomach.

retina the light-sensitive layer lining the inside of the posterior wall of the eye in vertebrates and some classes of molluscs. The inner nervous layer of the vertebrate retina is transparent and contains photoreceptive **cone cells** and/or **rod cells** together with their associated nerve cells and fibres, glia, and blood vessels; the outer layer, next to the choroid, is darkly pigmented and prevents back-reflection of light.

retinal 1 pertaining to the retina. 2 the *recommended trivial name* for (2E,4E,6E,8E)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1-en-1-yl)nona-2,4,6,8-tetraen-1-al; *other names*: retinal₁; retinaldehyde; vitamin A aldehyde; vitamin A₁ aldehyde; retinene; retinene₁; a compound that plays an important role in the visual process in most vertebrates (*compare* **3,4-didehydroretinal**). In the retina this (all-E)-isomer, commonly distinguished as all-*trans*-retinal, is converted by **retinal isomerase** to its (2E,4Z,6E,8E)-stereoisomer, generally termed 11-*cis*-retinal, which then combines (as a Schiff's base) with an opsin to form (in cone cells) **iodopsin** or (in rod cells) **rhodopsin**. Interaction with a photon brings about dissociation of iodopsin or rhodopsin, to regenerate all-*trans*-retinal and the opsin, thereby completing the visual cycle. Retinal is formed in the retina by enzymic oxidation of **retinol**; in addition, most vertebrates can convert dietary α -, β -, and γ -carotenes to retinal by oxidative cleavage of the polyprenyl chain at its midpoint. [Note: (1) The numbering system of the systematic name is different from that of the trivial name, the latter following the rules used for carotenoids. (2) The fuller name **retinaldehyde** is recommended if confusion with def. 1 may occur.]



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retinal₁

retinal₁ (sometimes) an alternative name for **retinal** (def. 2) (especially to distinguish it from **retinal₂**; not recommended).

retinal₂ (sometimes) an alternative name for **3,4-didehydroretinal** (especially to distinguish it from **retinal₁**; not recommended).

retinal cone see **cone cell**.

retinaldehyde see **retinal**.

retinal isomerase EC 5.2.1.3; systematic name: all-trans-retinal 11-cis-trans-isomerase; other name: retinene isomerase; an enzyme that catalyses the conversion of all-trans-retinal to 11-cis-retinal. Example from *Todarodes pacificus* (Japanese flying squid): database code REIS_TODPA, 301 amino acids (33.45 kDa). See also **retinal** (def. 2).

retinal rod see **rod cell**.

retinene or **retinene₁** an alternative name for **retinal** (def. 2) (not recommended).

retinene₂ an alternative name for **3,4-didehydroretinal** (not recommended).

retinene isomerase an alternative name for **retinal isomerase**.

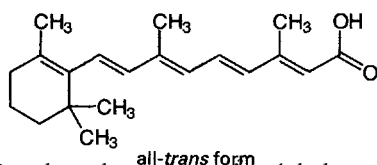
retinitis pigmentosa a relatively common genetic disorder which in humans affects 1 in 4000 people and is caused by the premature death of photoreceptors. The result is progressive night blindness, contraction of the visual field and finally complete loss of sight. One form of autosomal dominant disease results from an abnormality on chromosome 3. In this, there is a single amino-acid substitution in the **rhodopsin** molecule caused by a base transversion of C to A, which results in the change of a proline to a histidine residue at position 23 in the **opsin** molecule, causing the characteristic retinal degeneration. Another severe form of the disease, affecting up to around 20% of some populations, is X-linked; it usually becomes manifest within the first two decades of life and progresses to blindness within 10–20 years. The gene, named **RPGR**, for this form has now been found on the short arm of the X-chromosome. The predicted 90 kDa protein product of the **RPGR** gene bears significant homology to the **RCC1** family.

retinoate 1 the anion of **retinoic acid**. 2 any salt or ester of retinoic acid.

retinoblastoma a childhood cancer of the developing retina. The offspring of surviving victims have a high incidence of this disease as well as several other types of malignancies. Retinoblastoma is associated with the inheritance of a copy of chromosome-13 from which a particular segment has been deleted. It develops through a mutation in a retinoblast (a retinal precursor cell). The affected chromosomal segment contains the **RB** gene which specifies a factor that restrains uninhibited cell proliferation. Thus the **RB** gene is a **tumour suppressor gene**. See also **retinoblastoma protein**.

retinoblastoma protein abbr.: pRb; the product of the retinoblastoma gene, **RB**. pRb is a 105 kDa DNA-binding protein that is localized in the nucleus of normal retinal cells but absent in **retinoblastoma** cells. The protein is phosphorylated in the cell cycle by protein kinase **CDC2**. It appears that the interaction of pRb with various **transcription factors** suppresses cellular proliferation. Example: database code RB_HUMAN, 928 amino acids (106.04 kDa).

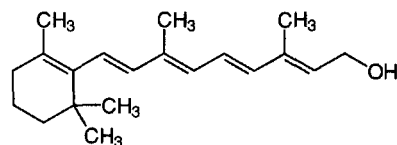
retinoic acid the recommended trivial name for (2E,4E,6E,8E)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1-en-1-yl)nona-2,4,6,8-tetraenoic acid; other names: **tretinoin**; vitamin A acid; vitamin A₁ acid; it is formed from **retinal** by enzymic oxidation in kidney and other tissues, and is active in preventing some, but not all, of the symptoms of vitamin A deficiency. It is a morphogen and regulator of differentiation during embryogenesis. See also **retinoic acid receptor**, **vitamin A**.



retinoic acid receptor a nuclear receptor for **retinoic acid** that mediates its effects on cells. These receptors bind retinoic acid and directly regulate gene expression. The proteins belong to the steroid/thyroid/retinoic acid family of nuclear receptors for hormones. The ligand-binding domain is in the C-terminal region, the zinc-finger DNA-binding domain in a central region, and a function-modulating domain in the N-terminal region. Binding of hormone induces changes in receptor conformation that control transcriptional activation and repression, and also regulates homo- or heterodimerization. In the absence of ligand, these receptors repress basal gene expression, probably through co-repressor proteins. α , β , γ and δ types of receptor are known. Example, α -1 receptor from *Mus musculus*: database code RRA1_MOUSE, 462 amino acids (50.73 kDa).

retinoid any member of a class of isoprenoids, whether naturally occurring or not, that contain or are derived from four prenyl groups linked in a head-to-tail manner. All retinoids may formally be derived from a monocyclic parent compound with 20 carbon atoms containing five carbon-carbon double bonds and a functional group at the terminus of its acyclic moiety.

retinol the recommended trivial name and international nonproprietary name for (2E,4E,6E,8E)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1-en-1-yl)nona-2,4,6,8-tetraen-1-ol; other names: **retinol₁**; vitamin A; vitamin A₁; vitamin A alcohol; vitamin A₁ alcohol; axerophthol; axerol; the predominant form of **vitamin A** in higher vertebrates and marine fishes (compare **3,4-didehydroretinol**). It is stored primarily in the liver, almost entirely as the ester, **retinyl palmitate**.



all-trans form

retinoyl the acyl group formally derived from **retinoic acid**.

retinyl the alkyl group formally derived from **retinol** by loss of its hydroxyl group.

retro+ prefix signifying backwards, or in the reverse direction.

retroendocytosis the process by which some particles, e.g. low-density lipoproteins, are released, after receptor-binding and endocytosis, back into the medium rather than being degraded within lysosomes.

retrogene a segment of DNA that has arisen by reverse transcription of the genetic material in an RNA virus (**retrovirus**) and incorporation of the DNA so formed into the host cell's genomic DNA.

retrogradation the spontaneous irreversible change of a liquid colloidal solution to an insoluble or gelled state. The term is used especially of aqueous amylose solutions in which microcrystalline precipitates of aligned, hydrogen-bonded amylose molecules occur on standing.

retrograde 1 moving backwards; reverse or inverse; tending towards a former state or condition; deteriorating. 2 to move backwards or in reverse, as in retrograde transport.

retrogress to move backwards or in a less favourable direction; to deteriorate or degenerate. —**retrogression** *n.*; **retrogressive** *adj.*

retroinhibition an alternative name for **feedback inhibition**.

retropepsin EC 3.4.23.16; a **pepsin**-type endopeptidase from **HIV** that catalyses the hydrolysis of Xaa-Y-Xbb bonds, where Xaa is a hydrophobic residue, and Xbb is variable, but often is Pro. It is part of the polypeptide product of the **pol** gene. This polypeptide also contains reverse transcriptase and ribonucle-



retroposon

ase H. Example from HIV-1: database code NRL_1HHP, 99 amino acids (10.78 kDa); 3-D structure known.

retroposon or **retrotransposon** a **transposable element** that uses a mechanism indistinguishable from part of a retroviral life cycle. Retroposons are present in a wide variety of organisms, such as yeast, *Drosophila*, and mammals. A good example is the Ty1 element of yeast: after the formation of a double-stranded DNA by **reverse transcriptase** it is integrated into a random site on the chromosome.

retroregulation the regulation of translation of messenger RNA (mRNA) by a sequence that is downstream of the region undergoing translation. For example, an mRNA of phage lambda carrying the sequence of gene *int* is not normally translated; mutation in a site called *sib*, located downstream of the *int* coding sequence, allows translation of the *int* mRNA.

retrotransposon an alternative name for **retroposon**.

Retrovir see **zidovudine**.

retroviral oncogene see **oncogene**, **retrovirus**.

retrovirus or **ribodeoxyvirus** or **RNA-DNA virus** any virus belonging to the family Retroviridae. This comprises single-stranded RNA animal viruses characterized by two unique features: firstly, their genome is diploid, consisting of two copies of the RNA; secondly, this RNA is transcribed by the virion-associated enzyme **reverse transcriptase** into double-stranded DNA. This double-stranded DNA, or **provirus**, can then integrate into the host genome and be passed from parent cell to progeny cell as a stably integrated component of the host genome. The virion is an enveloped, roundish particle, ≈ 100 nm in diameter, with a helical nucleocapsid containing 6 MDa of RNA, consisting of two identical molecules of ≈ 3 MDa each, joined to their 5' ends and hydrogen-bonded into loops (70S RNA), plus small (45S) RNA from the host. The genome is bounded at each end by long terminal repeat sequences and comprises at least three genes: *gag*, which encodes for the inner capsid protein; *pol*, which codes for a polyprotein containing **retropepsin**, ribonuclease H (see **ribonuclease**), and **reverse transcriptase**; and *env*, which codes for the spikes on the envelope. Members of a subfamily of retroviruses, known as the Oncovirinae (or the RNA tumour viruses), contain an additional **oncogene**, which confers on the virus the ability to form tumours in the host. These oncoviruses include the Rous sarcoma virus and other leukosis viruses of birds, and a number of viruses causing leukemia in various mammals (not in humans). Tumour-producing retroviruses have also been called **oncornaviruses**, **rousviruses**, and **leukoviruses**. Other subfamilies include the Spumavirinae (spumaviruses) and the Lentivirinae (lentiviruses); **HIV** is a member of the latter. The retroviruses are in class VI of the **Baltimore classification**.

reverse dialysis a technique for the concentration of macromolecules in solution. The (dilute) solution is placed in a bag made of a semipermeable material, through which the macromolecules cannot pass but water and other small molecules can. The bag is placed in a bed of dry water-soluble polymer (which cannot enter the membrane) such as polyethylene glycol. Water, together with other small molecules, is then drawn out of the bag into the external phase until equilibrium is reached, thereby concentrating the solution of macromolecules in the bag. *Compare* **dialysis**.

reversed phase or **reverse phase** a system used in **adsorption chromatography** (often thin-layer or high-pressure liquid chromatography), in which the **stationary phase** consists of a material with a hydrophobic surface. In such a system, molecules migrate in order of polarity, the most polar migrating fastest. In column chromatography, mobile phases are used in a sequence commencing with more polar solvents and proceeding in decreasing order of polarity, the reverse of the sequence used in normal phase chromatography. The stationary phase often consists of silicic acid in which all hydrophilic groups are masked by silanization.

reversed rocket immunoelectrophoresis a variant of

rocket immunoelectrophoresis in which antibodies are forced by electrophoresis through antigen-containing agarose gel, the heights of the 'rockets' so formed being related to the antibody concentrations. It is useful for titre determinations in large series of antibody samples.

reverse genetics another name for **positional cloning** a term coined to describe the study of the biological consequences to an organism of artificial alteration of its DNA. The changes may be either gross (e.g. deletion, insertion, or transposition of segments of DNA effected by recombinant DNA technology) or limited to specific loci (e.g. base substitution through site-directed mutagenesis).

reverse gyrase an ATPase enzyme with a **DNA helicase**-like domain and a type-I (**DNA**) **topoisomerase** in the same protein. It is able to induce positive supercoiling into DNA, a property of DNA that is found only in bacteria that express this enzyme. Example (deduced from DNA sequence) from *Sulfolobus acidocaldarius* (an archaeobacterium): database code SSOREVGYRA, 780 amino acids (87.38 kDa).

reverse mutation a heritable change in a mutant gene that restores a nucleotide sequence to that which obtained prior to mutation. *Compare* **back mutation**, **reversion**.

reverse osmosis the pressure-induced flow of solvent through a semipermeable membrane in a direction that enhances any existing difference in the activities (or concentrations) of the nonpermeant solutes on the two sides of the membrane (*compare* **osmosis**). The principle can be used in the preparation of purified water from tap water or sea water.

reverse phase see **reversed phase**.

reverse pinocytosis see **exocytosis**.

reverse transcriptase EC 2.7.7.49; *recommended name*: RNA-directed DNA polymerase; *systematic name*: deoxynucleoside-triphosphate: DNA deoxynucleotidyltransferase (RNA-directed); *other names*: DNA nucleotidyltransferase (RNA-directed); revertase. A DNA polymerase enzyme, found particularly in retroviruses and possibly in normal animal cells, that uses either DNA or RNA as a template. It catalyses a reaction between a deoxynucleoside triphosphate and DNA_n to form pyrophosphate and DNA_{n+1}; this protein also has **ribonuclease H** (RNase H) activity. When the single-stranded RNA of the retrovirus enters a cell, the enzyme first makes a DNA copy of one of the **long terminal repeats** of the viral genomic RNA strand using a tRNA as a primer. The stretch of RNA used as a template is then digested by the RNase H activity of the enzyme, releasing the newly synthesized stretch of DNA which then pairs with a complementary stretch of viral RNA to form a DNA-RNA hybrid. This is then used by the same enzyme to make a double-stranded DNA which circularizes and integrates into a host cell chromosome. Examples: (i) AZT-resistant variant of HIV enzyme, database code A47175, 559 amino acids (64.32 kDa); 3-D structures known for parts of the HIV enzyme (database codes NRL_1HRHA2, NRL_1HRHB2); (ii) from *Escherichia coli* retron EC67: database code RT67_ECOLI, 586 amino acids (67.16 kDa). Reverse transcriptase lacks a proof-reading exonuclease function and is highly error-prone. It can thus evolve even within a single patient, and this interferes with long-term therapy. Reverse transcriptase is used in cDNA technology to make cDNA from RNA. **Telomerase** is a template-bearing reverse transcriptase.

reverse transcription the process carried out by **reverse transcriptase**. See also **transcription**.

reverse turn (*in protein structure*) a structural feature that allows the peptide chain to turn back through about 180° by what is termed a **beta bend**.

reversible competitive antagonist (*in pharmacology*) an antagonist for which the rate of dissociation from its receptor is relatively high, such that by increasing the concentration of an agonist the receptor **occupancy** (def. 2) of the antagonist falls. At sufficiently high concentrations of agonist a maximal tissue response can be obtained, even in the presence of the an-



reversible inhibition

tagonist, such that the extent of antagonism is determined by the relative concentrations of agonist and antagonist.

reversible inhibition (of an enzyme or other system) a type of inhibition that occurs when the equilibrium between free inhibitor and the inhibitor–enzyme (or system) complex lies sufficiently far towards the free inhibitor state for the inhibitor to be readily removed, leaving an active enzyme (or system).

reversible reaction a reaction in which the **equilibrium constant** is such that the reaction can be made to proceed at a detectable rate in either direction under appropriate conditions.

reversion the restoration, by a second mutation, of the genomic base sequence of a mutant virus or organism to that of the parental strain. Less strictly, any return to the original **phenotype**. *See also* **back mutation**, **reverse mutation**.

reversion frequency the proportion of the total number of cells in a population that undergo **reversion**.

reversion spectroscopy a **spectroscopy** in which two spectra alongside each other, one of which is reversed by optical means, are formed by the entering light. The two spectra may be moved longitudinally relative to each other by moving a micrometer screw. If two absorption bands are aligned when a particular substance, e.g. oxyhemoglobin, is viewed, the amount by which the screw has to be turned to align the same band when a mixture of the same substance and another with a similar absorption spectrum, e.g. carbonmonoxyhemoglobin, is viewed is related to the proportion of the second substance present.

revertant 1 a mutant that has partially or completely regained the wild-type phenotype. 2 a gene that has undergone **reversion**.

revertase an alternative name for **reverse transcriptase**.

Reynolds number *symbol*: *Re*; a nondimensional parameter of fluid motion through a cylindrical tube or past an obstruction. It is given by the expression vl/ν , where *l* has the dimensions of length, *v* is the fluid's velocity, and ν is the kinematic viscosity. In general, low Reynolds numbers show that viscous forces predominate in controlling the flow, while high Reynolds numbers indicate that inertial forces are more important. [After Osborne Reynolds (1842–1912), Irish physicist and engineer.]

RF *abbr.* for 1 release factor. 2 replicative form.

R factor 1 *abbr.* for resistance factor (i.e. **resistance plasmid**). 2 (formerly) *abbr.* for release factor.

rfDNA *see* **DNA**.

RFLP *abbr.* for restriction fragment length polymorphism.

R form 1 *abbr.* for relaxed form (of a protein). 2 **R-form** (hyphenated) *abbr.* for rough form; describing bacterial colonies with a jagged perimeter.

RGD the amino-acid sequence Arg–Gly–Asp, which is present in **fibronectin** and some other extracellular matrix proteins and is recognized by some **integrins** that bind these proteins.

RGK-2 *see* **tonin**.

rH the negative logarithm of the concentration (expressed as pressure in atmospheres) of hydrogen gas in equilibrium with a redox system; i.e.

$$rH = -\log[H_2].$$

Also,

$$rH = 2(E_h F)/(RT \ln 10) + pH,$$

where *F* is the Faraday constant, *R* the gas constant, *T* the thermodynamic temperature, and *E_h* the **redox potential** at the pH of the system under study.

Rh 1 *abbr.* for Rhesus factor. 2 *symbol* for rhodium.

Rha *symbol* for rhamnose.

Rhabdoviridae a family of single-stranded RNA viruses infecting vertebrates, arthropods, amoebae, and plants, whose members are identified by their elongated rod or bullet-shaped appearance. The virions are 70–80 nm particles, covered with a lipid envelope. They contain negative-strand RNA together with RNA-dependent RNA polymerase, which transcribes the genome directly to mRNA – they are class IV viruses in the

Baltimore classification. Rhabdoviruses are usually transmitted by arthropods; they include the plant rhabdoviruses, the vesiculoviruses (e.g. the vesicular stomatitis virus of cattle), the lyssaviruses, which include the rabies virus, and sigma virus of *Drosophila*. [From Greek *rhabdos*, rod.]

rhabdovirus *see* **Rhabdoviridae**.

L-rhamnosamine 2-amino-2,6-dideoxy-L-mannose; an amino sugar that occurs in some bacterial lipopolysaccharides.

rhamnose *symbol*: Rha; 6-deoxy-L-mannose; a monosaccharide (hexose) that occurs commonly as a compound of plant glycosides, in polysaccharides of gums and mucilages, and in bacterial polysaccharides; it is also a component of some plant cell-wall polysaccharides, and frequently acts as the sugar component of flavonoids. Free rhamnose is also found in some plants. The D-enantiomer occurs in some bacterial capsular polysaccharides. *See also* **L-rhamnose isomerase**.

L-rhamnose isomerase EC 5.3.1.14; *systematic name*: L-rhamnose ketol-isomerase; an enzyme that catalyses the interconversion of L-rhamnose and L-rhamnulose. It initiates the catabolism of rhamnose. L-rhamnulose is then phosphorylated by rhamnulose kinase, EC 2.7.1.5, to L-rhamnulose-1-phosphate which is cleaved by rhamnulose-1-phosphate aldolase, EC 4.1.2.19, to glyceraldehyde phosphate and (S)-lactaldehyde. Example from *Escherichia coli*: database code RHAA_ECOLI, 419 amino acids (47.12 kDa).

rhamnulose 6-deoxyfructose; its formation by **L-rhamnose isomerase** initiates rhamnose catabolism.

rheology the science concerning the deformation and flow of matter. —**rheological** *adj.*

rheosome or **flow signal** a chemical messenger or transmitter substance (other than a neurotransmitter) carried from a site of generation to site of action within a cell (e.g. a neuron) by cytoplasmic flow; a translocated **biosem** or **second messenger**.

rheostat a device in which a controllable resistance is introduced into an electrical circuit to vary the output to permit an apparatus to be regulated within desired limits.

Rhesus factor *abbr.*: Rh; a protein associated with the rhesus (Rh) system of blood groups in humans. The system was so designated as a result of the finding that rabbit and guinea-pig antibodies raised against erythrocytes from *Macacus rhesus* monkeys agglutinated erythrocytes from 85% of the random human population. Initially an antigen, designated D, was isolated from the erythrocytes of Rh-positive individuals; those lacking D were classified Rh-negative. Subsequently, another antigen, LW (Landsteiner-Weiner), was identified, which, although phenotypically similar to D, is encoded by a gene that segregates independently from Rh. C and E antigens have also been found. The antigenic determinant is associated with the protein structure. In hemolytic disease of the newborn due to rhesus incompatibility, antibodies from a Rh-negative mother, often formed during the birth of a previous Rh-positive child whose blood has entered the mother's bloodstream, pass across the placenta and cause destruction of fetal red cells.

rhinovirus any of a group of small RNA viruses belonging to the Picornaviridae family, members of which are responsible for the common cold in humans, and foot-and-mouth disease in cattle.

Rhizobium a genus of soil bacteria whose members reside in specialized root nodules of leguminous plants and set up a symbiosis. The bacteria are active in **nitrogen fixation** providing the plant with organic nitrogen and obtaining carbon compounds in return. Many if not all *Rhizobium* species contain plasmids, which enable them to colonize their host plant species.

rhizomorph a stout, rootlike mass of hyphae, produced by some fungi, that serves to generate vegetative spread.

rho *symbol*: ρ (lower case) or P (upper case); the seventeenth letter of the Greek alphabet. For uses *see* **Appendix A**. *See also* **rho protein**, **Rho protein**.

Rhodamine any of a series of dyes based on 9-(2-carboxyphenyl)anthracene. Rhodamine 6G, N-[9-(2-carboxyethyl)phenyl]-

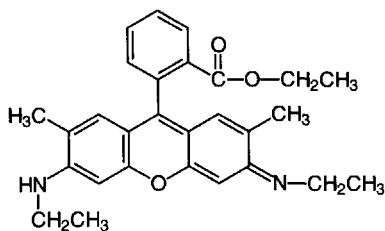
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rhodanese

6-(diethylamino)-3*H*-xanthen-3-ylidene]ethylethanamine, is used as a spray to reveal lipids as pink fluorescent spots after thin-layer chromatography.



Rhodamine 6G

rhodanese EC 2.8.1.1; *recommended name*: thiosulfate sulfur-transferase; *systematic name*: thiosulfate:cyanide sulfur-transferase; *other names*: thiosulfate cyanide transsulfurase; thiosulfate thiotransferase. An enzyme that catalyses the displacement of sulfite from thiosulfate by cyanide ion, forming sulfite and thiocyanate and thus detoxifying cyanide. It is of interest structurally in having two domains of closely similar conformation yet very different sequence. Example from *Gallus gallus* (chicken): database code THTR_CHICK, 289 amino acids (32.25 kDa). [From the use of rhodanate as a name for thiocyanate.]

rhodanid any of a group of **thiocyanates** produced by the metabolism of mustard-oil glycosides (glycosinolates) in damaged leaf tissues.

rhodium *symbol*: Rh; a silvery-white metallic transition element of group VIII of the (IUPAC) **periodic table**; atomic number 45; relative atomic mass 102.9. It occurs mainly in oxidation state III. A rare member of the platinum metals, it possesses marked properties of nobility, being, e.g., resistant to nitric acid and aqua regia.

rhodo+ or (before a vowel) **rhod+** *comb. form* indicating a red colour, or the odour of a rose. [From Greek *rhodos*, rose.]

rhodopsin or **visual purple** a brilliant purplish-red, light-sensitive visual pigment found in the rod cells of the retinas of most vertebrates (exceptions being freshwater and migratory fish and some amphibians). It is a seven-transmembrane-domain protein homologous to other members of the G-protein-linked receptor family. It consists of a rod-type **opsin** combined (as a Schiff's base) with 11-*cis*-retinal (see **retinal**). It has an absorption maximum at ≈ 500 nm. *Compare* **porphyropsin**. A single amino-acid substitution in the rhodopsin molecule causes one form of **retinitis pigmentosa** in humans. *See also* **bacteriorhodopsin**, **halorhodopsin**, **slow-cycling rhodopsin**.

rho factor an alternative name for **rho protein**.

rho protein or (sometimes) **rho factor** a bacterial protein, an ATP-dependent helicase, that facilitates transcription termination. Binding of rho to the nascent RNA is followed by activation of rho's RNA-dependent ATPase activity, and release of the mRNA from the DNA template; it recognizes and binds to C-rich regions of the RNA transcript. In *Escherichia coli* this protein is the site of action of the antibiotic bicyclomycin, which binds to rho protein and inhibits its ATPase activity. Example from *E. coli*: database code RHO_ECOLI, 419 amino acids (46.95 kDa). *Compare* **Rho protein**.

Rho protein a small Ras-related GTPase (see **RAS**), involved in controlling the polymerization of actin into filaments, and subsequent organization. **RHOB** is an early response gene transiently induced in fibroblasts by platelet-derived growth factor or epidermal growth factor; its overexpression can be tumorigenic. *Compare* **rho protein**.

RI *abbr.* for refractive index.

RIA *abbr.* for radioimmunoassay.

Rib *symbol* for ribose.

ribaric acid one of the **pentaric acids**. *See also* **aldaric acid**.

ribitol or (formerly) **adonitol** a **pentitol** derived formally by reduction of the -CHO group of either D- or L-ribose. It is a *meso* compound and does not require a D or L configurational descriptor. It occurs free in some plants and is a component of **riboflavin**.

ribitol teichoic acid a complex polymer of **ribitol** residues linked by phosphate groups, substituted by glucosyl or acetylglucosamine residues and ester-linked to D-alanine. It is found in the walls of Gram-positive bacteria.

ribo+ *comb. form* 1 of, relating to, or containing **ribose**, especially as opposed to deoxyribose. 2 containing a ribose residue linked through its anomeric carbon. It is used also in **riboflavin**, inappropriately, for a ribitol residue.

ribo- *prefix (in chemical nomenclature)* indicating a particular configuration of three contiguous >CHOH groups (as found in the acyclic form of D- or L-ribose). *See also* **monosaccharide**.

ribocharin an acidic protein specifically associated with the granular component of the nucleolus and nucleoplasmic 65S particles. These particles contain the nuclear 28S rRNA and may represent the precursor to the large ribosomal subunit.

ribodeoxyvirus (sometimes) any RNA virus that replicates via DNA; such viruses are now usually known as **retroviruses**.

Compare **deoxyribovirus**.

riboflavin an alternative name for **vitamin B₂**.

ribofuranosyl an alternative term for **ribosyl**.

ribonuclease *abbr.*: RNase or (sometimes) RNAase; any of a group of nuclease enzymes that cleave phosphodiester bonds in chains of RNA. With the exception of RNase P, which is a **ribozyme**, they are all proteins. As in the case of other enzymes that hydrolyse macromolecules, RNases are classified according to whether they are **exoribonucleases**, catalysing sequential cleavage of mononucleotides from either a free 5' or 3' terminus, or both, or **endonucleases**, cleaving bonds within the polynucleotide chain. Endonucleases, which unlike exonucleases can cleave circular molecules of RNA, fall into further categories according to whether they cleave only specific sites or at any phosphodiester linkage. Enzymes that cleave RNA fall into eight subclasses: EC 3.1.13, exoribonucleases producing 5'-phosphomonoesters; EC 3.1.14, exoribonucleases producing other than 5'-phosphomonoesters; EC 3.1.15, exonucleases active with either ribo- or deoxyribonucleic acids and producing 5'-phosphomonoesters; EC 3.1.16, exonucleases active with either ribo- or deoxyribonucleic acids and producing other than 5'-phosphomonoesters; EC 3.1.26, endoribonucleases producing 5'-phosphomonoesters; EC 3.1.27, endoribonucleases producing other than 5'-phosphomonoesters; EC 3.1.30, endonucleases active with either ribo- or deoxyribonucleic acids and producing 5'-phosphomonoesters; and EC 3.1.31, endonucleases active with either ribo- or deoxyribonucleic acids and producing other than 5'-phosphomonoesters. Ribonucleases include enzymes having mechanisms of (A) phosphotransferases, (B) phosphodiesterases, or (C) phosphorylases. In group C are polynucleotide phosphorylase and polynucleotide pyrophosphorylase. The differences between groups A and B are more than pedantic. Thus the feature of group A is the involvement of the 2'-OH group in an intramolecular attack at the adjacent phosphodiester bond, in addition to a property for conferring specificity for one or more of the four bases. The nucleoside 2':3'-cyclic phosphates are obligate intermediate products. In group B, a direct attack of water on the 3':5'-phosphodiester bond is catalysed. This latter group includes, therefore, the enzymes that hydrolyse both RNA and DNA and those that cleave diester bridges in RNA to form only 5'-nucleotide products. Thus group A enzymes may be classed as cyclizing and group B enzymes as non-cyclizing. The group A enzymes do not depend for their activity on divalent cations whereas the group B enzymes require Mg²⁺ or Ca²⁺. Within each group of enzymes there are many types, as described below.

Exoribonucleases catalyse the sequential cleavage of mono-

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nucleotides from one or other, or both, ends of the polynucleotide. **Exoribonuclease II** EC 3.1.13.1; *other name*: ribonuclease II; reaction: exonucleolytic cleavage in the 3'- to 5'-direction to yield 5'-phosphomononucleotides. **Exoribonuclease H** EC 3.1.13.2; reaction: exonucleolytic cleavage to 5'-phosphomonoester oligonucleotides in both 5'- to 3'- and 3'- to 5'-directions. Example from *E. coli*: database code RNB_ECOLI, 644 amino acids (72.4 kDa). **Poly(A)-specific ribonuclease** EC 3.1.13.4; reaction: exonucleolytic cleavage of poly(A) to 5'-AMP. **Yeast ribonuclease** EC 3.1.14.1; reaction: exonucleolytic cleavage to 3'-phosphomononucleotides.

Endoribonucleases cleave the polynucleotide internally. This first group (sub-subclass EC 3.1.26) produce exclusively 5'-phosphomonoesters. **Physarum polycephalum ribonuclease** EC 3.1.26.1; reaction: endonucleolytic cleavage to 5'-phosphomonoester. **Ribonuclease alpha** EC 3.1.26.2; reaction: endonucleolytic cleavage to 5'-phosphomonoester. **Ribonuclease III** EC 3.1.26.3; *other names*: RNase O; RNase D; reaction: endonucleolytic cleavage to 5'-phosphomonoester. Example from *Rhodobacter capsulatus*: database code RCLEPRNCG, 265 amino acids (28.8 kDa). **Ribonuclease H** EC 3.1.26.4; *other names*: endoribonuclease H (calf thymus); calf thymus ribonuclease H; reaction: endonucleolytic cleavage to 5'-phosphomonoester; required for removing RNA from **Okazaki fragments**; *see also* **RNA-directed DNA polymerase**. **Ribonuclease P** EC 3.1.26.5; reaction: endonucleolytic cleavage of RNA, removing 5'-sequence of extra nucleotides from tRNA precursor; it is a **ribozyme**; *see* **RNase P**. **Ribonuclease IV** EC 3.1.26.6; *other names*: endoribonuclease IV; poly(A)-specific ribonuclease; reaction: endonucleolytic cleavage of poly(A) to fragments terminated by 3'-hydroxyl and 5'-phosphate groups. Example from yeast: PAN1_YEAST, 1480 amino acids (160 kDa). **Ribonuclease P4** EC 3.1.26.7; reaction: endonucleolytic cleavage of RNA, removing 3'-sequence of extra nucleotides from tRNA precursor. **Ribonuclease M5** EC 3.1.26.8; *other name*: 5S rRNA maturation nuclease; reaction: endonucleolytic cleavage of RNA, removing 21 and 42 nucleotides, respectively, from the 5'- and 3'-termini of a 5S rRNA precursor. **Ribonuclease (poly(U)-specific)** EC 3.1.26.9; reaction: endonucleolytic cleavage of poly(U) to fragments terminated by 3'-hydroxyl and 5'-phosphate groups. **Ribonuclease IX** EC 3.1.26.10; reaction: endonucleolytic cleavage of poly(U) or poly(C) to fragments terminated by 3'-hydroxyl and 5'-phosphate groups.

The second group of endoribonucleases (sub-subclass EC 3.1.27) yield products other than 5'-phosphomonoesters. **Ribonuclease T2** EC 3.1.27.1; *other name*: ribonuclease II; reaction: two-stage endonucleolytic cleavage to 3'-phosphomononucleotides and 3'-phosphooligonucleotides with 2',3'-cyclic phosphate intermediates. Example (glycoprotein precursor) from *Aspergillus oryzae*: database code RTN2_ASPOR, 276 amino acids (30.71 kDa); residues 1-17 are the signal. **Bacillus subtilis ribonuclease** EC 3.1.27.2; reaction: endonucleolytic cleavage to 2',3'-cyclic nucleotides. **Ribonuclease T1** EC 3.1.27.3; *other names*: guanyloribonuclease; *Aspergillus oryzae* ribonuclease; ribonuclease N1; ribonuclease N2; reaction: two-stage endonucleolytic cleavage to 3'-phosphomononucleotides and 3'-phosphooligonucleotides ending in G-p with 2',3'-cyclic phosphate intermediates. Database code NRL_5RNT, 104 amino acids (11.08 kDa); 3-D structure known, including bound intermediate. **Ribonuclease U2** EC 3.1.27.4; reaction: two-stage endonucleolytic cleavage to 3'-phosphomononucleotides and 3'-phosphooligonucleotides ending in Ap or Gp with 2',3'-cyclic phosphate intermediates. Example from the smut fungus, *Ustilago spheerogena*: database code RNU2_USTSP, 113 amino acids (12.26 kDa). **Pancreatic ribonuclease** EC 3.1.27.5; *other names*: ribonuclease; ribonuclease I; ribonuclease A; reaction: endonucleolytic cleavage to 3'-phosphomononucleotides and 3'-phosphooligonucleotides ending in Cp or Up with 2',3'-cyclic phosphate intermediates. Example (bovine): database

code NRL_3RN3, 124 amino acids (13.68 kDa); 3-D structure known. **Enterobacter ribonuclease** EC 3.1.27.6; reaction: endonucleolytic cleavage to 3'-phosphomononucleotides and 3'-phosphooligonucleotides with 2',3'-cyclic phosphate intermediates. Example (precursor) from *E. coli*: database code RNI_ECOLI, 268 amino acids (29.58 kDa); residues 1-23 are the signal sequence. **Ribonuclease F** EC 3.1.27.7; reaction: endonucleolytic cleavage of RNA precursor into two, leaving 5'-hydroxyl and 3'-phosphate groups. **Ribonuclease V** EC 3.1.27.8; *other name*: endoribonuclease V; reaction: hydrolysis of poly(A), forming oligoribonucleotides and ultimately 3'-AMP. *Compare deoxyribonuclease*.

ribonuclease 1 *abbr.*: RNA; any anionic form of ribonucleic acid. **2** any salt of ribonucleic acid.

ribonucleic acid *abbr.*: RNA; *former names*: pentose nucleic acid (*abbr.*: PNA); ribose nucleic acid; yeast nucleic acid. *See* RNA.

ribonucleoprotein *abbr.*: RNP; any conjugated protein that contains RNA as the nonprotein moiety. Various specific kinds of RNP are listed with their abbreviations at **RNP**.

ribonucleoside any **nucleoside** (def. 1) in which the glycosyl moiety is β -D-ribofuranose. The link is from C-1 of ribose to N-9 of a purine or to N-1 of a pyrimidine. To distinguish between the numbering systems of the base and sugar, the numbers of the sugar atoms are characterized by the addition of 'prime', i.e. 1' to 5'. The various products have trivial names, e.g. adenosine, guanosine, cytidine, and uridine. Strictly, the term ribonucleoside should be confined to structures occurring in nucleic acids, but it can be used for any base-ribose compound. By extension, it can be applied to **pseudouridine** (which occurs in tRNA) with a C-C link between ribose and uracil.

ribonucleoside-diphosphate reductase *the recommended name for nucleoside-diphosphate reductase*.

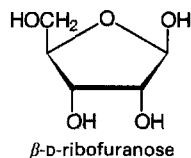
ribonucleoside-triphosphate reductase *the recommended name for nucleoside-triphosphate reductase*.

ribonucleotide any ribonucleoside in ester linkage to phosphate, commonly at the 5' position of the ribose moiety, although some 2'- and 3'-esters and some (cyclic) 2',3'-, or 3',5'-diesters are known. The 5'-ribonucleotides are metabolically important; they may be esters of [mono]phosphate, diphosphate, or triphosphate, e.g. guanosine 5'-[mono]phosphate, cytidine 5'-diphosphate, adenosine 5'-triphosphate. *See* **nucleotide**.

ribonucleotide reductase any enzyme that catalyses the reduction of ribonucleotides to deoxyribonucleotides during biosynthesis. *See* **nucleoside-diphosphate reductase**, **nucleoside-triphosphate reductase**.

ribophorin either of two types of protein, ribophorins I and II, that form two essential subunits of **N-oligosaccharyltransferase**. They were originally isolated as ribosome-binding proteins of rough endoplasmic reticulum. Examples from human; ribophorin I: database code RIB1_HUMAN, 607 amino acids (68.57 kDa); ribophorin II: database code RIB2_HUMAN, 631 amino acids (69.30 kDa).

ribose *symbol*: Rib; *ribo*-pentose. The D enantiomer, as β -D-ribofuranose, forms the glycosyl moiety of all ribonucleosides, ribonucleotides, and ribonucleic acids, and also of ribose phosphates, various glycosides, some coenzymes, and some forms of vitamin B₁₂.



ribose nucleic acid *a former name for ribonucleic acid*.

ribose-5-phosphate epimerase EC 5.3.1.6; *systematic name*: D-ribose-5-phosphate ketol-isomerase; *other names*: phospho-



ribose-phosphate pyrophosphokinase

pentoisomerase; phosphoriboisomerase. An enzyme that catalyses the reaction:

D-ribose 5-phosphate = D-ribulose 5-phosphate.

It is an enzyme of pentose metabolism, especially the **pentose phosphate pathway**. Example from *Escherichia coli*: database code RPIA_ECOLI, 219 amino acids (22.83 kDa).

ribose-phosphate pyrophosphokinase EC 2.7.6.1; *other name*: phosphoribosyl pyrophosphate synthetase (*abbr.*: PRS); an enzyme that catalyses the formation of 5-phospho- α -D-ribose-1-diphosphate (*abbr.*: PRPP; *other name*: phosphoribosyl pyrophosphate) from ATP and D-ribose 5-phosphate with release of AMP; it is Mg^{2+} -dependent. PRPP is an intermediate in the biosynthesis of histidine and tryptophan, and in the *de novo* biosynthesis of purines and pyrimidines. The enzyme has a well-conserved sequence. Examples: in human there are two isoforms: PRS-II (also in rat): database code KPR1_HUMAN, 317 amino acids (34.66 kDa); PRS-II: database code KPR2_HUMAN, 317 amino acids (34.60 kDa).

riboside the product of the formation of a glycosidic bond between ribose and another group, either through the nitrogen of an amine or a hydroxyl group, i.e. via N- or O-glycosidic linkages. *See also pseudouridine. Compare ribonucleoside.*

ribosomal DNA *abbr.*: rDNA; the DNA of the genes coding for ribosomal RNA (*see ribosome*).

ribosomal protein *abbr.*: r-protein; any of the protein components of the ribosomal subunits. They are typically small, and some are basic. Ribosomal proteins are referred to as S (small subunit) proteins and L (large subunit) proteins; in *Escherichia coli* there are 21 S proteins, designated S1–S21, and 34 L proteins, designated L1–L34. In eukaryotes (e.g. rat liver) there are 33 S proteins and 49 L proteins. *See also ribosome.*

ribosomal RNA *abbr.*: rRNA; *see ribosome*.

ribosome an intracellular organelle, about 200 Å in diameter, consisting of RNA and protein. It is the site of protein biosynthesis resulting from translation of messenger RNA (mRNA). It consists of two subunits, one large and one small, each containing only protein and RNA. Both the ribosome and its subunits are characterized by their sedimentation coefficients, expressed in **Svedberg units** (*symbol*: S). Hence, the prokaryotic ribosome (70S) comprises a large (50S) subunit and a small (30S) subunit, while the eukaryotic ribosome (80S) comprises a large (60S) subunit and a small (40S) subunit. Several ribosomes may bind to one mRNA to form a **polysome**. Two sites on the ribosomal large subunit are involved in translation, namely the **aminoacyl site** (A site) and **peptidyl site** (P site). Ribosomes from prokaryotes, eukaryotes, mitochondria, and chloroplasts have characteristically distinct **ribosomal proteins**. Prokaryotic ribosomes contain 35–45% protein, while eukaryotic ribosomes contain 50% protein. **Ribosomal RNA** (rRNA) is RNA that forms part of the ribosomal structure. In prokaryotes, a 30S ribosomal subunit contains one 16S RNA molecule with associated proteins, while a 50S subunit contains one 5S and one 23S RNA molecule. In eukaryotes, the 60S subunit contains one 28S, one 5.8S, and one 5S RNA molecule, while the 40S subunit contains one 18S RNA molecule. The structures are known. In *Escherichia coli*, the rRNAs are synthesized as part of a large transcript containing transfer RNA (tRNA) molecules, which is then cleaved during processing. In eukaryotes, the genes for the 5.8S, 18S, and 28S molecules reside in the **nucleolus**, and code for a 40S transcript, which is then cleaved during processing; this includes splicing that occurs in the absence of protein. The gene for the 5S rRNA is a separate transcription unit outside the nucleolus. *See also mRNA binding site, protein biosynthesis.*

ribosome-inactivating protein *abbr.*: RIP; any of a group of proteins, obtained from plants, that inactivate eukaryotic ribosomes. They are of two types. Type I proteins have a single chain, and include agrostin, dianthrin, dodecandrin, **gelonin**, luffin, **nomordin**, **saporin**, and **tritritin**. They are enzymes. EC

ribothymidylic

3.2.2.22, that catalyse endohydrolysis of the N-glycosyl bond at one specific adenosine on 28S rRNA. They are plant protein toxins and possibly have antiviral activity. Examples: **luffin**, from the smooth loofah, *Luffa cylindrica*, has two subunits; precursor sequences: database code RIPA_LUFCY, 277 amino acids (30.18 kDa), and database code RIPP_LUFCY, 250 amino acids (27.26 kDa); **saporin-2** from *Saponaria officinalis*: database code RIP2_SAPOF, 292 amino acids (32.77 kDa); the protein from wheat is called **tritritin**, database code S33631, 275 amino acids (29.58 kDa); that from *Phytolacca dodecandra* is **dodecandrin**, fragmentary data in database code S32971. Type II ribosome-inactivating proteins are enzymes with a similar action to the type I proteins, but have an additional covalently bound chain with lectin properties. They include **abrin**, **modeccin**, **ricin**, **viscumin**, and **volkensin**.

ribosome recycling factor *abbr.*: RRF; *formerly known as ribosome releasing factor*; a factor which, in bacteria is required for the recycling of ribosomes through decomposition of the termination complex following release-factor-mediated release of the completed polypeptide chain from the peptidyl tRNA. Example from *Escherichia coli*: database code RRF_ECOLI, 185 amino acids (20.61 kDa).

ribosyl or ribofuranosyl any **glycosyl** group formally derivable from the α - or the β -furanose forms of D- or L-ribose. (The pyranose forms of ribose are exhibited only by the uncombined sugar.)

ribosylate to add a ribosyl residue. *Compare ADP-ribosylation.*

ribosylthymine an alternative name for **ribothymidine**.

ribosylthymine monophosphate an alternative name for **ribothymidine monophosphate**.

ribosylthymine phosphate an alternative name for **ribothymidine phosphate**.

ribothymidine or **ribosylthymine** *symbol*: T or Thd; 5-methyluridine; thymine riboside; 1- β -D-ribofuranosyl-5-methyluracil; a rare ribonucleoside, found as a minor component of many species of transfer RNA, where it is present as a residue of its 5'-phosphate. *See also ribothymidine 5'-phosphate. Compare thymidine.*

ribothymidine monophosphate or **ribosylthymine monophosphate** *abbr.*: TMP; the common name for any ribothymidine phosphate, particularly for ribothymidine 5'-phosphate.

ribothymidine phosphate or **ribosylthymine phosphate** *abbr.*: TMP; any phosphoric monoester or diester of ribothymidine. Of the three possible monoesters – the 2'-phosphate, the 3'-phosphate, and the 5'-phosphate – and the two possible diesters – the 2',3'-phosphate and the 3',5'-phosphate – only **ribothymidine 5'-phosphate** is of any significance (the locant therefore being omitted if no ambiguity may arise).

ribothymidine 5'-phosphate or **ribosylthymine 5'-phosphate** or **5'-ribothymidylic acid** *symbol*: Thd5'P or pT; the alternative recommended names for ribothymidine monophosphate or ribosylthymine monophosphate (*abbr.*: TMP); 5'-ribothymidylyl phosphate; thymine riboside 5'-(dihydrogen diphosphate); thymine (mono)ribonucleotide. (The locant is commonly omitted if there is no ambiguity as to the position of phosphorylation.) A residue of ribothymidylic acid occurs in many species of transfer RNA, where it is formed *in situ* by methylation of the uracil moiety of an existing uridylic acid residue. This modification is effected by tRNA (uracil-5-)-methyltransferase (EC 2.1.1.35), the methyl group being derived from S-adenosyl-L-methionine.

ribothymidylate 1 either the monoanion or the dianion of **ribothymidylic acid**. 2 any mixture of ribothymidylic acid and its anions. 3 any salt or ester of ribothymidylic acid.

ribothymidylic acid the trivial name for any phosphoric monoester of ribothymidine. The position of the phosphoric residue on the ribose moiety of a given ester may be specified by a prefixed locant – *see ribothymidine phosphate*. However, 5'-ribothymidylic acid is the ester commonly denoted, its locant usually being omitted if no ambiguity may arise. 5'-ribo-



thymidylic acid is also an alternative recommended name for ribothymidine 5'-phosphate.

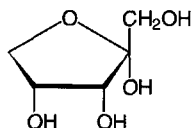
ribothymidyl the ribothymidine[mono]phospho group; the acyl group derived from **ribothymidylic acid**.

ribovirus an alternative name for **viroid**.

ribozyme an RNA molecule with catalytic activity. Such activity was first discovered by Sidney Altman in 1981 for **RNase P**. Thomas R. Cech in 1982 discovered the catalysis by RNA of the reactions involved in the **splicing** of ribosomal RNA from *Tetrahymena*. In this reaction a 414-nucleotide **intron** is removed from a 6.4 kb precursor to yield the mature 26S rRNA molecule. A guanosine nucleotide is essential in the reaction, binding to the RNA and then attacking the 5' splice site to form a phosphodiester bond with the 5' end of the intron. After further reactions the 414-nucleotide intron is released and undergoes two further splicing reactions to produce a linear RNA of 395 nucleotides which has lost 19 nucleotides and is named L19. L19 is catalytically active and is a true enzyme acting both as a nuclease and a polymerase. The rate of conversion of a pentacytidylate into longer and shorter oligomers is about 10^{10} times the uncatalysed rate. Mg^{2+} plays an essential role. Much smaller ribozymes have been demonstrated, e.g. the **viroids** which infect plants and undergo self-splicing after replication. Comparisons of nucleotide sequences in the vicinity of specific cleavage sites suggest that the active site has a **hammerhead** secondary structure consisting of three helical regions radiating from a central core of unpaired nucleotides. mRNA precursors in the mitochondria of yeast and fungi also undergo self-splicing as do some RNA precursors in chloroplasts. Such reactions can be classified according to the nature of the unit that attacks the upstream splice site. Group I are mediated by a guanosine cofactor. In group II the attacking moiety is the 2'-OH of a specific adenylate of the intron. Such reactions resemble those that occur in the **spliceosomes**. The work on ribozymes has a special interest in respect of theories on the origin of life because those interested in the origin of life on earth favour the idea that the first macromolecules were composed of RNA. The fact that RNA can catalyse reactions lends support to their views. See also **ribonuclease**.

(Rib5)ppA symbol for adenosinediphosphoribose (alternative to AdoPPRib or A5'pp5Rib).

D-ribulose a nonsystematic name for the ketopentose D-erythro-2-pentulose.



α -D-furanose form

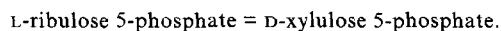
D-ribulose 1,5-bisphosphate abbr.: RBP or RuBP; former name: ribulose diphosphate (abbr.: RDP or RuDP); a key metabolite of the **reductive pentose phosphate cycle**. It is formed in a reaction in which ribulose 5-phosphate is phosphorylated by ATP, catalysed by phosphoribulokinase, EC 2.7.1.19, and is a substrate of **ribulose-bisphosphate carboxylase**.

ribulose-bisphosphate carboxylase EC 4.1.1.39; systematic name: 3-phospho-D-glycerate carboxy-lyase (dimerizing); other names: ribulose-bisphosphate carboxylase/oxygenase (abbr.: Rubisco); the key enzyme in the fixation of CO_2 in the **reductive pentose phosphate cycle**; a copper protein. One molecule of D-ribulose 1,5-bisphosphate reacts with one molecule of CO_2 to yield two molecules of 3-phospho-D-glycerate, which can be converted into a hexose product (fructose 6-phosphate). If, as is possible, D-ribulose 1,5-bisphosphate reacts with O_2 instead of CO_2 , the products are one molecule each of 3-phospho-D-glycerate and 2-phosphoglycolate. This

enzyme comprises up to 50% of leaf proteins and is the most abundant protein in the biosphere. Examples, all from tomato: large chain precursor: database code RBL_LYCES, 477 amino acids (52.62 kDa); small chain 1 precursor (less 17): database code RBS1_LYCES, 181 amino acids (20.28 kDa); small chain 2a precursor (less 5): database code RBS2_LYCES, 180 amino acids (20.25 kDa); small chain 3a/3c precursor: database code RBS3_LYCES, 180 amino acids (20.21 kDa); small chain 3b precursor: database code RBS4_LYCES, 180 amino acids (20.19 kDa).

D-ribulose 5-phosphate the 5-phosphate ester of D-ribulose; it is a component of the **pentose phosphate pathway**, the product of 6-phosphogluconate dehydrogenase. It is converted to ribulose 1,5-bisphosphate in the **reductive pentose phosphate cycle** of photosynthesis.

L-ribulose-phosphate 4-epimerase EC 5.1.3.4; an enzyme that catalyses the reaction:



Example: *araD* gene product from *Escherichia coli*: database code ARAD_ECOLI, 231 amino acids (25.49 kDa).

Richner-Ranhart syndrome see **tyrosine transaminase**.

ricin a highly toxic lectin obtained from the seeds of the castor oil plant, *Ricinus communis*. It is a 493 amino-acid glycoprotein dimer comprising an A chain (M_r 66 000) and a B chain (M_r 34 000) linked by a disulfide bond. Ricin binds to the cell via interaction between the B chain and galactose groups on cell-surface receptors; then the disulfide bond is cleaved allowing the A chain to enter the cell. The A chain then binds to the proteins of the 60S ribosomal subunit, apparently in the region of binding of elongation factors eEF-1 and eEF-2, thereby halting protein translation. Protein from *R. communis* (precursor): database code RIC1_RICCO, 576 amino acids (64.09 kDa). See also **abrin**, **ribosome-inactivating protein**.

ricketts or **rachitis** a disturbance of calcium (and phosphate) metabolism in young growing animals (including humans), resulting from a deficiency of **vitamin D** or of its activation. It is characterized by softening of the bones and skeletal deformities, particularly bowing of the legs.— **rachitic adj.**

RID abbr. for radial immunodiffusion.

RIEP abbr. for rocket immunoelectrophoresis.

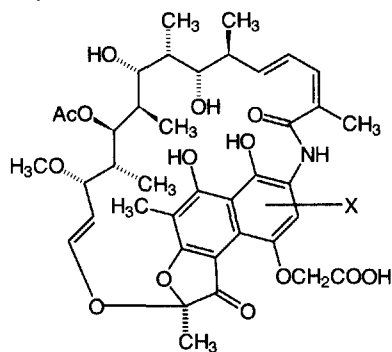
Rieske protein EC 1.10.99.1; recommended name: plastoquinol-plastocyanin reductase; systematic name: plastoquinol:oxidized-plastocyanin oxidoreductase; one of the four subunits of the cytochrome b_6-f complex. An enzyme that catalyses a reaction between plastoquinol-1 and two molecules of oxidized plastocyanin to form plastoquinone and two molecules of reduced plastocyanin. It is a high-potential 2Fe-2S protein. Example, precursor from tobacco: database code UCRI_TOBAC, 228 amino acids (24.12 kDa). Apart from the Rieske protein the subunits of the b_6f complex consist of **cytochromes b_6** and **f** and a 17 kDa peptide; it is similar to other iron-sulfur proteins (e.g. in mitochondria) and has three motifs.

rifampicin or **rifampin** an antibiotic, M_r 823, obtained as a semisynthetic derivative of rifamycin B, from which it differs only in the substituents on the ring marked X in the structure for that compound. It acts by inhibiting bacterial transcription, and is effective in sensitive bacteria (see **rifamycin**) at extremely low concentrations ($0.01 \mu g\ ml^{-1}$) while having no effect on eukaryotic transcription at 10^4 times that dose. Mitochondrial and chloroplast transcription may be affected at higher doses. It is stable when stored dry at $4^\circ C$ in the absence of light. In addition to its use as an antibiotic it has application in molecular biology as an agent for removing plasmids from bacteria in a process known as **plasmid curing**.

rifamycin any of a group of closely related anti-transcription antibiotics produced by *Streptomyces mediterranei* that are active against Gram-positive bacteria and *Mycobacterium tuberculosis* but are much less effective against Gram-negative organisms. Chemically and functionally they are closely related.



lated to the **streptovaricins**, and collectively the two groups are referred to as **ansamycins** – they are characterized by a chromophoric naphthohydroquinone nucleus spanned by a long aliphatic bridge. The mode of action of ansamycins involves inhibition of bacterial but not eukaryotic RNA synthesis via binding to the β subunit of bacterial DNA-dependent RNA polymerase and prevention of chain initiation; chain elongation is not affected once the initiation of RNA chain synthesis progresses beyond the third diester bond.



rifamycin B

rigid (in lipid biochemistry) a term sometimes used to describe the ordered phase of a lipid bilayer below the phase transition temperature. See **phase transition (of lipids)**.

Rimington, Claude (1903–93), British biochemist, distinguished for porphyrin research. With two coauthors he suggested (1968) that the British king, George III, suffered from variegate porphyria at a time preceding and during the American revolution. The hypothesis remains unproven and the cause of the king's deranged personality remains unknown.

rimorphin a 13-residue peptide that is a major Leu-enkephalin-containing peptide in tissues that contain **dynorphin** and α -neo-dynorphin. It yields three **enkephalin** sequences.

ring closure the formation of a covalent bond that converts a linear molecule into a ring structure.

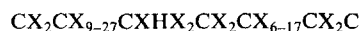
Ringer, Sydney (1835–1910), British physician and physiologist who devised **Ringer's solution**.

Ringer-Locke solution a less common name for **Locke's solution**.

Ringer's solution or **Ringer solution** any of a number of aqueous solutions of mixed inorganic salts, of composition determined empirically by **Ringer** or adapted therefrom, that have been useful as fluid replacements *in vivo* or in place of serum as perfusing or suspending media for organs, tissues, or cells *in vitro*. Ringer's own mixtures (reported over the period 1882–87) were devised mostly for maintaining normal contractility of perfused isolated frog heart. One of the most effective solutions contained (in g L⁻¹, approx.) NaCl (7.0), KCl (0.04), CaCl₂ (anhydrous, 0.09), and NaHCO₃ (0.5). See also **Krebs' mammalian Ringer**, **Locke's solution**, **Tyrode's solution**.

Ringer-Tyrode solution a less common name for **Tyrode's solution**.

RING finger a zinc finger motif defined as



and present in a family of nuclear proteins.

ring opening the breaking of a bond that converts a ring structure to an open-chain form.

RIP *abbr.* for 1 repeat-induced point (mutation); a phenomenon found in *Neurospora*; when more than one copy of a DNA sequence is present in a haploid genome that goes through a cross, both ectopic and endogenous copies are subject to point mutation at a high rate. RIP occurs in the dikaryotic ascogenous hyphae. Because transformation of *Neurospora* typically leads to gene amplification, RIP provides a highly efficient

and selective method of introducing mutations *in vivo* in any gene cloned *in vitro*. 2 ribosome-inactivating protein.

RISBASES proteins that have sequence similarity with members of the ribonuclease family but are thought not to function as ribonucleases. Examples are bovine seminal RNase and **angiogenin**. [From ribonuclease with special biological action.]

RIST *abbr.* for radioimmunosorbent test.

ristocetin a glycopeptide antibiotic complex from *Nocardia lurida*. Ristocetin A, *M_r* 2066, acts primarily on Gram-positive bacteria by inhibiting the biosynthesis of peptidoglycans, the mucopeptides of bacterial cell walls. Clinically, it is particularly useful against infections of the gut when administered by injection; however, it is ineffective when taken orally, and has toxic side effects including kidney damage.

Rittenberg tube an evacuable two-legged tube provided with a stopcock and used for the conversion of ¹⁵N-labelled ammonium salts into dinitrogen prior to mass-spectrometric analysis. It has been jocularly known, especially in Israel, as **Rittenberg's trousers**. [After David Rittenberg (1906–70), US biochemist.]

R loop the structure formed when an RNA of complementary sequence stably base pairs with one of the DNA strands of double-stranded DNA, as happens at the start of **transcription**.

r.m.m. *abbr.* for relative molecular mass.

RMS *abbr.* for root mean square.

Rn symbol for radon.

RNA *abbr.* and common name for **ribonucleate** (def. 1) or **ribonucleic acid**; one of the two main types of **nucleic acid**, consisting of a long, unbranched macromolecule formed from ribonucleotides, the 3'-phosphate group of each constituent ribonucleotide (except the last) being joined in 3',5'-phosphodiester linkage to the 5'-hydroxyl group on each ribose moiety of the next one. The presence of a free 2'-hydroxyl group on each ribose moiety renders these phosphodiester bonds susceptible to hydrolytic attack by alkali, in contrast to those of **DNA**. The RNA chain has polarity, with one 5' end and one 3' end. Two purines, adenine and guanine, and two pyrimidines, cytosine and uracil, are the major bases usually present. In addition, **minor bases** may occur; **transfer RNA**, however, contains unusual bases in relatively large amounts. The sequence of bases carries information, whereas the sugar and phosphate groups play a structural role. RNA is fundamental to **protein biosynthesis** in all living cells. **Messenger RNA** (mRNA) is responsible for carrying the coded genetic 'message', transcribed from DNA, to sites of protein assembly at the **ribosomes**. The latter are composed of **ribosomal RNA** (rRNA) plus proteins. A third species, transfer RNA (tRNA), is instrumental in importing amino acids to the assembly site, according to instructions carried by mRNA. In some viruses, RNA is the genetic material instead of DNA. Specific forms, functions, molecules, or preparations of RNA may be designated by prefixes or suffixes, thus: A-RNA or RNA-A, **A-form** of RNA; cRNA, **complementary RNA**; dsRNA, double-strand(ed) RNA; hnRNA or H-RNA or HnRNA, **heterogeneous nuclear RNA**; mRNA, **messenger RNA**; micRNA, **messenger-RNA-interfering complementary RNA**; mtRNA, **mitochondrial RNA**; nRNA, nuclear RNA; rRNA, **ribosomal RNA**; sRNA or S-RNA, **soluble RNA**; scRNA, small cytoplasmic RNA; snRNA, **small nuclear RNA**; snoRNA, **small nucleolar RNA**; ssRNA, single-strand(ed) RNA; tRNA, **transfer RNA**; tRNA, **translational-control RNA**; Z-RNA or RNA-Z, **Z-form** of RNA.

RNA adenylating enzyme see **polynucleotide adenylyltransferase**.

RNAase (sometimes) *abbr.* for ribonuclease; RNase is more usual.

RNA-dependent DNA polymerase another name for RNA-directed DNA polymerase (see **reverse transcriptase**).

RNA-dependent RNA replicase see **RNA-directed RNA polymerase**.

RNA-directed RNA polymerase EC 2.7.7.48; other name: RNA nucleotidyltransferase (RNA-directed); an enzyme that

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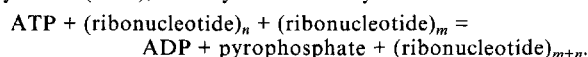


catalyses a reaction between a nucleoside triphosphate and RNA_n to form RNA_{n+1} and pyrophosphates; it is required for the replication of RNA viruses; in the case of eukaryotic viruses, the enzyme normally comprises several subunits. Example from influenza virus (strain Ann Arbor), subunit P1 (or PB1): database code RRP1_IAANN, 757 amino acids (86.34 kDa); subunit P2 (or PA): database code RRP2_IAANN, 716 amino acids (82.71 kDa); subunit P3 (or PB2): database code RRP3_IAANN, 759 amino acids (85.99 kDa).

RNA-DNA virus a former name for retrovirus.

RNA helicase an ATP-dependent enzyme that is similar to **DNA helicase** except that its substrate is RNA. Example *dbpA* gene product (*Escherichia coli*) specific for 23S rRNA: database code DBPA_ECOLI, 457 amino acids (49.13 kDa).

RNA ligase (ATP) EC 6.5.1.3; *other name*: polyribonucleotide synthase (ATP); an enzyme that catalyses the reaction:



Example from bacteriophage T4: database code RLIG_BPT4, 374 amino acids (43.46 kDa). Example, tRNA ligase from yeast: database code TRNL_YEAST, 827 amino acids (95.23 kDa).

RNA polymerase any enzyme that utilizes ATP, GTP, CTP, and UTP to synthesize RNA from a DNA or RNA template. DNA-dependent RNA polymerases include (i) the enzymes responsible for transcription of DNA and (ii) **primases**, which synthesize the RNA primers required for replication of DNA. An **RNA-directed RNA polymerase** (replicase or RNA synthetase) is involved in the replication of RNA viruses.

DNA-dependent (or DNA-directed) RNA polymerase (EC 2.7.7.6); *other name*: RNA nucleotidyltransferase (DNA-directed); it catalyses a reaction between a nucleoside triphosphate and RNA_n to form RNA_{n+1} and pyrophosphate. This EC class number includes the following examples.

In prokaryotes there is a single DNA-dependent RNA polymerase which transcribes all classes of DNA. The core *Escherichia coli* enzyme is composed of three types of subunit, α , β , and β' , and has the composition $\alpha_2\beta\beta'$; the holoenzyme contains an additional σ subunit or sigma factor. The core enzyme has a general affinity for DNA; sequence-specific DNA binding is conferred by the **sigma factor** which is responsible for promoter recognition and transcription initiation. Different sigma factors recognize different promoters. Following transcription initiation, the sigma factor dissociates from the core enzyme. The functions of the individual subunits are not fully understood. The β subunit is probably involved in binding ribonucleotide substrates; it is the target for **rifamycin** and **streptolydigin** antibiotics. The β' subunit may bind template DNA. Examples: α subunit from *E. coli* (identical in *Salmonella typhimurium*): database code RPOA_ECOLI, 329 amino acids (36.47 kDa); β' subunit from *E. coli*: database code RPOC_ECOLI, 1407 amino acids (154.98 kDa).

Phage RNA polymerases are much smaller and simpler than bacterial ones: the polymerases from phage T3 and T7 RNA, e.g., are single polypeptide chains of <100 kDa. Example from phage T3: database code RPOL_BPT3, 884 amino acids (98.68 kDa).

Eukaryotes have three types of DNA-dependent RNA polymerase in the nucleus, each producing a different type of RNA. RNA polymerase I, located in the nucleolus, synthesizes precursors of most ribosomal RNAs; RNA polymerase II, located in the nucleoplasm, synthesizes messenger RNA precursors; and RNA polymerase III, also located in the nucleoplasm, synthesizes the precursors of 5S rRNA, the transfer RNAs, and other small RNAs. These enzymes can be distinguished by their sensitivity to α -amanitin (see **amanitin**); class I enzymes are not inhibited by this octapeptide, whereas class II enzymes are rapidly inhibited by low concentrations; the sensitivity of class III enzymes varies depending on species. **RNA polymerase I** has 13 subunits totalling over 600 kDa; it recog-

nizes a single type of promoter and requires at least two ancillary factors, UBF1 and SL1; it accounts for 50–70% of the RNA polymerase activity in eukaryotic cells. **RNA polymerase II** from all sources contains two large subunits of 215 and 139 kDa and several smaller polypeptides. The 215 kDa subunit has homology with the prokaryotic β' subunit and contains a C-terminal domain (CTD) with multiple repeats of the consensus sequence YSPTSPS; the CTD is involved in initiation. The 139 kDa subunit has homology with the bacterial β subunit; the third largest subunit is related to the bacterial α subunit. Example of the latter: database code RPB3_YEAST, 318 amino acids (35.24 kDa). Three of the remaining subunits are common to RNA polymerases I, II, and III. Example: database code RPB5_YEAST, 215 amino acids (25.05 kDa). Most promoters used by RNA polymerase II contain the **TATA box**, which is recognized by the TATA-binding protein (TBP), a component of the **transcription factor** TFIID. Several other accessory factors are required for transcription by RNA polymerase II. **RNA polymerase III** has 14 subunits totalling 700 kDa. The promoters it uses may be upstream of the transcribed gene (for snRNAs) or within the gene (for tRNAs and 5S RNA). Three accessory factors, TFIIB (a **zinc-finger** protein), TFIIB, and TFIIC, are involved in transcription from internal promoters. Examples: 47 kDa subunit of RNA polymerase III: database code RPC4_YEAST, 424 amino acids (46.89 kDa); 40 kDa subunit common to polymerases I and III: database code RPC5_YEAST, 335 amino acids (37.64 kDa).

Mitochondria and chloroplasts also contain RNA polymerase activity that synthesizes the RNAs required for mitochondrial and chloroplast protein synthesis. Polymerases from these organelles are smaller than the nuclear enzymes and more like prokaryotic RNA polymerases.

[RNA-polymerase]-subunit kinase EC 2.7.1.141; *other name*: CTD kinase (*abbr.* for C-terminal domain kinase); an enzyme that catalyses a reaction between ATP and [DNA-directed RNA polymerase] to form phospho-[DNA-directed RNA polymerase] and ADP.

RNAse *usual abbr. for* ribonuclease; *compare* RNAase.

RNAse MRP a site-specific endoribonuclease for mitochondrial RNA processing (hence 'MRP'). It contains an essential RNA component and cleaves mitochondrial RNA (mtRNA) transcripts that are complementary to the D-loop region of vertebrate mtDNA.

RNAse P a catalytic activity responsible for the formation of the 5'-phosphate terminus of tRNA, by cleaving a sequence of bases from the 5' end of a primary transcript, as a first step in the formation of pre-tRNA. The activity resides in the RNA moiety of a complex consisting of 86% RNA and 14% protein by weight. It was the first nonprotein enzymic activity to be discovered, by **Altman** – and is an example of a **ribozyme**. See also **ribonuclease**.

RNA synthetase an RNA-directed **RNA polymerase**.

RNA tumour virus a common designation for **retrovirus**, as this is the only group of RNA viruses known to cause cancer.

RNP *abbr. for* ribonucleoprotein. Specific kinds of RNP may be designated by prefixes, thus: hnRNP, **heterogeneous nuclear ribonucleoprotein**; scRNP, **small cytoplasmic ribonucleoprotein**; snRNP, **small nuclear ribonucleoprotein**.

Roberts, Richard J. (1943–), British molecular biologist; Nobel Laureate in Physiology or Medicine (1993) jointly with P. A. Sharp 'for their discovery of split genes'.

Robinson, (Sir) Robert (1886–1975), British organic chemist; Nobel Laureate in Chemistry (1947) 'for his investigations on plant products of biological importance, especially the alkaloids'.

Robison ester an old name for D-glucose 6-phosphate. *Compare* **Neuberg ester**. [After R. Robison, British chemist (1883–1941).]

Rochelli salt potassium sodium tartrate tetrahydrate; it is used as a constituent of **Fehling's solution**. [After La Rochelle, French port.]

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rocket immunoelectrophoresis

rocket immunoelectrophoresis *abbr.*: RIEP; a one-dimensional **electroimmunodiffusion**, also known as **electroimmunoassay**, used for quantitative analysis of proteins. Exact volumes of protein (antigen) solution are applied in an agarose gel of uniform thickness containing the corresponding antibody. An electric field causes the migration of both antigen and antibody. They react to form elongated curved precipitation lines, known as rockets because of shape, which are proportional in size to the amount of antigen. *See also* **crossed immunoelectrophoresis**.

Rodbell, Martin (1925–98), US biochemist; Nobel Laureate in Physiology or Medicine (1994) jointly with A. G. Gilman 'for their discovery of G-proteins and the role of these proteins in signal transduction in cells'.

rod cell or **rod** a rod-shaped cell that is one of the two types of light-sensitive cell of the vertebrate retina. Rod cells contain **rhodopsin** or (in freshwater and migratory fish and some amphibians) **porphyropsin**, and are responsible for vision in dim light. They occupy all of the margin of the retina and are absent from the fovea. Each rod photoreceptor is a highly specialized cell with an outer and inner segment, a cell body, and a synaptic region where the rod passes a chemical signal to retinal nerve cells. The phototransduction apparatus is in the outer segment, which contains a stack of disks, each formed by a closed stack of membrane in which rhodopsin molecules are embedded. The plasma membrane surrounding the outer segment contains cyclic-GMP-gated Na^+ channels. These are kept open in the dark by cyclic GMP molecules bound to the channels. Metarhodopsin II, an activated form of rhodopsin resulting from interaction with a photon, activates **transducin**, a **G-protein**, which then activates a cyclic GMP phosphodiesterase, reducing the cellular level of cyclic GMP to initiate a nerve impulse in the optic nerve. *Compare* **cone cell**.

rod gel a gel used in a form of polyacrylamide gel electrophoresis, in which the gel is cast in cylindrical glass tubes. Now largely superseded by slab-gel electrophoresis.

roentgen a variant spelling of **röntgen**.

rolling circle a mechanism for the replication of circular double-stranded DNA. One strand is nicked, then, using the unbroken strand as a template, the DNA is replicated around the circle. Replication does not cease when one circumference has been replicated, but continues around the circumference several more times (hence the name 'rolling'). Consequently, multimers of the replicon are produced in a long single strand with the appearance of a tail; this may then be cut into unit replicons, or into multimers containing tandemly repeated copies of the original unit; these may remain as single strands or be transcribed to yield a complementary strand.

röntgen or **roentgen** *symbol* R; the unit of **exposure** (def. 3) to X- or gamma radiation, expressed in terms of the amount of ionization caused in air. Defined originally (1928, 1937) as that quantity of radiation having a corpuscular emission per 0.001293 gram of air (equivalent of 1 cm^3 of dry air at stp) producing, in air, ions carrying one electrostatic unit quantity of electricity of either sign. Expressed in SI units, 1 R causes the production in air of ions (of one sign) carrying a charge of $2.58 \times 10^4 \text{ C kg}^{-1}$; this corresponds to the formation of 1.61×10^{15} ion-pairs per kg of air and to an absorption of energy by air equal to 0.00869 J kg^{-1} . The energy absorption by water or tissue from 1 R of X- or gamma-radiation is about 0.0096 J kg^{-1} . Prior to 1956, the röntgen was used in clinical work to express both exposure and **absorbed dose**. *See also* **rad** (def. 2), **rem**, **rep**. [After Wilhelm Konrad von Röntgen (1845–1923), German physicist, discoverer of X-rays.]

röntgen equivalent man or **röntgen equivalent man** *see* **rem**.

röntgen equivalent physical *see* **rep**.

röntgen rays a former name for **X-rays**.

room temperature ambient temperature, often used for reactions that can be carried out on the bench without devices to maintain constant temperature. Normally within several degrees of 20 °C.

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root effect the lowering of ligand affinity and of cooperativity of ligand binding (in hemoglobin) brought about by a low pH. [After R. W. Root who reported it in 1931.]

root mean square *abbr.* RMS; the positive square root of the mean of the squares of a series of values of x_i :

$$\text{RMS value} = \sqrt{\frac{x_1^2 + x_2^2 + \dots + x_n^2}{n}}$$

Compare **standard deviation**.

ROS *abbr.* for rod outer segment; *see* **rod cell**.

ROS a gene family encoding receptor-like tyrosine kinases of unknown function; *v-ros* is the oncogene of the acutely transforming avian sarcoma virus UR2 (named after the University of Rochester). The vertebrate gene product, Ros, has close similarity to the *Drosophila* **Sevenless protein**; it appears to be associated with a phosphoinositide response (*see* **phosphatidylinositol cycle**); Ros-1 is a member of the Src family. Example from chick: database code KROS_CHICK.

rosetting a method of isolating cells by allowing them to associate with red blood cells. For example, lymphocytes become surrounded with red cells, forming a 'rosette', and may then be isolated by sedimentation through **Ficoll** gradients.

rota-evaporate to evaporate a solution in a rotating round-bottomed flask in a temperature-controlled water bath, using an apparatus that can at the same time apply a vacuum to the flask interior. —**rota-evaporation** *n*.

rotamer *see* **rotational isomer**.

rotary evaporator an apparatus used for **flash evaporation**; *see also* **rota-evaporate**.

rotational correlation time *see* **correlation time**.

rotational diffusion rotation of a solute molecule around its centre of mass as a result of **Brownian movement**.

rotational frictional coefficient *see* **frictional coefficient**.

rotational isomer molecules of identical **configuration** may be distinguished as having different **conformations** after rotation about a bond. If an assembly of atoms X–A–B–Y (X and Y not being collinear with A–B) is viewed along the axis A–B, the X–A and B–Y bonds will make an angle with each other which may be 0° (superposition), 180° or some intermediate angle. This angle is termed the torsion angle, designated θ or ψ . Conformations are then described as synperiplanar (*sp*), synclinal (*sc*), anticlinal (*ac*) or antiperiplanar (*ap*) according as the torsion angle is within $\pm 30^\circ$, $\pm 60^\circ$, $\pm 120^\circ$, or $\pm 180^\circ$, respectively. Formerly, the terms *cis*, *gauche* and *trans* were in use for synperiplanar, synclinal and antiperiplanar respectively.

rotational strength or **rotatory strength** *symbol*: R ; an index of the degree to which an **absorption band** exhibited by an optically active solute displays **circular dichroism**. It may be evaluated for the absorption band in question by means of the expression:

$$R = 2.296 \times 10^{-44} \int \left(\frac{\Delta \epsilon_\lambda}{\lambda} \right) d\lambda$$

where λ is a wavelength of observation and $\Delta \epsilon_\lambda$ is the difference between the molar (decadic) absorption coefficients of the left and right circularly polarized light beams at that wavelength. Rotational strength is thus essentially the absorption coefficient difference integrated over the entire absorption band.

rotation axis a crystal or an oligomeric protein molecule is said to possess an *n*-fold rotation axis if it presents exactly the same appearance after a rotation about that axis of $360^\circ/n$.

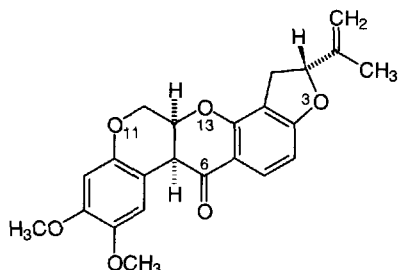
Compare **screw axis**.

rotation-reflection axis *see* **symmetry**.

rotenone an insecticidal material and the active principle in derris. It is present in plants of the genera *Derris*, *Lonchocarpus*, *Tephrosia*, and *Mundulea*; these plants were used by indigenous peoples as fish poisons (*Derris* in Asia, *Lonchocarpus* in South America, and *Tephrosia* more extensively). Rotenone inhibits mitochondrial respiration on the oxygen side of the nonheme iron of NADH dehydrogenase. Extracts of rotenone-



containing plants are used as insecticidal sprays and dusts (derris dust). Compare **picric acid**.



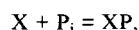
Rot value or **R_{0t} value** a measure of the kinetic complexity of RNA based on its hybridization ability. The notation is a modification of R_0 , where R_0 is the initial RNA concentration and t is time. It is analogous to the C_{0t} value (see **C_{0t}**) for DNA. $R_{0t_{1/2}}$ is the value corresponding to half complete association between complementary polynucleotides.

rough endoplasmic reticulum *abbr.*: RER; a region of the endoplasmic reticulum associated with ribosomes and involved in the synthesis of secreted and membrane-bound proteins. The nascent protein destined for secretion with its **signal peptide** associates with a receptor for the **signal recognition particle** and then translocates through the RER lipid bilayer by means of a group of proteins known as a **translocon**, which includes **TRAM**. The signal peptide is removed by the **signal peptidase**. The process is aided by **heat-shock proteins**, which serve as **chaperones**.

Round-up *see* **glyphosate**.

rousvirus *see* **retrovirus**.

rP the negative decadic logarithm of the molar concentration of inorganic phosphate; i.e. $rP = -\log[P_i]$. If, for the equilibrium



$$K = [X][P_i]/[XP]$$

then

$$rP = -\log K - \log([XP]/[X]).$$

RP-HPLC *abbr.* for reversed phase high-pressure liquid chromatography; *see* **reversed phase**.

R plasmid *abbr.* for resistance plasmid.

r-protein *abbr.* for ribosomal protein.

RP-TLC *abbr.* for reversed phase thin-layer chromatography; *see* **reversed phase**.

RQ *abbr.* for respiratory quotient.

RRM *abbr.* for RNA recognition motif; a conserved 80-amino acid sequence motif present in members of a family of proteins that bind RNA. It has a β - α - β - α structure.

rRNA *abbr.* for ribosomal RNA; *see* **ribosome**.

(RS)- prefix (in chemical nomenclature) denoting a racemate containing a single chiral centre. *See* **sequence rule**.

R segment *abbr.* for repeat segment; part of retroviral RNA consisting of direct repeats i.e. repeats having the same orientation; the size of the repeated segments varies in different strains of virus, from 80 to 100 nucleotides.

R/S convention the system of configuration specification of chiral compounds based on the **sequence rule**.

RSKG-5 *see* **tonin**.

R state *see* **R form**.

rtDNA (sometimes) *abbr.* for **recombinant DNA**; use not recommended.

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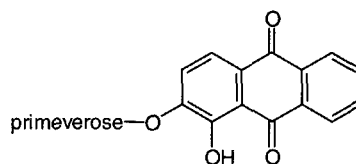
RTP symbol for any unspecified purine nucleoside 5'-triphosphate.

Ru symbol for ruthenium.

RU-486 or **RU-38486** mifepristone; 11 β -(4-dimethylamino-phenyl)-17 β -hydroxy-17 α -prop-1-ynylestra-4, 9-dien-3-one; a steroidal progesterone-receptor antagonist that prevents implantation of a fertilized ovum in the uterus. It is an effective contraceptive even when taken after coitus.

rubber a *cis*-1,4-polyisoprene with M_t varying from about 1×10^5 to 1×10^6 , representing polymers of about 1500 to 60 000 **isoprene** residues. It is found in the latex of about 300 different flowering plants, but only *Hevea brasiliensis* is a commercial source. A nonelastic rubber from *Manilkara bidentata* latex is known as **balata**. It is a *trans*-1,4-polyisoprene.

ruberythric acid the glycoside; 2-alizarin β -D-primeveroside, a major component of the plant dyestuff madder.



rubidium symbol: Rb; a silvery-white metallic element of group 1 of the IUPAC **periodic table**; atomic number 37; relative atomic mass 85.48. Highly reactive, it must be stored in an inert atmosphere. Rb^+ may be used experimentally as a K^+ analogue (sometimes as substitute, other times as competitor).

Rubisco *abbr.* for ribulose-bisphosphate carboxylase.

RuBP *abbr.* for ribulose 1,5-bisphosphate.

rubredoxin *abbr.*: Rd; any of a group of bacterial **iron-sulfur** proteins that are small, highly conserved, contain non-heme iron, and are nonenzymic. They lack acid-labile sulfur, and are characterized by having either one or two iron atoms each entirely in mercaptide coordination, i.e. surrounded by four cysteine residues or sulfur-containing ligands. They are electron acceptors for, e.g., hydrogenase-linked redox processes and hydroxylase systems. Rubredoxins of M_t 6000–19 000 have been found in a number of bacteria. The ferric form is intense red, the ferrous form is colourless. Example from *Alcaligenes eutrophus*; database code RUBR_ALCEU, 78 amino acids (8.64 kDa). Compare **ferredoxin**.

rudimentary a gene that in *Drosophila* is the equivalent of *pyr-1* in a number of other eukaryotes. It encodes a **CAD protein**.

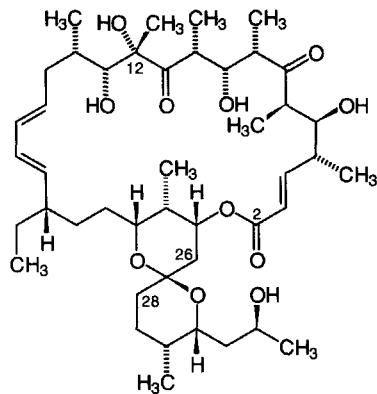
ruffled edge or **ruffling** the appearance of the leading edge of a moving cell in culture, caused by the extension of microspikes and lamellipodia, which when not attached to the substratum have the appearance of ruffles in flimsy dress fabric.

Russell's viper venom the venom of Russell's viper, *Vipera russellii*, a snake of SE Asia. It is used experimentally as a source of several enzymes especially phospholipases, proteases, and ribonucleases.

rusticyanin a blue copper-containing single-chain polypeptide obtained from a thiobacillus. It undergoes reduction during oxidation of Fe^{2+} with dioxygen. It is stable in acid, labile in alkali, and contains no arginine. Example from *Thiobacillus ferrooxidans*; database code RUS1_THIFE, 154 amino acids (16.38 kDa).

rutamycin or **oligomycin D** 26-demethyloligomycin A; an antifungal agent that has no significant activity against bacteria but is strongly inhibitory to a limited number of yeast and other fungal species. It acts by inhibiting mitochondrial ATPase F_1 , preventing phosphoryl group transfer, but has no effect on isolated F_1 units. (Illustrated on p. 581.) *See also* **oligomycin**.

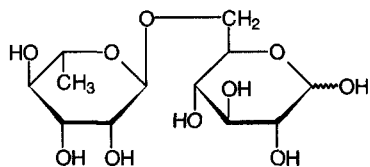
ruthenium red



rutamycin

ruthenium red ruthenium oxychloride ammoniated; $[(\text{NH}_3)_5\text{Ru}-\text{O}-(\text{NH}_3)_4\text{O}-\text{Ru}-(\text{NH}_3)_5]\text{Cl}_6 \cdot 4\text{H}_2\text{O}$. A brownish-red, water-soluble inorganic compound, used in microscopy and electron microscopy as a stain for acid mucopolysaccharides. It inhibits the influx of Ca^{2+} ions into mitochondria at extremely low concentrations.

rutinose the disaccharide, 6- $\text{O}-\beta$ -L-rhamnosyl-D-glucose; it occurs in some glycosides, e.g. rutin, the 3'-rutinoside of quercetin.



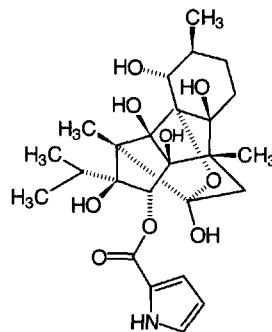
Ružička, Leopold (1887–1976), Croatian-born Swiss chemist; Nobel Laureate in Chemistry (1939) 'for his work on polymethylenes and higher terpenes' [prize shared with A. F. J. Butenandt].

R_B value or relative band rate or relative band speed (in chromatography) symbol: R_B ; the ratio of the distance travelled by a zone of the substance of interest to the distance simultaneously travelled by a reference substance B. See also **R_f value**.

R_f value or **R_F value** or band rate or band speed symbol: R_f or R_F ; (in chromatography) the ratio of the distance travelled by the

centre of a zone of the substance of interest to the distance simultaneously travelled by the mobile phase. In paper and thin-layer chromatography, it is generally determined from the distance travelled by the substance from the origin and the distance travelled by the eluent front from the same point. [From 'rate relative to front'.]

ryanodine an alkaloid with insecticidal activity, isolated from *Ryania speciosa*. See also **ryanodine receptor**.



ryanodine receptor *abbr.*: RYR; any of a group of receptor proteins that bind ryanodine. They were first discovered in muscle, but subsequently identified in other cell types. Their natural role is to function as intracellular Ca^{2+} -release channels. In muscle, they provide communication between transverse tubules (T-tubules) and sarcoplasmic reticulum (SR); contraction of skeletal muscle is triggered by release of Ca^{2+} from SR following depolarization of T-tubules. The Ca^{2+} channel is modulated by Ca^{2+} , Mg^{2+} , and calmodulin. Three types are known: RYR1, the skeletal muscle type; RYR2, the cardiac muscle type; and RYR3, a highly truncated protein found in some nonmuscle cells. Cyclic ADP-ribose activates the RYR2 but not the RYR1 receptor. In muscle the receptors are integral transmembrane glycoproteins, probably tetrameric; the C-terminal region is on the SR lumen side of the membrane, while the N-terminal region (the major part of the molecule for RYR1 and 2) protrudes into the cytosol. Deficiency is associated with malignant hyperthermia. Example, skeletal muscle type from human: database code RYNR_HUMAN, 5032 amino acids (563.79 kDa).

RYN method a method for determining the plausibility of an open reading frame. A computational analysis of the sequence determines the frequency of the favoured codons R (purine), Y (pyrimidine), and N (anything). If these are found to be in the proportions expected from that normally found, it lends support to the fact that the correct reading frame has been selected, and that the open reading frame is correct.

RYR *abbr.* for ryanodine receptor.

Ss

s symbol for 1 second (def. 1). 2 solid state (in a chemical equation). 3 atomic orbital corresponding to an orbital angular momentum quantum number (l) = 0. 4 singlet (in nuclear magnetic resonance spectroscopy). 5 strong absorption (in infrared spectroscopy).

S symbol for 1 sedimentation coefficient. 2 solubility. 3 (in monosaccharide nomenclature) the **septanose** form; it is suffixed to the symbol for the corresponding monosaccharide; e.g. glucoseptanose is symbolized as Glcs. 4 specific entropy (i.e. **entropy**/mass). 5 spin quantum number.

s- abbr.: for **sec-**.

S symbol for 1 sulfur. 2 siemens. 3 Svedberg unit ($S_{W,20}$ is the symbol under standard conditions). 4 a residue of the α -amino acid L-serine (alternative to Ser). 5 a residue of an incompletely specified base in a nucleic-acid sequence that may be either cytosine or guanine. 6 a residue of the ribonucleoside **thiouridine** when the position of the thiol group in its thiouracil moiety is unknown or unspecified (alternative to sU or Srd). 7 substitution (reaction mechanism).

2S symbol for a residue of the ribonucleoside 2-thiouridine (alternative to s^2U); see **thiouridine**.

4S symbol for a residue of the ribonucleoside 4-thiouridine (alternative to s^4U); see **thiouridine**.

S_N1 abbr. for unimolecular nucleophilic substitution.

S_N2 abbr. for bimolecular nucleophilic substitution.

S_{W,20} see **S** (def. 3).

S symbol for 1 entropy. 2 spin quantum number (of a system). 3 strangeness quantum number. 4 area (alternative to A).

S stereochemical descriptor denoting *sinister*; i.e. the absolute configuration of a chiral compound in which the priority (obtained using the Cahn-Ingold-Prelog sequence rules) of the substituent groups on the chiral centre decreases in an anticlockwise (left-handed) direction. For use as a prefix in chemical nomenclature, the descriptor is placed in parentheses and connected with a hyphen; e.g. (*S*)-alanine. For chiral compounds in which the relative but not the absolute configuration is known, *sinister* configuration is denoted by the prefix (*S**). See **sequence rule**. Compare **R**.

S- prefix (in chemical nomenclature) denoting substitution on a sulfur atom in a compound; a particular sulfur atom may be specified by a right superscript.

-S conformational descriptor designating the **skew conformation** of a six-membered ring form of a monosaccharide or monosaccharide derivative. Locants of ring atoms that lie on the side of the structure's reference plane from which the numbering appears clockwise are indicated by superscripts preceding the letter, and those that lie on the other side of the reference plane by subscripts following the letter; e.g. 1,2-*O*-ethylidene- α -D-glucopyranose- 1S_3 . See also **conformation**.

s.a. abbr. for specific activity.

SAA abbr. for serum amyloid A; see **amyloid A protein**.

SABP abbr. for secretory actin-binding protein.

saccharase the trivial name for any of a group of widely distributed hydrolases that attack the fructose (**fructosaccharases**) or glucose (**glucosaccharases**) end of appropriate oligosaccharides, liberating fructose or glucose respectively. The term is also used as an alternative trivial name for **invertase** (which is both a fructosaccharase and a glucosaccharase).

saccharate see **glucarate**.

saccharic acid 1 an old name for **aldaric acid**. 2 an old name for glucaric acid; see **glucarate**.

saccharide any carbohydrate; the term is applied especially to the simple sugars, i.e. the **monosaccharides**, **disaccharides**, and short-chain **oligosaccharides**. See also **polysaccharide**.

saccharification the hydrolysis of sugar derivatives or complex carbohydrates to soluble, fermentable sugars.

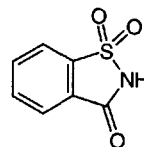
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saccharimeter an instrument for measuring the amount of sugar in a solution, especially a **polarimeter** used for such a purpose. Compare **saccharometer**.

saccharin 2,3-dihydro-3-oxobenzisulfonazole; *O*-sulfobenzimide; an artificial, nonnutritive sweetening agent that is several hundred times sweeter than sucrose. It is often used as its more soluble sodium salt. Its carcinogenicity has been extensively studied. Rats receiving high doses produced tumours of the urinary tract but there is no clear-cut evidence of such an effect in humans.



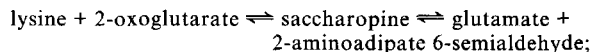
saccharogenesis the process of producing soluble, fermentable sugars from a complex carbohydrate. —**saccharogenic** *adj.*

saccharogenic method a method for assaying amylase and similar enzymes that hydrolyse polysaccharides. It depends on determination of the amount of simple sugar product formed by the enzyme.

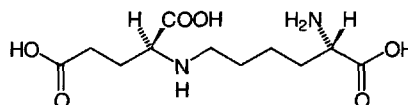
saccharometer an instrument for measuring the amount of sugar in a solution, especially a **hydrometer** specially calibrated for the purpose. Compare **saccharimeter**.

Saccharomyces a genus of budding yeasts that reproduce asexually by budding or sexually by conjugation. They are used especially as simple model organisms in the study of eukaryotic cell biology, and in genetic engineering. *Saccharomyces cerevisiae* is used commercially in bread-making and in the production of alcoholic beverages and industrial alcohol. It is able to divide vegetatively in either the haploid or diploid phase. This permits the isolation of recessive mutations in the haploids and complementation testing in the diploids. Transformation procedures that permit DNA uptake by either **protoplasts** or whole yeast cells have been developed and a wide range of cloning vectors constructed. In cases where a recombinant **glycoprotein** is desired, yeast has proved very useful since it can, unlike bacteria, effect post-translational **glycosylation**. It has been used for the production of a vaccine for hepatitis B.

saccharopine N^6 -[1,3-dicarboxypropyl]-L-lysine; an intermediate in the **aminoacidic pathway** for lysine biosynthesis (e.g. in *Saccharomyces* spp., hence the name) and for lysine catabolism (e.g. in mammalian liver). The reactions involved are:



they are catalysed by **saccharopine dehydrogenases**, which are characterized as lysine-forming or glutamate-forming, and require either $NAD^+/NADH$ or $NADP^+/NADPH$ as coenzyme; see **saccharopine dehydrogenase (NAD^+ , L-glutamate-forming)** and **saccharopine dehydrogenase (NAD^+ , L-lysine forming)**.



saccharopine dehydrogenase (NAD^+ , L-glutamate-forming) EC 1.5.1.9; an enzyme of the pathways for the synthesis and degradation of lysine. It catalyses the oxidation by NAD^+



saccharopine dehydrogenase

of N^6 -(L-1,3-dicarboxypropyl)-L-lysine to L-glutamate and 2-amino adipate 6-semialdehyde with formation of NADH. There is a second saccharopine dehydrogenase with the same reaction mechanism, in which the coenzyme is NADP⁺: EC 1.5.1.10, saccharopine dehydrogenase (NADP⁺: L-glutamate-forming). *See also* **saccharopine**.

saccharopine dehydrogenase (NAD⁺, L-lysine-forming) EC 1.5.1.7; *other name*: lysine-2-oxoglutarate reductase; an enzyme of the pathways for the synthesis and degradation of lysine. It catalyses the reaction between L-lysine, 2-oxoglutarate, and NADH to form N^6 -(L-1,3-dicarboxypropyl)-L-lysine (**saccharopine**), NAD⁺, and H₂O. Example from *Yarrowia lipolytica* (formerly *Candida lipolytica*): database code LYS1_YARLI, 369 amino acids (40.57 kDa). *See also* **saccharopine**.

saccharose an old name for **sucrose**.

safety cabinet a cabinet providing an area at bench level in which pathogenic agents may be handled or stored while preventing their entry into the atmosphere outside the cabinet; at the same time, sterility within the cabinet is normally ensured. Usually the cabinet is ventilated with a stream of air, which is extracted through a filter system that traps any pathogens. There are several grades of safety, the highest grades being for the more potentially hazardous materials; in the lowest grade, there is an inward flow of air through the front aperture; a higher grade cabinet is ventilated by a downward flow of filtered air, extracted so as to ensure also an inward flow through the front aperture, the extracted air being recirculated through the filter; in the highest grade, there is no front aperture, the operator manipulating the interior components using gloves mechanically sealed into the front of the cabinet.

SAH *abbr.* for S-adenosyl-L-homocysteine; *see* **S-adenosylhomocysteine**.

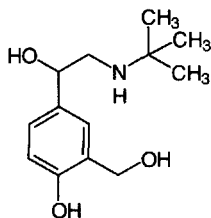
SAICAR or **SACAIR** *abbr.* for 1-(5-phosphoribosyl)-5-amino-4-(N-succinocarboxamide)-imidazole.

SAICAR synthetase *see* **phosphoribosylaminoimidazole-succinocarboxamide synthase**.

Sakaguchi test a test for guanidines; in alkaline solution they give a red colour with the **Sakaguchi reagent**, which contains α -naphthol and sodium hypochlorite. Arginine undergoes this reaction, as do arginine-containing proteins.

Sakmann, Bert (1942–), German physiologist; Nobel Laureate in Physiology or Medicine (1991) jointly with E. Neher 'for their discoveries concerning the function of single ion channels in cells'.

salbutamol *other name*: albuterol; 4-hydroxy-3-hydroxy-methyl- α -(*tert*-butylamino)methyl]-benzyl alcohol, α^1 -[[1,1-dimethylethyl]amino]methyl]-4-hydroxy-1,3-benzenedimethanol; a β -adrenergic agonist. The modification of the catechol ring by introduction of the hydroxymethyl group enhances the β_2 -adrenoceptor activity by a factor of about 60 compared with isoproterenol.



salicylic acid 2-hydroxybenzoic acid; a white, crystalline, sweet-tasting substance, slightly soluble in water; its esters are found in bark of sweet birch and wintergreen, the leaves of which are used as a topical keratolytic and antifungal agent. It serves as a natural inducer of heat production ('calorigen') in *Arum* lilies, and has a broad regulatory role in plants. Its O-acetyl derivative is **aspirin**.

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saline 1 of, concerned with, or containing common salt (sodium chloride). 2 of, or containing, salts of alkali metals or magnesium. 3 an aqueous solution of sodium chloride and (sometimes) other salts, of defined osmolarity, prepared for intravenous injection or perfusion; usually 0.9% w/v sodium chloride. *See also* **physiological saline**.

saliva the mixed secretions of the salivary glands and of the mucous membrane of the mouth. It contains an α -amylase (ptyalin), mucin, various inorganic ions, urea, buffers, lysozyme, etc., and has a pH near 7. It is secreted in response to food, which it serves to moisten and lubricate; the amylase may bring about some hydrolysis of ingested starch. *See also* **sialic**. —**salivary** *adj.*

salivary gland any of the various glands that discharge **saliva** into the mouth. In humans there are three pairs: the parotid, sublingual, and submaxillary glands. In snakes they include the venom glands.

salivary juice the collective secretions of the salivary glands; i.e. **saliva**. Its secretion was studied by **Pavlov** and led him to the discovery of the conditioned reflex.

Salkowski test a test for the presence of cholesterol. When concentrated sulfuric acid is added to a chloroform solution of cholesterol, the chloroform layer shows a red to blue colour and the acid layer shows a green fluorescence. [After Ernst Leopold Salkowski (1844–1923), German physiological chemist.]

salmine a 6–7 kDa **protamine**, found in salmon sperm.

salmon *common name* for a number of fish of the family Salmonidae which live in coastal waters of the North Atlantic and North Pacific and breed in rivers. The flesh of these fish is comparatively rich in oils with a high content of long-chain n -3 **fatty acids**; typical composition (major fatty acids, expressed as % of total fatty acids): 16:0, 10; 16:1(n -7), 5; 18:0, 4; 18:1(n -9), 24; 18:2(n -6), 5; 18:3(n -3), 5; 20:1(n -9), 1; 20:4(n -6), 5; 20:5(n -3), 5; 22:6(n -3), 17.

Salmonella a genus of rod-shaped, motile, aerobic bacteria including the agents responsible for typhoid fever (*S. typhi*), some types of food poisoning, and other enteric diseases.

Salmonella mutagenesis test or **Ames test** a method of screening chemicals for carcinogenicity, in which their mutagenic effect on selected strains of *Salmonella typhimurium* is evaluated – there being a high correlation between mutagenicity and carcinogenicity for substances of known action. The strains of *S. typhimurium* used are normally those that require histidine. They are plated out on histidine-deficient medium after being mixed with the potential mutagen. Mutations brought about by the test substance which reverse the defect in their ability to synthesize histidine are revealed by colonies growing on the plate.

salt 1 any member of a class of compounds formed, together with water, by reaction of an acid with a base (most commonly an inorganic acid and metallic base). 2 or **common salt** sodium chloride.

saltation 1 the action of jumping or leaping. 2 proceeding in a discontinuous manner by leaps interspersed by periods of rest. 3 an irreversible and inherited change in a cell or organism due to mutation. —**saltatory** *adj.*

saltatory conduction the passage of a nerve impulse across a gap, as between the nodes of Ranvier in a myelinated nerve fibre.

saltatory evolution evolution by sudden changes or by periods of active change with intervening inactive periods.

saltatory replication a hypothetical, sudden lateral replication of DNA thought to be responsible for the production of families of similar nucleotide sequences.

salt bridge 1 a tube, usually closed with porous plugs, that is filled with a solution of a salt (often potassium chloride, commonly saturated) and used to obtain electrical contact between two electrolytic half-cells without intermingling of their respective electrolytes. 2 any electrostatic bond between positively and negatively charged groups on amino-acid residues



salt fractionation

of a protein, that contributes to the stability of the protein structure.

salt fractionation the differential precipitation of proteins (or nucleic acids) from solution by the addition of neutral salts, often ammonium sulfate. *Compare* **salting out**.

salting in the increase in the solubility of certain proteins (e.g. globulins) and dipolar ions (e.g. cystine) produced in solutions of low ionic strength by an increase in the concentration of neutral salts.

salting out the decrease in the solubility of proteins, gases, uncharged molecules, and, in sufficiently concentrated solutions, electrolytes, produced in solutions of high ionic strength by an increase in the concentration of neutral salts.

salting-out chromatography a method of chromatography whereby water-soluble organic compounds are separated by ion-exchange chromatography by use of an aqueous salt solution as eluent.

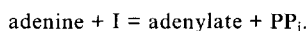
salting-out constant *symbol:* K'_s ; the slope of the straight line relating the logarithm of the solubility of a substance in neutral salt solutions and the ionic strength. It is given by

$$K'_s = -[\log(s/s^*)]/I,$$

where s is the solubility in mol dm^{-3} at ionic strength I , and s^* is the hypothetical solubility at zero ionic strength.

salt link or **salt linkage** an ionic bond.

salvage pathway any metabolic pathway that utilizes compounds formed in catabolism for biosynthetic purposes, despite these compounds not being intermediates in the corresponding normal biosynthetic pathways. Thus, free purine and pyrimidine bases may be converted to the corresponding ribonucleotides. In the case of purines, the ribose phosphate moiety of 5-phospho- α -D-ribose diphosphate (I) is transferred to the purine; thus



See also **purine biosynthesis**.

SAM *abbr.* for *S*-adenosylmethionine.

SAMase a trivial name for *S*-adenosylmethionine cleaving enzyme (*S*-adenosylmethionine hydrolase; EC 3.3.1.2).

same-sense mutation an alternative name for **silent mutation**.

sample a small, separated portion of the whole. In experimentation it is often essential that the sample be fully representative of the characteristics of the whole (and always so for a **statistical sample**).

sample applicator a device to aid application of samples to chromatography plates, etc.

sample well a hole cut in an electrophoresis gel in which the sample to be analysed is placed.

sampling error (*in statistical analysis*) the chance difference between a measurement obtained from a **sample** and the corresponding measurement obtained from the whole population.

Samuelsson, Bengt Ingemar (1934–), Swedish biochemist; Nobel Laureate in Physiology or Medicine (1982) jointly with S. K. Bergström and J. R. Vane 'for their discoveries concerning prostaglandins and related biologically active substances'.

sandwich assay or **sandwich technique** a type of immunoassay in which antibody against the antigen to be assayed is bound to a solid surface (e.g. plastic). After addition of solution containing antigen the fixed antigen is washed. A second antibody, which is radioactive or fluorescent, is added, sandwiching the antigen; after removal of excess, the amount of bound label is determined. The second antibody may be specific for a different epitope on the antigen, thus enhancing overall specificity, or for the first antibody bound to an antigen.

sandwich complex a 2:1 macrocycle:cation complex in which the metal ion is held between two parallel macrocyclic complexes.

sandwich technique an alternative name for **sandwich assay**.

sandwich test an immunohistochemical procedure employing a **sandwich assay** to detect an antibody within a cell.

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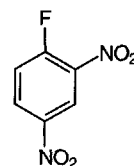
Sanfilippo syndrome A an alternative name for **mucopolysaccharidosis type IIIA**.

Sanfilippo syndrome B an alternative name for **mucopolysaccharidosis type IIIB**.

Sanger, Frederick (1918–), British biochemist, protein chemist, and nucleic-acid chemist; Nobel Laureate in Chemistry (1958) 'for his work on the structure of proteins, especially that of insulin'; Nobel Laureate in Chemistry (1980) jointly with W. Gilbert 'for their contributions concerning the determination of base sequences in nucleic acids' [prize shared with P. Berg].

Sanger's method **1** (for identifying and estimating N-terminal amino-acid residues of polypeptides and proteins) a method in which free unprotonated amino groups react with 1-fluoro-2,4-dinitrobenzene (FDNB). The dinitrophenylamino groups formed are stable to the acid used to hydrolyse the peptide bonds; the yellow arylated (DNP-) amino acids so released can then be identified by chromatography and estimated spectrophotometrically. **2** (for polypeptide and protein sequencing) a general method for deriving the primary sequence of a polypeptide chain, based on selective hydrolytic degradation of the chain into smaller peptides. The sequence of each smaller peptide may then be found by use of FDNB to label its N-terminal amino acid, together with total hydrolysis to give the amino-acid composition and further stepwise degradation by use of carboxypeptidase or by **Edman degradation**. By selection of different degradation procedures to ensure overlap of sequences between the various smaller peptides, the overall sequence of the starting material may be determined. **3** (for DNA sequencing) an alternative name for **chain-termination method**.

Sanger's reagent 2,4-dinitrofluorobenzene (*abbr.*: DNFB); 1-fluoro-2,4-dinitrobenzene (*abbr.*: FDNB); a substance used in structural protein chemistry to arylate free amino groups. *See* **Sanger's method** (def. 1, 2).



sapecin an insect **defensin** that is active against (mainly Gram-positive) bacteria. It has a high affinity for **cardiolipin**. Example (precursor includes presumed signal of unknown length) from *Sarcophaga peregrina*: database code SAPE_SARPE, 94 amino acids (9.90 kDa).

sapogenin any of the **aglycon** moieties of **saponins**; they may be steroids or triterpenes.

saponifiable fraction the portion of total lipid that, after treatment with hot alkali, is soluble in water and insoluble in ether.

saponifiable lipid any **lipid** that can be hydrolysed with alkali to give soap (fatty-acid salts) as one product; they include any **glycolipid**, **cholesteryl ester**, or **sphingolipid**.

saponification the hydrolysis of an **ester** by an alkali to give the alcohol and salt of the acid. Originally the term meant the hydrolysis of a neutral fat by alkali (e.g. potassium hydroxide) into glycerol and a soap. —**saponify** *vb*.

saponification equivalent the amount in grams of an ester that consumes one gram-equivalent of alkali in **saponification**. *Compare* **saponification number**.

saponification number the number of milligrams of potassium hydroxide consumed in the complete **saponification** of one gram of any particular fat, oil, or wax. *Compare* **saponification equivalent**.

saponin any member of a large group of glycosides, widely distributed in plants, that are powerful **surfactants**. Each saponin



saponin detoxifying enzyme

consists of an aglycon moiety (i.e. **sapogenin**), which may be a steroid or a triterpene, and a sugar moiety, which may be glucose, galactose, a pentose or methylpentose, or an oligosaccharide. All saponins foam strongly when shaken with water. They are membrane-active, powerful hemolytic agents and may be used at low concentrations to **permeabilize** cells. They are thus highly toxic on injection, but not on ingestion as they are not absorbed through the gut. *See also digitonin*.

saponin detoxifying enzyme an enzyme of unknown mechanism that acts, as the name implies, to prevent activity of antifungal **saponins**. Example from *Gaeumannomyces graminis avenacinase* (the so-called take-all fungus, an extreme pathogen of cereals): database code GGU17568, 637 amino acids (68.66 kDa).

sapor that which is sensed by taste; any quality relating to taste or savour. —**saporific** or **saporous** *adj.*

saporin a type I **ribosome-inactivating protein**, EC 3.2.2.22, from the soapwort *Saponaria officinalis*.

saposin or **sphingolipid activator protein** any of a group of peptide cofactors of enzymes for the lysosomal degradation of sphingolipids. They stimulate various enzymes, including glucosylceramidase, galactosylceramidase, cerebroside-sulfatase, α -galactosidase, β -galactosidase, and sphingomyelin phosphodiesterase. Saposins A to D are known. Deficiency of any may cause a mild form of a **sphingolipidosis** that normally results from deficiency of an enzyme the saposin activates. They contain characteristic domains known as saposin A-type or B-type domains. Example (precursor) from human: database code SAP_HUMAN, 524 amino acids (58.04 kDa). Specific polypeptides derived from this (amino acids in the precursor sequence in brackets) are as follows: saposin A (60–142) together with saposin C (311–391) stimulate hydrolysis of glucosylceramide by β -glucosylceramidase (EC 3.2.1.45) and galactosylceramide by β -galactosylceramidase (EC 3.2.1.46). Saposin C deficiency causes a variant of **Gaucher's disease**. Saposin B (195–274) is a homodimer and forms a solubilizing complex with the substrates of the sphingolipid hydrolases; deficiency causes a variant of **metachromatic leukodystrophy**. Saposin D (405–486) is a specific activator of sphingomyelin phosphodiesterase (EC 3.1.4.12) and deficiency causes a variant of **Tay-Sachs disease** (G_{M2} gangliosidosis). *See also swaposin*.

saprophyte any plant, fungus, or microorganism that lives on dead or decaying organic matter. —**saprophytic** *adj.*

sarafotoxin any of several toxic peptides that have been purified from the burrowing asp, *Atractaspis engaddensis*, and that cause cardiac arrest when injected into mice, probably as a direct result of coronary vasospasm arising from their strong vasoconstrictor activity. All have 21 amino acids, with two pairs of half-cystine residues in identical positions and a hydrophobic tail with a C-terminal tryptophan residue. They have strong homology with **endothelins** and have endothelin receptor agonist action. Sarafotoxins S6A–E are synthesized as a precursor from *A. engaddensis*: database code SRTX_ATREN, 543 amino acids (62.32 kDa). Sarafotoxin S6D has the sequence

CTCKDMTDKCELYFCHQDIHW.

See also **bibrototoxin**.

Saran Wrap *proprietary name* for a vinylidene chloride copolymer cling-film used for gel drying, food storage, and other purposes.

α -sarcin a cytotoxin produced by *Aspergillus giganteus*. It has ribonuclease activity, specific for purines in both single-stranded and double-stranded RNA. It inhibits protein synthesis, cleaving a single phosphodiester linkage between guanosine and adenosine residues in 28S rRNA in the 60S' subunit of the eukaryotic ribosome, resulting in loss of 393 nucleotides from the 3' end of 28S rRNA. This single site of action in the intact ribosome contrasts with its action on purified 28S rRNA, in which it causes extensive degradation. cut-

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ting on the 3' side of every adenine or guanine. Example (precursor): database code RNAS_ASPGI, 177 amino acids (19.72 kDa), residues 1–27 being the signal, and 28–177 the enzyme.

sarcolemma the outer membrane of a muscle fibre. It consists of the plasma membrane, a covering basement membrane (about 100 nm thick and sometimes common to more than one fibre), and the associated loose network of collagen fibres. The term was originally used by light microscopists to designate the line visibly marking the outer fibre edge.

sarcoma a malignant tumour of connective tissue or of its derivatives. —**sarcomatous** *adj.*

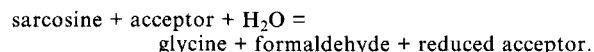
sarcomere the functional unit of a myofibril of vertebrate muscle, about 2.3 μ m long. In **striated muscle** the sarcomeres of many parallel myofibrils are positioned such that myosin **thick filaments** are aligned in register across the myofibril (having a dense appearance in microscopic preparations), as also are the actin **thin filaments** (which have a lighter appearance), resulting in the alternation of light, isotropic bands known as **I bands**, and dark, anisotropic bands known as **A bands**; the I band is bisected by a very dense, narrow **Z line**, while the central, less dense region of the A band is known as the **H zone**, which in turn is bisected by the dark **M line**, or midline, the location of specific proteins that link adjacent thick filaments to each other. The Z lines are due to attachment sites for thin filaments.

sarcoplasm all of the contents of a muscle fibre except nuclei. The sarcoplasm is pervaded by the **sarcoplasmic reticulum**. *Compare* **cytoplasm**. —**sarcoplasmic** *adj.*

sarcoplasmic reticulum *abbr.*: SR; a fine reticular network of membrane-limited elements that pervades the **sarcoplasm** of a muscle fibre. Like the endoplasmic reticulum the SR is continuous over large portions of the cell, it has a large surface-to-volume ratio, and is continuous with and/or homologous to the nuclear envelope. At the level of the I band (*see* **sarcomere**), the SR forms dilated sacs, or cisternae. Opposite the I bands and Z lines of every sarcomere there are always two SR terminal sacs that face each other across a narrow space, through which run **T tubules**. At the level of the A–I junction longitudinally orientated SR sacs run along the fibril, forming a palisade around the A band. The SR contains a Ca^{2+} -translocating ATPase that moves Ca^{2+} into the SR during ATP hydrolysis; ATP resynthesis is associated with movement of Ca^{2+} out of the SR vesicles.

sarcosine *N*-methylglycine; an intermediate in the conversion of choline, via betaine and dimethylglycine, to glycine, and in the synthesis of choline from glycine. *See also* **sarcosinemia**.

sarcosine dehydrogenase EC 1.5.99.1; a flavoprotein enzyme that catalyses the reaction:



The flavin is FMN.

sarcosinemia or (*esp. Brit.*) **sarcosinaemia** a rare inherited metabolic disease of humans, also found in ethylnitrosourea-induced mutants of mice, giving elevated plasma and urine levels of **sarcosine**. It may be accompanied by mental retardation or psychiatric disturbances. Primary sarcosinemia results from deficiency of **sarcosine dehydrogenase** (EC 1.5.99.1). Sarcosinemia may also be found in folate deficiency (*see* **follic acid**) in humans and in glutaric aciduria type II, but in primary sarcosinemia elevated levels of other organic acids are not found. Inheritance is probably autosomal recessive.

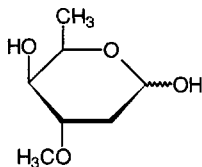
sarcosome a subcellular organelle found in muscle tissue that has essentially the same biochemical properties as a **mitochondrion** in other tissues. In muscles with intermittent activity, sarcosomes are found at the level of the I bands (*see* **sarcomere**); such muscles include skeletal muscles of vertebrates and invertebrates whose metabolism during activity is largely anaerobic. In muscles required to perform sustained activity, such as bird and insect flight muscles and vertebrate heart muscle, sarcosomes are found at the level of the A bands; these muscles have predominantly aerobic metabolism.



sarcotubule another name for **T tubule**.

sarin methylphosphonofluoridic acid 1-methylethyl ester; an **organophosphate** (def. 2) that inhibits acetylcholinesterase and other serine esterases. It is a military nerve gas but is useful in the treatment of leprosy and has some effect on AIDS.

D-sarmentose 2,6-dideoxy-3-O-methyl-D-xylo-hexose; a substance found as a component of some **cardiac glycosides**.



satellite DNA a fraction of DNA, present in most eukaryotic cells in amounts of up to 10% or more of total DNA, with a highly repetitive nucleotide sequence. Its base composition differs sufficiently from the bulk of the DNA to give rise to a 'satellite' band having a different buoyant density when DNA preparations, following **shear force degradation**, are centrifuged in cesium chloride density gradients. For instance, mouse satellite DNA contains about a million copies of a repeating sequence of approximately 300 base pairs with a lower (G+C) content than that of the moderately repetitive and unique DNA sequences. *See also* **minisatellite**.

satellite RNA small RNAs (≈ 350 bases) also called **virusoids**, found in plants; they are **encapsidated** by some plant viruses. *See also* **ribozyme**.

satellite virus a small virus that occurs in association with another virus, upon which it is dependent. An example is **satellite tobacco mosaic virus (STMV)**. This 172 Å diameter particle encloses a 1059 nucleotide RNA that encodes only two proteins, the 159-residue viral coat protein and a 58-residue protein of unknown function. STMV can only multiply in cells that have been coinfecting with tobacco mosaic virus. This is the only known example of such a parasitic relationship between a spherical virus and a rod-shaped virus.

saturable capable of undergoing **saturation** (def. 5). — **saturability** *n*.

saturated fatty acid any **fatty acid** in which the carbon-carbon bonds of the alkyl chain are exclusively single bonds. The predominant naturally occurring saturated fatty acids are **palmitate** and **stearate**; others include **laurate**, **myristate**, **arachidate** (see **arachidic acid**), **behenate** (see **behenoyl**), **lignocerate**, and **cerotate** (see **ceretoyl**). All saturated fatty acids can be derived metabolically from acetyl-CoA, both in animals and plants. *See also* **fatty acid nomenclature**.

saturates an informal term for **saturated fatty acids**.

saturation **1** (of a chemical compound) the point at which all the valence bonds of the component atoms are satisfied, especially the state of an organic compound that contains only single carbon-carbon bonds. **2** (of a solution) the state in which it has the greatest concentration of the solute that can remain in stable equilibrium with undissolved solute at a given temperature and pressure. **3** (of a gas) the state in which there is the greatest concentration of the vapour associated with a liquid or solid that can remain in stable equilibrium with unvaporized liquid or solid, respectively, at a given temperature and pressure. **4** (of an active site of an enzyme or transporter) the point at which the site is fully occupied with ligand. Saturable enzyme or receptor systems exhibit hyperbolic or sigmoid curves of activity versus concentration (see **saturation kinetics**). **5** the act or process of bringing to or towards saturation; especially the addition of hydrogen to triple and double bonds.

saturation analysis or **binding assay** a development of **isotope dilution analysis** used for the measurement of very low concentrations of biologically potent substances (e.g. a specified hor-

mone) in small samples of complex mixtures such as biological fluids. The essential steps of the technique are: (1) the substance, S, is allowed to react with a fixed amount of a specific binding agent under conditions such that the latter is always fully saturated with S; (2) the partitioning of S between the free and bound states is determined by the use of an indicator and by physical separation of the free from the bound form; and (3) the distribution of the indicator is compared with distributions obtained using a series of known amounts of S. The indicator is a labelled form of pure S, the label commonly being a radioisotope of high specific activity; alternatively, fluorescent, enzymic, or other labels may be used if they give the required sensitivity. The specific binding agent is generally an antibody (usually tailor-made), as in **enzyme immunoassay**, **fluoroimmunoassay**, or **radioimmunoassay**, or a naturally occurring macromolecular carrier or cellular receptor substance, as in **competitive protein-binding assay**.

saturation fraction *symbol:* y ; the ratio of the concentration of bound ligand to the total concentration of the binding agent. Hence, in a binding reaction where a compound P binds X to form PX, then $y = [PX]/[P_{total}]$. A plot of y against $[X]$ is known as the **adsorption isotherm** for this system.

saturation kinetics the kinetics of a reaction (or of **mediated transport**) when the velocity of the reaction (or of the transport) increases to a maximal value (plateau) as the concentration of the reactant (or of a component being transported) is increased.

saturation ratio or **cooperativity index** *symbol:* R_s ; (of an allosteric protein) the ratio of the concentration of the substrate (or other small molecule) necessary to produce 90% to that required to produce 10% saturation of the specific sites on a protein. It is a convenient measure of the extent of cooperativity. *See also* **cooperative ligand binding**.

saxitoxin *abbr.:* STX; a paralytic poison, produced by the marine dinoflagellates *Gonyaulax catenella* and *G. excavata*. It can be accumulated by shellfish, making them toxic when eaten. Its action is similar to that of **tetrodotoxin**, with which it competes for sodium channels.

Sb *symbol* for antimony.

SBH *abbr. for* sequencing by hybridization; a hypothetical method yet to be implemented for the very rapid sequencing of DNA. It is based on the idea of having all possible octamers (there are 63 536) on a small hybridization screen called a DNA chip. As the octamers form nested arrays, a DNA fragment (of a few hundred bases) would only hybridize to some of these and the sequence could be deduced computationally.

SBS model or **SBS** *abbr. for* side-by-side model.

sc *abbr. for* synclinal (def. 2).

s.c. *abbr. for* subcutaneous.

Sc **1** *symbol* for scandium. **2** *abbr. for* secretory component (see **secretory piece**).

SCAD *abbr. for* short-chain acyl-CoA dehydrogenase or butyryl-CoA dehydrogenase; see **beta-oxidation system**.

scaffolding protein any of a group of proteins that have specific binding sites and are instrumental in determining the structure and assembly of large three-dimensional structures such as a virus. Such proteins are integral to the **self-assembly doctrine** elaborated by Caspar and Klug.

scalar **1** a physical quantity, such as mass or time, that is characterized only by its magnitude and does not contain any concept of direction. **2** describable by a number representing a point on a scale undirected in space. *Compare* **tensor**, **vector** (def. 1).

scaler an electronic circuit or device that counts electrical pulses by aggregating the pulses and giving one output pulse for a predetermined number of input pulses.

SCAMP *abbr. for* secretory carrier membrane protein; it is a Ca^{2+} -binding protein with leucine zipper and zinc finger motifs. SCAMPs are widely distributed as components of post-Golgi membranes. They function in the mechanisms for recycling proteins between the Golgi and the cell surface. Ex-



scanner

ample from rat, SCAMP 37: database code S37395, 338 amino acids (37.96 kDa).

scanner an instrument or device used to measure or sample the distribution of some quantity or condition in a particular system, area, or region, e.g. to measure the distribution of colour or radioactivity on a chromatogram or electrophoretogram, or to measure the radial distribution of light absorbance in a cell in an ultracentrifuge.

scanning electron microscope *abbr.*: SEM; *see* **electron microscope**.

scanning proton microprobe an instrument in which a sample is bombarded by a high-energy (1–4 MeV) proton beam, causing emission of X-rays that can be characteristically attributed to certain elements. The technique permits the detection of elements down to about 1 ppm.

Scatchard plot a method of analysis of the binding of ligand to a macromolecule (or receptor), based on the **Scatchard equation**, $\bar{v}/[L] = (n - \bar{v})/K$, where $\bar{v}$ is the average number of ligand molecules bound per binding macromolecule, $[L]$ is the free ligand concentration, n is the number of binding sites on the macromolecule and K is the intrinsic association constant. A plot is made of $\bar{v}/[L]$ on the y -axis and $\bar{v}$ on the x -axis. If the binding sites are identical and independent, a straight line is obtained of slope $-K$ with intercepts on the x -axis of n and on the y -axis of Kn . If there is interaction between the binding sites, or there are several classes of independent sites, the plot is not linear. [After George Scatchard (1892–1973), US physical chemist.]

scatter to disperse in random direction(s) by impact or other interaction, especially light, electrons, neutrons, or X-rays. The measurement of scattered rays or particles may lead to useful analysis, as in **light scattering**.

scatter diagram a graph upon which two measured characteristics of each individual of a population are represented by one dot.

scavenger (*in chemistry*) a substance that reacts with (or otherwise removes) a trace component, such as a trace metal ion, or that traps a reactive **reaction intermediate**.

SCF *abbr.* for stem cell factor.

Schally, Andrew Victor (1926–), Polish-born US biochemist and physiologist; Nobel Laureate in Physiology or Medicine (1977) jointly with R. Guillemin 'for their discoveries concerning the peptide hormone production of the brain' [prize shared with R. Yalow].

Schardinger dextrin a former name for **cyclodextrin**. [After Franz Schardinger (1853–1920), who first described such compounds in 1908.]

Schardinger enzyme an alternative name for **xanthine oxidase**.

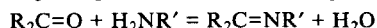
Schardinger reaction the decolorization of Methylene Blue by milk in the presence of formaldehyde; the reaction is catalysed by **xanthine oxidase** (or Schardinger enzyme) in the milk.

Schatz, Albert (1920–), PhD student of S. A. Waksman, legally recognized as the co-discoverer of **streptomycin**.

Scheie syndrome an alternative name for **mucopolysaccharidosis I S**. [After Harold G. Scheie (1909–).]

Schiff, Hugo (1834–1915), German chemist who devised the reagent for aldehydes known as **Schiff reagent**.

Schiff base or **Schiff's base** any **imine** having the more limited general structure $R_2C=NR'$ where R may be any organyl group or H and R' is any organyl group. It may be formed by condensation of the carbonyl group of an aldehyde or ketone with the amino group of a primary amine according to the equation:



Often considered to be synonymous with **azomethine**.

Schiff reagent a solution of leucofuchsin – fuchsin bleached with sulfurous acid – that produces a red colour when reacted with an aldehyde. The reagent is used in the **Feulgen reaction** and the **periodic acid-Schiff reaction**.

Schiff test 1 (for aldehydes) a test based on the red colour produced when aldehydes react with **Schiff reagent 2** (for uric

acid) a test based on the reduction of silver ions (silver nitrate solution) to metallic silver by uric acid in sodium carbonate solution. **3** (for urea) a test based on the formation of a purple colour when urea is treated with a concentrated solution of furfural in concentrated hydrochloric acid solution. Allantoin, but not uric acid, also gives a positive test.

Schild equation the expected relationship between the **concentration ratio**, r , and the concentration of a reversible competitive antagonist, B. It is given by $r = [B]/K_B + 1$, where K_B is the dissociation equilibrium constant for the combination of B with the receptor. *Compare* **Gaddum equation**. [After Heinz Otto Schild (1906–84), Austrian-born British pharmacologist.]

Schild plot a plot of $\log(\text{concentration ratio} - 1)$ versus $\log$ concentration of antagonist. For a reversible competitive antagonist the plot has a gradient of 1. The intercept on the abscissa then defines the concentration of antagonist required to reduce the response of the target tissue (or organism) by half (the dissociation equilibrium constant: $-\log K_B = pA_2$ when the slope is exactly unity and there are no complicating factors; *see* **pA₂**).

Schilling test a test for cobalamin (vitamin B₁₂) malabsorption that depends on the oral administration of ^{57}Co - or ^{58}Co -labelled **cyanocobalamin** and measurement of the urinary excretion of the radioactive label during the ensuing 24 hours. [After Robert Frederick Schilling (1919–), US hematologist and nutritionist.]

schizophrenia any of a group of mental disorders characterized by a special type of personality disintegration, random thought processes, and impaired relation to reality. The biochemical basis of the disease remains enigmatic. Antipsychotic (neuroleptic) drugs such as **chlorpromazine**, which are widely used in the symptomatic treatment of all forms of schizophrenia, are considered to act by blocking postsynaptic dopamine D₂ receptors, though most are not totally selective in this action. —**schizophrenic** *adj.*, *n.*

Schizosaccharomyces pombe the fission yeast that is used to brew the African beer called *pombe*. The rod-shaped cells grow by elongation, and the spores germinate and divide as haploids. The arrangement of the membranous organelles, e.g. the Golgi apparatus, in fission yeast is closely similar to that of higher eukaryotes, in marked contrast to that of budding yeast. It is especially suitable for analysing the relationship between cell size and the cell cycle for two reasons: (1) the cells divide symmetrically, unlike budding yeast; and (2) the yeast grows in length but not in diameter.

Schlesinger test a qualitative test for **urobilin** in urine. The urobilinogen in the urine is first oxidized by iodine to urobilin, then a zinc-urobilin complex is formed by addition of zinc acetate in ethanol; the complex has a yellow-green fluorescence and an absorption band at 507 nm.

Schlieren method an optical method of exhibiting inhomogeneities in refractive index in transparent media, dependent on the deflection of a ray of light from its undisturbed path when it passes through a medium in which there is a solute inducing a gradient of refractive index normal to the ray. The curvature of the ray is proportional to the refractive-index gradient in the direction normal to the ray. It is used in biochemistry particularly in measurements of diffusion, sedimentation, and electrophoretic mobility of proteins and other substances. [From German *Schliere*, streak.]

Schmidt-Thannhauser procedure a method for the extraction and determination of DNA, RNA, and phosphoproteins. Material insoluble in dilute, cold trichloroacetic acid is dried, digested with alkali, and subsequently precipitated with a strong acid. The total phosphorus in the precipitate represents DNA, the soluble inorganic phosphorus represents the phosphoproteins, and the soluble organic phosphorus represents the RNA. [After Gerhard Schmidt (1901–81) and Siegfried Joseph Thannhauser (1885–1962), German-born US biochemist.]

Schoenheimer, Rudolf (1898–1941), German-born US biochemist; a leader in early work with isotopic tracers (^2H and ^{15}N) that led him (with colleagues) to put forward the concept

of the **dynamic state of body constituents**, which became an important principle in metabolic regulation.

Schultz–Dale reaction an *in vitro* anaphylactic reaction in which the uterus or ileum of a sensitized animal (guinea pig) contracts specifically when a small amount of the sensitizing antigen is added to the bathing medium. The reaction is caused by histamine and similar substances that are released when the antigen combines with cellular antibody. [After H. H. Dale.]

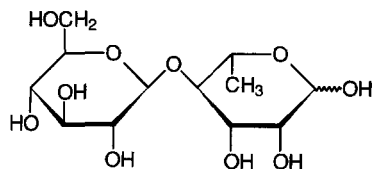
Schütz–Borrisow rule or **Schütz law** an empirical rule according to which the velocity of an enzyme reaction is proportional to the square root of the enzyme concentration. It was first developed for pepsin but applies only to the crude enzyme under certain limited conditions; similar results have been obtained with some other proteinases. The effect is due to the presence of a reversibly dissociating pepsin inhibitor in crude pepsin preparations.

Schwann cell a uninucleate cell of the **neurilemma** surrounding a myelinated neuronal axon that is responsible for the formation and maintenance of the **myelin sheath** in peripheral myelinated neurons. [After Theodor Schwann (1810–82), German anatomist and physiologist.]

Schwannomin *see* **merlin**.

SCIF *abbr.* for secondary cytotoxic T-cell-inducing factor (i.e. **interleukin 2**).

scillabiose the disaccharide 4-*O*- β -D-glucopyranosyl-L-rhamnose; it occurs in a **glycoside** of *Scilla maritima*.



scinderin or **adseverin** a Ca^{2+} -dependent cytosolic actin filament severing protein present in chromaffin cells, platelets, and a variety of secretory cells. Example (bovine): database code BBSCINDE, 715 amino acids (80.68 kDa).

scintillant 1 (of a phosphor) exhibiting scintillation. **2** a jargon term for **scintillator**.

scintillate 1 to give off sparks, to sparkle. **2** (of a phosphor) to emit quanta of light discontinuously, by fluorescence, when struck by a charged particle or a high-energy photon.

scintillation 1 the act of scintillating. **2** the flash of light produced in a phosphor when it is struck by a charged particle or a high-energy photon.

scintillation autoradiography an alternative term for **fluorography** (def. 2).

scintillation cocktail or **scintillation fluid** any mixture, containing a **scintillator**, to which **beta emitters** can be added for the purpose of **scintillation counting**; the radiation excites the scintillator to emit light. Many scintillation cocktails are based on hydrophobic solvents, since for the most part scintillators sparingly soluble in aqueous solutions have been used. The first scintillator to be widely used was 2,5-diphenyloxazole (PPO), often used in conjunction with 1,4-bis(5-phenyl-2-oxazolyl)benzene (PoPoP), which is excited by the light emitted by PPO to emit light of a higher wavelength more suitable for measurement. Subsequently many other scintillators, especially biodegradable compounds and others more compatible with water, have been developed. *See also* **biphenylphenyloxadiazole**, **bismethylstyrylbenzene**.

scintillation counter an instrument for the detection and measurement of the intensity of high-energy radiation. Instruments for detecting gamma radiation contain a detector, normally a crystal of sodium iodide containing a small amount of thallium, and this is encased in a metal 'top hat' lined with a

white reflector, the open end being sealed with Perspex. When the sodium iodide absorbs γ rays it emits a flash of light. For the counting of β^- emissions, the radioactive substance is dissolved or suspended in a **scintillation cocktail** contained in a glass or plastic vial, which is inserted into a chamber in the counter. Sometimes a solid scintillant is used for this also, e.g. for the effluent from chromatography columns. In the case of both types of instrument, the flashes of light produced are detected by a photomultiplier and converted into pulses of electric current which may be counted by an electronic counter.

scintillation fluid an alternative term for **scintillation cocktail**.

scintillation spectrometer another name for **scintillation counter**.

scintillator or **scintillant** any phosphor that fluoresces when struck by a charged particle or a high-energy photon. *See* **scintillation cocktail**.

scintillon a subcellular, crystal-like structure, found in bioluminescent organisms, that emits light on acidification in the presence of oxygen.

scissile capable of being split easily or cut smoothly, especially of an easily cleaved chemical bond.

scission a division or splitting; an act of dividing a nonisometric object orthogonally to its longest axis, especially a chemical bond or a fibrous molecule; **cleavage** (def. 2).

scleral pertaining to the sclera, the fibrous outer layer of the eyeball.

scleroprotein any of a group of simple, generally fibrous proteins that are insoluble in aqueous solutions; examples include **keratin** and **silk fibroin**.

sclerosis hardening of a tissue as a sequel to inflammation; especially an increase of connective tissue in the nervous system. *See also* **arteriosclerosis**. —**sclerotic** *adj.*

scopolamine *see* **hyoscyine**.

scorbutic of, pertaining to, or affected with **scurvy**.

scotophobin a 15-residue peptide that accumulates in the brains of rats trained to avoid the dark; administration is said to elicit a similar response in untrained animals. It is thought to be the first memory-directing substance isolated and identified.

scotopia adaptation to darkness; the adjustment of the eye to vision in dim light or the dark. It is considered to involve mainly retinal **rod cells**. *Compare* **photopia**. —**scotopic** *adj.*

scotopsin an **opsin** that is the apoprotein of a **rhodopsin**.

SCP *abbr.* for single-cell protein.

scrapie a **spongiform encephalopathy** of sheep and goats, and the first to be identified of a group once thought to be 'slow virus' infections occurring naturally and caused by an unconventional agent. Scrapie has similarities to two other animal diseases, transmissible mink encephalopathy and **bovine spongiform encephalopathy**, and three transmissible human dementias, **kuru**, **Creutzfeldt–Jakob disease**, and **Gerstmann–Sträussler syndrome**. All these diseases are now accepted as being caused by alterations in the **prion proteins** in the nervous system. The disease has been known since its first reported occurrence in merino sheep in Spain in 1732, which was rapidly followed by reports of other cases in Britain and Germany. The incubation period for the scrapie agent is usually 1–3 years. Different breeds of sheep exhibit markedly different susceptibilities to scrapie. As many as 20 different strains of scrapie agent, each with distinct properties in terms of incubation period in a given host and location of damage within the brain have been identified and transferred to mice. It would appear that the strains are passed to the progeny, so any explanation of the nature of the infective agent and process must take this into account.

screen 1 to protect from; esp. from harmful agents or radioactivity using sheets of lead, glass, or plastic; or from radiofrequency interference using a copper sheath or wire net. **2** to sift data, often using a limited or general assay, for selection of a particular phenomenon or result; samples selected by preliminary screening can be isolated for more detailed study. **3** (*in clinical chemistry*) to carry out any diagnostic procedure on a



screw axis

large cohort of patients with a view to detecting hitherto asymptomatic disease; *see also* **genetic screening**. **4** any device or material that acts to screen.

screw axis a crystal or an oligomeric protein molecule is said to possess an *n*-fold screw axis if it presents exactly the same appearance after rotation about that axis of $360^\circ/n$ coupled with a translation parallel to the axis. *Compare* **rotation axis**.

scRNA *abbr.* for small cytoplasmic RNA.

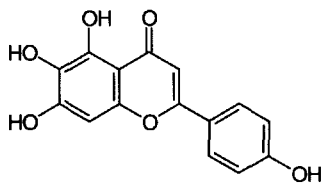
scRNP *abbr.* for small cytoplasmic ribonucleoprotein.

scurvy avitaminosis C; a deficiency disease caused by lack of **ascorbic acid** (vitamin C) in the diet. It is characterized by anemia, sponginess and ulceration of the gums, hemorrhages into the skin and subcutaneous tissues, and delayed wound healing.

scute a gene in *Drosophila* for **AS-C protein**.

scutellarin EC 3.4.21.60; *other name*: taipan activator (being from the venom of the taipan (snake), *Oxyuranus scutellatus*). An enzyme that catalyses the selective cleavage of Arg-[Ile and Arg]-Ile in prothrombin to form **thrombin** and two inactive fragments; its specificity is similar to that of factor Xa (*see* **blood coagulation**). *Compare* **scutellarin**.

scutellarein 4',5,6,7-tetrahydroxyflavone; the aglycon of scutellarin. *Compare* **scutellarin**.



scutellarin the glucuronide of **scutellarein**, from *Scutellaria* spp. *Compare* **scutellarin**.

scryp *colloquialism* for scRNP (i.e. **small cytoplasmic ribonucleoprotein**).

SD *abbr.* for standard deviation.

SDS *abbr.* for sodium dodecyl sulfate.

SD sequence *abbr.* for Shine-Dalgarno sequence.

SDS-PAGE *abbr.* for SDS-polyacrylamide gel electrophoresis.

SDS-polyacrylamide gel electrophoresis or **SDS gel electrophoresis** *abbr.*: SDS-PAGE; a rapid and inexpensive method for resolving a protein into its subunits and determining their relative molecular masses (M_r values), based on the ability of **sodium dodecyl sulfate** to cause dissociation of oligomeric proteins and to bind to the subunits to form complexes of constant charge-to-mass ratio. On **zone electrophoresis** in polyacrylamide gel, such complexes often have mobilities directly related to their M_r values, which, if unknown, may thus be obtained by comparison with standards. *Other name*: detergent gel electrophoresis.

Se *symbol* for selenium.

SE *abbr.* for standard error; *see* **standard error of estimate of the mean**.

SEA a gene family encoding membrane receptor tyrosine kinases having strong homology with the **insulin receptor** family. The acutely transforming virus, avian erythroblastosis virus (*abbr.*: AEV) strain 13, contains the oncogene *v-sea*; this transforms fibroblasts and erythroblasts, but not avian myeloid cells. Sea, the product of *c-sea*, the cellular homologue of *v-sea*, phosphorylates Shc (*see* **SHC**) on tyrosine residues. Example from AEV strain 13: database code KSEA_AVIET, 370 amino acids (41.70 kDa).

sealase a former name for DNA ligase (ATP); *see* **DNA ligase**.

sea-urchin-hatching proteinase *see* **envelysin**.

sebaceous gland any of the single or branched glands in the skin that produce the oily substance, **sebum**. They open into the hair follicles.

sebum the oily secretion of the sebaceous glands in the skin. It

contains lipids such as waxes, squalene, and triacylglycerols; the triacylglycerols are broken down by anaerobic bacteria during secretion to form fatty acids. Sebum helps to form an effective barrier against water loss.

sec *symbol* for **secant** (def. 1).

sec. *abbr.* and *obsolete symbol* for **second** (def. 1).

sec- *abbr.*: *s-*; *prefix* (in chemical nomenclature) signifying **secondary** (def. 1a).

secant **1** *abbr.* and *symbol*: sec; the reciprocal of the **cosine** of an angle. **2** a straight line cutting a curve at two or more points.

seco- *prefix* (in chemical nomenclature) denoting cleavage of a ring with addition of one or more hydrogen atoms at each terminal group thus created. The numbering of the skeletal atoms of the parent molecule is retained.

seco-carotenoid a **carotenoid** in which one of the rings has undergone fission by breakage of a carbon-carbon bond; e.g. semi- β -carotenone (5,6-seco- β - β -carotene-5,6-dione).

second **1** *symbol*: s; the SI base unit of time, equal to the duration of 9 192 631 770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the caesium-133 atom. **2** *symbol*: °; a unit of plane angle equal to 1/60 of a minute; i.e. 4.848 14 microradian ($\pi/648\,000$ rad). *See* **angle** (def. 1).

secondary (in chemistry) **1 a** *prefix*: *sec-* (*abbr.*: *s-*); describing an alkyl compound (e.g. an alkanol) in which the functional group (e.g. a hydroxyl group) is attached to a carbon atom linked to two others. **b** describing the carbon atom bearing the functional group in such a compound. **2** describing an amide or amine in which two appropriate groups are attached to a nitrogen atom. **3** describing a salt formed by replacing two of the ionizable hydrogen atoms of a tribasic acid by one or two other cations of appropriate valency. *Compare* **primary**, **tertiary**.

secondary bond a loose term to describe any of the weak attractive forces between atoms and/or molecules, especially when these stabilize a three-dimensional structure. It includes **hydrogen bonds**, **hydrophobic interactions**, and **van der Waals forces**.

secondary cell wall deposition (in plants) the deposition of microfibrils and noncellulosic polysaccharides on the primary cell wall. It provides additional conducting and supporting tissue and frequently makes up the greater part of the mature cell wall.

secondary charge effect an effect seen in the hydrodynamic study of charged macromolecules when the positive and negative ions of the supporting electrolyte have different sedimentation (or diffusion) coefficients, leading to the establishment of local electric fields due to charge separation. These local fields act as an additional force on the macromolecules, either accelerating or retarding their movement.

secondary cytotoxic T cell-inducing factor *abbr.*: SCIF; *an alternative name* for **interleukin 2**.

secondary deficiency a nutritional deficiency occurring when the dietary content of some essential nutrient is known to be adequate in normal conditions. It may be caused by a failure of absorption from the gut due to disease, diminished synthesis of a vitamin by the intestinal flora during treatment with an antibacterial agent, or increased requirement, e.g. in pregnancy.

secondary electron an electron emitted from a solid as a result of the impact of another electron. If the energy of the incident primary electron is sufficient, more than one secondary electron may be emitted, an effect exploited in an **electron multiplier**. *See also* **photomultiplier**.

secondary emission the phenomenon of emission of **secondary electrons**.

secondary F'-containing strain *see* **F' plasmid**.

secondary fluor a second fluorescent agent that absorbs the fluorescent light emitted by a **primary fluor** and is thus excited to emit light at a higher wavelength, which may be more convenient for detection.

secondary hyperlipidemia *see* **hyperlipidemia**.

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secondary immune response

secondary immune response an immune response to an **immunogen** to which an animal has previously been exposed and to which its lymphoid tissue has been primed. It consists characteristically of the rapid production of large amounts of **antibody** over a few days, following which the level of antibody slowly declines in an exponential manner. *Compare* **primary immune response**.

secondary lysosome or **phagolysosome** a **lysosome** that contains not only the specific acid hydrolases but also their substrates taken up from the environment or sequestered from the cell's own cytoplasm. Secondary lysosomes are formed by fusion of a **primary lysosome** with a **phagosome** or a vesicle containing endocytosed matter. *Compare* **telolysosome**.

secondary messenger a variant form of **second messenger**.

secondary metabolism the formation of end products of metabolism – **secondary metabolites** – that often have no readily apparent use in the producing organism.

secondary metabolite any of a group of chemically very diverse natural products that have a restricted taxonomic distribution, usually possess no immediately obvious function in cell growth, and are often (but not always) synthesized by cells that have stopped dividing; typically they exist as members of closely related chemical families, usually of $M_r < 1500$, but some bacterial toxins are considerably larger. An example of a secondary metabolite is penicillin.

secondary response *see* **secondary immune response**.

secondary structure 1 (of a protein) the arrangements of the polypeptide chain that form more-or-less regular hydrogen-bonded structures, in particular α -helical and β -pleated sheet structures; *see* **alpha helix**, **beta strand**. 2 (of a nucleic acid) the arrangement of one or two polynucleotide chains in bihelical structures, stabilized by hydrogen bonds between complementary bases (*see* **complementary base sequence**); *compare* **clover-leaf structure**. *See also* **primary structure**, **quaternary structure**, **tertiary structure**.

second law of thermodynamics *see* **thermodynamics**.

second messenger or **secondary messenger** a mediator that is caused to accumulate in an effector (target) cell by the action of a **hormone**, **growth factor**, or other agonist, and that brings about the action of that agonist on the cell. The major second messenger systems are currently recognized to include: synthesis of cyclic AMP by **adenylate cyclase** (the first of such systems to be discovered); synthesis of cyclic GMP by **guanylate cyclase**; opening of **ion channels**; the phosphoinositide system, (*see* **phosphatidylinositol cycle**) involving activation of phosphoinositide-specific phospholipase C (*see* **phospholipase**), or 1-phosphatidylinositol 3-kinase; and protein phosphorylation by serine/threonine-specific or tyrosine-specific **protein kinases**. An important component in several of these systems involves coupling of activated receptor to the effector system through a **G-protein**.

second-order rate constant *see* **rate constant**.

second-order reaction *see* **order of reaction**.

seco-steroid a **steroid** in which one of the rings has undergone fission by breakage of a carbon-carbon bond.

secretagogue or **secretogogue** an agent that stimulates secretion (usually of a named hormone, juice, or other glandular product). *See also* **+agogue**.

secretase a type of enzyme that cleaves the extracellular domain of a number of membrane proteins of type I or type II topology, releasing the extracellular part of the protein into the circulation. Cleavage normally occurs close to the extracellular face of the membrane. Proteins on which secretases act include Alzheimer's amyloid precursor protein, angiotensin converting enzyme, transforming growth factor- α , and the tumour necrosis factor ligand and receptor superfamilies. Secretase activity is inhibited by a number of metalloproteinase inhibitors, suggesting that secretases may be metalloproteinases.

secrete to elaborate and emit a **secretion** from a cell or, particularly, from a gland.

secretin a digestive hormone, discovered in 1902 by William

Maddock Bayliss (1860–1924, British biologist) and Ernest Henry Starling (1866–1927, British physiologist) (the first hormone to be discovered). It has the sequence (human):



(3.04 kDa), and is formed from a 134-amino-acid precursor, which is synthesized in specific endocrine cells, S cells, in the mucosa of the small intestine. It stimulates the release of bicarbonate, enzymes, and potassium ions from the pancreas and stimulates the secretion of enzymes and electrolytes in the gut; it also inhibits HCl production by the stomach. Secretin lends its name to a superfamily that on structural grounds includes **gastric inhibitory peptide**, **GHRH**, **glucagon**, **peptide II**, and **vasoactive intestinal polypeptide**. Example of precursor from *Mus musculus*: database code SECR_MOUSE, 133 amino acids (14.91 kDa).

secretin receptor any of several membrane proteins that bind **secretin** and mediate its intracellular effects. These are seven-transmembrane-domain, G-protein-coupled, adenylate cyclase stimulating receptors, not structurally related to other G-protein-coupled receptors. Example from human (precursor): database code SCRC_HUMAN, 440 amino acids (50.23 kDa).

secretion 1 the process of elaborating, segregating, and emitting a substance or juice from a cell, organ, or organism; e.g. the secretion of insulin by the B cells of the islets of Langerhans, or the secretion of urine by the kidneys. At the cellular level there are three basic types of secretion: **apocrine**, **holocrine**, and **merocrine**. 2 a product of **secretion** (def. 1) from a cell, organ, or organism.

secretogogue a variant spelling of **secretagogue**.

secretogranin I another name for **chromogranin B** (*see* **chromogranin**).

secretor any person who secretes water-soluble glycoproteins with A, B, or H blood-group specificities into their mucous secretions, e.g. saliva and gastric juices. Over 80% of humans are secretors, their status being determined by the allelic constitution of a **secretor gene** in their genomes.

secretor gene a gene determining **secretor** phenotype in humans. Whether a person is a secretor or nonsecretor is determined by a pair of allelic genes, *Se* and *se*. Secretors are either *Se/Se* or *Se/se* while nonsecretors are *se/se*. These alleles occur at a different locus from the ABO locus and do not affect the A, B, and H specificities of the red blood cells.

secretory describing a molecule or molecular fragment destined for secretion from a cell, such as secretory protein, polypeptide, etc.

secretory actin-binding protein *abbr.*: SABP; *also called* prolactin-inducible protein; gross cystic disease fluid protein 15; a glycoprotein that has been isolated in several different studies (as the alternative names indicate) but whose function remains unknown. As the prolactin-inducible protein, it was identified as a highly specific and sensitive marker of primary and metastatic breast cancer. Example from human (precursor): database code SABP_HUMAN, 146 amino acids (16.55 kDa).

secretory component another name for **secretory piece**.

secretory granule a small subcellular vesicle, surrounded by a single-layered membrane, that is formed from the Golgi apparatus and contains a highly concentrated protein destined for secretion. Secretory granules move towards the periphery of the cell, their membranes fuse with the cell membrane, and their protein load is exteriorized. Processing of the contained protein may take place in secretory granules. *Compare* **zymogen granule**.

secretory piece (*abbr.*: SP) or **secretory component** (*abbr.*: Sc) the 58 kDa polypeptide that joins together the two identical monomers comprising the secretory form of **immunoglobulin A**. It has strong affinity for mucus, thus prolonging retention of IgA on mucous surfaces, and may protect this class of immunoglobulin against proteolytic destruction in the digestive tract. *Other names*: T chain; transport piece.

secretory protein I *abbr.*: SP-I; a glycoprotein of the parathyroid gland that is co-secreted *in vitro* with **parathyrin**. It consists



of at least two homologous glycoproteins of similar size (M_r 72 000 and 70 000), related to chromogranin A (see **chromogranin**).

sector any portion of a circle bounded by two radii and the included arc. —**sectoral** *adj.*

sector cell or **sector-shaped cell** a cell used in an analytical ultracentrifuge to contain the sample. It is sector-shaped when viewed parallel to the rotation axis. Since the accelerating force is radial, sedimenting molecules move along radii and this cell shape minimizes the collision of sedimenting molecules with the cell walls.

sediment 1 to undergo **sedimentation**. 2 the accumulation of material resulting from sedimentation.

sedimentation the setting of solid particles (or solute) through a liquid (or solution) under the influence of a gravitational or centrifugal field.

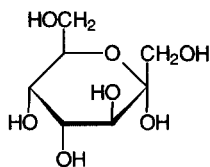
sedimentation coefficient symbol: s ; the rate of sedimentation of a particle in an ultracentrifuge or other system. For a particle or solute sedimenting in a system at constant applied field, temperature, and pressure, $s = v/a$, where a is the acceleration of free fall or centrifugation and v is the velocity of sedimentation. In centrifugation $s = v/\omega^2 r$, where ω is the angular velocity in rad s^{-1} and r is the distance of the particle or solute from the axis of rotation. It has the dimensions of time. The sedimentation coefficients of macromolecules and cellular particles are often expressed in **svedbergs**.

sedimentation equilibrium a technique in which compounds of interest are centrifuged for a period sufficiently long for the system to come to equilibrium. The technique has a number of applications. In one, used for molecules that have a density greater than the solvent, centrifugation is carried out at relatively slow speeds so that sedimentation of the molecule is slow enough to be counterbalanced by diffusion. At equilibrium, the concentration is lower at the meniscus and higher at the bottom of the cell, but is unchanging. The relative molecular mass, M_r , of the compound in question, often a protein, can be calculated from the expression:

$$M_r = 2RT \ln(c_2/c_1) / \omega^2 (1 - v\rho)(r_2^2 - r_1^2),$$

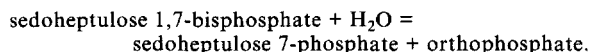
where R is the gas constant, T is the thermodynamic temperature, v is the specific volume of the component, ρ is the density of the solution, ω is the angular velocity of the rotor, and c_1 and c_2 are the concentrations of the component at distances r_1 and r_2 from the centre of rotation. Ideally, determinations should be done at several concentrations and extrapolated to zero concentration.

sedoheptulose a nonsystematic name for D-altrio-2-heptulose.



α -D-pyranose form

sedoheptulose-bisphosphatase EC 3.1.3.37; a chloroplast enzyme of the **reductive pentose phosphate cycle** that catalyses the reaction:



Example from *Arabidopsis thaliana* (precursor): database code S17P_ARATH, 393 amino acids (42.41 kDa).

sedoheptulose 7-phosphate the 7-phosphate ester of **sedoheptulose**; it is a component of the **pentose phosphate pathway**.

Seebeck effect the phenomenon whereby an electromotive force is generated in a circuit containing junctions between

dissimilar metals if these junctions are not all at the same temperature. See **thermocouple**. Compare **Peltier effect**. [After Thomas Johann Seebeck (1770–1831), Estonian-born German physicist.]

segmentation 1 (of an ovum) see **cleavage** (def. 3). 2 (of an animal's body) see **metamerism**.

segregation (*in genetics*) 1 the separation of homologous genetic elements (i.e. allele pairs) during meiosis in diploid cells. 2 the separation of sister chromatids during mitosis; by extension, the separation of any two independent genetic elements during cell division in prokaryotes. 3 (of plasmids) the partitioning of plasmids into daughter cells at cell division. Naturally occurring plasmids contain a partitioning function, *par*, which ensures correct partitioning into the two daughter cells.

segresome a membrane-bound, cytoplasmic aggregate of a dye, formed either by passage of the dye through the membrane of a pre-existing **phagosome** or **lysosome**, or by some process leading to the *de novo* formation of a membrane around the dye aggregate.

Seitz filter a filtration apparatus that removes microorganisms from a liquid. It formerly consisted of a flat pad of asbestos, or asbestos and cellulose, suitably mounted between the sample container and the receiving flask, filtration being assisted by reducing the pressure in the receiving flask. It has now been superseded by a sintered-glass filter. [After Seitz-Filter-Werke, T. and G. Seitz GmbH & Co., Bad Kreuznach, Germany.]

selachyl alcohol 3-(octadec-9-enyloxy)-1,2-propanediol; a 1-octadec-9-enyl ether of glycerol, and a hydrolysis product of **ether lipids**.

selected-ion monitoring or **mass fragmentography** or **molecular fragmentography** a type of **gas chromatography/mass spectrometry** in which the intensities of one or more molecular or fragment ions selected from the mass spectrum of a particular compound are monitored continuously as a means of quantifying that compound.

selectin any of a group of leukocyte surface molecules that act as leukocyte adhesion molecules; they are classed as CD62 in the **CD marker** system. All have **lectin** family carbohydrate-binding domains and EGF-repeats, with four motifs. They are expressed on the surface of platelets and endothelial cells as well as leukocytes. **E selectin** (CD62E), M_r 107–115 000 in various glycosylated forms, is expressed on endothelial cells, being induced by cytokines such as interleukin 1β and tumour necrosis factor α in 3–6 hours. It mediates endothelial cell binding to neutrophils, monocytes, and some memory T cells. Example from human (precursor): database code LEM2_HUMAN, 610 amino acids (66.58 kDa). **L selectin** (or lymph node homing receptor; CD62L), M_r 74 000 (on lymphocytes), is constitutively expressed on monocytes, lymphocytes (thymocytes, NK cells), neutrophils, eosinophils, and basophils. It mediates the adherence of lymphocytes, neutrophils, and monocytes to endothelium, including that of lymphocytes to endothelial cells of high endothelial venules in peripheral lymph nodes. It binds to **MadCAM** and CD34. Example from human (precursor): database code LEM1_HUMAN, 372 amino acids (42.14 kDa). **P-selectin** (or **PADGEM**; *abbr.* for platelet activation dependent granule external membrane protein), M_r 140 000, is expressed at the surface of activated platelets and endothelial cells, and is inducible within minutes by thrombin, histamine, or peroxides from α granules of platelets or Weibel–Palade bodies of neutrophils. It mediates the adhesion of neutrophils and monocytes to platelets and endothelial cells. Example from human (precursor): database code LEM3_HUMAN, 830 amino acids (90.74 kDa).

selection (*in microbiology*) a laboratory method in which a mixture of microorganisms is cultured under particular growth conditions that permit only cells with certain characteristics to survive, thereby enabling their isolation.

selection pressure the intensity with which natural selection operates; it is often measured by the change in **gene frequency** per generation due to the effects of natural selection.

selectively labelled indicating an isotopically labelled mol-



selective medium

ecule in which the position(s) of the labelling is known, but the number of labelling nuclides at that (those) position(s) is not known; e.g. selectively labelled ethanol is designated as $[1,2-^2\text{H}]\text{CH}_3\text{-CH}_2\text{-OH}$, which indicates that some, or all, of the hydrogen atoms on carbon atoms 1 and 2 are ^2H atoms.

selective medium a culture medium designed either to encourage or to inhibit the growth of certain types of organisms, especially microorganisms.

selective theory of immunity or **elective theory of immunity** any theory that ascribes to antigens a selective function in immunity by which antigens select cells genetically prepared to respond specifically and cause them to multiply and differentiate. They include the **clonal-selection theory** and the **natural-selection theory of immunity**. *Compare* **instructive theory of immunity**.

selenium *symbol:* Se; a metalloid element of group 16 of the (IUPAC) **periodic table**; atomic number 34; relative atomic mass 78.96. Its main oxidation states are -2, +2, +4, and +6, with properties similar to those of sulfur, but more metallic. It occurs in various allotropic forms (grey, red, black); the grey form shows some electrical conductivity, which is enhanced by light, and it is used in 'selenium cells' to measure light intensity. There are several isotopes, the most abundant being selenium-80 (mass 79.916; relative abundance 49.8%); others are selenium-74 (0.9%), selenium-76 (9.0%), selenium-77 (7.6%), selenium-78 (23.5%), and selenium-82 (9.4%). Radioactive isotopes are **selenium-72**, **selenium-73**, **selenium-75**, and **selenium-79**. Selenium is an essential trace element, required for the formation of **selenoproteins**, notably **glutathione peroxidase**. Gut absorption is poor, but selenium deficiency is rare (seen, e.g., in some parts of China with low soil selenium). The most obvious clinical feature of dietary deficiency is myopathy (especially cardiomyopathy). Measurement of red-cell glutathione peroxidase activity gives an index of selenium status.

selenium-72 a radioactive nuclide of selenium, $^{72}_{34}\text{Se}$, mass 71.927; it emits gamma radiation (0.046 MeV) and has a half-life of 8.4 days.

selenium-73 a radioactive nuclide of selenium, $^{73}_{34}\text{Se}$, mass 72.927; it emits a β^+ particle (1.32 MeV) and gamma radiation (0.361 MeV) and has a half-life of 7.1 h.

selenium-75 a radionuclide of selenium, $^{75}_{34}\text{Se}$, mass 74.923; it emits gamma radiation (0.481 MeV) and has a half-life of 118.5 days.

selenium-79 a radionuclide of selenium, $^{79}_{34}\text{Se}$, mass 78.918; it emits an electron (β^- particle, 0.16 MeV), no gamma radiation, and has a half-life of 6.5×10^4 years.

selenoamino acid an amino acid in which selenium replaces a sulfur atom; *see* **selenocysteine**, **selenomethionine**.

selenocysteine $\text{H-Se-CH}_2\text{-CH(NH}_2\text{)-COOH}$; an essential component of glutathione peroxidase and some other proteins (*see* **selenoprotein**). For its synthesis, a special tRNA charged with serine responds to a UGA codon. The serine is converted to selenocysteine just before its use on the ribosome. It is assumed that the neighbouring base sequence indicates a special sense for this codon, which is normally a 'stop' codon.

selenomethionine $\text{CH}_3\text{-Se-[CH}_2\text{]}_2\text{-CH(NH}_2\text{)-COOH}$; a selenoamino acid used as an antimetabolite, competing with methionine; it is also found in proteins; *see* **selenoprotein**.

selenoprotein a protein containing selenium, almost invariably as **selenocysteine**, although **selenomethionine** can occur, apparently as a random substitute for methionine. Well-known selenoproteins include the mammalian **glutathione peroxidases** and tetraiodothyronine 5'-deiodinase, and bacterial formate dehydrogenases and glycine reductase.

self **1** the individuality of a person or thing; the totality of components intrinsic to an individual or thing. **2** (*in immunology*) the components of an individual's own tissues that can be distinguished from foreign substances by the individual's immune system, and towards which **immunological tolerance** is shown.

self-absorption the absorption of any emission by the emitting molecules.

self antigen any (potentially) antigenic molecule originating in an individual that is recognized as nonforeign by the individual's immune system, and towards which **immunological tolerance** is normally shown. *See also* **autoimmunity**, **self-tolerance**.

self-assembly the formation of a complex entity from more simple, identical units without intervention from any external agency.

self-assembly doctrine a doctrine elaborated by Caspar and Klug to explain the assembly of small viruses and subcellular organelles. It is postulated that the structure assembles as a result of specific binding sites on proteins that interact to form a scaffold on which other components form.

self-catabolite repression repression of gene expression by a catabolite formed within the same organism.

self-cloning the cloning of a gene within the species from which it was derived, a technique that in some countries attracts less stringent government restrictions than when a gene is cloned in another species, being considered the less hazardous of the two approaches with respect to the genetic environment.

self determinant a **determinant** originating from the tissue of an organism and thus treated as **self** (def. 2) by the immune system of that organism (except in the case of autoimmune pathology; *see* **autoimmunity**).

self-fertilization the union of male and female gametes of the same organism.

self-hybridization hybridization between nucleic acids that exhibit complementarity.

selfing (*in microbiology*) the production of wild-type 'transductants' by infection of an auxotrophic strain with a transducing phage grown on the same auxotroph as the donor.

selfish DNA hypothesis a hypothesis put forward to account for the presence in eukaryotic genomes of large amounts of repetitive DNA (*see* **repetitive sequences**). Because there is no confirmatory evidence that much of such DNA serves any useful purpose for its host, this DNA has been termed **selfish DNA** or **junk DNA**, and is suggested to be a molecular parasite that, over many generations has disseminated itself throughout the genome. If this is true, it remains a mystery why natural selection has not eliminated 'selfish DNA'.

self tolerance the state in which the immune system of any particular individual lacks reactivity against potentially antigenic molecules originating within that individual. Self tolerance develops during perinatal life. *See also* **acquired tolerance**, **autoimmunity**, **immunological tolerance**.

Seliwanoff test a test for ketohexoses based on the production of a red colour when the sample is boiled with a solution of 0.05% w/v resorcinol in 3 M hydrochloric acid. The test relies on the fact that ketoses are dehydrated more rapidly than aldoses to yield furfural products. [After Feodor Feodorovich Seliwanoff (1859-1938), Russian chemist.]

SEM or **S.E.M.** **1** *abbr.* for standard error of estimate of mean. **2** *abbr.* for scanning electron microscope; *see* **electron microscope**.

sem-5 the gene for Sem-5, sex muscle abnormal protein 5, in *Caenorhabditis elegans*. Sem-5 acts both in vulval induction and sex myoblast migration, and is linked in a signalling chain from tyrosine kinase receptors (e.g. EGF-type Let-23). It contains SH2 and SH3 domains (*see* **SH domain**) characteristic of tyrosine kinase binding proteins. Mutations of the *sem-5* gene block signalling pathways from several tyrosine kinase receptors in the worm. Database code SEM5_CAEL, 228 amino acids (26.18 kDa).

semantide or **semantophoretic molecule** or **informational macromolecule** any of the different types of macromolecules that carry the information of the genes or a transcript (or a translation) thereof. The genes themselves are **primary semantides**, messenger RNA molecules are **secondary semantides**, and most polypeptides are **tertiary semantides**.

semantophoretic molecule *an alternative name for semantide*

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semen

semen the fecundating fluid of male higher animals, consisting of spermatozoa suspended in seminal plasma – secretions of the accessory glands.

semenogelin one of two major gel-forming proteins of semen. These proteins have 80% of their amino-acid residues in common, and their genes are localized to the long arm of chromosome 20. They are hydrolysed by a **kallikrein**-like proteinase. Example (precursor) of the predominant human protein of this type, semenogelin 1: database code SEM1_HUMAN, 462 amino acids (52.07 kDa).

semi+ prefix 1 half. 2 partial or partially; intermediate.

SEMI *abbr.* for specific endogenous mitotic inhibitor; see **chalone**.

semialdehyde any of a class of organic compounds formed by the reduction of one carboxyl group of a dicarboxylic acid to an aldehyde group, as in glutamic semialdehyde.

semiconductor any material, e.g. germanium or silicon, whose electrical conductivity lies between that of conductors and insulators and increases with increasing temperature. Its crystal structure has atomic bonds that allow the conduction of electric current by either positive or negative carriers when the appropriate additives (dopants) are present.

semiconservative replication of DNA *see* **replication of DNA**.

semiconstitutive mutant a mutant in which negative regulation is partially lost, so that a gene is expressed in the absence of inducer, but at a lower level than in the fully constitutive mutation, yet in the presence of inducer the expression reaches the fully constitutive level. *See also* **constitutive mutation**.

semi-micro 1 describing or concerning something moderately small; applied to a scale of operation, an item of equipment, a quantity of material, etc. 2 (*in analytical chemistry*) a term originally used to indicate analysis of amounts of substance in the range 10–100 mg. *Compare* **micro** (def. 1). *See also* **micro-analysis**.

seminal 1 a pertaining to seed. b of, or relating to, semen. 2 (of an experiment, idea, or publication) germinating new concepts; fundamentally original.

seminal plasma the suspending fluid in which semen is ejaculated.

seminalplasmin another name for **caltrin**.

semiochemical any chemical substance that delivers a message or signal from one organism to another. The two broad classes into which such substances are divided are known as **pheromones** and **allelochemicals**, depending on whether the information transfer occurs between, respectively, individuals of the same species (intraspecific) or between members of different species (interspecific). *See also* **allomone**, **kairomone**.

semiotics *functioning as sing.* 1 the practice of symptomatology; a term that became obsolescent, but has been reintroduced in connection with CT, NMR, and other imaging techniques. 2 the philosophical theory of signs and symbols, having three branches: syntactics, semantics, and pragmatics.

semipermeable membrane a membrane that allows the passage of only certain solutes, usually small molecules, but is freely permeable to solvent.

semipolar double bond an alternative name for **dipolar bond**.

semiquinone any free radical derived from a quinone or quinoid compound by the removal from it of a single hydrogen atom. *Compare* **quinone**.

semisolid medium (*in cell culture*) a medium of consistency between those of a solution and a full gel.

semisynthesis the partial synthesis of a chemical compound, especially a novel substance, by chemical or/and enzymic modification of one isolated from natural sources or produced biotechnologically (e.g. by fermentation). The procedure has been applied with conspicuous success in the manufacture of new pharmaceuticals such as artificial penicillins and cephalosporins. —**semisynthetic** *adj.*

semisystematic name or **semitrivial name** (*in organic-chemical nomenclature*) a name of a molecular entity that has only a part of it used in a systematic sense; e.g. (meth)ane, (but)ene.

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(calci)diol (in each of which only the part of the name outside the parentheses is systematic). Most names in organic chemistry are of this type. *Compare* **systematic name**, **trivial name**.

senescence the process of ageing. —**senescent** *adj.*

senile aged, to a point of severely decreased capability. —**senility** *n.*

senna a preparation of the dried leaves of *Cassia angustifolia* or *C. senna* used as a laxative. It contains glycosides of hydroxyanthracene (sennoside B), which are metabolized to purgative anthraquinones by colonic bacteria.

sense DNA *see* **coding strand**.

sense strand (of duplex DNA) *see* **coding strand**.

sensitivity 1 an index of the ability of any analytical method or other detection procedure to make quantitative determinations at very low levels. 2 (*in metabolic control theory*) the susceptibility of flux through a pathway to change in response to change in the level of one of the component enzymes, expressed quantitatively in terms of the **flux control coefficient**. An alternative meaning arises when an allosteric ligand acts at several points in a pathway, where a plot of the fractional modification in flux versus ligand concentration yields a sigmoidal curve for a positive effector; the sensitivity in this case can be derived from the slope of the sigmoidal curve.

sensitivity coefficient former name for **flux control coefficient**.

sensitization or **sensitisation** 1 enhanced reactivity to any treatment (such as exposure to an agonist) of any tissue, cell, or molecular system resulting from prior experience or treatment (often by the same agonist). 2 the conditioning of an animal by antigen in such a way that **hypersensitivity** reactions occur on subsequent exposure to the antigen. 3 the attachment of antibody to sheep red blood cells to make them susceptible to lysis by complement in the **complement-fixation test**. Red blood cells treated in such a way are commonly referred to as **sensitized cells**. —**sensitize** or **sensitise** *vb.*

sensitizer or **sensitiser** *see* **allergen**.

sensor any device that records the level of a substance of interest or under investigation, using physical parameters (heat, conductivity) or chemical methods. Sensors are often applied to continuous monitoring in flow systems; **biosensors** utilize the reactions of enzymes immobilized on a probe and often coupled with a second substrate that yields a coloured or fluorescent product.

sensory transducer any component of bacterial systems involved in signal transduction. An example is the *ntxB/ntxC* system, which occurs in several bacteria (the following examples are from *Escherichia coli*). Under nitrogen limitation the product of the *ntxB* gene, NtrB (EC 2.7.1.-), activates NtrC – the *ntxC* gene product – by phosphorylating it; in nitrogen excess it reverses this. Examples: NtrB, database code NTRB_ECOLI, 349 amino acids (38.51 kDa); NtrC, database code NTRC_ECOLI, 469 amino acids (52.19 kDa); NtrC contains an ATP-binding domain that interacts with **sigma factor** σ^{54} . Another instance of a sensory transducer is sensory rhodopsin I transducer from archaeobacteria. Example from *Halobacterium halobium*: database code HTRI_HALHA, 536 amino acids (56.61 kDa); this is a typical seven-transmembrane-domain opsin-type protein used for sensing light. *See also* **che**.

separation (*in genetics*) the break-up of a cointegrate into two or more **replicons** of similar sizes. *Compare* **excision**.

Sephacel proprietary name for a gel-filtration medium based on **cellulose** cross-linked using **epichlorohydrin** so that a mesh results, the size of the apertures in the mesh being determined by the extent of cross-linking.

Sephacryl proprietary name for a gel-filtration medium prepared from allyldextran covalently cross-linked using *N,N*-methylene bisacrylamide, to yield a mesh; the size of the apertures in the mesh depend on the degree of cross-linking.

Sephadex proprietary name for a gel-filtration medium based on **dextran** cross-linked using **epichlorohydrin** so that a mesh results, the size of the apertures in the mesh being determined by the extent of cross-linking.



Sephacrose *proprietary name* for a gel-filtration medium based on agarose cross-linked using epichlorohydrin so that a mesh results, the size of the apertures in the mesh being determined by the extent of cross-linking.

septanose the form of any monosaccharide, or monosaccharide derivative, in which the molecule contains a ring consisting of six carbon atoms and one oxygen atom. In monosaccharide symbolism the septanose form may be indicated by the suffix *s*; e.g. Glcs symbolizes glucoseptanose. *Compare* furanose, heptose, pyranose.

septate divided by, or involving, a septum or septa.

separation the process of forming the septum that cuts a cell into two at the end of the cell cycle in bacteria.

septum **1** (*in anatomy*) a wall of tissue, sheet of cells, or other membrane-like structure that partitions two compartments. **2** a disk of silicone rubber, PTFE, or similar material used in any apparatus to seal an injection port. The needle of a syringe can be punched through the septum for the purpose of injection, and on withdrawal of the needle the elasticity of the septum ensures that the hole reseals.

Sequenase *proprietary name* for a chemically modified T7 DNA polymerase with high processivity. It is the usual choice for DNA sequencing by the chain-termination method.

sequenator an apparatus for determining the sequence of monomeric residues (e.g. amino acids) in an ordered linear polymer (e.g. an oligopeptide) by repeating a chemical process.

sequence **1** the ordinal arrangement of the constituent parts of a biopolymer, e.g. the order of amino-acid residues in a polypeptide chain or of the nucleotide residues in a polynucleotide chain; the known arrangement of such units in any biopolymer or fragment. **2** any particular segment, occurring in or derived from a biopolymer, having a known order of its constituent parts; a synthetic polymer composed of parts equivalent to those occurring naturally and present in a particular known order. **3** to determine the order of residues in a biopolymer or fragment. —*sequential adj.*

sequence compression *see* band compression.

sequence homology similarity (often to an indicated degree) of sequences of residues in encoded macromolecules or specified segments of them. The term is often used synonymously with conservation (when used of primary structure) if the sequences occur in corresponding locations in specimens of any particular macromolecule obtained from genetically differing sources. Homology is discussed more fully in Appendix E.

sequence hypothesis the hypothesis, formulated by F. H. C. Crick in 1958, that 'the specificity of a piece of nucleic acid is expressed solely by the sequence of its bases, and that this sequence is a (simple) code for the amino-acid sequence of a particular protein.'

sequence ladder the ladder-like bands of separated components occurring in the chromatographic gel that result from DNA sequencing experiments.

sequencer an apparatus for automatically determining the sequence of monomeric residues in an ordered linear polymer.

sequence rule or **Cahn-Ingold-Prelog rule** or ***R/S* convention** (of configuration specification) a system for specifying the absolute molecular chirality of a chemical compound. This system considers the sequence of the proton numbers of the atoms directly attached to a chiral centre (*see* chirality); if two or more of these atoms have the same proton number, their substituents are then taken into account, the group with substituents of highest proton number taking precedence. When a multiple bond to a substituent is present, the substituent is arbitrarily considered to occur twice, or thrice; an isotope of higher nucleon number takes precedence over one of lower number. The priorities of some common ligands are tabulated in Appendix G. For a tetrahedral centre (e.g. a carbon atom) the substituent with the lowest priority is placed away from the observer (behind the chiral atom) with the other three projecting towards the observer; if these three substituents lie in a clockwise (right-handed) array of decreasing priority, then the

chiral centre is designated with the stereochemical descriptor *R*, denoting Latin *rectus*, right; if the substituents lie in an anti-clockwise (left-handed) array of decreasing priority, then the chiral centre is designated with the stereochemical descriptor *S*, denoting Latin *sinister*, left. For the name of a compound consisting of molecules with a single chiral centre the descriptor *R* or *S*, with a preceding locant if needed, is formed into a prefix by placing it in parentheses and adding a linking hyphen; a racemate with a single chiral centre in its molecules is labelled with the prefix (*RS*)-. When a molecule contains more than one chiral centre the prefix to the name is composed of multiple descriptors, each preceded by an appropriate locant; e.g. (2*R*,3*R*)-2,3-dihydroxybutanedioic acid (= *L_e*-tartaric acid). If only the relative, but not the absolute, configurations of two or more chiral centres in a molecule are known, these centres may be designated with the descriptors *R** and *S** (spoken as *R* star, *S* star), preceded when necessary by appropriate locants, on the arbitrary assumption that the chiral centre first cited (which usually is the one with the lowest locant) has an *R* configuration; for examples, compare the systematic names of allthreonine and threonine. *See also* *n/t* convention. Extensions of the sequence rule are the *E/Z* convention, the *pro-E/pro-Z* convention, the *pro-R/pro-S* convention and the *Re/Si* convention. [After Robert Sidney Cahn (1899–1981), British natural-products chemist, editor, and scientific semanticist, (Sir) Christopher Kelk Ingold (1893–1970), British physical organic chemist, and Vladimir Prelog, who jointly developed the system and described it in a series of three papers published in 1951, 1956, and 1966.]

sequence-tagged site *abbr.*: STS; a short, single-copy DNA sequence that characterizes a mapping landmark on the genome. Such sequences provide unambiguous identification of DNA markers generated by the Human Genome Project (*see* HUGO). STSs are short segments of DNA of known (and unique) sequence which bind to YACs. If two YACs contain the same STS they must overlap. The YACs can be built up into an overlapping series of clones (a *contig*) covering an entire gene.

sequencing the act or process of determining the sequence (def. 1) of proteins or nucleic acids. *See* chain-termination method, chemical cleavage method, solid-phase technique.

sequencing by hybridization *see* SBH.

sequencing gel a polyacrylamide gel run to resolve oligonucleotides produced in a DNA sequencing procedure. *See* chain-termination method, chemical cleavage method.

sequential analysis the analysis of a protein or nucleic-acid sequence (def. 1).

sequential mechanism an enzyme reaction involving two or more substrates in such a way that all the substrates must be bound to the enzyme to form the central complex before any products can be released. A sequential mechanism may be an ordered mechanism or a random mechanism.

sequestering agent *see* chelating agent.

sequestration **1** the process of removal or separation. **2** (*in chemistry*) the effective removal of ions from solution by co-ordination with another kind of ion or molecule to form a firm complex molecule with different properties from the original ion. *Compare* chelation. **3** (*in cell biology*) the intracellular enclosing of material in an organelle. —*sequester vb.*

Ser symbol for a residue of the α -amino acid L-serine (alternative to S).

SER *abbr.* for smooth endoplasmic reticulum (*see* endoplasmic reticulum).

sera a plural of serum.

SERCA *abbr.* for sarcoplasmic/endoplasmic reticulum calcium-transporting ATPase, EC 3.6.1.38; any of a family of Ca^{2+} pump integral membrane protein isoforms of sarcoplasmic or endoplasmic reticulum. They catalyse the hydrolysis of ATP coupled with Ca^{2+} transport. Example from *Drosophila melanogaster*: database code ATCB_DROME, 1002 amino acids (109.47 kDa).

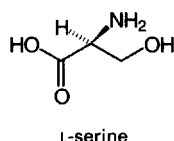
serial dilution a method of achieving dilution of a liquid by se-

serinate

quentially transferring a sample of predetermined volume into another, empty, vessel (usually a tube), which is then made up with diluent to the same total volume as the first. The overall dilution achieved is $(v/s)^x$ -fold, where v is the chosen total volume in each tube, s is the size of the sample, and x is the number of times the procedure is repeated.

serinate 1 serine anion; the anion, $\text{HO}-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COO}^-$, derived from **serine**. 2 any salt containing serine anion. 3 any ester of serine.

serine the trivial name for α -amino- β -hydroxypropionic acid; 2-amino-3-hydroxypropanoic acid; $\text{HO}-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COOH}$; a chiral α -amino acid. L-Serine (symbol: S or Ser), (S)-2-amino-3-hydroxypropanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: AGC or AGU; UCA, UCC, UCG, or UCU. In mammals it is a nonessential dietary amino acid and is glucogenic. D-Serine (symbol: D-Ser or D-Ser), (R)-2-amino-3-hydroxypropanoic acid, in both free and combined forms occurs in silkworms and earthworms; residues of D-serine occur in certain peptide antibiotics, e.g. polymyxins D₁ and D₂. [Note: The absolute configuration of D-serine (and hence that at the α -carbon atom of other α -amino acids of the D series) is identical with that of D-glyceraldehyde (and hence with that at the reference carbon atom of other monosaccharides of the D series).]



L-serine dehydratase EC 4.2.1.13; *systematic name*: L-serine hydro-lyase (deaminating); *other names*: serine deaminase; L-hydroxyaminoacid dehydratase. An enzyme that catalyses a reaction between L-serine and water to form pyruvate and ammonium ion. It is a pyridoxal-phosphate enzyme. The enzyme for gluconeogenesis from serine in *Escherichia coli* is a 4Fe-4S protein; database code SDHL_ECOLI, 454 amino acids (48.85 kDa).

serine hydroxymethyl transferase *see* glycine hydroxymethyltransferase.

serine-like carboxypeptidase *see* serine endopeptidase.

serine-C-palmitoyltransferase EC 2.3.1.50; an enzyme of the pathway for the biosynthesis of sphingoid bases. It catalyses the formation of 3-dehydro-D-sphinganine from palmitoyl-CoA and L-serine, releasing CoA and CO₂. Pyridoxal-phosphate is a coenzyme. Example from yeast (long-chain-base biosynthesis protein), two forms: database code LCB1_YEAST, 558 amino acids (62.16 kDa); and database code LCB2_YEAST, 561 amino acids (63.04 kDa).

serine peptidase any enzyme characterized by having a residue of serine (and one of histidine) at the active centre and by being irreversibly inhibited by organic fluorophosphates (which phosphorylate the serine's hydroxyl group). Serine peptidases are of two types: (1) **serine-type carboxypeptidases**, and (2) **serine endopeptidases**, sub-subclass EC 3.4.21. During catalysis by serine peptidases an acyl-enzyme intermediate is formed by esterification of the hydroxyl group of the reactive serine with the carbonyl group of a sensitive peptide bond of the substrate. The serine endopeptidase group includes **chymotrypsin**, most **elastases**, **kalikrein**, **thrombin**, **trypsin**, and a number of microbial proteinases. *See also* **serine-type carboxypeptidase**.

serine/threonine kinase a protein kinase that phosphorylates serines or threonines on its target protein.

serine/threonine-specific protein phosphatase EC 3.1.3.16; *recommended name*: phosphoprotein phosphatase; *other names*: protein phosphatase-1; protein phosphatase-2A; protein phosphatase-2B; protein phosphatase-2C. This name

includes a group of enzymes that hydrolyse the serine- or threonine-bound phosphate group from many phosphoproteins. Examples from yeast: PP2a-1 catalytic subunit: database code P2A1_YEAST, 369 amino acids (41.89 kDa); PP2a-2 catalytic subunit: database code P2A2_YEAST, 377 amino acids (43.00 kDa).

serine transhydroxymethylase *see* glycine hydroxymethyltransferase.

serine-type carboxypeptidase EC 3.4.16.1; this name includes many enzymes that hydrolyse protein with release of a C-terminal amino acid; they have a broad specificity. Examples (3-D structures) from wheat (chains a and b, respectively): database codes NRL_1WHA, 255 amino acids (28.38 kDa), and NRL_1WHSB, 153 amino acids (17.28 kDa).

serinium serine cation; the cation, $\text{HO}-\text{CH}_2-\text{CH}(\text{NH}_3^+)-\text{COOH}$, derived from **serine**.

serino+ prefix denoting the group, $\text{HO}-\text{CH}_2-\text{CH}(\text{NH}-)-\text{COOH}$, derived from **serine** by loss of hydrogen from the amino group.

sero+ comb. form denoting a connection within or origin in serum.

serological pertaining to, or originating in, serum; connected with the study of serum.

serology the study of serum; in particular, the immunological phenomena involving circulating antibody that may be observed *in vitro*.

serosa a serous membrane, especially the peritoneal covering. — serosal adj.

serotonergic 1 describing a nerve that is activated by serotonin. 2 describing a nerve that acts by the release of serotonin from the nerve endings. *Compare* adrenergic, aminergic, cholinergic, peptidergic, purinergic.

serotonin 5-hydroxytryptamine (abbr.: 5-HT); a monoamine neurotransmitter occurring in the peripheral and central nervous systems, also having hormonal properties; it is also found in mast cells, platelets, brain, and the enterochromaffin cells of the gastrointestinal tract, and is capable of causing vasoconstriction, increased vascular permeability, and contraction of smooth muscle. It is formed from tryptophan, after hydroxylation to 5-hydroxytryptophan, by tryptophan 5-monooxygenase (EC 1.14.16.4) followed by decarboxylation by aromatic-L-amino-acid decarboxylase (EC 4.1.1.28). It is a substrate for monoamine oxidase.

serotonin receptor *see* 5-hydroxytryptamine receptor.

serotransferrin *see* transferrin.

serous 1 relating to or resembling serum (or similar watery fluid). 2 producing a fluid resembling serum.

serous membrane or **serosa** a thin connective-tissue membrane lining certain closed body cavities and reflected over the viscera. The peritoneum of the abdomen, pleura of the chest, and pericardium of the heart are all serous membranes. Each consists of a **parietal layer**, lining the walls of the cavity, and a **visceral layer** covering the organs. The two are continuous, forming a closed sac. *Compare* mucous membrane.

serpentine domain any sequence of transmembrane segments of a single polypeptide chain of a membrane protein that traverses the membrane several times, as, e.g., in **seven-transmembrane-domain proteins**.

serpin any of a superfamily of proteins, many of which inhibit serine proteinases and exhibit a high degree of homology with classical serine proteinase inhibitors such as α_1 -antitrypsin or antithrombin. The superfamily is now considered to extend beyond compounds that are known to be proteinase inhibitors and, based on homology, includes proteins such as **ovalbumin** and **angiotensinogen**, which have no inhibitory function. [From serine proteinase inhibitor.]

Sertoli cell or **sustentacular cell** a supporting cell of the mammalian testis that surrounds and nourishes developing sperm cells. [After Enrico Sertoli (1842–1910), Italian histologist.]

serum (pl. sera or (esp. US) serums) the liquid extruded after blood has clotted and the clot has retracted (**syneresis** (def. 2)).

Compare plasma.

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serum albumin

serum albumin or **plasma albumin** the smallest and most abundant of the plasma proteins; M_r 65 000–68 000 for the bovine form, 69 000 for the human. It is involved in the regulation of osmotic pressure and in the transport of sparingly soluble metabolites. The serum albumins belong to a multigene family of proteins that includes **α -fetoprotein** and **vitamin D binding protein**. Human serum albumin has three homologous domains which assemble to form a heart-shaped molecule. It consists of 28 helices, ranging from five to 31 amino-acid residues each, formed into six subdomains; two each of these subdomains form one of the major domains. There is no beta structure. Bovine (serum) albumin (*abbr.*: BSA) and human (serum) albumin (*abbr.*: HSA) are widely used as standards in protein estimations. These include fatty acids, amino acids (such as tryptophan), steroids, metals (e.g. calcium, copper, and zinc), and hydrophobic drugs. *See also albumin*.

serum amyloid P-component *abbr.*: SAP; a precursor of amyloid component P. Amyloid component P is found in basement membranes and is associated with amyloid deposits.

serum converting enzyme a former name for **angiotensin converting enzyme**.

serum globulin any **serum protein** that is also a **globulin**. An early classification of globulins was according to their mobility on electrophoresis. **Serum albumin** moves the fastest; behind albumin on paper electrophoresis or cellulose acetate strips are two bands known as the α_1 - and α_2 -globulin bands, followed by a β -globulin band with the γ -globulins forming a diffuse band near the origin. The α_1 -globulins include thyroxine-binding protein, transcortin, α_1 -glycoprotein, α -lipoprotein, and antitrypsin. The α_2 -globulins include haptoglobin, α_2 -glycoprotein, macroglobulin and ceruloplasmin. The β -globulins include transferrin, β -lipoprotein and β -glycoprotein. The γ -globulins comprise the immunoglobulins.

serum protein any protein contained in serum; the term is usually applied to those proteins present in high quantities. The most prominent of these is **albumin**, which is present in the normal serum of human adults in the range 34–48 g L⁻¹. *See also serum globulin*.

serum response element *abbr.*: SRE; a stretch of DNA that regulates promoters for a group of genes that are activated by addition of serum to cell cultures. Such elements have the consensus sequence CCATATTAGG.

serum spreading factor *see vitronectin*.

seryl the acyl group, HO-CH₂-CH(NH₂)-CO-, derived from **serine**.

sesqui+ *comb. form 1* denoting one-and-a-half times; in a ratio of three parts to two. **2 (in chemistry)** indicating a compound having one-and-a-half molecular proportions of the specified element group, or unit, either in relation to a second component, as in sesquioxide (e.g. chromium sesquioxide, Co₂O₃), or in relation to a reference compound, as in **sesquiterpene**; it also sometimes denotes a compound intermediate in constitution between two others, as in sesquicarbonate (e.g. sodium sesquicarbonate, Na₂CO₃·NaHCO₃·2H₂O).

sesquiterpene any **terpene** formed from three isoprene units, i.e. having 15 carbon atoms per molecule as compared with a monoterpene which has 10 carbon atoms per molecule and is formed from two isoprene units.

sesquiterpene lactone any sesquiterpene (15 carbon atoms) containing a lactone structure. Several are derived from

Artemisia spp., including **artemesinin** (qinghaosu) from *A. annua* (qing hao). Qing hao has been used for almost 2000 years as an antimalarial; its mechanism of action is unknown.

sester+ *comb. form 1* denoting two-and-a-half times; in a ratio of five parts to two or 10 parts to 4. **2 (in chemistry)** indicating a compound having two-and-a-half molecular proportions of the specified group or unit in relation to a reference compound, as in **sesterterpene**.

sesterterpene any **terpene** formed from five isoprene units, i.e. having 25 carbon atoms per molecule as compared with a monoterpene, which has 10 carbon atoms per molecule and is formed from two isoprene units. An **ophiobolin** is an example.

sev symbol for **sevenless** gene; *see Sevenless protein*.

Sevag method a method for removing proteins from nucleic acids during purification of the latter. The proteins are precipitated after repeated denaturation by shaking with a solution of octanol in chloroform.

Sevenless protein *abbr.*: Sev; a protein, encoded by the *sevenless* (*sev*) gene, that is part of a system for differentiation of R7 photoreceptors in *Drosophila*. Sev is a receptor for an extracellular signal; it has **tyrosine kinase** activity and belongs to the family of insulin receptors. Sev has a ligand, **Boss**, the product of the *bride of sevenless* (*boss*) gene; it is expressed by R8 photoreceptor cells and internalized (as a transmembrane protein) in a Sev-dependent manner. Examples, Sev: database code 7LES-DROME, 2554 amino acids (286.79 kDa); Boss: database code BOSS_DROME, 896 amino acids (99.81 kDa).

seven-minute phosphate *see labile phosphate*.

seven-transmembrane-domain protein or **seven-transmembrane-segment protein** or **seven-transmembrane-helix protein** a type of receptor protein that contains, in a single polypeptide chain, seven hydrophobic domains that traverse the lipid bilayer. This structural characteristic is typically used to describe the structures of receptors that couple to effector systems through **G-proteins**. These are also sometimes referred to as serpentine receptors. The structures of only a few membrane proteins have been determined at high resolution, and this structural characteristic has been attributed to other proteins mainly on the basis of sequence comparison. Thus, for the majority of proteins, the name 'seven-transmembrane-domain protein' is a tentative one. The suggestion that the transmembrane domains are helical is even more tentative.

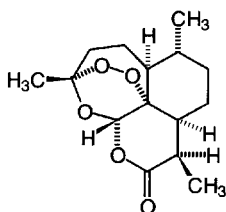
seven-up a receptor required for photoreceptor cells during eye development. It is a **zinc-finger** DNA-binding protein with two forms (1 and 2) generated by alternative splicing. Example (type 1) from *Drosophila melanogaster*: database code 7UPI-DROME, 543 amino acids (57.92 kDa).

sex chromatin a condensed mass of **chromatin** that represents an inactivated X chromosome. In mammalian nuclei, each X chromosome apart from the activated one forms a sex chromatin body or **Barr body**.

sex chromosome a chromosome involved in sex determination. In many species, these are the **X chromosome** and **Y chromosome**. They are distinguished from the **autosomes** i.e. all the other chromosomes of that species. In some species, however, e.g. fish of the genus *Xiphophorus*, sex determinants are spread over a number of chromosomes.

sex determination the specification of the sex of an individual organism by genes. In many animal species, these genes are located on the **X chromosome** and the **Y chromosome**. Their effects are manifested as a result of secretion of **androgens** and **estrogens**.

sex factor or **fertility factor** or **F factor** or **F agent** an episomal genetic element (*see episome*) in bacteria that enables the cell to act as a genetic donor. Cells of such donor strains are able to form stable unions with cells devoid of such factors and to transfer to them extrachromosomal material, including the sex factor, and sometimes segments of bacterial chromosomes (*see conjugation* (def. 4)). The **F plasmid** is the prototypical sex factor, responsible for conjugation in certain strains of *Escherichia coli*.





sex hormone

sex hormone any steroid hormone that is responsible for controlling sexual characteristics and reproductive function. **Estrogens** and **progesterone** are the female sex hormones, while **androgens** are the male sex hormones. They are produced mainly by the ovaries or testes, with contributions from the adrenals and placenta.

sex linkage a special type of gene linkage occurring when a gene producing a certain phenotype (often unrelated to primary or secondary sexual characters) is located on a **sex chromosome**. Such a gene, or the character it determines, is said to be **sex linked**. One consequence of a mutation of genes on the X chromosome is that they may often be apparent only in male offspring, which have only one X chromosome. The mutation may be masked in the female because of the presence of a second, unaffected X chromosome.

sexual conjugation *see* **conjugation** (def. 4).

S-form *abbr.* for smooth form; describing bacterial colonies with a **smooth** appearance. *Compare* **R form** (def. 2).

SGF *abbr.* for skeletal growth factor.

SGOT *abbr.* for serum glutamic-oxaloacetic transaminase (name not now recommended; *see* **aspartate transaminase**).

SGPT *abbr.* for serum glutamic-pyruvic transaminase (name not now recommended; *see* **alanine transaminase**).

shadow casting a procedure used in the preparation of specimens for electron microscopic examination. The specimen is placed in a vacuum and a heavy metal is applied by evaporation – the metal is evaporated with an electrically heated tungsten filament. In high vacuum the metal atoms travel in straight lines and so, if evaporation is from an acute angle, metal will condense on only one side of the features of the specimen leading to variations in electron-density over the specimen, creating a 'shadow' effect.

sham operated describing a control procedure in which all the operations employed in a surgical technique are imitated with the exception of the experimental lesion that is the subject of study.

Shardinger enzyme *see* **xanthine oxidase**.

shark's tooth comb a plastic device for creating wells in an electrophoretic gel. It is shaped like a comb with the teeth coming to a point.

Sharp, Phillip Allen (1944–), US molecular biologist; Nobel Laureate in Physiology or Medicine (1993) jointly with R. J. Roberts 'for their discovery of split genes'.

Sharple's super centrifuge a, now superseded, efficient centrifuge with a flow-through rotor in which the sedimentation path was short. It is used for harvesting bacteria, blood cells, etc. The version powered with an electric motor turned at 21 000–23 000 rpm while the air-turbine-driven version turned at 50 000 rpm. The instruments of this type have been superseded by others that can operate continuous flow rotors.

SHC a gene family that was identified by screening cDNA libraries for genes encoding SH2-containing proteins (*see* **SH domain**). The mammalian gene encodes two overlapping, widely expressed proteins with a C-terminal SH2 domain that binds activated growth factor receptors, and which are phosphorylated on tyrosine in growth factor activated cells. These Shc proteins are transforming when overexpressed in fibroblasts. Phosphorylated Shc forms a complex with **Grb2**, and may form a link between certain tyrosine kinases and the **Grb2/Ras** signalling pathway. Example of gene product from human: database code SHC_HUMAN, 473 amino acids (51.61 kDa).

SH domain *abbr.* for Src homology domain; the **SH2 domain** is a protein module consisting of about 100 residues found in many proteins involved in signal transduction. Apart from Src, which contains the prototype SH2 domain (*see* **src**), these include among many others phospholipase C, products of proto-oncogenes such as **ABL**, **FYN**, and **FGF**, GTPase-activating protein, and protein tyrosine phosphatases. The SH2 domain has also been found in structural proteins such as **fodrin**. One function of the domain is to bind to the autophosphorylated site of growth factor receptors e.g. PDGF and EGF recep-

tors. Proteins with SH2 domains are frequently found also to have SH3 domains. The **SH3 domain** is composed of approximately 50 amino acids and was originally identified in Src, but has since been found within a number of other proteins. SH3 domains mediate protein-protein interactions through their ability to recognize and bind proteins containing specific amino-acid sequences rich in proline and hydrophobic residues. To date, there are known to be at least two general classes of ligand to which SH3 domains can bind: class I, which comprises proteins containing the consensus sequence PXLPPZP; and class II, with the sequence XPPLPXR (where X is any amino acid and Z appears to be a residue specific to the type of SH3 domain involved). Given the nature of the proteins possessing SH3 domains and SH3-binding domains, it is clear that the protein interactions mediated by these motifs are important to the operation and maintenance of cytoskeletal architecture and intracellular signalling pathways.

shear 1 to disrupt or fracture (an entity) using forces acting parallel to a plane, rather than perpendicularly, as with tensile or compressive forces. For example, shear forces can be used to disrupt DNA or other macromolecules by enclosing the molecules between two closely applied plane surfaces moving rapidly in opposite directions. Homogenization techniques depend on shear forces. 2 the deformation or fracture produced by such forces.

shear gradient *see* **viscosity**.

shear rate *see* **viscosity**.

shear stress *see* **viscosity**.

shikalkin the racemic mixture of **alkannin** and its (+)(R) enantiomer, **shikonin**.

shikimate the anion of shikimic acid; [$-3\alpha,4\alpha,5\beta$ -trihydroxy-1-cyclohexene-1-carboxylic acid; (3R,4S,5R)-trihydroxy-1-cyclohexenecarboxylic acid; an important intermediate, in microorganisms and plants, in the pathway for the synthesis of phenylalanine, tyrosine, tryptophan, and many other substances (e.g. tocopherol, vitamin K, folic acid, ubiquinone, enterobactin, and some antibiotics); this is known as the **shikimate pathway**. The enzyme **3-phosphoshikimate 1-carboxyvinyltransferase** (EPSP synthase) in higher plants is specifically inhibited by the compound **glyphosate**, which inhibits growth and can thus be used as an effective and readily degraded broad-spectrum herbicide. Shikimic acid was first isolated from the plant *Illicium religiosum* (star anise) in 1885. The name is derived from the Japanese name for this plant, *shikimi-no-ki*. *See also* **shikimate kinase**.

shikimate 5-dehydrogenase EC 1.1.1.25; *systematic name* shikimate: NADP+ 5-oxidoreductase; an enzyme of the **shikimate pathway** that catalyses the reduction by NADPH of 3-dehydroshikimate to shikimate with formation of NADP+. The description of the enzyme as a 5-dehydrogenase derives from the older numbering system for shikimate. With the now-accepted chemical numbering, it is in fact a 3-dehydrogenase. *See also* **pentafunctional arom polypeptide**.

shikimate kinase EC 2.7.1.71; *systematic name*: ATP:shikimate 3-phosphotransferase; an enzyme of the **shikimate pathway** that catalyses the phosphorylation by ATP of shikimate to shikimate 3-phosphate with formation of ADP. The product, shikimate 3-phosphate, is the precursor of 5-enolpyruvylshikimate 3-phosphate (5-O-(1-carboxyvinyl)-3-phosphoshikimate) from which **chorismate** is formed. Example, shikimate kinase I from *Escherichia coli*: database code AROK_ECOLI, 172 amino acids (19.39 kDa). *See* **pentafunctional arom polypeptide** for another example. *See also* **shikimate**.

shikimate pathway the 'classical' shikimate pathway in plants and microorganisms led to the production of the aromatic amino acids (phenylalanine, tyrosine, tryptophan) from carbohydrate precursors, with a requirement for **chorismate** as well as **shikimate**. Chorismate was regarded as a (single) branch point, being converted to phenylalanine and tyrosine via prephenate (*see also* **arogenate**) and to tryptophan via anthranilate. Chorismate can also be converted to ubiquinones,

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plastoquinones, and tocopherols. Moreover, on isomerization it yields isochorismate, from which are derived many naphthoquinones (including the vitamins K), 3-carboxy aromatic amino acids, salicylate, and dihydroxybenzoate (for enterobactin formation). Certain non-aromatic cyclohexane compounds are produced from shikimate itself. The pathway is now recognized to contain multiple branch points at many (possibly all) of the pathway intermediates, and has been termed a 'metabolic tree with many branches'. The carbohydrate precursors are phosphoenolpyruvate and D-erythrose 4-phosphate; they yield 2-dehydro-3-deoxy-D-arabino-heptonate 7-phosphate (I, more commonly termed 3-deoxy-D-arabino-heptulosonate 7-phosphate, *abbr.*: DAHP) by action of EC 4.1.2.15, 2-dehydro-3-deoxyphosphoheptonate aldolase (DAHP synthase). I is converted to 3-dehydroquininate (II) by 3-dehydroquininate synthase, EC 4.6.1.3, and II is then converted to 3-dehydroshikimate (III) by 3-dehydroshikimate dehydratase, EC 4.2.1.10. Shikimate is then formed from III by **shikimate 5-dehydrogenase**, EC 1.1.1.25, and converted to shikimate 3-phosphate (IV) by **shikimate kinase**, EC 2.7.1.71. Phosphoenolpyruvate is used again to convert IV to 5-enolpyruvylshikimate 3-phosphate (V, *abbr.*: EPSP) by 3-phosphoshikimate 1-carboxyvinyltransferase, EC 2.5.1.19, and V yields chorismate (VI) by action of **chorismate synthase**, EC 4.6.1.4. Chorismate leads to prephenate by action of **chorismate mutase**, EC 5.4.99.5, and in another branching undergoes isomerization by isochorismate synthase, EC 5.4.99.6, to form isochorismate. Organization of the various enzymatic activities varies considerably in plants, bacteria, yeasts, and other fungi. In yeasts and other fungi, and also in *Euglena gracilis*, the conversion of DAHP to EPSP requires a complex pentafunctional polypeptide specified by the *arom* gene cluster (see **pentafunctional arom polypeptide**).

shikinin the (+)-(*R*) enantiomer of **alkannin**.

Shine-Dalgarno sequence a sequence for the formation of the correct preinitiation complex between a 30S ribosomal subunit and a messenger RNA (mRNA) molecule. It is suggested that a seven-nucleotide, pyrimidine-rich sequence near the 3' terminus of 16S (prokaryotic) or 18S eukaryote ribosomal RNA, found in many organisms, can form base pairs with a complementary three- to seven-nucleotide, purine-rich sequence (the Shine-Dalgarno sequence) preceding the initiation codon in many mRNA molecules, thereby allowing the initiating ribosomal subunit to bind the mRNA so as to discriminate between many AUG or GUG (i.e. initiation) triplets and to select only the one at the beginning of a cistron. [After John Shine (1946–) and Lynn Dalgarno (1935–).]

shockate the preparation that is obtained when (bacterial) cells are suspended in an osmotically stabilizing medium and lysed by osmotic shocking through the sudden addition of water, followed by immediate centrifugation.

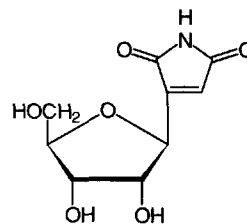
short-chain describing an aliphatic compound of any type with a chain length of less than 10 carbons.

short-patch repair a type of **excision repair** of DNA in which the average size of excised DNA is about 20 nucleotides. It contrasts with **very-short-patch repair**, which deals with mismatches between specific bases, and **long-patch repair**, in which relatively long stretches of DNA are involved, mostly about 1500 nucleotides long, but in some cases up to 9000 nucleotides or more. See also **DNA repair enzyme**, *uvr*.

shotgun cloning a method, first used for cloning an entire genome, in which the DNA is randomly broken into fragments of approximately the average size of genes. The fragments are then cloned into suitable vectors, to give a **genome library**.

shotgun collection see **genomic library**.

showdomycin 2-(β-D-ribofuranosyl)-maleimide; an antibiotic produced by *Streptomyces showdoensis* and active against a number of Gram-positive bacteria. It is sometimes used in biochemical studies as a thiol-reactive agent. Compare **N-ethylmaleimide**.



shuttle (*sometimes*) an intracellular metabolic cycle concerned in the transfer of, e.g., reducing equivalents or carbon chains across membranes. Different carriers operate each way across the membrane. Examples include the **3-hydroxybutyrate cycle**, the **glycerol-3-phosphate cycle**, and the **malate-aspartate cycle**.

shuttle vector a DNA molecule, e.g. a plasmid, that is able to replicate in two different host organisms and can therefore be used to 'shuttle' or convey genes from one to the other.

shuttle vesicle an endocytotic vesicle that transfers proteins across an epithelial cell from one face to another e.g. from the sinusoidal to the bile-canalicular face of a hepatocyte.

sl symbol for a residue of the ribonucleoside thioinosine; mercaptopurine ribonucleoside (alternative to S_{no}).

Si symbol for silicon.

Si- stereochemical descriptor denoting *sinister* (i.e. counter-clockwise, or 'left-handed') configuration of face of a trigonal atom. See **Re/Si convention**.

SI *abbr.* for *Système International* (d'Unités); International System (of Units); a rationalized coherent system of metric units, in which the magnitude of any physical quantity may be expressed, that is recommended for use by scientists throughout the world. The system is founded on seven **SI base units**: **ampere**, **candela**, **kelvin**, **kilogram**, **metre**, **mole**, and **second**. There are also 21 **SI derived units**, which are expressed in terms of the base units, and two **SI supplementary units**, **radian** and **steradian**, which are now regarded as dimensionless derived units. A series of prefixes signify decimal multiples and submultiples of any SI unit.

Sia symbol for the sialoyl group.

sial+ a variant of **sialo+** (before a vowel).

sialagogue or **sialogogue** any agent that stimulates the flow of saliva. —**sialagogic** or **sialogogic** *adj.*

sialate the anion of **sialic acid**.

sialic of, or relating to, saliva or a salivary gland.

sialic acid any of the *N*- or *O*-substituted derivatives of **neuraminic acid** found as components of complex carbohydrates, e.g. *N*-acetylneuraminic acid (5-(acetylamino)-3,5-dideoxy-D-glycero-D-galacto-non-2-ulonic acid, (*abbr.*: Neu5Ac; see also **NeuNAc**). Linkage of sialic acids is through C-2 to monosaccharides at the ends of glycosylation chains of glycolipids and glycoproteins. Neu5Ac can also be linked at its C-8 and C-9 hydroxyl groups to another Neu5Ac residue, often forming multiple repeats as in mammalian **NCAM**. Hydroxyl groups at C-4, -7, -8, and -9 can be *O*-acylated (see also **NeuOAc**). Other common derivatives are *N*-glycolylneuraminic acid (in mammals, but not normally in humans; *abbr.*: NeuGc), and **N-ketodeoxyoctanooylneuraminic acid** (e.g. in bacteria).

sialidase EC 3.2.1.18; *recommended name*: *exo-α-sialidase*; *former name*: *neuraminidase*; a glycosidase enzyme that hydrolyses 2,3-, 2,6-, and 2,8-glucosidic linkages joining terminal nonreducing *N*- or *O*-acylneuraminyl residues to galactose, *N*-acetylhexosamine, or *N*- or *O*-acylated neuraminyl residues in oligosaccharides, glycoproteins, or glycolipids. Example from *Clostridium perfringens*: database code NANH_CLOPE, 382 amino acids (42.81 kDa).

sialidosis another name for **mucopolidosis I**.

sialo- or (before a vowel) **sial+** *comb. form* 1 of, or relating to, saliva or a salivary gland; **sialic-** 2 denoting the presence in a

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sialoadhesin

compound of one or more residues of sialic acid. *Compare asialo-.*

sialoadhesin a macrophage-restricted sialic-acid-dependent glycoprotein receptor of several forms. It contains 17 immunoglobulin-like domains. It is an adhesion molecule that recognizes oligosaccharides terminating in NeuAc(α 2-3)Gal in N-linked and O-linked glycans. Example, 16-domain form from the mouse: database code MMSIAL1, 1598 amino acids (172.68 kDa).

sialogogue a variant spelling of **sialagogue**.

sialosyl the glycosyl group derived by removal of the exomeric hydroxyl group from sialic acid or sialate.

sialoyl the acyl group derived from sialic acid by removing the hydroxyl group from its carboxyl group.

sickle cell an abnormal red blood cell obtained from an individual exhibiting **sickle-cell anemia**. At low oxygen tensions such cells adopt a crescent shape reminiscent of the blade of a sickle. The defect originates in a mutation leading to replacement of Glu by Val at position 6 in the hemoglobin β chain. In its deoxygenated state the hemoglobin forms long rodlike helical fibres, which cause the deformation of the red cells because the β 6 Val can pack into a pocket at the EF corner (between helices E and F) of another β chain, a pocket which is lost in conformational changes resulting from oxygenation. Sickling can also result from β 121 Glu $\rightarrow$ Lys replacement, a site that lies close to β 6.

sickle-cell anemia an inherited disease of humans arising in individuals homozygous for a mutation resulting in replacement of Glu by Val at position 6 of the hemoglobin β chain. It is particularly common in populations indigenous to or originating from subtropical areas, including the Mediterranean region, parts of Saudi Arabia and India, and in those of African origin. The resultant abnormal red blood cells, called **sickle cells**, are removed from the circulation, leading to anemia. There is no satisfactory treatment, and early death often results. Less serious consequences occur in heterozygotes i.e. individuals with **sickle-cell trait**; these survive longer and benefit from greater resistance against the malaria parasite, which is adversely affected by sickle-cell trait during the red-cell phase of its life cycle. This results from the fact that, when the heterozygous sickle cells pass through capillaries, they sickle, causing them to lose K^+ and killing the malaria parasite, which needs high levels of K^+ . *See also glucose-6-phosphate dehydrogenase.*

side-by-side model *abbr.*: SBS model or SBS; a model of duplex DNA consisting of an alternating arrangement of right-handed and left-handed segments of duplex DNA, or other double-stranded polynucleotide, with five base pairs per segment. This produces a structure similar to the **Watson-Crick model of DNA** except that the two antiparallel strands are side by side rather than intertwined in a double helix.

side chain 1 any part of an organic molecule based on an aliphatic carbon chain that can be regarded as branching from a main chain. 2 the part of an amino acid that extends from the α -carbon atom and is unique for that amino acid.

sideramine one of the two main categories of **siderochrome**. *See also sideromycin.*

siderochrome or **siderophore** or **ironophore** any of various low-molecular-mass Fe(III)-chelating substances elaborated by aerobic or facultatively anaerobic microorganisms, especially when growing in iron-deficient media. Fe^{3+} -siderochrome complexes have very high stability constants (10^{30} – 10^{50} dm³ mol⁻¹). They are taken up by specific transport systems in the microorganisms; subsequent intracellular release of iron requires enzymic action. A siderochrome is thus an iron **ionophore**. Siderochromes are of two main categories, one or both of which may be formed by any particular organism: (1) catechol derivatives, e.g. **enterobactin**; and (2) hydroxamic acid derivatives (or **sideramines**), e.g. **mycobactin** (*see also ferriochrome*). It is curious that the transport systems for ferriochrome or ferric enterobactin in certain strains of bacteria

appear to be identical with the membrane receptors for colicin M or B respectively (*see colicin*), and in the former case for the antibiotic albomycin and various phages as well.

sideromycin any of a class of iron-containing antibiotic substances elaborated by certain actinomycetes. They are structurally related to sideramines (*see siderochrome*), by which their antibiotic action can be antagonized.

siderophore an alternative name for **siderophore D**.

siderophilin an alternative name for serotransferrin, serum transferrin.

siderophore 1 or **siderophore** a macrophage containing granules of **hemosiderin**. 2 an alternative name for **siderochrome**.

siderosome a membrane-bound organelle found in spleen macrophages and containing aggregates of **hemosiderin**. Siderosomes may be derived from **lysosomes**.

siemens *symbol*: S; the SI derived unit of **electric conductance**, defined as the electric conductance between two points on an electric conductor when an electric potential difference of one volt causes an electric current of one ampere to flow; i.e. $1\text{ S} = 1\text{ }\Omega^{-1}$. It was formerly called the **mho** or the reciprocal ohm. [After (Ernst) Werner von Siemens (1816–92), German electrical engineer, industrialist, inventor, and metrologist.]

sievert *symbol*: Sv; 1 the SI derived unit of (radiation) **dose equivalent**, equal to one joule per kilogram of living tissue; i.e. $1\text{ Sv} = 1\text{ J kg}^{-1}$. Also, $1\text{ Sv} = 100\text{ rem}$. *Compare gray*. 2 or **intensity millicurie** an obsolete non-SI unit of intensity of gamma radiation, numerically equal to 8.4 röntgens per hour. [After Rolf Maximilian Sievert (1896–1966), Swedish radiation physicist.]

sievorptive chromatography a form of **chromatography** in which absorption chromatography and molecular-sieve chromatography are combined within a single separative system.

Si-face (of an atom or molecule) *see Re/Si convention*.

SIF cell *abbr.* for small intensely fluorescent cell.

sigma *symbol*: σ (lower case) or Σ (upper case); the eighteenth letter of the Greek alphabet. For uses *see Appendix A*.

sigma bond *see orbital*.

sigma factor an RNA polymerase initiation factor occurring in bacteria; a protein that promotes attachment of the RNA polymerase, of which it is a subunit, to specific initiation sites and is then released. The main sigma factor of *Escherichia coli* is σ^{70} : database code RP70_ECOLI, 613 amino acids (70.18 kDa). An unrelated family is σ^{54} (four motifs), with associated regulatory proteins; members of this are specific to certain operons; example from *E. coli*: database code RP54_ECOLI, 477 amino acids (53.93 kDa). Other sigma factors are involved in response to heat shock and respiratory stress; examples (*E. coli*), σ^E : database code RPOE_ECOLI, 191 amino acids (21.67 kDa); and σ^{32} : database code RP32_ECOLI, 284 amino acids (32.43 kDa). Major switches in transcriptional control in bacteria, e.g. sporulation in bacilli, and transcriptional control in certain phages, involve modification to or replacement of sigma factors.

sigma orbital *see orbital*.

sigma receptor any of a class of receptors formerly regarded as opioid receptors but now generally regarded as nonopioid receptors because morphine has a very low affinity for them, and naloxone, the classical opioid antagonist, does not block the action of morphine. They are noted for their activation by powerful psychotomimetic drugs, e.g. **phencyclidine**. These receptors appear to mediate disorientated and depersonalized feelings.

sigma subunit a component of bacterial RNA polymerase; *see sigma factor*.

sigmoid or **sigmoidal** S-shaped; especially as in a sigmoid flexure (e.g. of the large intestine); or a sigmoid plot, i.e. one similar to a skewed letter S, that is with slope increasing initially slowly, progressively increasing more rapidly, and then decreasing towards a plateau. —**sigmoidicity** *n*.

sigmoid kinetics (*in enzymology*), a type of kinetics of en-



zyme-catalysed reactions characterized by a sigmoid plot of rate versus substrate concentration. Such kinetics are characteristic of **allosteric** enzymes, in that progressive activation occurs with increasing substrate concentration due to **cooperativity** in binding of substrate. The presence of allosteric modifiers will alter the inflection of the curve, activators shifting it towards that found in Michaelis–Menten kinetics (see **Michaelis kinetics**).

signal 1 any variable property or parameter that serves to convey information. **2** the message potentially to be transmitted by any agonist interacting with a cell membrane receptor. **3** a **signal peptide**.

signalase see **signal peptidase**.

signal averaging a method used to improve **signal-to-noise ratios**, particularly in spectroscopy, by summing repeated scans of the information. The signals are thereby reinforced while the background noise tends to average out.

signal hypothesis a model proposed for the synthesis of secretory proteins on ribosomes bound to the rough **endoplasmic reticulum**. In 1971 the US cytologist Gunter Blobel (1936–) and the US cell biologist David Domingo Sabatini (1931–) suggested that secretory proteins would have a common sequence of amino-acid residues near the N terminus of the nascent chains and that the peptide, or a modification of it, would then be recognized by a factor mediating the binding to the membrane. César Milstein and coworkers in 1972 proposed from experiments in which immunoglobulin light chain mRNA was translated in a reticulocyte lysate system, that light chains are initially synthesized as a precursor of slightly higher molecular weight, which is subsequently converted into the authentic product. From these experiments they proposed, independently of Blobel and Sabatini, that a short peptide at the N terminus of a precursor protein would be a simple way to provide a signal to ensure that secretory proteins were synthesized on membrane-bound ribosomes. This hypothesis has been largely substantiated (see **signal peptidase**, **signal peptide**, **signal recognition particle**).

signalling molecule any extracellular or intracellular molecule, e.g. **hormone**, **cytokine**, or **second messenger**, that cues the response of a cell to the behaviour of other cells or objects in its environment.

signal peptidase any of a group of endopeptidases that remove the **signal peptide** after insertion of membrane proteins into the membrane. In prokaryotes there are two types: 1 **signal peptidase I**, EC 3.4.99.36; *recommended name*: leader peptidase; *other name*: prelipoprotein signal peptidase; an endopeptidase that cleaves N-terminal signal peptides from secreted proteins and periplasmic membrane proteins. It is a protein of the inner membrane with a single transmembrane domain. Example from *Salmonella typhimurium*: database code LEP_SALTY, 324 amino acids (35.78 kDa). 2 **signal peptidase II**, EC 3.4.99.35; *recommended name*: premurein-leader peptidase; an endopeptidase that selectively cleaves the signal peptide from bacterial membrane lipoproteins, hydrolysing the protein on the N-terminal side of a specific Cys residue. Example from *Haemophilus influenzae*: LSPA_HAEIN, 171 amino acids (19.36 kDa). In eukaryotes, signal peptidases, EC 3.4.–, are endoplasmic integral membrane proteins with five subunits, numbered according to the approximate M_r , SPC25, SPC22/23, SPC21, SPC18 and SPC12. Example, SPC21 from *Canis familiaris*: database code SPC3_CANFA, 191 amino acids (21.47 kDa).

signal peptidase II see **prelipo-leader peptidase**.

signal peptide or **leader peptide** any sequence of amino-acid residues which when linked to a newly synthesized protein directs the protein to a location among the organelles of a eukaryotic cell and from the cytoplasm to the periplasmic space of prokaryotic cells. The term was originally applied to the N-terminal linked peptide that is responsible for the attachment of ribosomes which translate the mRNA for secretory proteins to the endoplasmic reticulum. Such peptides usually comprise

20–30 amino-acid residues and are characterized by a central region of hydrophobic amino acids preceded by one or more basic amino-acid residues. An interesting feature is that at the junction of the signal peptide and the mature protein there is no consensus sequence of amino-acid residues, so the basis for the specificity of the protease at this position remains unclear (see **signal peptidase**). The term signal peptide is now more generally applied to any peptide that is responsible for the location of a newly synthesized protein within the organelles of eukaryotic cells (see **protein kinesis**). Thus it applies to the mitochondrial proteins synthesized in the cytoplasm, nuclear proteins and luminal proteins that are retained within the cisternae of the endoplasmic reticulum. In addition to the signal peptides described, which may be located either at the N terminus or C terminus of the nascent protein and which are subsequently cleaved, single-spanning membrane proteins also contain a signal sequence within the protein known as a **stop-transfer sequence**. This serves to anchor the protein within the membrane. The orientation of the protein within the membrane may be either type I or type II depending on whether the N terminus of the resulting protein is extracytoplasmic (type I) or cytoplasmic (type II). See also **signal recognition particle**.

signal recognition particle *abbr.*: SRP; a ribonucleoprotein particle of 325 kDa composed of a 7S (300 nucleotide) RNA molecule and a complex of six different polypeptides. This binds both to the N-terminal signal peptide for proteins destined for the endoplasmic reticulum as they emerge from the large ribosomal subunit and also to the ribosome. This binding arrests further translation thereby preventing the proteins from being released into the cytosol. The SRP–ribosome complex then diffuses to the endoplasmic reticulum where it is bound to the **signal recognition particle receptor**, which allows resumption of protein synthesis and facilitates the passage of the growing polypeptide chain through the **translocon**. Through a process involving GTP hydrolysis, the SRP–SRP receptor complex dissociates and SRP returns to the cytosol. Of the six polypeptides of SRP the 54 kDa subunit (SRP54) is the central player. It contains an N-terminal GTPase domain and a C-terminal domain that binds directly to the signal peptide and the SRP RNA.

signal recognition particle receptor *abbr.*: SRP receptor; a transmembrane heterodimeric protein (α subunit, 69 kDa; β subunit, 30 kDa), also known as **docking protein**, located in the membrane of the rough endoplasmic reticulum. Both subunits contain GTPase domains with which **signal recognition particle** interacts. In the presence of GTP and SRP receptor, SRP is released from the ribosome–nascent chain complex. Example, α subunit (dog): database code SSRA_CANFA, 286 amino acids (31.94 kDa). The α subunits have an unusual charge distribution with a highly negatively charged N-terminus and a highly positively charged but phosphorylated cytoplasmic tail (C terminus). See also **signal peptide**, **signal recognition particle**.

signal-to-noise ratio the proportional difference between the average level of values observed for a specific property under investigation (signal) measured against the average of nonspecific values (noise) in any record. See also **signal averaging**.

signal transduction the transfer of a signal from the outside to the inside of a cell by means other than the introduction of the signal molecule itself into the cell. Typically, interaction of the extracellular signal – a hormone, growth factor, or other agonist – with a specific membrane receptor leads to synthesis within the cell of one or more **second messengers**, or to activation of other downstream cascades, e.g. by phosphorylation of proteins.

signature an alternative term for **motif** (in protein sequences); the word is used in the PRINTS database. See **Appendix E**.

significant (in statistics) reaching a degree of **probability** that experimental values lie on a distribution different from that of control values, such that the investigator will conclude that experimental and control samples differ, using tests such as Student's *t*-test or the chi-square test. The degree of probability



ity that will be accepted as indicating a significant difference depends on the type of investigation being conducted. If the tests indicate that the probability that the results are due to chance is $\geq 5\%$ (for biological or medical studies) or $\leq 1\%$ or, in some cases, $\leq 0.1\%$ (for biochemical studies), differences are considered significant. —**significance** *n*.

silane **1** monosilane; silicon tetrahydride; SiH_4 ; the silicon-containing analogue of methane. **2** any of a class of saturated silicon hydrides. Such compounds are silicon-containing analogues of alkanes or cycloalkanes; linear or branched silanes are of general formula $\text{Si}_n\text{H}_{2n+2}$, and cyclic silanes, termed cyclosilanes, are of general formula $[\text{SiH}_2]_n$ ($n > 2$). The value of n , when > 1 , may be indicated by a specific or general numerical prefix; e.g. disilane, cyclohexasilane, oligosilane, polysilane. Silane and disilane are gases with a repulsive odour; higher silanes are liquids or solids. Silanes are spontaneously flammable in oxygen or air. **3** a term often loosely applied to any hydrocarbyl or other derivative of a silane.

silanization treatment of a hydrophilic surface with a reactive **silane** (def. 3) such as dimethyldichlorosilane, trimethylchlorosilane, octyltrichlorosilane, or an octadecyltrialkoxysilane in order to render it more hydrophobic. Such a treatment is useful in particular for converting hydrophilic **silanol** (def. 4) groups on the surfaces of glassware or dried silica-gel particles into lipophilic alkylsiloxane moieties. When applied to glass, it reduces adsorption of polar compounds and minimizes the activation of lymphocytes, platelets, or other cells that can occur when untreated glass vessels are used; other benefits include promotion of drainage of aqueous liquids and minimization of electrical leakage (see also **siliconize**). Silanization has been applied to particles of dried silica gel for use in reversed-phase adsorption chromatography (see also **end capping**) and to diatomaceous earth used as the support for the stationary phase in reversed-phase liquid-liquid partition chromatography or in some variants of gas-liquid (partition) chromatography. Compare **silylation**. —**silanize** *vb.*; **silanized** *adj.*

silanol **1** silyl hydroxide; H_3SiOH ; the silicon-containing analogue of methanol. **2** any monohydroxy derivative of a **silane** (def. 2); such compounds are silicon-containing analogues of alkanols and are of general formula $\text{Si}_n\text{H}_{2n+1}\text{OH}$. **3** a commonly used term for any Si-hydrocarbyl derivative of silanol, silanediol, or silanetriol; such derivatives are respectively of general formula R_2SiOH , $\text{R}_2\text{Si}(\text{OH})_2$, or $\text{RSi}(\text{OH})_3$. **4** the trivalent group >Si-OH occurring in hydrated silica (so-called silicic acid). Such groups occur also on the surfaces of dried silica-gel particles and of silica-based glasses under normal conditions (i.e. in equilibrium with undried air).

sildenafil see **Viagra**.

silent mutation or **synonymous mutation** or **same-sense mutation** a mutation in a codon of a DNA sequence that does not cause an amino-acid change in the translation product. The term 'silent' is also applied to a protein gene product arising from such a mutation, e.g. **silent hemoglobin**. Other names: silent site mutation; silent nucleotide change.

silica silicon(IV) oxide; silicon dioxide; SiO_2 ; a colourless or white solid used as the basis of many chromatographic procedures; silicic acid contains **silanol** (def. 4) groups to some of which water is bound. For chromatographic use, it is heated to 100°C to drive out this water. Heating above 170°C causes formation of oxide linkages (Si-O-Si), and when heated to very high temperatures ($800\text{--}1000^\circ\text{C}$) it forms glass. Silicic acid is prepared for different uses in a variety of particle sizes. For chromatography, these range from 10 to $500\ \mu\text{m}$, the larger sizes being used for column chromatography at normal pressures, the smaller sizes for thin layer chromatography and high-pressure liquid chromatography. Silicic acid precipitated as granules, about $3\ \text{mm}$ diameter, is known as **silica gel**; in its dry state it takes up water and can be used as a drying agent, which can be re-used after heating to drive out water. A powder of very fine silica particles, $0.007\text{--}0.014\ \mu\text{m}$, known as **fumed silica**, has thixotropic properties and was formerly em-

ployed to create a gel in organic solvents, e.g. in scintillation cocktails to maintain hydrophilic materials in suspension during counting (this use is now superseded by cocktails that themselves form a gel with water). Silica is found in certain algae, notably in the hard shell, or test, of diatoms; it also serves to reinforce the stems of the primitive vascular plants known as horsetails.

siliceous or **silicious** of, pertaining to, or containing **silica**.

silicone the common name for any oligomeric or polymeric **siloxane** containing the repeating structural unit $-\text{O-SiR}_2-$, where R is a hydrocarbyl (e.g. methyl or phenyl) group. The molecules of an individual silicone are usually linear, but sometimes branched, crosslinked, or cyclic. Silicones are odourless and generally colourless, and are characterized by high hydrophobicity and by remarkably high chemical and thermal stability. Depending on their molecular size and configuration and on the nature of their R groups, silicones may be oils, greases, gums, elastomers, or resins. By virtue of their unusual and varied properties they have acquired a correspondingly wide range of clinical, electrical, engineering, industrial, and laboratory applications. In commercial usage, the term may be extended to include organosilicon compounds used for **silanization** and **silylation**.

siliconize to render (filter-paper, glassware, textiles, etc.) water-repellent by application of a coating of an appropriate silicone. See also **silanization**. —**siliconized** *adj.*; **siliconizing** *n*.

siloxane any saturated silicon-oxygen hydride with unbranched or branched chains of alternating silicon and oxygen atoms in the molecule. Unbranched siloxanes are of general formula $\text{H}_3\text{Si}[\text{O-SiH}_2]_n\text{SiH}_3$; each is named after its respective parent **silane** (def. 2), e.g. disiloxane ($n = 0$), tetrasiloxane ($n = 2$). By extension, hydrocarbyl derivatives are commonly included. See also **silicone**.

silyl **1** strictly, $\text{H}_3\text{Si-}$, the group derived from **silane** (def. 1) by loss of a hydrogen atom. **2** a commonly used term for any hydrocarbyl derivative, $\text{R}_3\text{Si-}$, of the **silyl** (def. 1) group.

silylation the process of converting organic compounds in general into their **silyl** (def. 2) derivatives by attachment of trimethylsilyl (or similar organosilicon) groups, or an instance of this process applied to a particular compound or mixture of compounds. Reagents frequently used for the purpose are trimethylchlorosilane and 1-(trimethylsilyl)imidazole, although a number of others have particular applications. The classes of reactive compounds include those possessing amino, carboxyl, hydroxyl, or thiol groups, for which silylation is a means of selective protection in the course of synthetic work (see **protecting group**). Silylation is widely used also for conversion of mixtures of related compounds of such classes into derivatives capable of separation and analysis by gas chromatography/mass spectrometry. Compare **silanization**. —**silylate** *vb.*; **silylator** *n.*; **silylated**, **silylating**, *adj.*

simian characteristic of, resembling, or belonging to the apes.

simian immunodeficiency virus *abbr.*: **SIV**; a retrovirus, one of the lentiviruses, common in the sooty mangabey (a type of monkey) and many African green monkeys. **SIV_{SM}**, a virus of the sooty mangabey, is closely related to – and probably gave rise to – **HIV-2** (see **HIV**). **HIV-1** is closely related to **SIV** from chimpanzees. Infection of sooty mangabeys and African green monkeys does not cause disease, but **SIV** infection has been found in macaques in captivity and is pathogenic.

simian virus 40 *abbr.*: **SV40**; an icosahedral virus containing circular duplex DNA of about $5.2\ \text{kb}$ associated with four histones (**H4**, **H2A**, **H2B**, **H3**) in a minichromosome. It infects monkeys and apes and is capable of inducing tumours in other species or their cells, its DNA being integrated into the host genome. It can enter two types of life cycle: in permissive cells (usually monkey permanent cell lines) replication occurs in normal infection; in nonpermissive cells there is no lytic infection, but growth transformation can occur. The virus is useful as an **expression vector** in recombinant DNA technology, having good promoter properties: it can be used either as a virion

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simple diffusion

or as a construct maintained transiently in host cells at high copy number in unintegrated plasmid-like DNA molecules. See also **small T-antigen**.

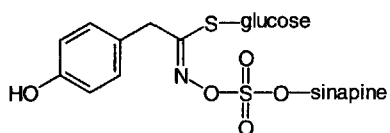
simple diffusion or passive diffusion a type of **diffusion** (def. 2) across a membrane in the absence of any specific transporting agent or carrier. Compare **facilitated diffusion**, **active transport**.

simple radial immunodiffusion see **single radial immunodiffusion**.

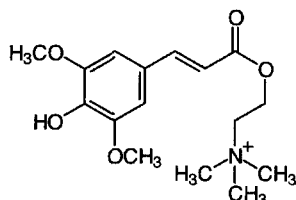
SIMS *abbr.* for secondary ion mass spectrometer.

sin *abbr.* for sine.

sinalbin a glucoside found in white or yellow mustard. See also **sinapine**.



sinapine 2[[3-(4-hydroxy-3,5-dimethoxyphenyl)-1-oxo-2-propenyl]oxy]-N,N,N-trimethylethanaminium; a component of **sinalbin**; it is also found in black mustard and other seeds.



sinaptobrevin a variant spelling of **synaptobrevin**.

sine *abbr.*: sin; a trigonometric function, being the ratio of the side of a right-angled triangle opposite a given angle to the hypotenuse; the sine of an obtuse angle is numerically equal to that of its supplement.

SINE *abbr.* for short interspersed element; a highly repeated sequence of DNA occurring interspersed in a mammalian genome; it is similar to a **LINE** but is shorter – typically less than 500 bp long – and is generated by **RNA polymerase III**. A family of closely related SINEs has of the order of 10^5 members, consisting of DNA segments that are similar if not identical in length and sequence. The function of SINEs is unknown. They appear to be processed **pseudogenes** derived from genes that encode small cytoplasmic RNAs including tRNAs. Thus the sequences of SINEs are often homologous to the sequences of such RNAs.

Singer-Nicholson model an alternative name for **fluid-mosaic model**.

single-cell protein *abbr.*: SCP; a protein-rich material produced by the culture of bacteria, yeasts, other fungi, or algae and extracted for use as a protein supplement in animal or human foods.

single-copy plasmid a plasmid that replicates at the same pace as the bacterial chromosome, and thus maintains equality in numbers with the chromosome.

single nucleotide polymorphism *abbr.*: SNP; the occurrence of single-base variations in the genetic code that occur about once every 1000 bases along the human genome.

single radial immunodiffusion or Mancini method an immunodiffusion technique in which holes (wells) are cut in a sheet of antibody-containing agar in a glass dish or on a slide. The wells are filled with antigen solution, which is then left to diffuse radially outwards; in doing so, it reacts with antibody to form a ring of precipitate. The diameter of the circle of precipitate is proportional to the concentration of antigen.

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single-spanning describing an integral membrane protein anchored in the membrane by means of a single hydrophobic sequence embedded in the membrane lipid bilayer.

single-strand binding protein *abbr.*: SSB protein; any of a group of proteins found in bacteria and mitochondria that have a greater affinity for **single-stranded DNA** than for double-stranded species. They are essential for replication, recombination, and repair. They bind tightly and cooperatively and do not catalyse associated activities such as the DNA-dependent ATPase activities of **helicases** and **topoisomerases**. They have been called 'unwinding', 'melting', and 'helix-destabilizing' proteins, but each of these names is somewhat misleading since these proteins have multiple functions. Example from *Escherichia coli*: database code SSB_ECOLI, 177 amino acids (18.82 kDa).

single-strand conformation polymorphism analysis

abbr.: SSCP analysis; a method for comparing mutations by the rate at which DNA migrates on denaturing gels, under conditions in which the conformation adopted by the DNA is critical. The technique involves amplification of a DNA sequence of interest using the **polymerase chain reaction** (PCR); the rate of migration of this DNA is then compared with that of a known mutation on gels of differing composition, such that DNA conformation (and thus rate of migration) will depend on the gel conditions (e.g. varying glycerol concentration, ionic strength). If the DNA of unknown composition migrates the same as the known DNA under all conditions, it is an indication that a mutation of a similar character to the known mutation may have occurred.

single-stranded DNA *abbr.*: ssDNA; a form of **DNA** consisting of a single chain of deoxyribonucleotides instead of the normal two chains. Single-stranded DNA occurs naturally in certain circumstances: (1) transiently, during replication (see **replicate**), when the duplex helical molecule of double-stranded DNA is unwound in the replication fork, initiated by **helicase**; (2) during replication of circular DNA, when in some circumstances only one strand is replicated, yielding a single-stranded product; as replication proceeds, the synthesized DNA unwinds as a single-stranded 'tail'. Replication may proceed around the unbroken circular DNA strand for more than one cycle, giving rise to the term **rolling circle**; (3) in certain filamentous **coliphages**, such as M13, f1, and fd, which contain a single-stranded circular DNA molecule.

singlet oxygen a dioxygen molecule in which two 2p electrons have similar spin. It is more highly reactive than the form in which these electrons are of opposite spin. Singlet oxygen is produced in chloroplasts that, due to mutation, lack carotenoids; excited triplet chlorophyll reacts with oxygen to yield singlet oxygen, which is normally quenched by carotenoids. In the absence of carotenoids, lethal photodynamic damage occurs.

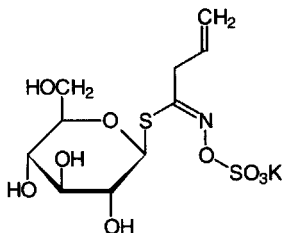
single-turnover kinetics the study of the kinetics of an enzyme-catalysed reaction in conditions under which formation of the enzyme-substrate complex can be measured. Such conditions are required for the detection and analysis of transient enzyme-substrate and enzyme-intermediate forms in the investigation of the pre-steady state (see **pre-steady state kinetics**), and contrast with measurement of the multiple recycling of the enzyme in the steady state. Although the enzyme reaction is proceeding normally, only the first part of the reaction is studied, leading to the expression 'single-turnover conditions'. In this kind of study, a substrate is chosen such that the enzyme-substrate complex can be distinguished from substrate alone, e.g. in having a different absorption spectrum. The rate of formation of the complex can then be determined, often by stopped-flow methods.

singly labelled describing a **specifically labelled** compound containing only one isotopically modified atom in a uniquely specified position, e.g. $\text{CH}_3\text{-CH}_2^2\text{H-OH}$.

sinigrin 1-thio- β -D-glucopyranose 1-[N-(sulfo-oxy)-3-butenimidate] monopotassium salt; a β -D-thioglucoside occurring in

sinister

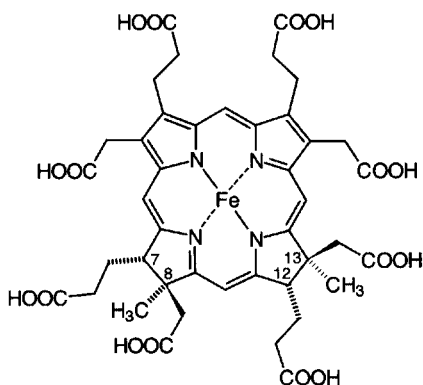
mustard seeds and horseradish roots; it is used as a substrate for thioglucosidase.



sinister Latin left; used to describe the left, or counterclockwise, configuration of one of the two faces of a compound containing a trigonal carbon atom. It is designated by the prefix *Si-*. Compare *rectus*. See also *Re/Si* convention.

sintered glass or frit small particles of glass that have been heated so that they just fuse but without extensive melting. The resultant material is porous, and is often used in glass filtration apparatus or as a support to retain packing materials in chromatographic columns.

siroheme or (esp. Brit.) **sirohaem** a heme molecule discovered in bacterial sulfite reductase, EC 1.8.99.1. It is a tetrahydroporphyrin, having adjacent, reduced pyrrole rings. It is also a component of nitrite reductase (EC 1.6.6.4) from *Nitrobacter*.



siroheme synthase or (esp. Brit.) **sirohaem synthase** a multifunctional bacterial enzyme that contains **uroporphyrin-III C-methyltransferase**, **ferrochelatase** and precorrin-2 oxidase. The latter activity results in the conversion of precorrin-2 into **siroheme** by oxidation and Fe²⁺ chelation. Example from *Escherichia coli*: database code CYSG_ECOLI, 457 amino acids (49.90 kDa).

sirup a variant spelling (esp. US) of **syryp**.

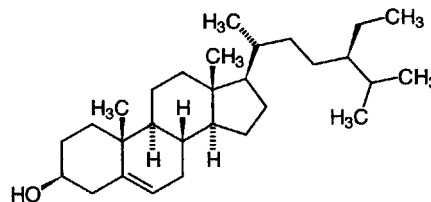
sis an oncogene (*v-sis*) from simian osteosarcoma virus that is also expressed in human disease. The related protooncogene, *c-sis*, encodes the B chain of **platelet-derived growth factor** (PDGF), and *v-sis* encodes a protein almost identical to the B chain, differing only in the amino acids at its N and C termini. Cells are transformed by *v-sis* and produce the PDGF-related molecules, which stimulate the PDGF receptor in an autocrine fashion. Example, PDGF-related transforming protein p28^{sis} of simian sarcoma virus: database code TSIS_SMSAV, 226 amino acids (25.42 kDa).

site-directed mutagenesis or **site-specific mutagenesis** the use of cDNA technology to generate a **point mutation** at a predetermined position in a genome. It may be achieved by stepwise synthesis of a strand of genomic nucleic acid with substitution of a mutagenic analogue of the base naturally pre-

sent (e.g. guanine for adenine) at a specific position in the nucleotide sequence. After purification this is replicated *in vitro* to allow the generation of base substitutions at the corresponding site in the complementary strand.

site-specific DNA methyltransferase see **DNA methylase**.

sitosterol 22,23-dihydrostigmasterol; a widely distributed plant sterol related to **stigmasterol**.



β-sitosterol

SI units see **SI**.

SIV abbr. for simian immunodeficiency virus.

sizing column a **gel-filtration** column that is used to provide information about the *M_r* (size) of proteins.

sizing gel a **gel-filtration** medium, especially one used in a sizing column.

sizukacillin see **peptaibolph**.

S2 kallikrein see **tonin**.

SKALP abbr. for skin-derived antileukoprotease; other name **elafin**; an inhibitor of serine proteinases; it probably inhibits elastase-mediated tissue proteolysis. Example, human protein sequence (precursor): database code ELAF_HUMAN, 117 amino acids (12.26 kDa).

skatole 3-methylindole; a bacterial metabolic product of indoleacetic acid that contributes to the odour of feces.

skeletal growth factor a protein that stimulates bone growth. Skeletal growth factor from human bone (hSGF) stimulates growth in human bone cells and embryonic chick bones in culture. The activity comprises **insulin-like growth factors** (IGFs) in association with IGF-binding proteins (IGFBPs). Example, precursor of human IGFBP-5: database code IBP5_HUMAN, 272 amino acids (30.53 kDa).

skeletin (formerly) an alternative name for **desmin**.

skeleton 1 (in zoology) any structure that provides support and protection for the soft tissues and organs of the body. An **exoskeleton**, such as the cuticle in arthropods, lies external to the body tissues, while an **endoskeleton**, such as the bony or cartilaginous skeleton of vertebrates, lies within the body. 2 (in chemistry) the carbon chain of a molecule, including any noncarbon elements that necessarily link carbon moieties of the chain. —**skeletal adj.**

skew 1 asymmetric, biased, distorted. 2 oblique, slanting. 3 (of a statistical distribution) not symmetrical about the mean. 4 to become or make skew; to distort. See also **skew conformation**. —**skewness n.**

skew conformation 1 any conformation of a six-membered ring form of a monosaccharide or monosaccharide derivative when three adjacent ring atoms and the remaining nonadjacent ring atom are coplanar. Such compounds may be designated by the conformational descriptor *-S* added to the name. The reference plane is so chosen that the lowest-numbered carbon atom in the ring, or failing that the atom numbered next above it, is exoplanar. Skew conformation is the median conformation through which one **boat conformation** passes during conversion to another boat conformation in the cycle of flexible forms; it is similar to **twist conformation** (def. 1). See also **conformation**. 2 a less common alternative term for **synclinal conformation**; see **conformation**.

SKI a gene encoding a protein of unknown function but related to *v-ski*, the oncogene of Sloan-Kettering viruses; it has no marked homology with other oncogenes.

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slab gel

slab gel a gel, usually polyacrylamide or starch, that is formed as a sheet rather than a rod.

slaty a coat colour mutation in mice allelic for the *tyrp-2* gene. The mice have a defect in melanin formation that causes dark-grey (slate-coloured) melanin to be formed. The enzyme affected is **dopachrome Δ -isomerase**.

sleep peptide any peptide that brings about sleep when infused into the cerebrospinal fluid (CSF) of animals. Delta-sleep-inducing peptide, isolated from CSF, has the structure Trp-Ala-Gly-Gly-Asp-Ala-Ser-Gly-Glu. A large number of endogenous substances isolated from the brain can likewise induce sleep, but it is not known if this is their physiological function.

sliding filament model a model of muscle contraction that envisages contraction occurring as a result of **thin filaments** sliding past **thick filaments**. The mechanism of activation involves enhanced binding of Ca^{2+} by troponin C (see **troponin**), one of the three troponin peptide components of the thin filament. In the resting state, troponins I and T inhibit the actin-tropomyosin complex of the thin filament from binding to myosin; increase in Ca^{2+} concentration causes troponin C (each molecule of which binds up to four Ca^{2+} ions) to release this inhibition, allowing the myosin headgroups to bind actin. Then, in the presence of ATP, the thin filaments slide over the thick filament myosin molecules to cause contraction. The myosin heads contain an active ATPase that hydrolyses the ATP; the latter must be continuously resynthesized for contraction to continue.

sliding (of repressor) the mode of action of **lac repressor** in *Escherichia coli*. It first binds at a non-operator site, then slides along the DNA until it finds the target site, i.e. the **operator**.

slime mould or (*US*) **slime mold** any of a group of organisms that exhibit features of both fungi and protozoa. Characteristically they exist as slimy masses on decaying wood or in damp soil, and have the ability to switch from an amoeboid single-celled phase, called a **myxamoeba**, to an aggregated or multicellular phase from which fruiting bodies may arise. In the so-called **true slime moulds**, the myxamoebae form an acellular mass called a plasmodium, whereas in the **cellular slime moulds** the myxamoebae aggregate to form a multicellular pseudoplasmodium, or 'slug'. A well-known example of a cellular slime mould is *Dictyostelium discoideum*; when the food supply is exhausted, its myxamoebae aggregate to form a small (1–2 mm) slug, containing up to 10^5 cells, which migrates to seek an area of food abundance. When nutritional conditions are favourable the slug undergoes differentiation to form fruiting bodies, from which spores are released. Germination of a spore leads to the formation of a myxamoeba. *D. discoideum* has a relatively small genome (about 10 times larger than that of a bacterium) and is popular with geneticists, providing insights into the nature of the inducible behaviour leading to aggregation, and the rapid cell division and differentiation that precede the formation of a fruiting body.

slip a term used in regard to situations in which the mitochondrial enzyme, **cytochrome c oxidase**, may carry out its redox reaction without pumping protons into the intermembrane space.

slope $\delta y/\delta x$, where results are expressed graphically as a continuous line, the ordinate (vertical axis) representing y , the abscissa (horizontal axis) representing x .

slow-cycling rhodopsin a **retinal**-containing energy-transducing pigment present in small amounts in some of the strains of *Halobacterium halobium* in which purple membranes occur. It is distinct from **bacteriorhodopsin** and **halorhodopsin** and is believed to play a role in the characteristic variation in the phototactic response of such organisms at different wavelengths. Illumination with yellow light or with blue or near-ultraviolet light results respectively in accumulation of forms with absorption maxima at 375 nm and 590 nm.

slow reacting substance (of anaphylaxis) or **slow reacting substance A** *abbr.*: SRS or SRS-A; one of a number of sub-

stances that mediate anaphylaxis; it was identified as a polar, lipid-like compound not normally present in mammalian tissues but synthesized in them as a consequence of the combination of antibody with antigen. It increases vascular permeability and induces contraction of some types of smooth muscle, especially bronchial muscle. It is slow in action relative to histamine – hence its name – and is not antagonized by antihistamines. **Leukotrienes C₄, D₄, and E₄** are now known to be the active components.

slurry a fluid suspension of particles having the consistency of cream.

Sm *symbol for samarium*.

small cytoplasmic ribonucleoprotein *abbr.*: scRNP (hence often colloquially termed 'scryp'); any of various particulate complexes of proteins with molecules of **small cytoplasmic RNA**, found in the cytoplasm. Their function is unknown.

small cytoplasmic RNA *abbr.*: scRNA; any of various low-molecular mass (100–300 nucleotides) RNA molecules found in the cytoplasm as components of **small cytoplasmic ribonucleoprotein**.

small intensely fluorescent cell *abbr.*: SIF cell; a cell found in autonomic ganglia and containing dopamine. The name derives from the appearance of such cells after formaldehyde condensation in fluorescence microscopy.

small nuclear ribonucleoprotein *abbr.*: snRNP (hence often colloquially termed 'snorp'); any of various particulate complexes of proteins with molecules of **small nuclear RNA** (snRNA), found in the eukaryotic nucleus; the particles are ≈ 250 kDa. They are named according to the snRNAs they contain: the U4/U6 snRNP, e.g., contains U4 and U6 snRNA. They are involved in RNA splicing (see **spliceosome**). Patients with systemic lupus erythematosus make antibodies directed against one or more snRNPs.

small nuclear RNA *abbr.*: snRNA; any of various low-molecular-mass (100–300 nucleotides) RNA molecules, designated U1, U2, ... U12 RNA, found in the eukaryotic nucleus as components of **small nuclear ribonucleoprotein**. See also **spliceosome**.

small nucleolar ribonucleoprotein *abbr.*: snoRNP (hence often colloquially termed 'snorp'); any of various particulate complexes of proteins with molecules of **small nucleolar RNA**, found in the eukaryotic nucleolus. They are possibly the same as **processosomes**.

small nucleolar RNA *abbr.*: snoRNA; any of a class of small RNAs that are associated with the eukaryotic nucleus as a component of **small nucleolar ribonucleoproteins** and participate in the processing of ribosomal RNA. They contain a long stretch of sequence complementarity to conserved sequences in mature mRNA. Six have been identified in metazoan cells (87–280 bases) and 12 in yeast (125–605 bases). They exist as ribonucleoprotein (RNP) particles, which appear to assemble with a large multi-RNA RNP complex.

small T-antigen see **T-antigen** (def. 1).

S₁ mapping or **S₁ nuclease mapping** a commonly used technique depending on the property of **S₁ nuclease** to remove single-stranded DNA or RNA from double-stranded hybrids of DNA and RNA. It may be used to detect the coding region of genomic DNA that contains **introns** by a procedure in which either cDNA for a particular mRNA or the mRNA itself is hybridized to the genomic DNA, and the single-stranded DNA removed by the S₁ nuclease leaving the hybridized coding DNA (the exons). The enzyme can also be used in a similar way to identify the 5' end of a mRNA.

smectic (of a substance) being or having a **mesomorphic** state in which the molecules are arranged in parallel layers. The molecules in any layer can move in the plane of that layer but not into adjacent layers. Compare **nematic**. See also **liquid crystal**.

smectic mesophase see **liposome**, **liquid crystal**.

Smith 1 Hamilton Othanel (1931–), US molecular biologist; Nobel Laureate in Physiology or Medicine (1978) jointly with W. Arber and D. Nathans 'for the discovery of restriction enzymes and their application to problems of molecular genet-

smooth

ics'. **2 Michael** (1932–), British-born Canadian biochemist; Nobel Laureate in Chemistry (1993) 'for his fundamental contributions to the establishment of oligonucleotide-based, site-directed mutagenesis and its development for protein studies' [prize (awarded 'for contributions to the developments of methods within DNA-based chemistry') shared with K. B. Mullis].

smooth describing a bacterial strain having shiny (smooth) colonies. *See* **smooth-rough variation**.

smooth endoplasmic reticulum *abbr.*: SER; *see* **endoplasmic reticulum**.

smooth muscle or **involuntary muscle** muscle that lacks striations, hence giving a 'smooth' appearance under the microscope; it produces slower, longer lasting contractions than **striated muscle** and is responsible for the contractile function of visceral organs such as stomach, bladder, intestine, and uterus, and the walls of arteries. The cells are mononucleate, and contain both **thick filaments** and **thin filaments** although these are not arranged in the pattern that leads to striations in skeletal muscle and cardiac muscle. The thin filaments are composed mainly of actin, polymerized into a double-helical strand, and associated with tropomyosin arranged along the length of the actin filament. The thick filaments are composed of aggregated myosin molecules. The thick and thin filaments comprise the contractile elements and are thought to function in an analogous manner to their striated muscle counterparts (*see* **sliding filament model**). Muscle contraction is regulated by the sarcoplasmic (cytosolic) free Ca^{2+} concentration, which increases from resting levels of 120–270 nM to 500–700 nM in response to stimuli such as membrane depolarization, or the action of α -adrenergic or muscarinic agonists. Ca^{2+} then binds to calmodulin, which activates **myosin-light-chain kinase**, which phosphorylates myosin at Ser¹⁹ of each of the two 20 kDa light chains. This causes cycling of myosin cross-bridges along actin filaments and the development of force. **Calponin**, **caldesmon**, and **protein kinase C** are also thought to play a role in fine-tuning the contractile state.

smooth-rough variation or **S→R variation** any of a number of types of cell-surface compositional variation that occur in bacteria, usually giving rise to changes in the appearance of colonies from that of the original strain. In the Enterobacteriaceae the colonies are typically shiny (smooth) and change to matt (rough), an S→R variation that may be accompanied by loss of surface antigens, reduction or loss of virulence, and changes in susceptibility to certain bacteriophages.

sn- prefix (*in chemical nomenclature*) signifying stereospecifically numbered; it is used in designating the configuration of derivatives of certain classes of prochiral compounds (especially glycerol and its derivatives) to ensure differentiation from conventional numbering. Thus sn-1 denotes C-1 of the glycerol moiety of a phospholipid under the **stereospecific numbering** system. C-2 and C-3 are similarly denoted sn-2 and sn-3 under the same system.

Sn symbol for tin.

SNAP *abbr.* for 1 *S*-nitroso-*N*-acetylpenicillamine. **2** synaptosome-associated protein; a number of these exist. The nomenclature has become somewhat confusing, since SNAPs exist with different but closely related roles. Thus SNAP-25 functions as a SNAP receptor, or **SNARE**, while other SNAP proteins such as α -SNAP (*see* def. 3) function to bind NSF proteins (*see* ***N*-ethylmaleimide**) to SNAREs such as SNAP-25. Example, SNAP-25 from chicken and mouse neurons: database code SN25_MOUSE, 206 amino acids (23.29 kDa). **3** soluble NSF attachment proteins, such as α -SNAP, β -SNAP, etc. These are cytoplasmic proteins that mediate binding of ***N*-ethylmaleimide-sensitive fusion protein** to membranes. *See also* **SNARE**, **SNARE hypothesis**.

SNARE *abbr.* for SNAP receptor; **syntaxin**, **synaptobrevin**, and SNAP-25 (*see* **SNAP** (def. 2)) act as SNAREs; they bind in an ATP-dependent manner to NSF-SNAP complexes (SNAP in this case being the NSF attachment protein (*see* ***N*-ethyl-**

maleimide)). Syntaxin, synaptobrevin, and SNAP-25 form a complex that is much more effective than the single proteins acting alone, leading to the concept of a core complex that bridges the synaptic vesicle and plasma membranes, and which may function in the docking and fusion process. *See also* **SNARE hypothesis**.

SNARE hypothesis a suggested model of vesicular fusion, in which two types of **SNARE** are involved; v-SNAREs are found on vesicles and t-SNAREs are found on the target membranes, each transport vesicle having its own v-SNARE which recognizes a cognate t-SNARE on a target membrane and mediates docking; fusion is then mediated by *N*-ethylmaleimide-sensitive fusion protein and an attachment protein (**SNAP** (def. 3)).

Snell, George Davis (1903–96), US geneticist; Nobel Laureate in Physiology or Medicine (1980) (jointly with B. Benacerraf and Jean Dausset 'for their discoveries concerning genetically determined structures on the cell surface that regulate immunological reactions').

Snell's law *see* **refraction**.

Sno symbol for a residue of the ribonucleoside thioinosine; mercaptopurine ribonucleoside; alternative to sl.

snoRNA *abbr.* for small nucleolar RNA.

snoRNP *abbr.* for small nucleolar ribonucleoprotein.

snoRP *colloquialism* for snoRNP, i.e. **small nucleolar ribonucleoprotein**.

snRNA *abbr.* for small nuclear RNA.

snRNP *abbr.* for small nuclear ribonucleoprotein.

S₁ nuclease EC 3.1.30.1; *recommended name*: *Aspergillus* nuclease S₁; a single-strand endonuclease, purified from *Aspergillus oryzae*, that degrades single-stranded RNA or DNA to 5' mononucleotides; it has no apparent base specificity. It is useful for the removal of unpaired regions in hybridization technology (*see also* **S₁ mapping**). The enzyme is a monomer, and contains three zinc atoms. Database code NUS1_ASPOR, 267 amino acids (29.02 kDa). Similar enzymes occur in *Penicillium* spp. and barley.

snurp *colloquialism* for snRNP, i.e. **small nuclear ribonucleoprotein**.

snurposome any of at least three (A, B, and C) morphologically and biochemically distinct granules in the amphibian oocyte nucleus that contain one or more types of small nuclear ribonucleoproteins (snRNP).

soap any salt formed between a hydroxide of a metal (usually an alkali or alkaline earth) and a higher fatty acid; such compounds are useful as **detergents** (def. 2). The meaning is sometimes extended to include the salt of any ionized lipid **amphipath**, e.g. of a long-chain amine or sulfate or of a sterol carboxylate. The common soaps of commerce were traditionally the mixed sodium salts (giving **hard soaps**) or the mixed potassium salts (giving **soft soaps**) of oleic, palmitic, and stearic acids, prepared by **saponification** of animal or vegetable fats or oils with sodium or potassium hydroxide respectively. Modern soap compositions are more complex, with solid or liquid soaps containing various additional detergents.

soapstone *see* **talc**.

SOD *abbr.* for superoxide dismutase.

soda any of various compounds of sodium, especially sodium hydroxide (NaOH; **caustic soda**), sodium hydrogencarbonate (sodium bicarbonate, NaHCO₃; **baking soda**), or sodium carbonate decahydrate (Na₂CO₃·10H₂O; **washing soda**). The term may also indicate sodium monoxide (Na₂O).

soda glass common glass, made from a mixture of silica, sodium oxide (soda), and lime, in approximate proportions SiO₂ 70%, Na₂O 15%, CaO 10%. Its low melting point led to the production of more heat-resisting glasses, e.g. Pyrex. *See also* **borosilicate (glass)**.

Soddy, Frederick (1877–1956), British inorganic and physical chemist; Nobel Laureate in Chemistry (1921) 'for his contributions to our knowledge of the chemistry of radioactive substances, and his investigations into the origin and nature of isotopes'.

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sodium

sodium *symbol:* Na; a soft, silvery-white, highly reactive alkali metal of group 1 of the (IUPAC) **periodic table**; atomic number 11; relative atomic mass 22.997. Sodium is widely distributed and is one of the most abundant elements of the Earth's crust; it exists in great quantity as sodium chloride (salt), especially in sea water. Apart from the nonradioactive, and most abundant, isotope, sodium-23, there are two radioactive isotopes, **sodium-22** and **sodium-24**. In biochemistry, sodium (Na^+) is the principal extracellular cation, and the one that largely determines the extracellular fluid volume. In humans the normal range for plasma sodium is 135–145 mmol L^{-1} . Sodium ions are actively pumped out of cells by **sodium/potassium ATPase**. Their concentration in the blood is regulated by the mineralocorticoid hormone **aldosterone**, which increases sodium reabsorption in the kidney distal tubule. In scientific and medical literature sodium was formerly given the Latin name 'natrium', hence the symbol Na and the term **natriuretic**.

sodium-22 a radioactive nuclide of sodium, $^{22}_{11}\text{Na}$. It emits a β^+ particle (1.83 MeV) and gamma radiation (1.275 MeV). Its half-life is 2.605 years.

sodium-24 a radioactive nuclide of sodium, $^{24}_{11}\text{Na}$. It emits an electron (β^- particle, 1.39 MeV) and gamma radiation (2.75 MeV). Its half-life is 15.02 h.

sodium-alanine symporter *see* **sodium/neutral amino acid cotransporter**.

sodium-calcium ion exchanger a plasma membrane protein **antiporter** of muscle and nerve cells that promotes the exchange diffusion of sodium and calcium ions. Influx of Na^+ to the cytosol drives the efflux of Ca^{2+} from the cell. The affinity for Ca^{2+} is low, so that it only functions at high intracellular Ca^{2+} concentrations, transporting Ca^{2+} , e.g., during excitation-contraction coupling. Example from human cardiac sarcolemma, type III membrane glycoprotein (precursor): database code NACA_HUMAN, 973 amino acids (108.42 kDa).

sodium channel any of several types of structure that permit controlled (gated) passage of sodium ions through membranes. Two important types of gated sodium-ion channels are the voltage-gated channel and the transmitter-gated channel. Voltage-gated channels are involved in conduction and transmission of nerve impulses, whereas transmitter-gated channels respond only to neurotransmitters; a well-studied example of the latter type of channel is the **acetylcholine receptor**. Voltage-gated sodium-ion channels consist of integral proteins in the plasma membranes of excitable cells. The proteins sense the membrane electric field, opening channels to permit an inward flux of sodium ions, thereby causing the membrane depolarization that is essential for the propagation of the **action potential**. Primary sequence data for the *Electrophorus* electroplax protein indicate that this channel consists of a single polypeptide chain of about 2000 amino-acid residues and contains four homologous repeats (domains I, II, III, and IV), each of which has six putative transmembrane segments (designated S1 to S6) with extensive nonmembrane cytosolic and extracellular loops. This sequence shares significant homology to other ion channels, particularly the $\alpha 1$ subunit of the Ca^{2+} channel, segment S4 showing strong structural similarity to sequences in the Ca^{2+} and K^+ channels. The cytoplasmic loop between repeats III and IV has been implicated in the rate of inactivation of the channel. As normally expressed the protein bears about 500 sugar residues (mostly *N*-acetylglucosamine and sialic acid) and 50 fatty-acyl chains (palmitoyl and stearoyl). Structural data are also available for enzymes of *Drosophila* and mammalian skeletal muscle and brain. The mammalian channels are heterooligomeric, consisting of a large α subunit similar to the *Electrophorus* protein, and several other smaller peptides; most of the functional properties reside in the α subunit. Example from rat brain (large subunit): database code CIN1_RAT, 2009 amino acids (228.51 kDa); seven motifs. The α subunit is the target for polypeptide neurotoxins, e.g. from scorpions; other sodium-channel blockers include **saxitoxin**, **tetrodotoxin**, and **veratridine**.

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sodium dodecyl sulfate or (*esp. Brit.*) **sodium dodecyl sulphate** *abbr.:* SDS; sodium lauryl sulfate; $\text{CH}_3(\text{CH}_2)_{10}\text{CH}_2\text{OSO}_3\text{Na}$; the sodium salt of the 1-dodecanol half-ester of sulfuric acid (1-dodecanol was formerly known as lauryl alcohol). It is an anionic detergent and wetting agent; aggregation number 62 (0–0.1 M Na^+), 101 (>0.1 M Na^+); CMC 7–10 mM (0–0.05 M Na^+), 1–2 mM (0.1–0.2 M Na^+). The amphipathic dodecylsulfate anion is a powerful hydrophobic-bond breaker, hence it is much used in protein chemistry as, e.g., a denaturant, a dissociant of oligomers, and a solubilizer of membrane components. *See also* **SDS-polyacrylamide gel electrophoresis**.

sodium-hydrogen ion exchanger a plasma membrane protein that functions as a Na^+/H^+ **antiporter**. Influx of Na^+ drives the efflux of H^+ . It functions to maintain intracellular pH, to which it is sensitive, its activity increasing as pH falls. Example from the gills of the crab, *Carcinus maenas*: database code CMU09274, 672 amino acids (75.89 kDa).

sodium-ion channel *see* **sodium channel**.

sodium/neurotransmitter transporter any of a family of integral membrane glycoproteins, having eight motifs and 12 putative transmembrane helices, that are involved in sodium-dependent transport of neurotransmitters. Examples (precursors) include sodium-dependent choline transporter from rat: database code NTCH_RAT, 635 amino acids (70.55 kDa); sodium- and chloride-dependent GABA transporter 1 from human: database code NTG1_HUMAN, 599 amino acids (66.94 kDa). This latter protein is the target for cocaine and other psychomotor stimulants.

sodium/neutral amino-acid cotransporter an integral membrane protein responsible for the Na^+ -dependent uptake of the neutral amino acids alanine, serine, glycine, cysteine, and proline. Example from pig: database code NAAA_PIG, 660 amino acids (72.66 kDa). Transporters with specificities for different types of amino acid are **system A**, **system L**, and **system N**.

sodium/potassium ATPase EC 3.6.1.37; *recommended name:* Na^+/K^+ -exchanging ATPase; *systematic name:* ATP phosphohydrolase (Na^+/K^+ -transporting); *other names:* sodium pump; Na^+/K^+ -ATPase. An integral membrane protein that catalyses the hydrolysis of ATP to ADP and orthophosphate. Na^+/K^+ -ATPases are **integral (cell membrane) proteins**, and pump out approximately 3 Na^+ ions for every 2 K^+ ions that enter, with hydrolysis of 1 ATP. They are tetrameric proteins, consisting of two large α subunits and two smaller β subunits. The α subunits bear the active site and penetrate the membrane, while the β subunits carry oligosaccharide groups and face the cell exterior only. **Ouabain** is a specific inhibitor. Examples from rat; α -1 chain: database code ATN1_RAT, 1023 amino acids (112.92 kDa); β chain: database code ATNB_RAT, 304 amino acids (35.16 kDa); γ chain: database code ATNG_RAT, 58 amino acids (6.49 kDa).

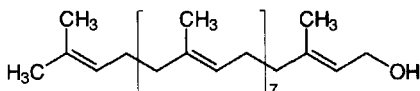
sodium pump any of various **transporters** that transport sodium ions across membranes, especially enzymes of the Na^+/K^+ -ATPase type (*see* **sodium/potassium ATPase**). The Na^+/K^+ ratio across the plasma membrane of eukaryotic cells (outside: Na^+ 140 mM, K^+ 10 mM; inside: Na^+ 10 mM, K^+ 100 mM) requires that sodium ions be continually pumped out of the cell in exchange for potassium ions if the gradient is to be maintained; this is an energy-requiring process. *See also* **active transport**.

software (*in computing*) programs used by computer hardware to execute commands keyed or otherwise supplied by the computer operator to bring about the completion of desired tasks. Most software is not permanently resident in the computer but is accessed from disks as instructed by the operator, **operating systems**, or other software.

sol any fluid colloidal system composed of two or more components. The dispersed phase may be either particles of solid or droplets of liquid, and the continuous phase either a liquid or a gas (*see* **aerosol**).



solanesol a 45-carbon polyprenol present in tobacco leaves.



solution the (usually reversible) act or process of forming a sol from a gel. *Compare gelation.* —**solute** *vb.*

solid a physical state of matter in which the constituent atoms, ions, or molecules have no translatory motion though they vibrate about fixed positions. In **crystalline solids** the constituents are arranged in a definite array, whereas in **amorphous solids** they are not. Solids, especially crystalline solids, commonly have characteristic melting temperatures at which they change to liquids.

solid-phase peptide synthesis *see* peptide synthesis.

solid-phase technique any technique in which reagents are immobilized on a support or in some other way insolubilized. In **radioimmunoassay** the insolubility or binding properties of an antigen-antibody complex are commonly exploited to remove labelled antigen. Enzymes immobilized on columns (**immobilized enzymes**) have many applications. In **peptide synthesis**, the growing peptide chain may be immobilized on a column. So-called solid-phase extraction systems are in effect extractions using batch column chromatography techniques, in which the material to be extracted is bound to a column, then removed by **batch elution**. In peptide **sequencing** a peptide immobilized on a column may be degraded residue by residue, facilitating recovery of the undegraded remaining peptide at each cycle.

solid scintillation fluorography a technique for visualizing radioactive materials in a specimen (e.g. a chromatogram) by covering the specimen with a solid **fluor** and allowing the ejected photons to darken an applied photographic emulsion. *Compare fluorography* (def. 2).

solid-state (*in electronics*) describing any system in which current flow takes place entirely through solid materials, such as semiconductors and transistors, rather than through vacuum devices such as thermionic valves.

solubility 1 the attribute of being soluble, of being able to dissolve. 2 the amount of a substance (**solute**) that can dissolve in a given amount of another substance (**solvent**) at a given temperature and pressure.

solubility product *symbol:* K_{sp} ; for a solute A_xB_y , that ionizes in a stated solvent according to the equilibrium $A_xB_y \rightleftharpoons xA + yB$, at a given temperature and pressure, $K_{sp} = [a_A]^x[a_B]^y$, where $[a_A]$ and $[a_B]$ are the activities of the two ions when the solution is saturated by the solute. In unsaturated solutions the product $[a_A]^x[a_B]^y$ can have any value less than K_{sp} , but if further amounts of either ion are added from whatever source, such that K_{sp} is exceeded, precipitation from solution will occur to the extent that the relationship $[a_A]^x[a_B]^y = K_{sp}$ is restored. In dilute solutions the concentrations C_A and C_B of A and B may be used as a good approximation.

solubilize or **solubilise** to bring into solution any material, especially complexes that normally exist in the cell as part of membrane structures. Such materials usually have hydrophobic domains, and solubilization depends on masking these with detergents or similar substances, to bring the material into micellar suspension.

soluble 1 capable of being dissolved in or as if in a fluid. 2 (of, e.g., a problem or mathematical equation) capable of being solved.

soluble collagen *see* tropocollagen.

soluble RNA *abbr.:* S-RNA or sRNA; *an old term for transfer RNA* (a 'soluble' RNA fraction from disrupted liver cells that did not sediment after centrifuging at 100 000 g for several hours led to the discovery of transfer RNA).

solute a dissolved substance, particularly a component of a solution that is present in a smaller amount than the solvent.

solution 1 a homogeneous molecular mixture of two or more substances, usually of dissimilar molecular structures, i.e. of a solid in a liquid. 2 the action or process of solving a problem. 3 the result of solving a problem, equation, etc.

solvation the (loose) combination of solvent with molecules or ions of the solute.

solvation effect the effect of solvent molecules on the behaviour of (especially) macromolecules in solution, especially during centrifugation.

solvent 1 a substance or mixture, usually a liquid, that is able to dissolve other substances to form a solution. Typically it has the same physical state as the solution itself, and is the component of greatest extent in the solution. 2 (*in chromatography*) the liquid used to pass through a chromatography column, paper, or other support to 'develop' the chromatogram.

solvent-perturbation method a technique in which some physical property of a macromolecule is measured in a polar and in a nonpolar solvent. For example, it can be determined whether an amino-acid residue in a protein is internal or external by measuring the absorption spectra of the protein in the two types of solvents. Difference spectroscopy is conveniently applied in such instances.

soma (*pl.* **somata** or **somas**) 1 the whole of any plant or animal organism but excluding the germ cells. 2 the body, as opposed to the mind. *See also somatic cell.* —**somatic** *adj.*

somat+ a variant form of **somato+** (before a vowel).

somatic cell any cell of an organism other than a germ cell.

somatic cell genetic disorder a category of genetic disorders in which defects are restricted to specific somatic cells, e.g. cancer cells. This contrasts with the other types of genetic disorder — single gene, multifactorial, and chromosome — in which the abnormality is present in all cells, including the germ cells.

somatic cell hybrid a hybrid cell formed from the fusion of different cells, usually from different species. Somatic cell hybrids obtained by the fusion of human and rodent cells are frequently used for gene mapping. *See cell fusion.*

somatic crossing over an alternative name for **mitotic recombination**.

somato+ or (*before a vowel*) **somat+** *comb.* for denoting the body, or soma.

somatocrinin another name for growth hormone releasing hormone (GHRH).

somatogenic 1 originating in the **soma** (def. 1), or body tissues. 2 of organic, rather than mental, origin.

somatoliberin the recommended name for growth hormone releasing hormone (GHRH).

somatomammotropin human chorionic somatomammotropin; *see* **choriomammotropin**.

somatomedin *generic name for insulin-like growth factors* (*abbr.:* IGF); in the human they comprise IGF-IA, IGF-IB, and IGF-II. Somatomedins were isolated on the basis of several properties: (a) activity on cartilage (incorporation of sulfate and thymidine), (b) insulin-like activity in adipose and muscle, and (c) mitogenic activity in cell culture. Somatomedin A is IGF-II, and somatomedin C is IGF-I (A and B). The term somatomedin should be used only when a generic term is needed, IGF-I (A or B) and IGF-II being used for the specific peptides. Somatomedin B is released from **vitronectin**, is a serum factor of unknown function, is increased in response to **somatotropin**, and has proteinase-inhibitory activity.

somatostatin or **somatotropin-release inhibiting factor** (*abbr.:* SRIF) or (*formerly*) **growth hormone-release inhibiting hormone** (*abbr.:* GH-RIH) a hypothalamic cyclic 14-residue peptide discovered as an inhibitor of the release of the pituitary growth hormone **somatotropin** and of the release of glucagon and insulin from the pancreas of fasted mammals; it also inhibits the release of gastrointestinal hormones and other secretory proteins. It has the sequence Ala-Gly-Cys-Lys-Asn-Phe-Phe-Trp-Lys-Thr-Phe-Thr-Ser-Cys, with a disulfide link

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somatostatin receptor

between cysteines 3 and 14; this is called **somatostatin 14**. There also exists a bioactive 28-residue peptide – **somatostatin 28** – in which the 14-residue peptide is extended from the N terminus (the N-terminal residues are SANSNPALAPRE-RKAG–). Both forms are derived from the 92 amino-acid precursor, **prosomatostatin**, by tissue-specific proteolytic processing. Somatostatin is widely distributed, occurring not only in the central nervous system, but also in peripheral tissues such as stomach, intestine, and pancreas. It has diverse effects that are tissue specific, and can function as a neurotransmitter. Example (precursor) from human and *Macaca fascicularis* (a crab-eating monkey): database code SMS1_HUMAN, 116 amino acids (12.72 kDa). See also **somatostatin receptor**.

somatostatin receptor any of several membrane proteins that bind different forms of **somatostatin** and mediate its effects. Pharmacological studies indicate that there are at least two subtypes of receptor. In appropriate systems somatostatin suppresses receptor-induced stimulation of adenylate cyclase by activating the inhibitory **G-protein** G_i ; it can decrease intracellular Ca^{2+} concentrations and inhibit the voltage-dependent current, I_{Ca} , by a G-protein-dependent cyclic AMP-independent mechanism. Somatostatin 14 enhances and somatostatin 28 reduces potassium current, I_K , also by G-protein-dependent cyclic AMP-independent mechanisms. There are several different types of somatostatin receptor; all are G-protein-coupled seven-transmembrane-domain proteins. The examples below are all from *Rattus norvegicus*. Type 1 has higher affinity for somatostatin 14 than 28 and is coupled to phosphotyrosine phosphatase and sodium-hydrogen-ion exchanger; example, database code SSR1_RAT, 391 amino acids (42.75 kDa). Types 2 and 3 are receptors for somatostatins 14 and 28 and inhibit adenylate cyclase; examples, database code SSR2_RAT, 369 amino acids (41.20 kDa) and database code SSR3_RAT, 428 amino acids (47.15 kDa). Type 4 is a receptor for somatostatin 14 and inhibits adenylate cyclase; example, database code SSR4_RAT, 384 amino acids (42.09 kDa). Type 5 is a receptor for somatostatin 28 and inhibits adenylate cyclase; example, database code SSR5_RAT, 363 amino acids (39.97 kDa). Hybridization studies indicate these are members of a larger family of somatostatin receptors; they are G-protein-coupled adenylate cyclase activators.

somatotrope or **somatotroph** a type of cell found in the lateral wings of the anterior pituitary and comprising about 50% of the hormone-producing cells of that part of the gland. Somatotropes are responsible for the production of **somatotropin**. Somatotrope adenomas cause **acromegaly**.

somatotroph see **somatotrope**.

somatotrophic 1 stimulating and maintaining the growth and development of any cell or tissue. 2 (*sometimes*) of or pertaining to **somatotropin**.

somatotropic 1 of, or relating to, **somatotropin**. 2 stimulating the mechanisms connected with growth or development of a cell or tissue. Compare **somatotrophic**.

somatotropic hormone see **somatotropin**.

somatotropin or **growth hormone** (abbr.: GH); also called **somatotropic hormone** (abbr.: STH); a hormone secreted by the anterior pituitary whose main effect is to stimulate growth. The human hormone (hGH) is mostly synthesized as the precursor of a single-chain, 191-residue protein, M_r 22 000, but 5–10% is a form of M_r 20 000 resulting from alternative splicing that deletes codons for amino acids 32–46 from the mRNA. Genes encoding hGH, **prolactin**, and **choriomammotropin** form a family with high nucleotide homology with five coding exons. Somatotropin secretion is pulsatile, especially during fasting, when cycles of about four hours' duration occur; it is stimulated by fasting or hypoglycemia, and hormonally by glucagon, GHRH, estrogens, and arginine vasopressin, and during the first phase of sleep; secretion is inhibited by hyperglycemia, somatostatin, the insulin-like growth factor IGF-I, and glucocorticoids. It is secreted from

storage granules. *In vivo*, somatotropin tends to promote synthesis of protein, and lipolysis or reduced fat deposition, and growth and mitosis of cartilage. Deficiency leads to pituitary dwarfism types I and IV. Its actions *in vitro* include stimulation of amino-acid uptake in muscle, and lipolysis in adipose tissue, but many of its actions *in vivo* may depend on its action in stimulating synthesis and secretion of **somatomedins**. hGH has 85% homology with human choriomammotropin and 35% homology with human prolactin. Example (precursor) from human: database code SOMA_HUMAN, 217 amino acids (24.82 kDa). See also **GHRH**, **GHRH receptor**, **hypopituitarism**.

somatotropin-release inhibiting factor abbr.: SRIF; an alternative name for **somatostatin**.

somatrem methionyl human **somatotropin** (growth hormone); somatotropin with an additional N-terminal methionine. It results from the production of somatotropin using cDNA technology in bacterial clones; methionine is involved in the initiation of bacterial protein synthesis.

+**some comb.** form denoting body, esp. intracellular particle.

somite any of a series of paired blocks of mesoderm that form during early development in vertebrate embryos and lie on either side of the notochord. —**somital** or **somitic** adj.

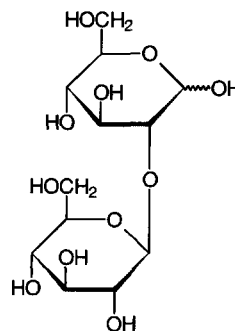
sonicate to expose any material or sample to ultrasonic pressure waves (at or around 20 kHz). The immersion of vessels in a water bath and their exposure to ultrasound is used as a means of cleaning (see **sonication bath**). The insertion of a **sonicator** probe into a vessel containing a suspension of membranes or other large assemblies causes their dispersion to smaller aggregates or assemblies.

sonication bath a water bath that is equipped with an oscillator, sited beneath the bowl, to produce ultrasound waves that permeate the water. It may be used for cleaning glass or other vessels or for degassing solvents.

sonicator a device that produces **ultrasonic** pressure waves. It usually consists of a stainless-steel rod, of diameter from 1–2 mm to about 2 cm, that is activated by an oscillator to produce ultrasound waves and is inserted into a vessel to **sonicate** a sample. See also **sonication bath**.

sonic hedgehog proteins analogues of **hedgehog** protein found in **zebrafish** and chickens and involved (in the latter case) in limb-bud pattern formation. The hedgehog proteins thus appear to be vital to pattern formation in several metazoan groups. The name sonic hedgehog is an allusion to a children's animated cartoon character.

sophorose the disaccharide 2-O- β -D-glucopyranosyl-D-glucose; it occurs in a glycoside of *Sophora japonica*.



sorbin a heat-stable peptide isolated from porcine intestine that increases water and sodium absorption in the intestine and gall-bladder. Example, database code SORB_PIG, 153 amino acids (17.46 kDa).

sorbitol a nonsystematic name for D-glucitol, one of the ten stereoisomeric hexitols. It can be derived from glucose chemically or metabolically (by NADPH-requiring aldehyde reductase; other name: aldose reductase, EC 1.1.1.21) by reduction of the aldehyde group, and is converted to D-fructose by L-idid-



sorbose

tol 2-dehydrogenase (EC 1.1.1.14). It is widely distributed in algae and higher plants, and was first discovered (1872) in the juice of berries from mountain ash (*Sorbus aucuparia*). It has a sweetening power about half that of sucrose, and is often used in sweeteners in conjunction with saccharin. Industrially it is of great importance in the food, pharmaceutical, paper, and textile industries. Its formation from excessive glucose concentrations in diabetics has been regarded as having pathological implications in lens tissue, which cannot convert it to fructose.

sorbose trivial name for the ketohexose *xylo*-2-hexulose; the enantiomer, L-sorbose, is present in fermented juice of mountain-ash berries, being formed by the bacterial oxidation of **sorbitol**. It is produced commercially by fermentation and used as an intermediate in the manufacture of **ascorbic acid**.

sorcina a protein, encoded in multidrug-resistant cells (see **MDR**), that binds calcium with high affinity. Example from human: database code SORC_HUMAN, 198 amino acids (21.68 kDa). See also **grancalcin**.

Soret band an absorption band at about 400 nm, characteristic of porphyrins. See also **cytochrome absorption bands**. [After Louis Soret (1827–90), Swiss physicist.]

sorocarp see **Dictyostelium discoideum**.

sos symbol for *son of sevenless*, a gene encoding a guanine-nucleotide releasing protein involved in neuronal development in *Drosophila*. When coupled to the activated Sev receptor (see **Sevenless protein**) the *sos* product, Sos, mediates activation of *ras* by stimulating release of GDP, thereby promoting uptake of GTP. Database code SOS_DROME, 1595 amino acids (177.64 kDa).

SOS repair or **error-prone repair** a metabolic alarm system, occurring in bacteria, that helps the cell to save itself in the presence of potentially lethal stresses such as UV irradiation, thymine starvation, or the inactivation of genes essential for replication. Responses include mutagenesis, filamentation (cell elongation in the absence of division), activated excision repair, and activation of latent phage genomes. Mutagenesis occurs because under SOS conditions the gaps formed opposite **thymine dimers** can be filled by replication rather than by daughter-strand transfer. Replication of the dimer template is extremely inaccurate. The proteins involved include RecA protein (see **recA**) and the products of two other RecA-inducible genes; expression is regulated by the *lexA* gene product.

Southern blotting a procedure for transferring denatured DNA from an agarose gel to a solid support membrane such as nitrocellulose or nylon. The procedure takes advantage of the property of nitrocellulose that it tenaciously binds single-stranded but not duplex DNA. The gel is soaked in 0.5 M NaOH, which converts the DNA to the single-stranded form. The gel is then overlaid with a sheet of nitrocellulose paper, which in turn is covered by a thick layer of paper towels and the entire assembly is compressed by a heavy weight. The liquid in the gel is forced (blotted) through the nitrocellulose so that the single-stranded DNA binds to it at the same position it had in the gel. The transfer to nitrocellulose can alternatively be accomplished by **electrophoretic transfer**. Nylon is now usually used in preference to nitrocellulose. DNA bands may be located using a radioactive or otherwise labelled probe. The name led to a nomenclature for other types of blotting (see also **Northern blotting**, **South-western blotting**, **Western blotting**). [After E. M. Southern (1938–), British molecular biologist.]

South-western blotting a technique in recombinant DNA technology for detecting plaques expressing **fusion proteins** where the foreign sequence encodes a DNA-binding protein that binds specifically to a particular DNA sequence. After treatment of the plaques to ensure release of the protein, **plaque-lift** is carried out onto a nitrocellulose (or other) **transfer membrane**, which is then incubated with a radiolabelled duplex DNA oligonucleotide that binds to the protein, which can then be detected by **autoradiography**. The term was derived by extension through the cardinal points from **Southern blotting**.

Soviet gramicidin an alternative name for **gramicidin S**.

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Soxhlet extractor an apparatus for exhaustive extraction of soluble components from a sample of solid material by recycling a solvent through it, whereby solvent vapour from a boiler is continually condensed and the condensate allowed to percolate through the sample in a solvent-permeable thimble, from which it is returned, usually periodically via a siphon, to the boiler, where the eluted substances accumulate. [After Franz Soxhlet (1848–1926), Belgian chemist, who devised it.]

soybean or (*esp. Brit.*) **soya bean** a leguminous plant, *Glycine max*, originating in China (known from at least 2500 BC) and now cultivated widely in Asia, Europe, and America for forage and for its seeds. The bean has an oil content of 18–22%, and is also rich in protein — soybean protein is used as a human food and is the basis of tofu. See also **soybean oil**, **soybean trypsin inhibitor**.

soybean oil an edible oil obtained from the seeds of the soybean plant and valued for its high content of linoleic acid. The composition is typically as follows: 18:2 ω6, 44–62%; 18:3 ω3, 4–11%; 18:1 ω9, 19–30%; 16:0, 7–14%.

soybean trypsin inhibitor *abbr.*: STI; either of two different types of trypsin inhibitor, the Bowman–Birk and the Kunitz inhibitors. Both have given their names to a class of proteinase inhibitor. The two soybean inhibitors differ markedly from each other in size, structure, and properties. The **Bowman–Birk inhibitor** is a 71 amino-acid, 7.873 kDa protein: database code IBB1_SOYBN. It has seven disulfide bridges. There are two active sites, one specific for trypsin and the other for chymotrypsin. Bowman–Birk type inhibitors are found in many cereals and legumes, and ancestrally may represent storage proteins. The **soybean Kunitz inhibitors** comprise three allelic proteins, A, B, and C, together with two Kunitz-type trypsin inhibitors, named KTI1 and KTI2. The A, B, and C proteins have 181 amino acids (20 kDa): database code (of A and C) ITRA_SOYBN. They have two disulfide bonds. They are heat-stable and resistant to proteases, and form a 1:1 stoichiometric complex with trypsin, that is highly stable ($K_d \times 10^9$). They inhibit chymotrypsin to a much smaller degree and are inactive against other endopeptidases, but modification of Arg⁶³ to Trp converts them to strong chymotrypsin inhibitors with little action on trypsin. The Kunitz family of inhibitors embraces trypsin, chymotrypsin, cathepsin D, and subtilisin inhibitors (see **Kunitz inhibitor**).

sp *abbr.* for synperiplanar; see **conformation**.

sp. (*pl. spp.*) *abbr.* for species (singular).

Sp *abbr.* for isomer of a thionucleotide.

SP *abbr.* for 1 secretory piece. 2 substance P.

SP-1 *abbr.* for secretory protein 1.

[S]pA symbol for adenosine 5'-thiophosphate (alternative to AdoP[S]).

space or **volume of distribution** (*in tracer kinetics*) an apparent volume obtained as the amount of radioactivity retained divided by the concentration of the tracer in the plasma. The concept of space has the limitations discussed under **apparent exchangeable mass**. See also **compartment**.

spacer 1 (*in molecular biology*) a sequence of bases of unknown function occurring in a DNA molecule and lying between sequences of transcribed DNA; spacer DNA is not usually transcribed. The term has also been used to refer to some stretches of **primary transcript**, e.g. a region in the mammalian primary transcript for rRNA, lying between 18S RNA and 28S RNA, that is not found in mature rRNA. 2 (*in chromatography*) an alternative term for **spacer arm**.

spacer arm or **spacer** the (usually) hydrocarbon chain interposed, by covalent linkages, between the specific ligand and the supporting matrix in **affinity chromatography**.

spacer gel (*in electrophoresis*) a small section of gel, polymerized above a polyacrylamide (or other) resolving gel, that lacks resolving power and serves to concentrate the material being electrophoresed as a tight band at the top of the resolving gel.

SPARC *abbr.* for secreted protein acidic and rich in cysteine; also called **osteonectin**; a protein expressed in cells of tissues



spare receptors

undergoing morphogenesis or wound repair and secreted into the basement membrane, where it interacts with other proteins, e.g. collagen and thrombospondin, and appears to have a role in regulating cell growth. It contains an **EF-hand** that binds one calcium ion and an acidic region that binds other calcium ions more loosely. Example from human (precursor): database code **SPRC_HUMAN**, 303 amino acids (34.63 kDa).

spare receptors receptors that are in excess of those required to evoke a given response to a drug, hormone, or other agonist. For a given agonist, the size of the population of spare receptors (**receptor reserve**) can vary with the tissue and with the magnitude of the response being measured. For different drugs, the receptor reserve varies with drug efficacy.

sparging the bubbling of gas into a liquid medium. This technique may be applied to fermenters to introduce air directly into a culture, or it may be used to replace one dissolved gas by another, e.g. in displacing oxygen with an inert gas in liquid chromatography solvents.

spatula a (usually) small, flattened, sometimes curved, utensil made of metal (often stainless steel), bone, or wood and used to dispense crystalline or powdered material.

Spd *abbr. for* sphingoid.

species (*pl. species*) *abbr.:* sp. (*sing.*) or spp. (*pl.*); a fundamental taxonomic category ranking below a genus and consisting of a group of closely related individuals that can interbreed freely to produce fertile offspring.

species-specific confined to the one or several species referred to.

specific 1 relating to a particular or specified thing. 2 (as a modifier of certain quantitative terms) indicating a relationship of the quantity to mass and dimension in terms of standard (now normally SI) units; e.g. **specific catalytic activity**, **specific enzyme activity**, **specific optical rotation**. 3 of, or relating to, a species. *See also* **specificity**.

specific acid catalysis a **homogeneous catalysis** in which the catalysts are free hydrons, H^+ (or hydronium ions, H_3O^+). The catalysis is not affected by other acidic species present. *Compare* **general acid-catalysis**.

specific activity the **activity** (def. 1, 2) of, e.g., a radionuclide or an enzyme, expressed as a function of mass.

specifically labelled describing an isotopically labelled compound in which both the position of labelling and the number of labelling nuclides at that position are known. Such a molecule is designated by a convention exemplified by $CH_3^2[H_2]-CH_2-O^2[H]$.

specific base catalysis a **homogeneous catalysis** in which the catalysts are hydroxide ions, OH^- . The catalysis is not affected by other basic species present. *Compare* **general base-catalysis**.

specific catalytic activity the **catalytic activity** of an enzyme under specified conditions divided by the mass of an enzyme or enzyme preparation. It is a quantity useful for denoting the extent of purification of an enzyme; it may be expressed in katals per kilogram.

specific dynamic action the increase in heat production above the basal level that occurs in animals after the ingestion of food, particularly of protein.

specific enzymic activity the phenomenological coefficient that relates **enzymic activity** under specified conditions to the mass of an enzyme or enzyme preparation. It may be expressed in katals per kilogram of protein. Now obsolete, it has been superseded by **specific catalytic activity**. *Compare* **molar enzymic activity**.

specific gravity a former name for **relative density**.

specific heat capacity *symbol:* c_p (at constant pressure) or c_v (at constant volume); the quantity of **heat** required to raise the **temperature** of unit mass of a substance by one degree. It is expressed (usually) in $J K^{-1} kg^{-1}$.

specificity an index of the degree to which an association between two molecular units or assemblies may be considered unique; in many applications it is judged by the magnitude of an effect (enzyme-substrate, ligand-receptor interaction, or

drug action). It is quantified in suitable units under defined conditions, in relation to the magnitude of effect of other substances with similar action, amplified by studies made when the substance under consideration is in a direct competitive interaction with other such substances. The success of other substances in competing with the subject of study is normally quantified by kinetic analysis (*see* **competitive inhibition**). The problem of quantifying specificity may be examined by referring to **specificity constant**.

specificity constant the ratio of the catalytic constant, k_{cat} , to the Michaelis constant, K_m . It can aid in comparing the activity of an enzyme in its action on different substrates. From **Michaelis kinetics**, using the Michaelis-Menten equation,

$$v = (k_{cat}/K_m)[E]_0[S]$$

when $[S]$ is very small (since then, little enzyme has bound substrate, so $[E] \approx [E]_0$, and $[S]$ can be neglected in relation to K_m). Thus for substrate A,

$$v_a = (k_{cat}/K_m)_A[E]_0[A],$$

and for substrate B,

$$v_b = (k_{cat}/K_m)_B[E]_0[B].$$

So

$$v_a/v_b = \{(k_{cat}/K_m)_A \times [A]\} / \{(k_{cat}/K_m)_B \times [B]\},$$

and, at the same substrate concentrations, the activity of the enzyme against the different substrates reduces to a comparison of the respective specificity constants (k_{cat}/K_m); K_m is not of itself a sufficient index of specificity.

specific optical rotation *symbol* $[\alpha]_D^{25}$; the optical rotation of light of a specified wavelength (or frequency) achieved by a solution at a specified temperature. It is given by the relation $[\alpha]_D^{25} = \alpha/l\rho$ where α is the observed rotation in degrees, l is the length of the sample tube (in dm), ρ is the density of the liquid (in $g mL^{-1}$), λ is the wavelength of the light (or other electromagnetic radiation) used, and θ is the Celsius temperature. For a solution of concentration c (in g solute per 100 mL solution), $[\alpha]_D^{25} = \alpha/lc$ in degrees $dm^2 g^{-1}$. *Compare* **molecular optical rotation**.

specific refractivity *see* **refractivity**.

specific viscosity *symbol:* η_{sp} ; the fractional change in **viscosity** produced by adding a solute to a solvent. It is given by the equation:

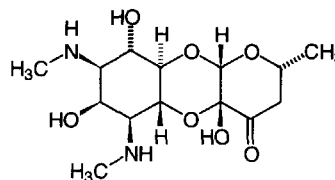
$$\eta_{sp} = (\eta/\eta_0) - 1 = Avc + Bv^2c^2 + Cv^3c^3 + \dots$$

where η_0 is the viscosity of the solvent, η is the intrinsic viscosity of the solution, A , B , and C are shape-dependent constants, c is the concentration of the solute, and v is the specific volume of one solute molecule.

specific volume *symbol:* v ; for a pure substance, the volume, V , of a sample, divided by the mass, m , of the sample; $v = V/m = 1/\rho$, where ρ is the **mass density** of the substance. The SI unit is $m^3 kg^{-1}$. *See also* **partial specific volume**.

'specifier' protein a protein that, when it combines with an enzyme or enzyme-substrate, modifies the catalytic specificity of the system. An example is α -lactalbumin, which modifies the substrate (acceptor) specificity of *N*-acetylglucosamine synthase (EC 2.4.1.90; *other name:* UDPgalactose:*N*-acetylglucosamine β -D-galactosyltransferase) from *N*-acetylglucosamine to glucose.

spectinomycin an aminocyclitol antibiotic found in the growth medium of *Streptomyces spectabilis*. It has a wide



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spectra

range of activity against Gram-negative bacteria, with a more selective activity against Gram-positive bacteria. It is highly active against *Neisseria gonorrhoeae*. It inhibits protein synthesis by binding to the bacterial 30S ribosomal subunit.

spectra the plural of **spectrum**.

spectrin a protein that is the major constituent of the erythrocyte cytoskeletal network. It associates with band 4.1 (see **band protein**) and **actin** to form the cytoskeletal superstructure of the erythrocyte plasma membrane. It is composed of nonhomologous chains, α and β , which aggregate side-to-side in an antiparallel fashion to form dimers, tetramers, and higher polymers. Anchorage to the cytoplasmic face of the plasma membrane is mediated by another protein, **ankyrin**, which links β spectrin to transmembrane protein band 3, and by the binding of band 4.1 to **glycophorin**. Interaction between spectrin and other proteins is thought to be responsible for the maintenance of the biconcave shape of human erythrocytes, for the regulation of plasma membrane components, and for the maintenance of the lipid asymmetry of the plasma membrane. Spectrin-like proteins have five domains. Example from human, α chain: database code SPCA_HUMAN, 2429 amino acids (280.74 kDa); β chain: database code SPCB_HUMAN, 2137 amino acids (246.04 kDa).

spectrofluorometer an instrument used for measuring the fluorescence emitted by compounds excited by incident radiation produced within the instrument. The excitation radiation is produced by a lamp, and filters or diffraction grating monochromators are employed to isolate both the incident and emitted radiation to narrow wavebands. The emitted radiation is detected by photomultipliers.

spectrogram a photographic record of a **spectrum** produced by a **spectrograph**.

spectrograph a **spectrometer** or **spectroscope** with a facility for producing a photographic record of the spectrum under consideration. *Compare spectroscope. —spectrographic adj.; spectrographically adv.; spectrography n.*

spectrometer any instrument used to produce and examine a **spectrum**, such as a **spectrophotometer** or a **spectroscope**. *—spectrometric adj.; spectrometry n.*

spectrophotometer an apparatus to measure the proportion of the incident light (or other electromagnetic radiation) absorbed (or transmitted) by a substance or solution at each wavelength of the spectrum. *—spectrophotometric adj.; spectrophotometry n.*

spectroscope an instrument for studying spectra, either qualitatively or quantitatively. *—spectroscopic or spectroscopical adj.; spectroscopically adv.*

spectroscopy the investigation of the chemical composition, molecular structure, or atomic structure of a substance or solution by observation of its interaction with electromagnetic radiation using a spectroscope.

spectrum (*pl. spectra*) (*in physics*) an arrangement of the components of a complex electromagnetic radiation (light, etc.) or sound in order of frequency or energy, thereby showing a distribution of energy among the components. *—spectral adj.*

Spemann's organizer an embryonic signalling centre located in the dorsal lip of the blastopore. It plays a crucial role in the organization of the formation of the main body axis of a developing embryo. [After Hans Spemann (1869–1941), German zoologist and embryologist.]

spermaceti or **cetaceum** a white, translucent, waxy solid, m.p. 42–50 °C, consisting mainly of the ester cetyl palmitate and obtained from oil in the head of the sperm whale (*Physeter catodon*) and related cetaceans. It is used in the manufacture of cosmetics, ointments, soaps, etc.

spermadhesin or **AQN1** a secretory protein of male accessory glands that is found on the sperm surface and mediates sperm binding to the zona pellucida. Example from *Sus scrofa*: database code AQN1_PIG, 111 amino acids (11.88 kDa).

spermatogenesis the formation of spermatozoa in the male reproductive system. The sperms are formed from male germ

cells, **spermatogonia**, which line the inner wall of the seminiferous tubules in the testis. A single spermatogonium divides by mitosis to form **primary spermatocytes**, each of which undergoes the initial division of meiosis to form two **secondary spermatocytes**. Each of these then undergoes the second meiotic division to form two **spermatids**, which mature into spermatozoa without further cell division by a process called **spermiogenesis**. All these cell types are nourished and supported by neighbouring **Sertoli cells**. Spermatogenesis also requires the presence of testosterone-secreting **Leydig cells**.

spermidine *N*-(3-aminopropyl)-1,4-diaminobutane; $\text{NH}_2(\text{CH}_2)_3\text{NH}(\text{CH}_2)_4\text{NH}_2$; one of the polyamines.

spermidine synthase EC 2.5.1.16; *other names*: putrescine aminopropyltransferase; aminopropyl-transferase; an enzyme that catalyses the formation of **spermidine** from *S*-adenosylmethionine and putrescine with formation of 5'-methylthioadenosine. Example from human: database code SPEE_HUMAN, 302 amino acids (33.79 kDa).

spermine *N,N*-bis(3-aminopropyl)-1,4-diaminobutane; $\text{NH}_2(\text{CH}_2)_3\text{NH}(\text{CH}_2)_4\text{NH}(\text{CH}_2)_3\text{NH}_2$; one of the polyamines.

spermine synthase EC 2.5.1.22; *other name*: spermidine aminopropyltransferase; an enzyme that catalyses the formation of **spermine** from *S*-adenosylmethionine and spermidine with formation of 5'-methylthioadenosine.

sperm tail see **cilium**.

SPF *abbr. for* specific-pathogen-free.

Sph *symbol for* sphingosine.

S phase the phase of the eukaryotic cell cycle in which DNA is synthesized. *See cell-division cycle.*

spherocyte a rounded red blood cell that has lost its normal biconcave shape owing to damage to the surface membrane or to some disorder of metabolism. Spherocytes appear smaller and darker than normal red cells.

spherocytosis the condition of having **spherocytes** in the blood.

spheroid 1 a solid figure that is similar to but not identical with a sphere. 2 an **ellipsoid of rotation**, especially one in which the major axis of its parent ellipse is only slightly longer than its minor axis. *—spheroidal adj.*

spheroplast the globular form of a bacterial cell in which the cell-wall structure has been modified (e.g. by growth in the presence of penicillin) rather than totally removed. The term also denotes a form of yeast cell obtained by partial hydrolysis of the cell wall. A prefix may be used to indicate the method of induction, e.g. penicillin-spheroplast. *Compare protoplast.*

spherosome an organelle, generally spherical or oblate and 0.4–3 μm in diameter, that occurs in lipid-storage tissues in plants, e.g. endosperm of castor-bean seeds. The nature of the surrounding membrane is still in doubt.

sphinganine *D*-erythro-2-amino-1,3-octadecanediol; *D*-erythro-dihydrosphingosine; the metabolic precursor of **sphingosine**. It is formed from palmitoyl-CoA and serine by, first, the action of **serine C-palmitoyltransferase**, which forms 3-dehydro-D-sphinganine, which is then reduced to sphinganine by 3-dehydrosphinganine reductase, EC 1.1.1.102. *See also sphingolipids, sphingoid.*

sphingoid or **sphingoid base** *symbol*: Spd; any of a class of compounds comprising **sphinganine** and its homologues and stereoisomers, and derivatives of these compounds.

sphingolipid any of a class of lipids containing the long-chain amino diol, **sphingosine**, or a closely related base (i.e. a **sphingoid**). A fatty acid is bound in an amide linkage to the amino group and the terminal hydroxyl may be linked to a number of residues such as a phosphate ester or a carbohydrate. The predominant base in animals is sphingosine while in plants it is **phytosphingosine**. The main classes are: (1) phosphosphingolipids (also known as sphingophospholipids), of which the main representative is **sphingomyelin**; and (2) **glycosphingolipids**, which contain at least one monosaccharide and a sphingoid, and include the **cerebrosides** and **gangliosides**. Sphingolipids play an important structural role in cell membranes, and may be involved in the regulation of protein kinase C.

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sphingolipid activator proteins

sphingolipid activator proteins *see* **saposins**.

sphingolipidosis (*pl.* **sphingolipidoses**) any of several inherited pathological conditions in humans in which one or more sphingolipids accumulate in the tissues because of a deficiency of the corresponding degradative enzyme. They include **ceramidase** deficiency, **Fabry's disease**, **galactosylceramide lipidosis**, **G_{M1}** and **G_{M2}** **gangliosidoses**, **Gaucher's disease**, **metachromatic leukodystrophy**, and **sphingomyelin lipidosis**.

sphingomyelin *N*-acyl-4-sphingeny-1-*O*-phosphorylcholine; any of a class of phospholipids in which the amino group of **sphingosine** is in amide linkage with one of several fatty acids, mostly C₂₀ or higher and either saturated or monounsaturated, while the terminal hydroxyl group of sphingosine is esterified to phosphorylcholine. Sphingomyelins are found in large amounts in brain and other nervous tissue. They do not occur in plants.

sphingomyelin lipidosis or **Niemann–Pick disease** a genetic disease of humans in which **sphingomyelin** accumulates in the tissues due to a deficiency of the enzyme **sphingomyelin phosphodiesterase**. It is associated with mental retardation.

sphingomyelin phosphodiesterase EC 3.1.4.12; *other names*: acid sphingomyelinase; neutral sphingomyelinase; a lysosomal enzyme that catalyses the hydrolysis of sphingomyelin to *N*-acylsphingosine and choline phosphate. Deficiency of the enzyme is the cause of Niemann–Pick disease (**sphingomyelin lipidosis**), in which there is an accumulation of sphingomyelin. Example (precursor) from human: database code ASM_HUMAN, 629 amino acids (69.78 kDa).

sphingophospholipid *see* **sphingolipid**.

sphingosine *symbol*: Sph; *trans*-D-erythro-2-amino-4-octadecene-1,3-diol; $\text{CH}_3(\text{CH}_2)_{12}\text{CH}=\text{CH}-\text{CHOH}-\text{CHNH}_2-\text{CH}_2\text{OH}$; a long-chain amino diol sphingoid base that occurs in most **sphingolipids** of animal tissues. *See also* **phytosphingosine**.

sphingosine *N*-acyltransferase EC 2.3.1.24; an enzyme of the pathway for synthesis of sphingolipids that catalyses the acylation of sphingosine by acyl-CoA to form *N*-acylsphingosine (ceramide) and CoA.

spicule a roughly cone-shaped structure or tissue element, such as that projecting from the membrane of a red blood cell. Calcareous or siliceous spicules are found in the skeletons of sponges and corals.

spin (*in physics*) **1** the quantized angular momentum of an elementary particle, in the absence of orbital motion. **2** the quantized angular momentum of an atomic nucleus, including contributions from the orbital motions of nucleons. **3** the quantized angular momentum of an electron that adds to the orbital angular momentum, thus producing fine structure in line spectra.

spinal cord the part of the central nervous system that is enclosed in the vertebral column. Its function is to connect the innervation of most parts of the body via the spinal nerves to the brain.

spinal nerves the pairs of nerves that arise from the spinal cord and are distributed to various peripheral parts of the body. In humans there are 31 pairs.

spin coupling or **spin–spin coupling** (*in nuclear magnetic resonance spectroscopy*) the influence of one nucleus on another, such that the nuclear spin of one nucleus is coupled via the bonding electrons to the coupled nucleus, which experiences two different microenvironments and thus resonates at two slightly different frequencies. This results in the splitting of the resonance line into two lines (a doublet).

spin density the fraction of unpaired electron spin that is in the vicinity of a given atomic nucleus in a molecule.

spindle the array of microtubules and associated molecules that forms between the opposite poles of a eukaryotic cell during mitosis or meiosis and serves to move the duplicated chromosomes apart. During mitosis the spindle starts to form outside the nucleus while the chromosomes are condensing during prophase. At the start of M phase, the two **centrosomes**

that have been formed move to opposite sides of the nucleus, and form the two poles of the mitotic spindle. Microtubules grow away from the centrosomes towards the chromosomes, the minus ends being attached to the centrosome, and the plus ends growing towards the **kinetochore**. These microtubules eventually bind to kinetochores and are thus known as kinetochore microtubules. Other microtubules, known as polar microtubules, grow towards the chromosomal region, but eventually bind directly to other polar microtubules growing from the opposite centrosome. These push the poles of the spindle apart. Each centrosome eventually becomes attached through the kinetochore microtubules to one of the sister chromatids of each chromosome, and pulls these away to effect the chromatid separation that occurs at anaphase. A similar series of events occurs at the corresponding phases of meiosis.

spin hamiltonian a formulation in quantum mechanics used in the determination of the energy levels of the spin of an electron or a nucleus in **electron spin resonance spectroscopy** or, less commonly, in **nuclear magnetic resonance spectroscopy**. *See also* **Hamiltonian operator**.

spin immunoassay or **free-radical assay technique** (of immunoassay) (*abbr.*: FRAT) a technique in which a **spin label** is attached to a specimen of the substance to be assayed and an antibody is raised against the substance. On mixing the antibody with the labelled substance there is a change in the electron spin resonance (ESR) spectrum because of the inhibition of tumbling caused by the antibody 'immobilizing' the labelled substance. On adding some of the unlabelled substance there is competition for the antibody, and the changes in the ESR spectrum are partially reversed to an extent dependent on the concentration of the added, unlabelled, substance.

spin label 1 a synthetic paramagnetic organic free radical, usually having a molecular structure and/or chemical reactivity that facilitates its attachment or incorporation at some particular target site in a macromolecule, or assemblage of macromolecules. **2 spin-label** to effect labelling of a substance or structure with a spin label (*def.* 1).

spin-lattice relaxation time or **longitudinal relaxation time** (*in nuclear magnetic resonance spectroscopy*) *symbol*: T_1 ; the time taken for the upper electron spin state to dissipate its excess energy. Its reciprocal is the spin-lattice relaxation rate (*see* **correlation time**).

spinner culture the culture of cells in bottles or tubes that are continuously rotated.

spin polarization or **spin polarisation** a phenomenon in which the unpaired electron spin on one atom or part of a molecule is transferred to another atom, brought about by interaction between the unpaired electron on the first atom and the paired electrons on the second atom (which lacks unpaired electrons) causing one of the paired electrons to have a lower energy than the other. *See also* **chemically induced dynamic nuclear polarization**.

spin–spin broadening an increase in line width in **electron spin resonance spectroscopy**, caused by interactions between neighbouring dipoles.

spin–spin coupling *see* **spin coupling**.

spin–spin coupling constant (*in nuclear magnetic resonance spectroscopy*) a measure of the specific interaction of one nucleus with another that gives rise to a splitting of the resonance bands produced by each nucleus.

spin–spin splitting splitting in the lines of a **nuclear magnetic resonance** spectrum brought about by the interaction of the magnetic moment of the nucleus of a given atom in a molecule with those of neighbouring nuclei. It is used in the assignment of particular nuclear resonances and in the determination of molecular conformations. *See also* **hyperfine splitting**.

spin state the orientation, in a strong magnetic field, of an unpaired electron or nuclear spin, e.g. parallel or antiparallel to the field direction.

spin trapping a technique in which a reactive free radical is

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Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

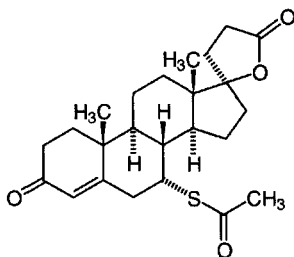


spiro compound

trapped by an addition reaction to produce a more stable radical, detectable by **electron spin resonance spectroscopy**, whose hyperfine coupling parameters permit identification of the initial radical trapped.

spiro compound any compound containing two rings joined by one common atom, designated the **spiro atom**, as in the case of the γ -lactone ring of **spironolactone**.

spironolactone 7-(acetylthio)-17-hydroxy-3-oxo-pregn-4-ene-21-carboxylic acid γ -lactone; a synthetic steroid lactone that is used as a diuretic. It is structurally similar to **aldosterone**, and competitively inhibits the renal tubular action of this hormone. It is metabolized to canrenone, the active compound, by removal of the acetylthio group and insertion of a 6,7 double bond.



spleen exonuclease *see* **phosphodiesterase II**.

splenin *see* **thymopoietin**.

splenopentin a synthetic peptide that corresponds to residues 32–36 of splenin (*see* **thymopoietin**).

spliceosome a **ribonucleoprotein** complex, containing RNA and **small nuclear ribonucleoproteins** (snRNPs) that is assembled during the **splicing** of the messenger RNA **primary transcript** to excise an **intron**. snRNPs designated U1, U2, U4, U5, and U6 are variously involved; U1 may bind the GU consensus sequence of the 5' splice site and U5 with U4/U6 probably binds to the 3' splice site consensus sequence, AG. U2 may be part of a component binding the UACUA[A]C consensus sequence ([A] is invariant) at a site slightly upstream from the 3' splice site, known as the branch site. The mechanism appears to involve cutting at the 5' side of the GU consensus sequence, which curls back to form a G5'–2'A bond with the invariant A of UACUA[A]C. This linkage generates the loop of the so-called **lariat**. Cutting out the 3' splice site follows on the 3' side of AG to release the complete intron (in the form of the lariat), and the cut ends of the exons are then ligated.

splice site the sequence of bases at each end of an **intron** that determines the point of **splicing**. The 5' splice site (site at the 5' end of the intron) is termed the donor site, consensus sequence GU, while the 3' splice site is termed the acceptor, consensus sequence AG.

splicing 1 (of DNA) the covalent linkage, by the action of a **DNA ligase** (EC 6.5.1.1, requires ATP, or EC 6.5.1.2, requires NAD⁺), of two fragments of duplex DNA at complementary single-stranded terminations. **2** (of RNA) the enzymic process in eukaryotic cells by which **introns** are excised from heterogeneous nuclear RNA (hnRNA) following its transcription from DNA, and the cut ends are rejoined to form messenger RNA (mRNA), in which the message is continuous, before its translation into polypeptide. In yeast, the splicing of transfer RNA involves cutting at the splicing junctions by an endonuclease followed by a process involving ATP whereby the two exon sequences are joined. Group I mitochondrial RNA is spliced by an autocatalytic mechanism, the RNA itself having the ability to catalyse its own splicing; nuclear mRNA precursor – the primary transcript – is spliced by a complex assembly incorporating snRNP and called a **spliceosome**; it involves the formation of a **lariat**. Splicing of group II mitochondrial RNA also involves the formation of a lariat but without formation

of a spliceosome. Splicing in which the cut ends are joined directly within the same RNA molecule is called **cis-splicing**; **trans-splicing** refers to situations in which the cut end from the splicing site rejoins to a splicing site on another RNA molecule. **3** (of protein) a post-translational modification of protein that includes two concerted proteolytic cleavages and one ligation that results in excision of an inner sequence of the original polypeptide chain to form one protein, and ligation of the two outer sequences to form a second protein. It is thought to be an autocatalytic process. The excised inner sequence is known as an **intein**, the ligated outer sequences as an **extein**. *See also* **gene splicing**.

splicing factor 2 *abbr.*: SF2; a protein, present in nuclear extracts, that functions in an early step in splicing pre-mRNA. It is resistant to mild heat treatment, but inactivated by *N*-ethylmaleimide. It consists of two peptides, P32 and P33, and is involved in ensuring accuracy of splicing. Example, P32 from human: database code SP32_HUMAN, 282 amino acids (31.36 kDa). *See also* **pre-mRNA processing protein**.

split gene a gene that is not represented by a continuous DNA sequence; an interrupted gene.

spoke protein a protein associated with the filamentous structure of kinetochore microtubules in the mitotic spindle. Soluble and cytoplasmically localized during interphase, it becomes primarily associated with the mitotic spindle in mitosis. Example, spoke protein 4 (of five different ones) from *Chlamydomonas reinhardtii*: database code RSP4_CHLRE, 465 amino acids (49.74 kDa).

spongiform encephalopathy any of a group of diseases of human and other animals characterized by the presence of many vacuoles in brain tissue, giving it a spongy appearance. The group includes **bovine spongiform encephalopathy**, **scrapie** in sheep, and **Creutzfeldt-Jakob disease**, **Gerstmann-Sträussler syndrome**, and **kuru** in humans. *See also* **prion**, **prion protein**.

spontaneous generation *see* **abiogenesis**.

spontaneous hypoglycemia *see* **glycemia**.

spore a small, often microscopic, reproductive unit consisting of one or several cells. Spores are produced by fungi, bacteria, certain plants, and protozoa. They may variously serve as a means of rapid vegetative propagation, or as a dormant stage in the organism's life cycle. Spores have very low metabolic activity and give rise to a vegetative cell upon germination; they are usually more heat resistant than vegetative cells and are adapted for dispersal. Spores may be sexual or asexual.

sporulation the formation of **spores**.

spot 1 (in chemistry) or **spot test** to test for a substance by applying a small amount of detection reagent to a surface. **2** (in molecular biology) to detect, using radioactive oligonucleotide probes, oligonucleotides applied in spots onto hybridization filters (**spot blots**).

spoT a gene in *Escherichia coli* for the enzyme guanosine-3',5'-bis(diphosphate) 3'-pyrophosphohydrolase; EC 3.1.7.2; *other name*: (ppGpp)ase; it catalyses the hydrolysis of guanosine 3',5'-bis(diphosphate) to guanosine 5'-diphosphate and pyrophosphate. The SpoT protein cycles ppGpp formed as a part of the **stringent response**. The name is jocular, ppGpp and pppGpp originally having been referred to as 'magic spot nucleotides'. Example from *E. coli*: SPOT_ECOLI, 702 amino acids (79.34 kDa). *See also* **relA**, **stringent factor**.

spp. *abbr.* for species (plural).

SPP-1 *see* **osteopontin**.

[S]ppA *symbol* for adenosine 5'- β -thiodiphosphate (alternative to AdoPP[S]).

[S]pppA *symbol* for adenosine 5'- γ -thiotriphosphate (alternative to AdoPPP[S]).

spreading factor *see* **hyaluronidase**.

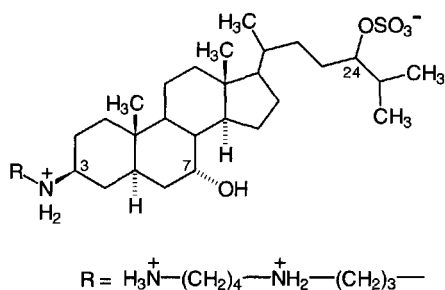
SQDG *abbr.* for sulfoquinovosyldiacylglycerol (*see* **sulfonolipid**).

SQL *abbr.* for symbolic query language; a **query language** used with related databases, e.g. **SYBASE**. It is often pronounced 'sequel'.

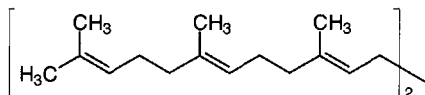
squalamine a water-soluble aminosterol isolated from the



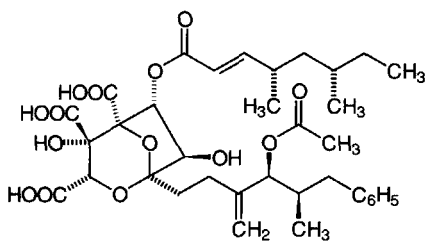
dogfish shark, *Squalus acanthias*. It has antibiotic properties against Gram-negative and Gram-positive bacteria and protozoa. It is an adduct of **spermidine** and a sulfate bile salt.



squalene 2,6,10,15,19,23-hexamethyl-2,6,10,14,18,22-tetra-cosahexaene; the linear triterpene precursor of all cyclic triterpenoids and the sterols. Although first isolated from shark-liver oils and later found in many other fish-liver and plant oils, squalene is widely distributed in plant and animal cells as a metabolic intermediate. It is formed from acetyl-CoA, via a pathway whose intermediates include **mevalonate**, **isopentenyl diphosphate**, **geranyl diphosphate**, and **farnesyl diphosphate**. The latter is converted to presqualene diphosphate, which is then reduced to squalene by NADPH with loss of pyrophosphate; both steps are catalysed by **farnesyltransferase**. [From Latin *squalus*, marine fish, later specifically used to mean shark.]



squalestatin a fermentation product of *Coelomycete* spp. that is a potent selective inhibitor of squalene synthase, and lowers serum cholesterol *in vivo*.



squalestatin 1

sr symbol for steradian.

Sr symbol for strontium.

SR abbr. for sarcoplasmic reticulum.

src an oncogene (*v-src*), originally discovered in the Rous sarcoma virus, the product of which is pp60^{v-src}, a 60 kDa protein that undergoes phosphorylation/dephosphorylation. A normal cellular gene (or **protooncogene**), *c-src*, codes for a related protein; both proteins have **tyrosine kinase** activity. The viral oncogene product and its normal counterpart differ in amino-acid residues, especially at the C terminus where the last 19 residues of pp60^{c-src} are replaced by 12 other residues in

pp60^{v-src}. The Src family of tyrosine kinases is the largest of the nonreceptor tyrosine kinase families and is formed from a group of proteins that all share a high homology with the oncoprotein, *v-Src*, and the cellular counterpart, *c-Src*. Members of the Src family include Src, Yes, Fyn (see **FYN**), Lck (see **LCK**), Lyn (see **lyn**), Blk (see **blk**), Hck, Fgr (see **FGR**), and Yrk, which range in size from 53 to 64 kDa. They all share a number of features including an N-terminal myristoylation site (a mechanism for promoting their association with membranes), a unique domain, two conserved segments known as Src homology 2 domain (SH2 domain) and Src homology 3 domain (SH3 domain (see **SH domain**)), a tyrosine kinase domain, and a C-terminal regulatory domain. In vertebrates, the kinase activity of Src family members is normally inhibited by the phosphorylation of a tyrosine residue within the kinase domain. In *v-src* the region of the gene encoding the negative regulatory C-terminal tyrosine has been deleted, leading to *v-Src* having a higher constitutive tyrosine kinase activity than *c-Src*. Other mutations associated with oncogenic activation map to the SH2 and SH3 domains, supporting the proposal that in the resting cell Src is kept in an inactive state by the binding of its SH2 domain to the phosphotyrosine residue in its negative regulatory domain so keeping the kinase in an inactive conformation. SH3 domains may participate in this repression, possibly by binding proline-rich regions in the kinase domain. Activation of Src family members occurs by two potential mechanisms: (1) the co-presence of another protein containing a specific phosphotyrosine residue that has a higher affinity for the SH2 domain than the negative regulatory phosphotyrosine within the Src member; or (2) the dephosphorylation of the negative regulatory phosphotyrosine by a specific protein tyrosine phosphatase. Example from human: database code KSRC_HUMAN, 536 amino acids (59.77 kDa).

Src homology domain see **SH domain**.

Srd symbol for a residue of the ribonucleoside thiouridine (when the position of the thiol group in its thiouracil moiety is unknown or unspecified; alternative to S or sU).

SRE abbr. for serum response element.

SRF abbr. for serum response factor; a nuclear, dimeric transcription factor. It forms a complex with **Elk-1** that is constitutively bound to the **serum response element**. Example from human: database code SRF_HUMAN, 508 amino acids (51.54 kDa).

SRIF abbr. for somatotropin release-inhibiting factor (i.e. **somatostatin**).

sRNA or **S-RNA** abbr. for soluble RNA (an early name for **transfer RNA**).

SRP abbr. for signal recognition particle.

SR protein any protein belonging to a family of pre-messenger RNA splicing factors possessing highly related primary amino-acid sequences. They share an N-terminal RNA recognition motif and a C-terminal domain that varies in length and consists almost exclusively of alternating serine (S) and arginine (R) residues, from which the proteins are named. Many animal cells possess a series of SR proteins of similar molecular mass: 20, 30, 40, 55, and 70–75 kDa; as yet these have undetermined specificities.

SRS or **SRS-A** abbr. for slow reacting substance (of anaphylaxis).

S→R variation see **smooth-rough variation**.

SSB protein abbr. for single-strand binding protein.

SSCP analysis abbr. for single-strand conformation polymorphism analysis.

ssDNA see **DNA**.

St symbol for stokes (a cgs unit of kinematic viscosity).

Sta symbol for statine.

stability constant symbol: K_s ; an equilibrium constant for the reversible formation of a complex chemical compound from two or more simpler entities; the reciprocal of the **dissociation constant**. In some cases, an **apparent stability constant** (symbol:



stable factor

K'_s), constrained with respect to certain variables (e.g. pH) is determined (for the distinction see **equilibrium constant**). The stability constant is also known as the **association constant** (symbol: K_{ass}) in cases where ions associate into a substance, e.g. for the reaction $A^+ + B^- = AB$, the (concentration) stability or association constant is given by:

$$K_s = [AB]/[A^+][B^-] = K_{\text{ass}}$$

stable factor an alternative name for factor VII; see **blood coagulation**.

stable isotope an isotope that is not radioactive.

stachyose or **lupeose** the tetrasaccharide, $O\text{-}\alpha\text{-D-galactopyranosyl-(1}\rightarrow\text{6)-}O\text{-}\alpha\text{-D-galactopyranosyl-(1}\rightarrow\text{6)-}O\text{-}\alpha\text{-D-glucopyranosyl-}\beta\text{-D-fructofuranoside}$. It is widely distributed, usually associated with **raffinose** (see that entry for synthesis) and **sucrose**; it is isolated from *Stachys tuberosa* rhizomes.

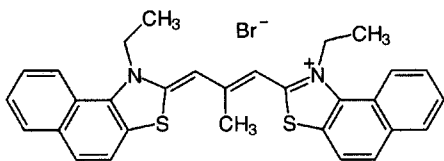
stacking gel an alternative name for **spacer gel**.

Staden, R. US molecular biologist; the author of a much used suite of computer programs for sequence analysis of nucleic acids and proteins, available for both **VMS** and **UNIX**.

staggered conformation see **conformation**.

stainless steel steel that is resistant to corrosion due to its high content of chromium and, often, nickel. It is highly resistant to all organic acids and weak mineral acids but is slowly attacked by concentrated mineral acids. It does not rust.

Stains-all a proprietary name for 1-ethyl-2-[3-(3-ethylnaphtho[1,2-*d*]thiazolin-2-ylidene)-2-methylpropenyl]naphtho[1,2-*d*]thiazolium bromide; a cationic carbocyanine metachromatic dye used as a semiselective stain for locating nucleic acids, proteins, and polysaccharides. RNA stains bluish-purple, DNA blue, protein red, and phosphoprotein blue.



stamp collecting according to Lord Rutherford (1871–1937), one of the two branches of science (the other being physics).

standard (in analytical chemistry) **1 primary standard** a chemical that can be obtained in an unvarying state (e.g. of purity and/or stability) and used as an approved example against which other examples of the same chemical can be compared. **2 secondary standard** a chemical that can be used for comparative purposes following calibration against a primary standard. Although its properties are less stringent than a primary standard, its ready availability makes for convenience of use. See also **standard atmosphere**, **standard state**, **standard temperature and pressure**.

standard atmosphere an internationally established reference for pressure, defined as 101 325 pascals. The **atmosphere** (def. 3) (symbol: atm), equal to 101 325 pascals, was formerly used as a unit of pressure; its use is now discouraged. See also **standard temperature and pressure**.

standard curve a plot of any appropriate parameter against known amounts (or concentrations) of detectable substance. Such a curve can be used to determine the amount (or concentration) of the same substance in an unknown solution given a reading for the parameter in the unknown solution.

standard deviation abbr.: SD; symbol: s (for a sample) or σ (for a distribution); the common measure of dispersion of a series of observations of x that is normally distributed. It is the root-mean-square average of the deviations of the observations from their mean and is in the same units as those of the observations. The standard deviation, s_x , is given by:

$$s_x = \sqrt{\left[\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \right]}$$

Where $\bar{x}$ is the arithmetic average of deviations of x and n is the number of observations; s_x^2 is called the variance of the sample of observations of x . Some authorities prefer the definition:

$$s_x = \sqrt{\left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right]}$$

so that s_x^2 is then the best estimate of the variance, σ^2 , of the population from which the sample of observations of $\bar{x}$ was made. See also **degrees of freedom**, **standard error of estimate of the mean**.

standard error of estimate of the mean or **standard error of the mean** abbr.: SEM; a measure of the reliability of an estimate of the mean of a population. It is equal to the **standard deviation**, s_x , of the observations divided by the square root of the number of observations, n ; i.e. $\text{SEM} = s_x/\sqrt{n}$.

standard free energy see **Gibbs energy**.

standardize or **standardise** to relate any instrument, vessel, or reaction to an absolute standard. —**standardization** or **standardisation** *n*.

standard pressure symbol: p ; an established reference for pressure. Since 1982 the recommended standard pressure for reporting thermodynamic data has been set at 10⁵ pascals (1 bar). Before 1982 it was taken to be 101 325 pascals (1 atmosphere).

standard state a set of conditions, established by convention, under which a standard reference is determined or calculated. See **chemical potential**, **enthalpy**, **entropy**, **Gibbs energy**, **hydrogen electrode**, **standard pressure**, **standard temperature and pressure**.

standard temperature and pressure abbr.: stp or s.t.p. or STP; the standard conditions for the comparison of properties of gases. Standard temperature is now 298.15 K (formerly 273.15 K, i.e. 0 °C) and **standard pressure** is 10⁵ pascals. Former name: normal temperature and pressure (abbr.: ntp or n.t.p. or NTP).

Stanley, Wendell Meredith (1904–71), US biochemist; Nobel Laureate in Chemistry (1946) jointly with J. H. Northrop 'for their preparation of enzymes and virus proteins in a pure form' [prize shared with J. B. Sumner].

Staphylococcus a genus of Gram-positive bacteria, commonly associated with wounds and generally causing localized infections. *Staphylococcus aureus* forms yellow or orange pigments and yields plasmids that are useful in recombinant DNA experimentation: *S. aureus* plasmids coding for resistance to various antibiotics can be transfected into *Escherichia coli* or into *Bacillus subtilis* where they are stable, replicate, and express antibiotic resistance. They have provided the basis from which improved vectors have been developed.

starch the most important reserve polysaccharide found in plants. It is a glucan consisting of two components: **amylose**, which is a homopolymer of glucose units linked only by $\alpha(1 \rightarrow 4)$ bonds; and **amylopectin**, also a glucose homopolymer (branched) that contains $\alpha(1 \rightarrow 6)$ bonds as well as $\alpha(1 \rightarrow 4)$ bonds. Starch is synthesized as a temporary storage form of fixed carbon in chloroplasts during active periods of photosynthesis, accumulating as starch grains. The starch is subsequently converted to sucrose and translocated to storage organs, such as seeds and tubers, where it is re-formed. Compare **glycogen**.

starch (bacterial glycogen) synthase EC 2.4.1.21; recommended name: starch synthase; systematic name: ADP-glucose:1,4- $\alpha\text{-D-glucan 4-}\alpha\text{-D-glucosyltransferase}$; other name: ADPglucose-starch glucosyltransferase. An enzyme that catalyses a reaction between ADPglucose and (1,4- $\alpha\text{-D-glucosyl})_n$ to form (1,4- $\alpha\text{-D-glucosyl})_{n+1}$ and ADP. It uses ADPglucose instead of the UDPglucose of **glycogen synthase**. Example from *Escherichia coli*: database code GLGA_ECOLI, 477 amino acids (52.48 kDa).

starch phosphorylase see **α -glucan phosphorylase**.

starch synthase EC 2.4.1.11; recommended name (example): glycogen (starch) synthase (the nature of the synthetic product should be included in the name); systematic name: UDP-glucose:glycogen 4- $\alpha\text{-D-glucosyltransferase}$; other name:



starch syrup

UDPglucose–glycogen glucosyltransferase. An enzyme that catalyses a reaction between UDPglucose and $(1,4\text{-}\alpha\text{-D-glucosyl})_n$ to form $(1,4\text{-}\alpha\text{-D-glucosyl})_{n+1}$ and UDP. Example (precursor) from maize chloroplasts: database code UGST-MAIZE, 605 amino acids (65.89 kDa). See also **glycogen synthase**. Compare **starch (bacterial glycogen) synthase**.

starch syrup an alternative name for **corn syrup**.

Start an important checkpoint in the G_1 phase of the eukaryotic cell-division cycle. Passage through Start commits the cell to enter S phase.

start codon the trinucleotide **codon** AUG that codes for *N*-formylmethionine, the first amino-acid residue in the synthesis of all prokaryotic and mitochondrial proteins.

start kinase a complex of Cdc2 (see **cdc gene**) and a G_1 cyclin that functions as a serine/threonine protein kinase; its activation is necessary for the cell to pass **Start**. The G_1 cyclin that interacts with Cdc2 is thought to be any of several. The identity of the G_1 cyclin modifies the specificity of the complex.

start point the base pair in DNA at which the first nucleotide is incorporated into an RNA transcript. It is most often a purine, and in many cases is the central base of the sequence CAT. It is sited at the downstream end of the **promoter**. Compare **start codon**.

start protein any protein that is involved uniquely at the start of the cell cycle. Example, protein Cdc10 from *Schizosaccharomyces pombe*: database code CC10_SCHPO, 767 amino acids (85.42 kDa). This protein (which has a close homologue in *Saccharomyces cerevisiae*) is probably a transcriptional regulator for *cdc10* and also for *cdc2*, which encodes a component of **start kinase**.

starve to deprive of nutrients. In animal experiments a minimum of 24 h deprivation is generally understood. Cultures may be starved of specified single nutrients.

stasis arrest or stagnation of flow.

+stasis *comb. form* denoting lack of movement; cessation of flow. — **+static** *adj.*

STAT *abbr. for* signal transducer and activator of transcription; a family of eukaryotic transcription factors that mediate the response to many cytokines and growth factors. Upon receptor activation, STAT proteins dimerize, translocate to the nucleus, and bind to specific promoter sequences on target genes. Seven different genes have been so far identified in mammals. Example: STA1_HUMAN, 750 amino acids (87.33 kDa).

+stat *comb. form* indicating a device for maintaining a particular property at a constant value, e.g. **cryostat**, **thermostat**. — **+static** *adj.*

stathmin *other names:* phosphoprotein p19; oncoprotein p18; prosolin; an ubiquitous phylogenetically conserved cytoplasmic protein that serves as a phosphorylated intermediate in diverse second-messenger pathways. Stathmin undergoes phosphorylation; both phosphorylation and expression of the protein are regulated throughout development, in response to extracellular signals that control cell proliferation and differentiation. It is a major phosphorylation substrate in neurons. Example (precursor) from *Rattus norvegicus*: database code STHM_RAT, 148 amino acids (17.14 kDa).

+static see **+stasis**, **+stat**.

statics *functioning as sing.* the branch of applied mathematics dealing with the interaction of forces in nonmoving systems.

statin 1 a phosphoprotein present specifically in the nuclei of nonproliferating cells. It forms a complex with a 45 kDa serine/threonine protein kinase, and has structural similarity to the **elongation factor** EF-1 α . Example from rat: database code STTN_RAT, 463 amino acids (50.28 kDa). 2 a variant spelling of **statine**.

+statin *word ending (in hormone nomenclature)* denoting a hypothalamic release-inhibiting factor (or hormone). Compare **+liberin**.

statine or **statin** *symbol:* Sta; [3S,4S]-4-amino-3-hydroxy-6-methylheptanoic acid; a novel amino acid contained in positions 3 and 5 of **pepstatin A**.

stationary phase 1 (*in chromatography*) the immobile phase to which the analyte adsorbs or partitions from the mobile phase. 2 (*in fermentation*) the phase of an *in vitro* culture of microorganisms or eukaryotic cells that follows the exponential growth phase and in which there is little or no growth.

statistically significant see **significant**.

statistical mechanics the theoretical analysis of molecular properties based on quantum theory.

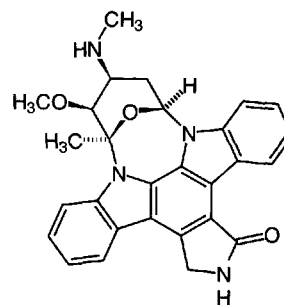
statistical sample a finite selection of items from a population of which the measurement of a variable will give a close approximation to the distribution of that variable in the whole population. See also **random sample**, **sample**.

statistics 1 *functioning as sing.* the science of collecting, analysing, interpreting, and presenting quantities of numerical data. 2 *functioning as pl.* a collection of quantitative data. — **statistical** *adj.*

statyl the acyl group derived from **statine**.

Staudinger, Hermann (1881–1965), German chemist; Nobel Laureate in Chemistry (1953) 'for his discoveries in the field of macromolecular chemistry'. Hew was particularly distinguished for his work on polymers such as polystyrene.

staurosporine an antibiotic from *Streptomyces* spp., M_r 466.2, best known as a potent inhibitor of protein kinase C.



Ste *symbol for* the stearyl group, $\text{CH}_3\text{-(CH}_2\text{)}_{16}\text{-CO-}$.

steady state a condition in which the properties of any part of a system are constant during a process or reaction; i.e. where the rate of formation or increase of a particular quantity is balanced by its rate of removal or decrease.

steady-state fluorescence anisotropy a method in which a sample containing a fluorescent compound is irradiated with plane-polarized light, the emitted fluorescence being analysed to determine its degree of polarization both in the plane of the excitation radiation and in a plane at right angles to this. Only molecules of the fluorescent compound that are aligned in the plane of the incident radiation can absorb radiation and be excited to emit fluorescence. If, in the interval between absorption of the radiation and the emission of fluorescence, the molecule moves out of the plane of polarization of the incident radiation, the emitted radiation will lose polarization (be depolarized) to a degree that depends on the extent to which the molecule had moved. The technique is used to measure the degree to which fluorescent probes are tumbling in a sample. If the probes are held in a relatively rigid position, the extent of depolarization will be small, if tumbling freely it will be high.

steady-state kinetics the analysis of an enzyme reaction in the steady state, i.e. after the pre-steady state and when the intermediates have reached a steady concentration. The duration of the steady state is relatively short, and this state is in any case an approximation since the accumulation of products and the depletion of substrate occur continuously from the start of the reaction. See **Michaelis kinetics**.

steapsin an old term for **pancreatic lipase**.

stearate 1 a trivial name for octadecanoate, $\text{CH}_3\text{-(CH}_2\text{)}_{16}\text{-COO-}$, the anion derived from stearic acid, or octadecanoic acid, a saturated straight-chain higher fatty acid. 2 any

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stearin

mixture of free stearic acid and its anion. **3** any salt or ester of stearic acid. *See also* **stearoyl**.

stearin 1 an old name for any of the glyceryl esters of stearic acid, especially the triester, **tristearin** (tristearoylglycerol). **2** a commercial name for a grade of stearic acid containing other fatty acids, especially palmitic acid.

stearoyl *symbol:* Ste; *numerical symbol* 18:0; *a trivial name for* octadecanoyl, $\text{CH}_3\text{--}[\text{CH}_2]_{16}\text{--CO--}$, the acyl group derived from stearic acid (*see* **stearate**). It is one of the major fatty-acid components of plant and animal lipids; tallow and other hard animal fats contain up to 35%. Together with palmitate it represents a high proportion of the saturated fatty-acid content of dietary and other lipids. It is synthesized from palmitoyl-CoA by a microsomal chain-elongation enzyme. It was formerly and sometimes still is (incorrectly) referred to as **stearyl**.

stearoyl-CoA desaturase EC 1.14.99.5; *systematic name:* stearoyl-CoA:hydrogen donor:oxygen oxidoreductase; *other names:* acyl-CoA desaturase; fatty-acid desaturase; Δ^9 -desaturase. An enzyme that catalyses a reaction between dioxygen, an electron donor and stearoyl-CoA to form oleoyl-CoA. It is an iron protein with a short half-life, and is the only inducible component of the endoplasmic reticulum fatty acyl-CoA desaturase system. It utilizes electrons from reduced cytochrome *b*₅. Example from rat: database code ACOD_RAT, 358 amino acids (41.44 kDa).

stearyl 1 a common name for octadecyl, $\text{CH}_3\text{--}[\text{CH}_2]_{16}\text{--CH}_2\text{--}$, the alkyl group derived from octadecane; octadecyl is recommended. **2** an old trivial name for octadecanoyl; **stearoyl** now used instead.

steatite *see* **talc**.

steato+ *comb. form* indicating presence or relation to fatty substances.

steatorrhoea the presence of abnormally large amounts of fat in the feces.

steatosis fatty degeneration of tissue.

steel factor *see* **stem cell factor**.

steel factor receptor *see* **KIT**.

stefin *see* **cystatin**.

Stein, William Howard (1911–80), US protein chemist; Nobel Laureate in Chemistry jointly with S. Moore (1972) 'for their contribution to the understanding of the connection between chemical structure and catalytic activity of the active centre of the ribonuclease molecule' [prize shared with C. B. Anfinsen]. In addition to this he, together with Moore, was responsible for devising the means for automatic analysis of the amino-acid constituents of protein hydrolysates, and free amino acids in body fluids, separated by ion-exchange chromatography.

Stelazine *see* **trifluoperazine**.

stellacyanin a blue copper-containing mucoprotein of low molecular weight obtained from the Japanese lacquer tree, *Rhus vernicifera*. It functions as an electron-transfer protein, accepting electrons from the cytochrome *b*₆–*f* complex and donating them to **photosystem I**, and has been much studied for the structure of the copper-binding site.

STEM *abbr. for* scanning transmission electron microscope.

stem cell any member of the various groups of reserve cells whose role is to replace cells that are destroyed during the normal life of the animal, e.g. blood cells, epithelial cells, spermatogonia, and skin cells. Stem cells can divide without limit; after division, the stem cell may remain as a stem cell or proceed to terminal differentiation. Although appearing morphologically unspecialized, the stem cell may be considered as having reached a state of differentiation where the possibilities for further differentiation are very limited.

stem cell factor (*abbr.:* SCF) or **mast cell growth factor** or **steel factor** a **cytokine** that stimulates the proliferation of mast cells. It is produced by endothelial cells, fibroblasts, bone marrow cells, and Sertoli cells. A homodimer glycoprotein with type I membrane and soluble forms, it is able to augment the proliferation of both myeloid and lymphoid hemopoietic progenitors. It acts as a ligand for the product of the protooncogene

c-kit (*see* **KIT**). Example from human precursor: database code SCF_HUMAN, 273 amino acids (30.86 kDa).

stem name any generic name of a category of similar compounds, enzymes, organisms, etc. that is used also as an affix in forming the specific names of individual members of such a category.

steno+ or (*before a vowel*) **sten+** *comb. form* indicating narrowness.

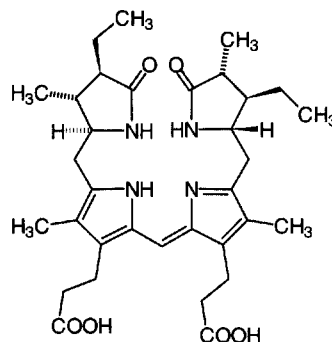
stenohaline describing an organism that is unable to tolerate wide variations in environmental osmotic pressure.

stenothermous describing an organism that is unable to tolerate wide variations in environmental temperature.

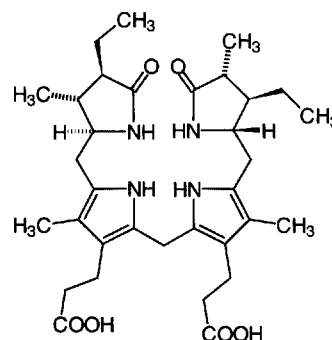
step gradient a discontinuous gradient, formed in a tube by layering solutions of different densities carefully one on the other. In column chromatography the same effect is achieved by pumping buffers of different composition sequentially through the column. *See* **gradient**.

steradian *symbol:* sr; the SI supplementary unit of solid angle. It is equal to the angle subtended at the centre of a sphere by an area of surface equal to the square of the sphere's radius. The surface of a sphere subtends an angle of 4π sr at its centre.

stercobilin or **stercobilin IX α** a linear tetrapyrrole formed in heme catabolism with a red brown colour. Structurally it is an oxidation product of **stercobilinogen**, comparable to the **urobilinogen**→**urobilin** conversion. Metabolically, it is a major product formed from urobilinogen by intestinal bacteria and is the major pigment of feces.



stercobilinogen an alternative name for 10,23-dihydrostercobilin; *former name*, stercobilinogen IX α . A product of heme catabolism; it is a linear tetrapyrrole and structurally is a **urobilinogen** molecule in which the double bonds between the methyl and ethyl substituents on the two end rings have been reduced.



sterculic acid *see* **cyclic fatty acid**.

stereo+ *comb. form* denoting solid or three-dimensional.

stereobase unit the smallest set of one, two, or more succes-



sive **configurational base units** that prescribes repetition in a polymer molecule.

stereochemistry the branch of chemistry dealing with the three-dimensional properties of molecules; the study of the properties of **stereoisomers**. See *cis-trans isomerism*, *Re/Si convention*, *sequence rule*, *stereospecific numbering*.

stereoheterotopic a term describing chemically-like ligands (or faces of double bonds) whose separate replacement with an achiral ligand (or for faces, addition of an achiral reagent) gives rise to two stereoisomers. If the stereoisomeric products are **enantiomers**, the ligands (faces) are enantiotopic; if they are **diastereoisomers**, the ligands (faces) are diastereotopic. for more extensive descriptions, see **diastereotopic**, **enantiotopic**.

stereoisomer any of two or more **isomers** that have the same molecular constitution and differ only in the three-dimensional arrangement of their atomic groupings in space. Stereoisomers may be **diastereoisomers** or **enantiomers**. Compare **constitutional isomer**. —**stereoisomeric** *adj.*; **stereoisomerism** *n.*

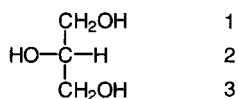
stereochemistry the body of mathematical methods dealing with the interpretation of the structure and dimensions of three-dimensional objects from information contained in two-dimensional representations, such as photomicrographs of tissue sections.

stereoregular polymer a regular polymer composed of molecules having a regular repetition of identical **stereobase units** at all stereoisomeric sites of the main chain.

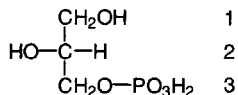
stereoselective polymerization the formation of a polymer from a mixture of stereoisomeric monomer molecules by preferential incorporation of one species into a growing polymer chain.

stereospecific (of an enzymatic or other reaction system) acting on or producing a particular **stereoisomer**; stereochemically specific. —**stereospecificity** *n.*

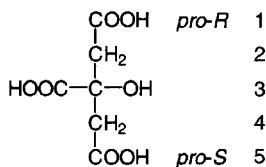
stereospecific numbering a numbering system finding most application for prochiral compounds, *Caabc*, with two chemically-like atoms, or groups of atoms, *a*. It provides identifying numbers for the prochiral *a* groups and, by extension, a numbering system for their chiral derivatives. A stereospecifically numbered compound is designated by the prefix *sn-* to differentiate it from a conventionally numbered compound. Two systems have been described. That of Hirschmann was



(1) glycerol



(2) *sn*-glycerol-3-phosphate



(3) citric acid

devised particularly so that the ^{14}C -labelled glycerol giving rise to $[3,4-^{14}\text{C}_2]\text{glucose}$ (in rats) could be termed $[1-^{14}\text{C}]$ glycerol and thus be related, biosynthetically and by nomenclature, to $[1-^{14}\text{C}]\text{lactate}$. It uses the basic *R/S* system (see **sequence rule**) to set up a model, which is viewed away from the group with the lowest sequence priority. If this is *c* in *Caabc*, a counterclockwise ordering is examined for the remaining three groups, *baa*, starting with *b*. The first *a* group to be encountered receives the lower number. The numbering for glycerol is shown in structure 1. By extension, the naturally occurring glycerol phosphate formed by the action of **glycerol kinase** becomes *sn*-glycerol 3-phosphate (structure 2). This structure has also been described as *D*-glycerol 3-phosphate, *L*-glycerol 3-phosphate, and *L*- α -glycerophosphate. The Hirschmann nomenclature has found wide acceptance in the lipid field. The other method, the Hanson and Hirschmann system, expands the *R/S* system in a logical fashion (rather than using the arbitrary counterclockwise ordering described for glycerol). The two chemically-like groups, *a*, of *Caabc* are first given *pro-R* and *pro-S* assignments. Then, following the established '*R* precedes *S* rule', *pro-R* precedes *pro-S*, and the *pro-R* group is assigned the lower number. This system has been applied to citric acid as shown in structure 3. Application of this system to glycerol unfortunately leads to the product of glycerol kinase being *sn*-glycerol 1-phosphate. Although this system has a logical basis in terms of established nomenclature, it seems unlikely that the *sn* system for glycerol and derivatives will be brought into line.

stereospecific polymerization a polymerization in which a **tactic polymer** is formed.

steric relating to effects involving arrangements of atoms in space.

steric hindrance 1 prevention of the free rotation of an atom or group in a molecular entity with respect to a connected one due to the relative sizes and/or spatial disposition of the atoms or groups in question. 2 restriction or prevention of the reaction of a molecular entity with another due to the relative sizes and/or spatial disposition of atoms or groups in either or both of the reactants.

sterile 1 (of inanimate objects) free from microbiological or other life forms. 2 (of a living organism) incapable of sexual reproduction.

sterilize or **sterilise** 1 to treat in such a way as to kill undesired microorganisms without damaging the material under treatment. Compare **disinfection**. 2 to render incapable of sexual reproduction. —**sterilization** or **sterilisation** *n.*

Stern-Volmer equation an expression relating **fluorescence quenching** to the concentration of the quenching substance, *c*, and the fluorescence lifetime, τ . It is: $F_0/F = 1 + k\tau c$, where F_0 is the fluorescence intensity when no quencher is present, F is the fluorescence intensity in the presence of quencher, and k is a constant that depends on the system; the concentration of the absorbing molecule must be the same in both quenched and unquenched systems. The **Stern-Volmer plot** is a plot of F_0/F against concentration of quenching substance, from the slope of which, if the fluorescence lifetime is known, the quenching constant can be obtained. [After Otto Stern (1888–1969) and M. Volmer].

steroid any of a large group of substances that have in common a ring system based on 1,2-cyclopentanoperhydrophenanthrene. The group includes such natural products as **bile acids**, **corticosteroids**, **sex hormones**, **sterols**, and various plant steroids. See also **vitamin D**.

steroid diabetes a condition involving decreased glucose tolerance, fasting hyperglycemia, and glucosuria, found in patients exposed to excessive quantities of **glucocorticoids**. It may thus accompany **Cushing's syndrome**.

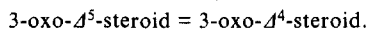
steroid-hormone receptor any member of a family of intracellular receptors that bind steroid hormones and mediate their effects; the family also includes receptors for thyroid



steroid Δ -isomerase

hormones, retinoids, and vitamin D. After binding steroid that has entered the cell, the steroid–receptor complex is translocated to the nucleus and binds to a specific DNA sequence, a **response element**. The receptors are composed of three domains: a modulating N-terminal domain, a DNA-binding domain, and a C-terminal steroid-binding domain. Their zinc fingers are characteristic (four motifs). Examples (human), steroid-hormone receptor ERR1: database code ERR1_HUMAN, 521 amino acids (55.68 kDa); estrogen receptor: database code ESTR_HUMAN, 595 amino acids (66.18 kDa).

steroid Δ -isomerase EC 5.3.3.1; *systematic name*: 3-oxo-steroid Δ^5 - Δ^4 -isomerase; *other name*: Δ^5 - Δ^4 -ketosteroid isomerase. An enzyme (dimer) involved in microbial transformation of steroids; it catalyses the reaction:



Example from *Comamonas testosteroni* (*Pseudomonas testosteroni*): database code SDIS_COMTE, 125 amino acids (13.38 kDa).

steroid 21-monooxygenase EC 1.14.99.10; *other names*: steroid 21-hydroxylase; cytochrome P450 21A1; an enzyme that catalyses a reaction between dioxygen, an electron donor, and a steroid to form a 21-hydroxysteroid. It is a heme–thiolate protein. A deficiency underlies congenital adrenal hyperplasia: there is a failure to synthesize cortisol, and this results in overproduction of **corticotropin**, as well as of 17-hydroxyprogesterone and androstenedione. Example from human: database code CPS1_HUMAN, 494 amino acids (55.82 kDa).

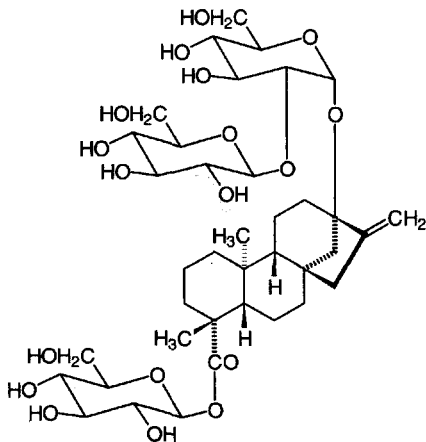
steroidogenesis the biosynthesis of steroids, but often referring to the genesis of other steroids from cholesterol.

sterol any of a group of **steroids** characterized by the presence of one or more hydroxyl groups and a hydrocarbon side-chain in the molecule.

sterol O-acyltransferase EC 2.3.1.26; *other names*: cholesterol acyltransferase; sterol-ester synthase; an intracellular enzyme that catalyses the formation of cholesterol ester from acyl-CoA and cholesterol. It is involved in the storage of cholesterol esters in phagocytic cells, a process that plays a part in producing lipid-laden cells in arterial walls. Example from human: database code ACAT_HUMAN, 550 amino acids (64.77 kDa).

steryl-sulfatase EC 3.1.6.2; *other names*: steroid sulfatase (sulfatase); steryl-sulfate sulfhydrolase; arylsulfatase C; an enzyme that catalyses the hydrolysis of 3- β -hydroxyandrost-5-en-17-one 3-sulfate to 3- β -hydroxyandrost-5-en-17-one and sulfate. Example (precursor) from human: database code STS_HUMAN, 583 amino acids (65.42 kDa).

stevioside 13-[(2-O- β -D-glucopyranosyl)- α -D-glucopyranosyl]oxy]kaur-16-en-18-oic acid β -D-glucopyranosyl ester; a glycoside extracted from the leaves of *Stevia rebaudiana* (yerba dulce). It is used as a sweetening agent, being about 300 times as sweet as sucrose.

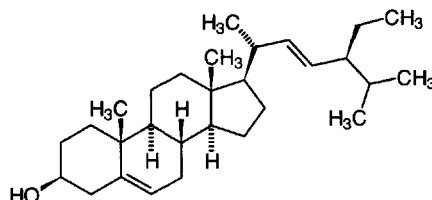


STH *abbr.* for somatotrophic hormone; *see* **somatotropin**.

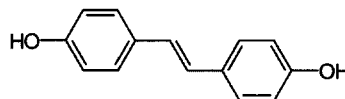
STI *abbr.* for soybean trypsin inhibitor.

sticky end *see* **cohesive end**.

stigmasterol 3 β -hydroxy-24-ethyl- $\Delta^{5,22}$ -cholestadiene; a **phyto-sterol** structurally related to β -sitosterol (*see* **sitosterol**) and found in plant oils.



stilbestrol or (*esp. Brit.*) **stilboestrol** 4,4'-dihydroxystilbene; a synthetic nonsteroid compound, first prepared in 1938, that proved to be the mother substance of a range of compounds with estrogenic activity, of which α , β -diethylstilbene (*see* **diethylstilbestrol**) is the most active. Stilbestrol was formerly used as a growth promoter in farm animals but is now banned in most countries. However, it is still sometimes used to treat hormonal disorders in small animals. It was formerly used to treat prostatic cancer but there is some evidence for its carcinogenicity.



trans-stilbestrol

still an apparatus for the distillation of liquids, including the preparation of distilled water. It consists of a heated vessel, a condenser, and a receiver. A continuous apparatus incorporates an automatic feed for the liquid in question, a heating element, and a thermoregulator.

stimulate (*in physiology*) to provide a **stimulus** to any tissue or cell leading to its excitation or to initiation of a sequence of events with some defined endpoint, involving an electrical event, activation of a metabolic pathway, or initiation of cell division or growth. —**stimulation** *n.*; **stimulative** *adj.*

stimulus (*in physiology*) any event or phenomenon, such as radiation, electrical potential, or addition of molecules that leads to excitation of a tissue or cell.

stimulus–response coupling a molecular mechanism, often involving **signal transduction**, that transmits a stimulus from the exterior of a cell's plasma membrane, in many cases through **second messenger** pathways, to effector systems that bring about a response. For example, in **stimulus–secretion coupling** the effector system brings about a secretion event such as degranulation of mast cells or release of hormone (e.g. insulin) from storage granules to the cell exterior.

stimulus–secretion coupling *see* **stimulus–response coupling**.

stochastic arrived at by skilful conjecture; e.g. a stochastic model, a stochastic process.

stock solution a solution of a reagent, at a stable or convenient concentration, from which appropriate dilutions can be made at the time of use.

Stock system (*in inorganic nomenclature*) a system of designating the **oxidation number** of an element by placing a Roman numeral in parentheses immediately following the name of the element or, where symbols are used, in superscript to the right of the symbol; e.g. manganese(IV) oxide, $\text{Mn}^{\text{IV}}\text{O}_2$.

stoichiometry or (*rarely*) **stoicheiometry** the quantitative relationship of the reactants and products of a chemical reaction



stokes

in the proportions that they appear in the chemical equation describing the reaction. —**stoichiometric or stoicheiometric adj.**

stokes *symbol:* St; the cgs unit of **kinematic viscosity**; its use is now discouraged. In SI units, kinematic viscosity is measured in square metres per second; $1 \text{ St} = 10^{-4} \text{ m}^2 \text{ s}^{-1}$. [After (Sir) George Gabriel Stokes (1819–1903), British mathematician.]

Stokes' law of fluorescence a law stating that the wavelength of the emitted fluorescent light, λ_{em} , is greater than that of the light exciting the fluorescence, λ_{ex} . This relationship is true for most fluorescent materials; those for which it does not hold are termed **anti-Stokes**.

Stokes' law of viscosity a law stating that the frictional coefficient, f , of a spherical particle of radius r , moving through a liquid of viscosity η , is given by $f = 6\pi\eta r$.

Stokes' loss the loss of excitation energy available for fluorescence due to collision of molecules in the first excited state, S_1 , with their neighbours, which results in a lower vibration level of S_1 .

Stokes' shift the difference in wavelength between the excitation and emission maxima for a particular fluorescent substance. In quantitative form, Stokes' shift is $10^7(1/\lambda_{\text{ex}} - 1/\lambda_{\text{em}})$, where λ_{ex} and λ_{em} are the corrected maximum wavelengths for excitation and emission expressed in nanometres.

stoma (*pl.* stomata) any of the pores occurring in the epidermis of plants, particularly in leaves, through which gaseous exchange takes place.

stop codon any of the trinucleotide **codons**, UGA, UAG, and UAA, that signal the termination of translation of a messenger RNA molecule and the release of the nascent polypeptide chain. *See also* **termination codon**.

stopped-flow technique a method in which two solutions are caused to flow into a mixing chamber and then into an observation chamber, after which the flow is caused to stop abruptly and the time course of the chemical reaction can be followed, usually spectroscopically, as the mixture in the observation chamber ages. Critical features include the quality of mixing, the speed with which the mixed solution fills the observation volume, and the geometry of the observation volume in relation to the sensitivity of observation. *See also* **rapid-reaction kinetics**.

STOP protein *abbr. for* stable tubule only polypeptide; a protein that blocks the endwise dissociation of microtubules.

stop-transfer protein *see* **signal peptide**.

storage granule any small **organelle**, bounded by a membrane having a single lipid bilayer, that contains stored material. For example, mast-cell storage granules contain histamine, while those of pancreatic B cells contain insulin; in both cases the stored material is destined for secretion. This involves a stimulus applied to the cell that causes fusion of the granule membrane with the plasma membrane, followed by release of the granule contents by **exocytosis**.

storage polysaccharide any polysaccharide that serves as a form of stored energy in living organisms. Storage polysaccharides include starch, phytoglycogen (e.g. in maize), and fructosans (e.g. inulin) in plants, and glycogen in animals.

stp or s.t.p. or STP *abbr. for* standard temperature and pressure.

straight-chain describing any chain of carbon atoms in which none of the carbon atoms is directly bonded to more than two other carbon atoms.

strain **1** a group of related individuals having certain characters that distinguish the members from other such groups within the same species or variety; a race. **2** a line of organisms descended or derived from a particular ancestral individual. **3** (*in physics*) the temporary or permanent deformation of a body resulting from an applied stress.

streptavidin a tetrameric biotin-binding protein (subunit M_r 14 500) produced by *Streptomyces avidinii*, capable of binding up to four molecules of **biotin** per molecule. It is useful in techniques such as enzyme-linked immunosorbent assay (ELISA), radioimmunoassay, immunocytochemistry, and protein blotting. It is also used with DNA probes as a detection reagent.

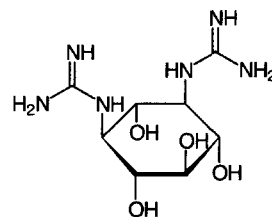
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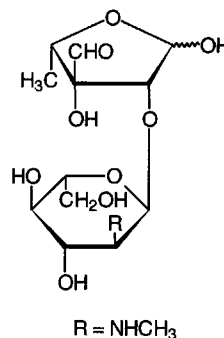
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where biotin can be incorporated into the molecule under study; streptavidin can then be used to bind to the biotin. The streptavidin may be labelled with **FITC**, gold, peroxidase, or another agent directly as a detection reagent. Alternatively, a biotin-labelled detection system (such as alkaline phosphatase) can be bound to streptavidin through another of its biotin-binding sites. Streptavidin has the advantage over **avidin** (also used in such techniques) of having a near-neutral isoelectric point (7.25–7.45); consequently its use results in less nonspecific binding. *See also* **biotinylation**.

streptidine 1,3-diguanido-2,4,5,6-cyclohexanetetrol; the non-carbohydrate component of **streptomycin**.



streptobiosamine 5-deoxy-2-*O*-[2-deoxy-2-(methylamino)- α -L-glucopyranosyl]-3-*C*-formyl-L-lyxose; a disaccharide component of **streptomycin**, containing **streptose** and *N*-methylglucosamine.



Streptococcus a genus of Gram-positive, facultatively or obligately anaerobic cocci or coccoid bacteria, of biochemical significance for their production of **streptokinase**, **streptolysin**, and **streptodornase**.

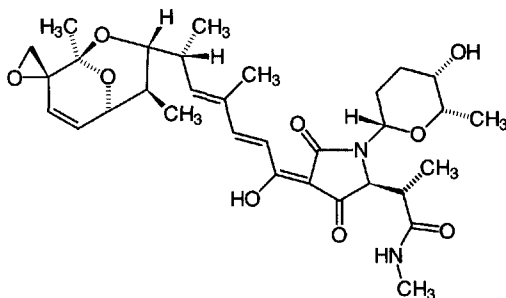
streptodornase any of the extracellular deoxyribonucleases produced by streptococci. At least four such serologically distinct nucleases are known. Some or all have the characteristics of **deoxyribonuclease I** (EC 3.1.21.1), bringing about endonucleolytic cleavage of DNA to 5'-phosphodinucleotide and 5'-phosphooligonucleotide end-products. [From streptococcal deoxyribonuclease.]

streptokinase any of a number of extracellular proteins produced by certain streptococci; they exhibit no enzymic activity, despite the name. In human blood streptokinase forms a tight 1:1 complex with **plasminogen** that catalyses the activation of plasminogen to plasmin, and this in turn lyses fibrin clots. It has been used therapeutically after heart attacks and strokes. It is assumed that streptokinase assists the invasiveness of pathogenic streptococci. Example, streptokinase A precursor from *Streptococcus pyogenes*: database code STRP_STRPY, 440 amino acids (49.84 kDa); residues 1–26 are the signal, 27–440 the protein.

streptolydigin an antibiotic, derived from *Streptomyces lydicus*, that possesses potent antibacterial action, particularly against anaerobes and some Gram-positive aerobes. This is probably due to its inhibitory activity against bacterial **RNA**

streptolysin

polymerase; it reduces the rate of transcription, slowing the rate of phosphodiester bond formation without affecting the fidelity of transcription. This contrasts with **rifampicin**, which inhibits initiation. Compare **rifampicin**.

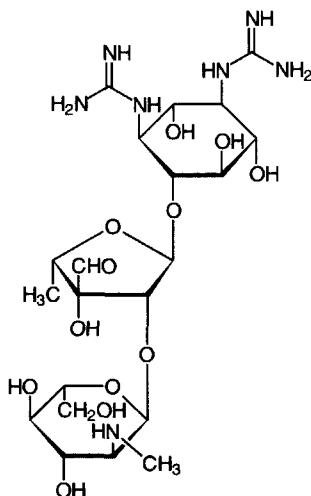


streptolysin either of two dissimilar **hemolysins** produced by certain streptococci (particularly of group A). **Streptolysin-O** (oxygen-sensitive) is produced by culture in serum-free medium. Its hemolytic activity is destroyed by oxygen (but restored by certain mild reducing agents, e.g. 2-mercaptoethanol) and inhibited by free cholesterol. It is immunogenic, the level of circulating antibody (**antistreptolysin-O**, ASO) being useful clinically in the diagnosis of streptococcal infections. Hemolysis by streptolysin-O is not complement-dependent. Streptolysin-O can also lyse lysosomes. Biochemically and serologically it resembles O-toxin of *Clostridium perfringens*. It can be used as an agent to permeabilize cells. **Streptolysin-S** (serum-dependent) is oxygen-insensitive and its production is promoted by serum proteins. Its hemolytic activity is inhibited by certain phosphatides, e.g. phosphatidyl ethanolamine, at low concentration. It is nonimmunogenic, and is also toxic to leukocytes.

Streptomyces a genus of filamentous, spore-forming, Gram-positive bacteria that are largely responsible for the 'earthy' smell of soil but are chiefly remarkable for the diversity of secondary metabolites that they produce, including clinically useful antibiotics, e.g. **streptomycin**. They are amenable to genetic manipulation with many plasmid and phage vectors; this provides a unique opportunity for overexpression and for the design and synthesis of novel and 'hybrid' antibiotics. See also **streptomycete**.

streptomycete strictly, any member of the bacterial genus **Streptomyces**. However, the term may sometimes be extended to include certain related types, including *Streptoverticillum* spp.

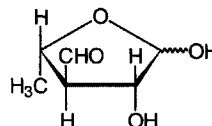
streptomycin an **aminoglycoside antibiotic** produced by the soil



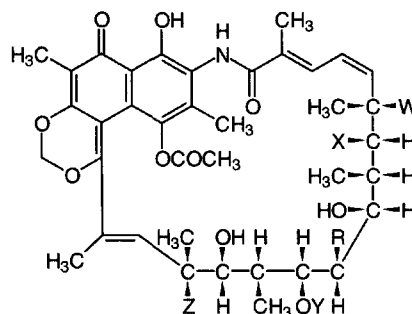
striated m

bacterium *Streptomyces griseus*, having a relatively broad spectrum of activity against bacteria; it is particularly active against mycobacteria (especially against *Mycobacterium tuberculosis*), enterobacteria, and staphylococci. It acts at the 30S ribosomal subunit and distorts the A site (**aminoacyl site**), thereby causing miscoding or prevention of binding of aminoacyl-tRNA. The components of streptomycin are **streptidine**, **streptose**, and *N*-methyl-L-glucosamine.

streptose L-streptose; 5-deoxy-3-C-formyl-L-lyxose; a component of **streptomycin** and **streptobiosamine**.

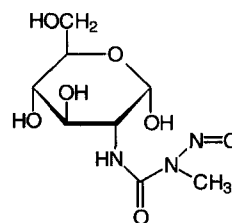


streptovaricin any one of a group of **ansamycin** antibiotics obtained from *Streptomyces spectabilis*. Compare **rifampicin**.



Streptovaricin	W	X	Y	Z	R
A	OH	OH	Ac	OH	COOCH ₃
B	H	OH	Ac	OH	COOCH ₃
C	H	OH	H	OH	COOCH ₃
D	H	OH	H	H	COOCH ₃
E	H	O=	H	OH	COOCH ₃
G	OH	OH	H	OH	COOCH ₃
J	H	OAc	H	OH	COOCH ₃
K	OH	OAc	H	OH	COOCH ₃

streptozocin 2-deoxy-2-(3-methyl-3-nitrosoureido)-D-glucopyranose; a broad-spectrum antibiotic, antineoplastic, and diabetogenic substance isolated from *Streptomyces achromogenes*. It has a rapid and specific cytotoxic effect on the B cells in pancreatic islets, and is used to induce experimental diabetes in animals and for chemotherapy of insulinoma or other islet-cell neoplasms in humans. It acts as a methylating agent for DNA.



α -D-streptozocin

striated muscle muscle in which the **thick filaments** and **thin fila-**

strict aerobe

ments are arranged in microscopically visible bands producing a striped appearance. Such an arrangement is typical of skeletal muscle and cardiac muscle. *Compare smooth muscle. See also sarcomere.*

strict aerobe *see aerobe.*

strict anaerobe *see anaerobe.*

stringency (*in molecular biology*) the rigour with which the ability of complementary DNA sequences to recognize each other (and thus to hybridize) is tested. As temperature is increased it becomes less likely that more distantly related sequences will hybridize; hence, high stringency is associated with high temperature.

stringent very rigorous. *See also stringency, stringent control, stringent plasmid, stringent response.*

stringent control a mechanism of control of plasmid replication whereby the plasmid is replicated only once for each time the chromosome is replicated, thus regulating the plasmid copy number.

stringent factor a protein associated with ribosomes, and encoded by the gene *relA*, that is associated with the **stringent response**. The stringent factor is associated with enzyme activity that donates a pyrophosphate group to 3' of either 5'-GTP or GDP for the production of pppGpp and ppGpp, which probably act as allosteric modulators mediating the stringent response. *See also relaxed mutant, spot.*

stringent plasmid a plasmid that has a limited number of copies per cell.

stringent response a response by bacteria to poor growth conditions (e.g. lack of amino acids) in which the cell shuts down protein synthesis and other metabolic activity. *See also relaxed mutant, stringent factor.*

strip **1** to remove adherent, admixed, or combined substances or material from a fundamental, underlying, or required substance or structure. **2** (*in molecular biology*) **a** to remove polysomes from rough endoplasmic reticulum *in vitro* to produce **stripped membranes** (*see degranulation* (def. 2)). **b** to remove ribosomal proteins from a ribosome to produce a **core particle**. **c** to remove the amino acid from an aminoacyl-transfer RNA molecule to produce **stripped transfer RNA**. **d** to remove hybridized **probe** (def. 3) from a **transfer membrane** prior to reprob-
ing. **3** (*in biochemistry*) to remove endogenous 2,3-bisphosphoglycerate from a hemoglobin solution to produce **stripped hemoglobin**. **4** to remove liquid components from a gaseous mixture.

stripped hemoglobin a hemoglobin solution from which endogenous 2,3-bisphosphoglycerate has been removed, e.g. by gel filtration or dialysis at high NaCl concentrations.

stripped membrane rough endoplasmic reticulum from which polysomes have been removed *in vitro*. This technique has been applied mainly to membranes from dog pancreas, which lack ribonuclease; the ribosomes are removed by treatment with EDTA.

stripped transfer RNA a molecule of aminoacyl-transfer RNA from which the amino acid has been hydrolytically removed.

stroma (*pl. stromata*) **1** fibrous connective tissue or other intercellular material that forms the structural framework of an organ, as opposed to the functional tissue (parenchyma). **2** the spongy framework of protein strands within a red blood cell in which the hemoglobin is packed. **3** (*in botany*) the space enclosed by the double membrane of a **chloroplast** but excluding the thylakoid space. It contains DNA, ribosomes, and some temporary products of photosynthesis. **4** (*in mycology*) a mass of interwoven hyphae in which fruiting bodies (perithecia) develop. —**stromal adj.**

stromelysin EC 3.4.24.17; *other names*: matrix metalloproteinase-3 (*abbr.*: MMP-3); transin-1; an enzyme that catalyses the degradation of fibronectin, laminin, and some (but not all) gelatins and collagens. Zn²⁺ and Ca²⁺ are cofactors. It is a glycoprotein of the extracellular matrix. Example, stromelysin 1 precursor from human: database code COG3_HUMAN, 477 amino acids (53.92 kDa).

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strong acid an **acid** that remains ionized at all except the lowest pH values, i.e. one that has a large **acid dissociation constant**.

strong base a **base** that remains ionized at all except the highest pH values, i.e. one that has a large **basic dissociation constant**.

strophanthin a highly toxic glycoside mixture produced by *Strophanthus kombe*, with actions similar to those of **digoxin** and **ouabain**.

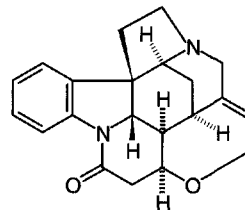
structural gene any gene that carries the genetic information determining the structure, and therefore the specificity of function, of a protein. One or more structural genes may be contained in an **operon**. *Compare regulator gene.*

structural polysaccharide any of several components of a plant cell wall that perform a structural function. They can be divided into three solubility classes: **pectins**, extractable with dilute acid; **hemicelluloses**, extractable with aqueous alkali; and **microfibrillar polysaccharides**.

structure factor a parameter describing the scattering of electromagnetic radiation (e.g. X-rays) by an electron. It is the ratio of the radiation scattered by any real sample to that scattered by a single electron at the origin.

Strutt, John William, Lord Rayleigh (3rd Baron Rayleigh) (1842–1919), British mathematician and physicist noted *inter alia* for his contributions to the theory of sound, for his investigations of physical optics, e.g. **Rayleigh scattering**, and for his discovery (with W. Ramsay) of argon; Nobel Laureate in Physics (1904) 'for his investigations of the densities of the most important gases and for his discovery of argon in connection with these studies'.

strychnine a bitter alkaloid obtained from seeds of the Indian tree *Strychnos nux-vomica*. It is highly toxic, and is sometimes used to poison moles, rodents, and other vermin. Its lethal convulsant action is a result of antagonism of the inhibitory transmitter, glycine, in the spinal cord (*see glycine receptor*). *See also brucine.*



STS *abbr. for* sequence-tagged site.

Stuart factor or **Stuart-Prower factor** an alternative name for factor X; *see blood coagulation*. [After a Mr Stuart, in whom the deficiency was first described.]

Student's *t* test a statistical test, based on a distribution due to William Sealy Gosset (1876–1937) writing in 1908 under the pseudonym 'Student', that can be applied to small samples of data, *x*. In this distribution *t* depends only on the sample size, *N*, and not on the variance of the population; as *N* increases, the distribution curve of *t* approaches a normal distribution curve more closely.

STX *abbr. for* saxitoxin.

sU *symbol for* a residue of the ribonucleoside thiouridine (when the position of the thiol group in its thiouracil moiety is unspecified or unknown; alternative to S or Srd).

s²U *symbol for* a residue of the ribonucleoside 2-thiouridine (alternative to ²S).

s⁴U *symbol for* a residue of the ribonucleoside 4-thiouridine (alternative to ⁴S).

sub *jargon term for* subculture (def. 1).

sub+ *prefix* **1** beneath, under, at the bottom (of). **2** immediately below in order, rank, or status; secondary. **3** less than, partial(ly), imperfect(ly). **4** in place of. **5** denoting a subordinate part or subdivision of a whole. **6** (*in chemical nomenclature*) designating a compound that contains less than the usual pro-



subcellular

portion of a specified element; e.g. carbon suboxide, C_2O_3 .
Compare *infra-*, *super-*.

subcellular describing organelles, functional units, or other components that are found within cells or derived from cells.

subcellular organelle a pleonasm that is commonly used to describe **organelles** (which by definition are subcellular).

subclone to clone into a new vector a subfragment of part of a larger cloned DNA, after excising the required fragment using restriction endonucleases.

subculture **1** to prepare a fresh culture from a small sample of an existing culture. **2** a culture prepared in such a way.

subcutaneous *abbr.*: s.c.; below the skin.

suberin the polymeric substance that covers the epidermal cell layer in the underground parts of plants (e.g. roots and tubers). It is a polyester containing ω -hydroxy fatty acids and aromatic fatty acids, e.g. **ferulic acid**.

sublime to vaporize from the solid without passing through a liquid phase. — **sublimation** *n*.

submaxillary protease an endopeptidase isolated from the submaxillary gland of mice. It acts specifically at Arg-|-Xaa bonds.

submaxillary relating to the region beneath the upper jaw (maxilla).

submaximal below the maximal.

submitochondrial particle any particle derived by disruption of mitochondria. See **H⁺-transporting ATPase, complex I, II, III, IV**.

submolecular describing any chemical or chemical component that exists or functions at a lower level of complexity than that of intact individual molecules. The term may be used in reference to electrons, ions, free radicals, or other molecular fragments. Compare **supramolecular**.

subnatant **1** lying under. **2** a liquid lying under a **supernatant** or under solid material. Compare **infranatant**.

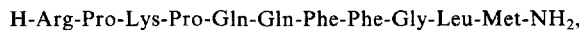
suboptimal below or failing to reach the optimal condition.

subsite a site, e.g. on a protein, that has a specific function within the overall function of a larger site.

substance K see **neurokinin A**.

substance K receptor or **neurokinin A receptor** or **NK-2 receptor** a membrane protein that binds substance K and mediates its intracellular effects. It is a G-protein-coupled, seven-transmembrane-domain protein that activates the **phosphatidylinositol cycle**. Example from *Bos taurus*: database code NK2R_BOVIN, 384 amino acids (43.07 kDa).

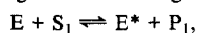
substance P *abbr.*: SP; an 11-residue amide,



found in most tissues of the mammalian body but at highest concentrations in nervous tissue (both central and peripheral) and in gut tissues. It also occurs in the central nervous system of many other vertebrates. Derived from **protachykinin β precursor**, it is a **tachykinin**; its main pharmacological actions are to cause hypotension and vasodilatation, and to stimulate contraction of the intestine and secretion of saliva. However, its physiological function remains to be fully determined. See also **tachykinin receptor**.

substituent group or **substituent** (*in organic-chemical nomenclature*) any atom or **group** (def. 1) that has replaced or can be considered to have replaced (i.e. been substituted for) a hydrogen atom in a parent molecular entity, which may be either real or hypothetical; in the special cases of bivalent or trivalent groups there are corresponding replacements of two or three hydrogen atoms.

substituted mechanism a mechanism of an enzyme reaction in which the first substrate reacts with the original enzyme to produce a modified enzyme and one product. The modified enzyme then reacts with the second substrate to produce the second product and regenerate the original enzyme, e.g.



then



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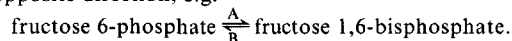
where E is the enzyme, E* the modified enzyme, S₁ and S₂ the substrates, and P₁ and P₂ the products.

substitution reaction the replacement of a substituent (normally hydrogen), usually at a carbon atom, by any other group.

substrate **1** a substance that is acted upon, especially by an enzyme; a molecule or structure whose transformation is catalysed by an enzyme. **2** or **substratum** **a** the base upon which an organism grows or lives, e.g. soil or rock. **b** the material upon which a microorganism grows or is placed to grow.

substrate activation the increase of an enzyme's activity by its substrate by an **allosteric** mechanism. Compare **substrate inhibition**.

substrate cycle interconversion of reactant and product by two enzymes, one of which (A) catalyses the reaction in one direction while the second enzyme (B) catalyses the reaction in the opposite direction, e.g.



As **flux** (def. 3) (i.e. the rate of reaction catalysed by A minus rate of reaction catalysed by B) through a system varies, the rate of each enzyme reaction may vary independently with concomitant changes in cycling rate. The existence of a substrate cycle increases sensitivity of control, since even if rates of forward and reverse reaction are high, changes in one or both may make relatively larger changes to flux; e.g. if rate of A = 9 and rate of B = 8, then flux = 1; if rate of A doubles to 18 and rate of B remains 8, then flux = 10 (i.e. increases tenfold).

substrate dissociation constant or (*formerly*) **substrate constant** *symbol* K_s ; the parameter of an enzymatic reaction that is the equilibrium (dissociation) constant for the reaction $ES = E + S$; i.e. $K_s = [E][S]/[ES]$.

substrate inhibition the inhibition of an enzyme's activity by its substrate by an **allosteric** mechanism. Compare **substrate activation**.

substrate-level phosphorylation the formation of ATP (or GTP) in a coupled reaction that is not linked to an electron transport system, e.g. coupled to the deacylation of succinyl coenzyme A or the conversion of 3-phosphoglyceroyl phosphate to 3-phosphoglycerate.

substrate-site hypothesis a hypothesis for the control of insulin release from pancreatic islet B cells by glucose. It postulates that the glucoreceptor is either a carrier transporting glucose into the cell or an enzyme phosphorylating glucose, and the signal for insulin release is a metabolite in the pathway of metabolism of glucose in the B cell. Compare **regulator-site hypothesis**.

substratum (*pl.* **substrata**) **1** any layer (stratum) or structure that lies beneath another, or that forms a base or underlying support for something else. **2** an *alternative term* for **substrate** (def. 2).

subtiligase one of several engineered derivatives of **subtilisin** that efficiently ligate peptides. The two mutations converting the peptidase, subtilisin, to a ligase involve Ser²²¹ to Cys and Pro²²⁵ to Ala.

subtilin or **subtilin C** an antibiotic produced by *Bacillus subtilis* and active against Gram-positive bacteria, *Neisseria* spp. (*N. catarrhalis*, *N. gonorrhoeae*), and some pathogenic fungi.

subtilisin EC 3.4.21.62; a bacterial endopeptidase enzyme isolated from *Bacillus subtilis* and other *Bacillus* spp. (e.g. commercially from *B. licheniformis*). It is a **serine endopeptidase** that catalyses the hydrolysis of proteins with broad specificity for peptide bonds, with preference for a large uncharged residue at P1 (see **peptidase P-sites**) and low activity for glutamyl residues; it hydrolyses peptide amides. Example from *B. subtilis*: database code NRL_1SBC, 274 amino acids (27.26 kDa); 3-D structure is known; the enzyme has similar active-site geometry to the trypsin family but otherwise no homology. See also **Nagarse proteinase**.

subtrahend (*in mathematics*) the number to be subtracted from another number (the *minuend*).



subunit

subunit any polypeptide component within a protein or other distinct biochemical entity, whether identical to or different from any other components, that is separable from such other components without rupture of covalent bonds. The term is sometimes used in a more restricted sense as referring to any chemically or physically identifiable submolecular entity within a protein, whether identical to, or different from, other components.

subunit-exchange chromatography a form of chromatography in which protein subunits are immobilized on a solid matrix and allowed to interact with protein subunits in solution. Quantification of subunit exchange between the matrix and solution may be used in analysing the association–dissociation properties of the protein system. The method also provides a powerful, specific purification method.

Suc- symbol for **succinyl** (def. 1).

-Suc- or **Suc<** symbol for **succinyl** (def. 2).

succinamic acid a trivial name for 3-carbamoylpropanoic acid, the monoamide of succinic acid.

succinamide a trivial name for butanediamide, the diamide of succinic acid.

succinate **1** a trivial name for butanedioate; ethanedicarboxylate; the dianion of succinic acid. It is an important intermediate in metabolism and a component of the **tricarboxylic-acid cycle**. **2** any mixture of free succinic acid and its mono- and dianions. **3** any salt or ester of succinic acid.

succinate-CoA ligase (GDP-forming) EC 6.2.1.4; *other name*: succinyl-CoA synthetase (GDP-forming); an enzyme of the tricarboxylic-acid cycle that catalyses the formation of succinyl-CoA from GTP, succinate, and CoA with release of GDP and orthophosphate. Examples: α -chain precursor from rat: database code SUCA_RAT, 333 amino acids (34.99 kDa); β chain (from pig): database code A44529, 417 amino acids (45.10 kDa).

succinate dehydrogenase EC 1.3.99.1; *systematic name*: succinate:(acceptor) oxidoreductase; *other names*: fumarate reductase/dehydrogenase; fumaric hydrogenase. A bacterial enzyme, or a degraded entity from **succinate dehydrogenase (ubiquinone)**, a major component of mitochondrial complex II (see **respiratory complex**). It catalyses the oxidation of succinate by an acceptor to form fumarate and a reduced acceptor; it does not react with ubiquinone. It is a flavoprotein (FAD) and plays an important role during anaerobic growth. Example from *Escherichia coli*; there are four subunits: (1) cytochrome *b*₅₅₆; (2) an iron–sulfur protein: database code DHSB_ECOLI, 238 amino acids (26.74 kDa); (3) a flavoprotein related to succinate dehydrogenase (ubiquinone): database code DHSA_ECOLI, 588 amino acids (64.33 kDa); and (4) a hydrophobic anchor protein.

succinate dehydrogenase (ubiquinone) EC 1.3.5.1; *systematic name*: succinate:ubiquinone oxidoreductase; *other name*: succinic dehydrogenase. An enzyme that catalyses the oxidation of succinate by ubiquinone to form fumarate and ubiquinol. It is an iron–sulfur flavoprotein (FAD), containing at least four different subunits: a flavoprotein, an iron–sulfur protein, and two hydrophobic anchor proteins. It is located in the mitochondrial inner membrane, where it is a major component of complex II (see **complex I, II, III, IV**) of the **respiratory chain**. Example, FAD, subunit (precursor) from bovine: database code DHSA_BOVIN, 664 amino acids (73.63 kDa). Two other examples (subunits, precursors, from yeast): flavoprotein subunit: database code DHSA_YEAST 640 amino acids (70.15 kDa); iron-sulfur protein subunit: database code DHSB_YEAST 266 amino acids (30.20 kDa).

succinate-semialdehyde dehydrogenase EC 1.2.1.24; an enzyme that catalyses the reaction:



Succinate semialdehyde can undergo transamination to form γ -aminobutyric acid. However, the transaminase primarily

functions to convert γ -aminobutyrate to succinate-semialdehyde, which is then converted to succinate by succinate-semialdehyde dehydrogenase in the catabolism of γ -aminobutyrate. See also **4-aminobutyrate transaminase**.

succinic dehydrogenase see **succinate dehydrogenase (ubiquinone)**.

succinimide a trivial name for butanimide; 2,5-pyrrolidinedione; the imide of succinic acid. For structure see **imide**.

succinoglycan an acidic exopolysaccharide needed for symbiosis between *Rhizobium meliloti* and alfalfa. Also synthesized by other bacteria, it is a polymer of octasaccharide subunits (one galactose and seven glucose residues) with acetyl, succinyl, and pyruvyl modifications.

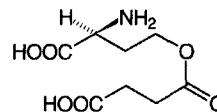
succinyl **1** (in biochemistry) symbol: Suc-; the univalent acyl group 3-carboxypropanoyl, HOOC-[CH₂]₂-CO-, derived from succinic acid by loss of one hydroxyl group. **2** (in chemistry) symbol: -Suc- or Suc<; the bivalent acyl group butanedioyl, -CO-[CH₂]₂-CO-, derived from succinic acid by loss of both hydroxyl groups.

succinylation the introduction of one or more **succinyl** (def. 1) groups into a substance by acylation, either chemically, e.g. with succinic anhydride, or enzymically, e.g. with succinyl-coenzyme A. Succinylation of a protein with succinic anhydride is used to acylate its free lysine (and other) amino groups in order to change their charge at neutral pH from positive to negative and to render adjacent peptide bonds resistant to hydrolysis by trypsin; it can also lead to disaggregation of a multimeric protein. The reaction occurs readily at pH 7–10 (at 0–25 °C), but it is irreversible, unlike the otherwise similar processes of **citraconylation** and **maleylation**, and so is of limited usefulness in protein sequence work.

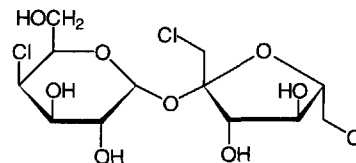
succinylcholine chloride see **suxamethonium**.

succinyl-CoA synthetase (GDP-forming) see **succinate-CoA ligase (GDP-forming)**.

O-succinylhomoserine an intermediate in methionine biosynthesis in plants and bacteria. It is formed by succinylation (utilizing succinyl-CoA) of homoserine. It reacts with cysteine with loss of succinate to form cystathionine.



sucralose 4,1',6'-trichloro-4,1',6'-trideoxy-galacto-sucrose; a chlorinated sucrose, tasting sweeter than sucrose itself.



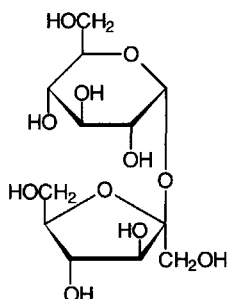
sucrase **1** *recommended name*: sucrose α -glucosidase; EC 3.2.1.48; *systematic name*: sucrose α -D-glucohydrolase; an α -D-glucosidase-type enzyme from intestinal mucosa that hydrolyses sucrose and maltose. It is also known as **sucrase-isomaltase** by virtue of its activity as an oligo-1,6-glucosidase, EC 3.2.1.10, displayed towards isomaltose. **2** a former alternative trivial name for **invertase**. **3** any of several bacterial or plant enzymes within sub-subclass EC 2.4.1 (hexosyltransferases) that synthesize various homopolysaccharides from sucrose by transfer of either its fructose moiety or its glucose moiety respectively to a fructan or to a glucan, with release of the opposing moiety as free D-glucose or D-fructose. The recommended names are formed by adding 'sucrase' to a prefix

sucrase-isomaltase

indicating the polysaccharide product. These enzymes include: (1) **amylsucrase**; EC 2.4.1.4; *other name*: sucrose-glucan glucosyltransferase; *systematic name*: sucrose:1,4- α -D-glucan 4- α -D-glucosyltransferase; it synthesizes amylose; (2) **dextran-sucrase**; (3) **inulosucrase**; EC 2.4.1.9; *other name*: sucrose 1-fructosyltransferase; *systematic name*: sucrose:2,1- β -D-fructan 1- β -D-fructosyltransferase; it synthesizes inulin; and (4) **levan-sucrase**; EC 2.4.1.10; *other name*: sucrose 6-fructosyltransferase; *systematic name*: sucrose:2,6- β -D-fructan 6- β -D-fructosyltransferase; it synthesizes levan.

sucrase-isomaltase *see* **oligo-1,6-glucosidase**.

sucrose or **saccharose** the trivial name for *O*- β -D-fructofuranosyl-(2 $\rightarrow$ 1)- α -D-glucopyranoside; a sweet-tasting, non-reducing disaccharide. The sugar of commerce, it is usually isolated industrially in crystalline form from sugar cane, *Saccharum officinarum*, or from sugar beet, *Beta vulgaris*.



sucrose gradient a density gradient made by mixing different concentrations of sucrose and layering them in a centrifuge tube. It is commonly used to separate molecules or particles on the basis of their sedimentation coefficients. *See also* **isopycnic centrifugation**.

sucrose-phosphate synthase EC 2.4.1.14; *systematic name*: UDPglucose:D-fructose-6-phosphate 2 α -D-glucosyltransferase; *other names*: UDPglucose-fructose-phosphate glucosyltransferase; sucrosephosphate-UDP glucosyltransferase. An enzyme that catalyses a reaction between UDPglucose and D-fructose 6-phosphate to form sucrose 6-phosphate and UDP. Example from *Zea mays* (maize): database code SPS_MAIZE, 1068 amino acids (118.44 kDa).

sucrose phosphorylase EC 2.4.1.7; *systematic name*: sucrose:orthophosphate α -D-glucosyltransferase; *other name*: sucrose glucosyltransferase. An enzyme that catalyses the phosphorylysis of sucrose to D-fructose and α -D-glucose 1-phosphate. Example from *Leuconostoc mesenteroides*: database code LEUSUCP, 490 amino acids (55.68 kDa).

sucrose synthase EC 2.4.1.13; *systematic name*: UDP glucose:D-fructose 2 α -D-glucosyltransferase; *other names*: UDPglucose-fructose glucosyltransferase; sucrose-UDP glucosyltransferase. An enzyme that catalyses a reaction between UDPglucose and D-fructose to form sucrose and UDP. Example, sucrose synthase 1 from *Hordeum vulgare* (barley): database code SUS1_HORVU, 807 amino acids (92.11 kDa).

Sudan the family name of a group of dyes that are soluble in organic solvents and are used for staining lipids. Sudan Red is the 2,2'-demethyl analogue of Oil Red O. Sudan Black B is used for staining chromosomes, Golgi apparatus, and leukocyte granules.

sugar a general and imprecise name used for any, usually sweet, soluble disaccharide or small oligosaccharide carbohydrate. More specifically it is applied to **sucrose**, the sugar of commerce. The name is also sometimes used as a synonym for **carbohydrate**. *See also* **beet sugar**, **blood sugar**, **cane sugar**, **corn sugar**, **fruit sugar**, **grape sugar**, **malt sugar**, **milk sugar**, **mushroom sugar**, **reducing sugar**.

sugar acid any **aldaric acid**, **aldonic acid**, or **uronic acid**.

sugar alcohol an old name for **alditol**.

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sugar beet the plant *Beta vulgaris*, which has a large root with a high sugar content, for which it is extensively cultivated.

sugar transporter any of various transmembrane proteins responsible for sugar transport into cells. There are several mechanisms, and different superfamilies of proteins are involved, characteristically with five motifs. For example, **GTR1** causes facilitated diffusion of glucose, whereas XylE protein in *Escherichia coli* acts as an H⁺-xylose symporter (*see* **xylE**). Both belong to the 12-transmembrane domain superfamily. Other types include the **ABC transporter** type and **PTS** type. *See* **glucose transporter**.

suicide enzyme *see* **suicide repair (of DNA)**.

suicide inhibitor a type of enzyme inhibitor that is distinguished from a classical active-site-directed enzyme inhibitor in two key respects: (1) the inhibitor itself is relatively unreactive and becomes reactive only after interaction with the enzyme's active site; and (2) conversion to the active form depends on the specific catalytic capabilities of the active site, i.e. some part of the enzyme's normal catalytic capability is required. *See also* **photosuicide inhibitor**.

suicide repair (of DNA) a response in both bacteria and mammalian cells to a situation in which exposure to a nontoxic alkylation dose causes the rapid induction of enzymes that counter the mutagenic effect by removal of **O⁶-methylguanine** formed by the alkylating agent. In the course of this repair reaction the methyl group is transferred to the enzyme, which is thereby inactivated (i.e. it is a **suicide enzyme**).

suicide substrate 1 a substrate that, in undergoing conversion by an enzyme, causes the enzyme to be modified in a way that permanently inactivates it. Hence the enzyme, by acting on the substrate, 'commits suicide'. 2 a compound that is converted by a metabolic pathway into an inhibitor of the pathway, as with the conversion of fluoroacetate to fluorocitrate. If the pathway is of sufficient importance, the cell thereby 'commits suicide'. *See also* **lethal synthesis**.

sulfa drug any member of a group of **sulfonamide** compounds that are clinically useful in the treatment of bacterial infections.

sulfamethoxazole 4-amino-N-(5-methyl-3-isoxalolyl)benzenesulfonamide; a **sulfonamide** antibacterial agent. As a mixture with **trimethoprim** it has the name **co-trimoxazole** (*proprietary names*: Bactrim; Septrim).

sulfane a chain of divalent sulfur atoms, as in hydrogen persulfide (HSSH) or polysulfanes (H₂S_x; e.g. H₂S₅ is pentasulfane).

sulfanilamide *p*-aminobenzenesulfonamide; *see* **sulfonamide**.

sulfatase *see* **arylsulfatase**.

sulfate or (*esp. Brit.*) **sulphate** the anion SO₄²⁻.

sulfate adenylyltransferase EC 2.7.7.4; *other names*: sulfate adenylyl transferase; ATP-sulfurylase; sulfurylase; an enzyme that catalyses the formation of **adenosine 5'-phosphosulfate** (adenylylsulfate) from ATP and sulfate with release of pyrophosphate. Example from *Escherichia coli*, two subunits: database code CYSD_ECOLI, 302 amino acids (35.16 kDa); database code CYSN_ECOLI, 475 amino acids (52.50 kDa).

sulfatide a **galactosyl cerebroside** with a sulfate radical esterified at the 3' position of the galactose moiety. Sulfatides are formed by **galactosylceramide sulfotransferase**.

sulfatide lipidosis an alternative name for **metachromatic leukodystrophy**.

sulfation factor a factor originally identified as promoting sulfate uptake by cartilage *in vitro*; it is now known as **somatomedin**.

sulfhemoglobin any of a group of poorly characterized derivatives of **hemoglobin**, often formed *in vivo* together with ferrihemoglobin. They are incapable of carrying oxygen and cannot be converted back to hemoglobin.

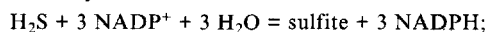
sulfhydryl or **thiol** the chemical group -SH; it is found in the amino acid **cysteine** and in other molecules. Two sulfhydryls can be oxidized to a disulfide bond, -S-S-.

sulfite or (*esp. Brit.*) **sulphite** the anion SO₃²⁻.

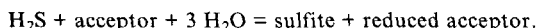
sulfite reductase 1 **sulfite reductase (NADPH)** EC 1.8.1.2;



systematic name: hydrogen sulfide:NADP⁺ oxidoreductase; the enzyme responsible for interconversion of hydrogen sulfide and sulfite in plants and most bacteria. It catalyses the reduction of sulfite, e.g. released from **3'-phosphoadenosine 5'-phosphosulfate** by thioredoxin. The reaction is:



cofactors are FAD, FMN, and siroheme. In plants and most microorganisms the H₂S reacts with *O*-acetylserine to form cysteine and acetate. The enzyme also oxidizes H₂S to sulfite. Example from *Thiocapsa roseopersicina*, two subunits; database code S34190, 522 amino acids (57.72 kDa); database code S34191, 561 amino acids (62.38 kDa). **2 sulfite reductase** EC 1.8.99.1; **systematic name:** hydrogen-sulfide:(acceptor) oxidoreductase. An iron-protein enzyme that catalyses the reaction:



Example from *Archaeoglobus fulgidus*, two subunits; database code S27478, 418 amino acids (47.47 kDa); database code S27479, 366 amino acids (41.52 kDa).

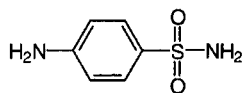
sulfmethemoglobin the complex formed by combination of sulfide with the ferric ion of **ferrihemoglobin** in poisoning by sulfide.

sulf(o)+ or (esp. Brit.) **sulph(o)+** of, or pertaining to, or containing a sulfur atom or group.

sulfokinase or **aryl sulfotransferase** EC 2.8.2.1; **other name:** phenol sulfotransferase; an enzyme that catalyses transfer of a sulfate radical from 3'-phosphoadenylylsulfate to a phenol to form an aryl sulfate.

sulfolipid any lipid containing sulfur, especially a **sulfatide**. See also **sulfonolipid**.

sulfonamide the amide of a sulfonic acid; the sulfonamide group is -SO₂-NH₂. In biology and medicine the name is used especially for any member of a large group of antibacterial substances – also known as **sulfa drugs** – derived from **sulfanilamide** (*p*-aminobenzenesulfonamide), many of which also have a substituent on the amide nitrogen.



sulfanilamide

sulfone the oxidation product that results when one oxygen atom is added to sulfur bonded to two alkyl chains.

sulfonolipid or **plant sulfolipid** 1,2-diacyl-3-(6-sulfo- α -D-quinovosyl)-sn-glycerol; sulfoquinovosyldiacylglycerol (**abbr.:** SQDG); the only sulfolipid present in higher plants and the only lipid with a sulfonic-acid linkage reported so far; the predominant fatty acid is α -linolenic acid. Sulfonolipid is localized in the chloroplasts, mainly in the membrane of the thylakoid system.

sulfoxide the oxidation product that results when two oxygen atoms are added to sulfur bonded to two alkyl chains.

sulfur or (esp. Brit.) **sulphur** **symbol:** S; a yellow nonmetallic element of group 16 of the IUPAC **periodic table**; atomic number 16; relative atomic mass 32.066. Its main oxidation states are -2, +4, and +6. Sulfur exists in a number of forms including crystalline (rhombic, monoclinic, rhombohedral) and amorphous forms. The relative abundance of stable isotopes is: sulfur-32 (mass 31.972) 95.02%; sulfur-33 (mass 32.971) 0.75%; sulfur-34 (mass 33.968) 4.21%. There are two radioactive isotopes, **sulfur-35** and **sulfur-38**. Sulfur is important as a constituent of the amino acids methionine and cysteine, and as a constituent of **glutathione**. It also occurs in **sulfolipids**. Sulfate is the terminal electron acceptor for anaerobic respiratory metabolism in certain bacteria e.g. *Desulfovibrio* spp.

sulfur-35 the radioactive nuclide, ³⁵S, with a mass of 34.969. It

emits an electron (β^- particle, 0.167 MeV), no gamma radiation, and has a half-life of 87.4 days.

sulfur-38 the radioactive nuclide, ³⁸S, with a mass of 37.971. It emits an electron (β^- particle, 3.0 MeV) and gamma radiation (1.94 MeV), and has a half-life of 2.87 h.

sulfurylase see **sulfate adenylyltransferase**.

sulph(o)+ a variant spelling (esp. Brit.) of **sulf(o)+**.

sum 1 the total value of a set of individual values. 2 to determine a sum.

sumatriptan 3-[(2-dimethylamino)ethyl]-*N*-methyl-1*H*-indole-5-methanesulfonamide; a preferential agonist for 5-HT_{1D} receptors (see **5-hydroxytryptamine receptor**). It causes constriction of cerebral blood vessels, and is used in the treatment of migraine and cluster headaches. **Proprietary name:** Imigran (succinate).

summation the determination of the **sum** (def. 1) of a series.

Sumner, James Batcheller (1887–1995), US biochemist; Nobel Laureate in Chemistry (1946) 'for his discovery that enzymes can be crystallized' [prize shared with J. H. Northrop and W. M. Stanley].

SUMT **abbr. for** *S*-adenosyl-L-methionine:uroporphyrin-III C-methyltransferase (see **uroporphyrin-III C-methyltransferase**).

super+ **prefix** 1 above, over, at the top (of). 2 of greater size or extent, or of higher quality. 3 beyond the usual or normal standard; exceeding. 4 (in chemical nomenclature) designating a compound that contains more than the usual proportion of a specified element. **Compare** **sub+**, **supra+**.

superantigen an antigen with the ability to activate a large proportion (up to 20%) of T lymphocytes in a given individual through interaction with the variable domain of the β chain of the lymphocyte's antigen receptor. This contrasts with conventional foreign antigens the peptides of which, attached to the major histocompatibility complex (MHC), have to be specifically recognized by the T-cell receptor. With superantigens the specificity resides in the β chain of the receptor not the MHC peptides. This accounts for the fact that many more than usual T lymphocytes are activated. Superantigens are present in a number of bacteria and viruses; the best example is the *Staphylococcus* enterotoxin. Superantigens have been implicated in **autoimmunity**.

supercoil or **superhelix** or **coiled coil** the form of DNA that can be envisaged as resulting if a **duplex** were turned through a hairpin bend and the two duplex rails twisted one around the other. DNA *in vivo* generally is supercoiled, and as it has a closed structure, a supercoil may be envisaged as a rubber band twisted to a helical shape. In **negative supercoiling** the DNA twists around its axis in the opposite direction from the clockwise turns of the right-handed double helix; this allows relief of torsional pressure, and the DNA is said to be **underwound**. Twisting in the other direction (**positive supercoiling**) increases torsional pressure, winding the double helix more highly; the DNA is said to be **overwound**. **Superhelix density**, σ , is defined as the number of turns of the superhelix, τ , per turn of the duplex, β ; i.e. $\sigma = \tau/\beta$ (β represents about 10 base pairs). **Linking number** is the number of times the duplex strands cross over each other (in projection onto a plane surface). The number of superhelical turns is called the **writting number** (W).

supercritical fluid chromatography a technique that can be regarded as an extension of **gas-liquid (partition) chromatography** in which the mobile phase is a relatively dense gas such as CO₂ (or SF₆, NH₃, or Xe). The gas must be pumped (at pressures up to 6000 psi) through a capillary column that is coated with stationary phases similar to those used in gas chromatography. The system depends on the partition of the compounds being separated between the stationary and mobile phases and has the advantage that compounds with high boiling points partition appreciably into the gas phase at temperatures well below their boiling points so that such compounds can be separated at lower temperatures than with less dense gases.

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superfamily

superfamily any group of genes and their cognate proteins that can be held to be related by sequence homology. The main criterion determining homology is conservation of residues that can be matched, if necessary, by alignment of the sequence. Normally, similarity of function is an additional criterion, but proteins apparently not related in function may be included if the degree of sequence homology is high. *See also* **Appendix E**.

superfolding (of proteins) the formation of small domains, beyond the level of simple secondary structures, that are energetically stable and contribute as units to the full tertiary structure of a protein. They may consist of **alpha helices** or **beta sheets** or both. Nine such protein folds have been identified in proteins having neither sequence nor functional similarities. In the folding of a protein, they may be formed initially before the protein fully folds. **Beta barrels** represent an example.

superfusion **1** the pouring of a liquid onto or over something. **2** (in physiology and medicine) a technique in which blood, plasma, or other fluid is allowed to drip onto, or flow over, the surface of an organ or piece of tissue.

supergene a group of linked genes (*see* **linkage** (def. 1)) that are usually inherited as a unit.

superhelix an alternative name for **supercoil**.

superhelix density *see* **supercoil**.

superinfection the proliferation of one or more pathogenic microorganisms arising from the disturbance of a mixed population of microorganisms normally sustained at non-pathogenic levels by secretion of endogenous antibiotics. Such disturbance can be induced by administration of exogenous antibiotic agents.

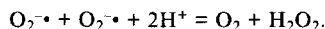
superinfection inhibition interference with the entry or establishment of an entering plasmid by a resident plasmid.

supernatant **1** floating on the surface of a liquid or lying over something. **2** (or **supernate**) a liquid lying above another liquid, a sediment, or a settled precipitate. **b** the fluid phase that remains after removal of solid material by centrifugation. A cell supernatant is the fluid phase that remains after centrifugation of a cell homogenate at high speed (characteristically at 100 000g for 30 min). *Compare* **infranatant**.

supernate an alternative name for **supernatant** (def. 2).

superoxide **1** the anion, O_2^- , formed by addition of one electron to **dioxygen**. **2** any compound containing the superoxide anion.

superoxide dismutase *abbr.*: SOD; EC 1.15.1.1; *systematic name*: superoxide:superoxide oxidoreductase; *other names*: cytocuprein; erythrocuprein; hemocuprein. Any of a group of metal-containing enzymes that bring about the **dismutation** of superoxide radicals to form dioxygen and hydrogen peroxide as follows:



It is important in removing the highly toxic superoxide radical. Some (e.g. those from *Neurospora*, yeast, erythrocytes, heart, peas) contain copper and zinc; others (e.g. those from *Escherichia coli*, *Streptomyces mutants*) contain manganese or iron. Eukaryotic cells have copper-zinc types in cytoplasm and manganese types in mitochondria. The copper-zinc types comprise a well-conserved family with four motifs. Examples: (1) copper-zinc type (bovine): database code NRL_2SODO, 151 amino acids (15.53 kDa); 3-D structure known; (2) manganese type (heterodimer) from human; chain A: database code NRL_IMSDA, 198 amino acids (22.18 kDa); 3-D structure known; chain B: database code NRL_IMSDB, 198 amino acids (22.18 kDa); 3-D structure known; (3) iron type (heterodimer) from *Pseudomonas putida*; chain A: database code NRL_3SDPA, 186 amino acids (20.47 kDa); 3-D structure known; chain B: database code NRL_3SDPB, 186 amino acids (20.47 kDa); 3-D structure known.

supported reagent any reagent linked to a **support medium**.

support medium any (usually solid or insoluble) medium to which a cell, reagent, or functional group can be attached.

suppression **1** (in molecular biology) the cancellation, or reversal of the effects, of one mutation by another mutation, except where the second mutation causes a single base change at the same point as an earlier **point mutation** (in which case the term **reversion** is used). In **intergenic suppression**, the effects of a mutation in one gene are reversed by a mutation in another gene, usually because the second gene codes for a mutant tRNA (*see* **frameshift suppressor**). In **intragenic suppression** a mutation is cancelled by a second mutation close to the first in the same gene, normally within the same triplet, giving rise to a codon that is compatible with translation to a functional protein product. As an example, if the first mutation changed AGT (leading to the RNA codon for Ser) to ATT (giving a nonsense codon), a suppressor mutation might change the triplet to ATA (giving the RNA codon for Tyr); if the change Ser to Tyr led to a functional protein, suppression would be achieved. *See also* **amber suppressor**, **ochre suppressor**, **opal suppressor**. **2** (in immunology) any attenuation of an immune response. The existence of a subpopulation of T cells, **suppressor cells**, is held to be responsible for attenuation of the effect of **helper cells**. A different mechanism is probably responsible for the reduction in antibody production arising from injection of antibody prior to administration of antigen. Allotype suppression refers to suppression of the production of one allotype (e.g. expression of the allotype inherited from the mother, and suppression of the allotype of the father).

suppressor cell or **suppressor T cell** or **T-suppressor cell** any of a subset of **T lymphocytes** that exert a negative regulatory effect (*see* **suppression** (def. 2)) on the stimulation of the immune response by **helper cells**. Classically they belong to the $CD8^+$ subset in the human. Induction of the suppressor function may derive from helper cells.

suppressor gene a gene that by its presence can reverse the effect of a specific type of mutation in another gene. *See* **suppression** (def. 1).

suppressor mutation a secondary mutation that totally or partially restores a function lost due to a primary mutation at another locus. *See* **suppression** (def. 1).

suppressor tRNA any species of **transfer RNA** (tRNA) that recognizes a **termination codon** in messenger RNA. It adds an amino-acid residue to what is normally the C terminus of a nascent polypeptide chain; hence termination of synthesis of the peptide is suppressed and a **readthrough** protein is consequently formed. Suppressor tRNA may occur naturally or be produced by mutagenesis.

supra+ *prefix* above, over, greater than. *Compare* **infra+**, **super+**.

supramolecular describing any chemical system or component thereof that exists and functions at a higher level of complexity than intact individual molecules. In biochemistry the term may refer to multienzymes, microtubules, organelles, biological membranes, viral envelopes, cell walls, ribosomes, or other cellular elements; hence, e.g., **supramolecular assembly**, **supramolecular complex**, **supramolecular structure**. *Compare* **submolecular**.

suprarenal gland an obsolete term for **adrenal gland**.

suprarenin a former name for **adrenaline** (i.e. **epinephrine**).

Sur *symbol* for a residue of the thiopyrimidine base **thiouracil**.

surface-active agent an alternative term for **surfactant**.

surface concentration *see* **concentration** (def. 1).

surface-stress theory a theory that considers the problem of mechanical stress in relation to enlargement of the bacterial cell wall during growth. The cell wall withstands stress from hydrostatic pressure by a stress-bearing, two-dimensionally linked peptidoglycan sacculus that surrounds the cell. During enlargement, enzymes that enlarge the wall must at the same time maintain the hydrostatic integrity of the cell by withstanding stress.

surface tension *symbol*: γ or σ or γ ; the property that makes a liquid behave as if it had an elastic skin on its surface – at an interface with a gas or an immiscible liquid – as a result of intermolecular forces tending to pull the molecules in the sur-



surfactant

face into the interior of the liquid. It is defined as the force acting tangential to the surface on one side of a line of unit length in the surface, and is expressed in newtons per metre. It can also be defined as the work required to produce unit increase in surface area, expressed in joules per square metre.

surfactant or **surface-active agent** any substance, such as a **detergent** or an **emulsifier**, that can bring about a reduction of the **surface tension** of a liquid, allowing it to foam, to penetrate porous solids more easily, and to wet surfaces of nonporous solids or of immiscible liquids.

surfactin a surface-active antibiotic and nonimmunogenic hemolysin produced extracellularly by *Bacillus subtilis*. It consists of a nonapeptide (containing *inter alia* two residues each of L- and D-leucine) in cyclic depsipeptide linkage with β -hydroxy- ω -methylmyristic acid.

surrogate genetics a branch of molecular genetics in which DNA is modified *in vitro* and then reintroduced into a genetic system – the ‘surrogate’ system – e.g. by injection into *Xenopus* oocytes, by transfection of cells, or by the creation of transgenic animals. The effect of the modification on the expression of the gene provides insights into the role in the genetic system of particular base sequences.

surveyor substrate one of a system of fluorescent substrates designed to measure distances within the active centre of an enzyme and also to survey other aspects of the topography of the active centre during the catalytic event.

survivin a protein expressed in the G₂/M phase of the cell-division cycle that associates with microtubules of the mitotic spindle. It inhibits **apoptosis**.

Sushi domain a polypeptide domain of approximately 60 amino acids that contains disulfide bonds between the first and third and second and fourth cysteines, as found in β_2 glycoprotein-1 of plasma, in complement proteins, and in **blood coagulation** factor XIII.

suppension culture the culture of cells in medium without adherence to a stationary substrate. Relatively few animal cell types are amenable to suspension culture (classically **HeLa cells** can be so grown). An intermediate situation utilizes beads to which the cells adhere; the beads can be maintained in suspension and collected by sedimentation.

Sutherland, Earl Wilbur Jr (1915–74), US biochemist; Nobel Laureate in Physiology or Medicine (1971) ‘for his discoveries concerning the mechanisms of the action of hormones’; these included the discovery of cyclic AMP.

suxamethonium succinylcholine chloride, 2,2′-[(1,4-dioxo-1,4-butanediyl)bis(oxy)]bis[*N,N,N*-trimethylethanaminium] dichloride; a nicotinic receptor ligand that causes sustained activation of the receptors leading to blockade of synaptic transmission (‘depolarizing block’). It is used as a neuromuscular blocking agent in surgery. It acts by mimicking acetylcholine at the neuromuscular junction, but disengagement from the receptor site and subsequent breakdown are slower than for acetylcholine. It is hydrolysed by serum cholinesterase and recovery is spontaneous. **Decamethonium** was the first drug of this type to be developed. Both of these compounds depend for their action on the N^+-N^+ distance of 1.1 nm. *Compare* **curare**, **tubocurarine**.

Sv symbol for 1 sievert. 2 svedberg (alternative to S).

SV40 abbr. for simian virus 40.

svedberg or **Svedberg unit** symbol: S or Sv; a coherent non-SI unit used as a measure of **sedimentation coefficient**; 1 S = 10^{-13} s.

Svedberg, Theodor (1884–1971), Swedish colloid chemist noted for his invention (in 1923) of the ultracentrifuge; Nobel Laureate in Chemistry (1926) ‘for his work on disperse systems’.

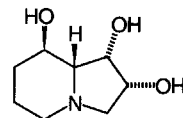
Svedberg centrifuge a centrifuge, designed and built by **Svedberg** in 1923, that was the forerunner of the **ultracentrifuge**.

Svedberg equation an equation relating the relative molecular mass, M_r , of a solute to its sedimentation velocity in an applied centrifugal field. If s is the sedimentation coefficient, D is the diffusion coefficient, $\bar{v}$ is the partial specific volume of

the solute, ρ is the density of the solution, R is the gas constant, and T is the thermodynamic temperature, then

$$M_r = sRT/D(1 - \bar{v}\rho),$$

swainsonine 8 α , β -indolizidine-1 α ,2 α ,8 β -triol; an inhibitor of a number of α -mannosidases, particularly Golgi mannosidase II (see **Golgi mannosidase**), an important enzyme in the synthesis of the core oligosaccharide of a number of membrane glycoproteins. It inhibits the growth of tumours and prevents metastases in mice. [After its source, *Swainsona canescens*].



swaposin a form of protein homology in which domains, characteristically found in one protein, are found arranged in other proteins in a different order. The phenomenon was first commented on in relation to certain plant aspartic peptidases, which contain saposin-like domains swapped into a different order to that found in **saposins** (hence the name). In another example some **pleckstrin homology domains** are found to be interrupted by inserts of **SH domains**.

sweetener any compound intended as a substitute for sucrose in sweetening food or drink, and aimed at avoiding the possible harmful effects of sucrose. Such compounds therefore should be nonnutritive but with a taste and aftertaste similar to those of sucrose; they should also be nonharmful. They include **aspartame**, **lactitol**, **mannitol**, **monellin**, **saccharin**, **stevioside**, **sucralose**, and **volemitol**. Sodium cyclamate was used as a sweetener until it was banned in the USA in 1970 because of its carcinogenicity.

sweetness one modality of the sense of taste, characterized by the effect of certain sugars such as sucrose and fructose.

swinging-bucket rotor a type of centrifuge head in which the centrifuge tubes are placed in buckets that swivel on trunnions so that at speed the tube is oriented with its axis parallel to the centrifugal force (i.e. horizontal). Such a head is essential for zonal centrifugation through density gradients as angled tubes induce mixing. *Compare* **angle head**.

SwissProt a protein sequence database maintained by the University of Geneva. It contains sequences of many proteins and references to articles in which the sequences were reported, together with a limited amount of general information and the name of the gene that encodes the protein.

switch peptide the flexible covalent connection between the **variable region** and **constant region** of an immunoglobulin molecule.

switch recombinase see **homothallic switching endonuclease**.

swivelase see **DNA swivelase**.

SYBASE a relational database used, e.g., for **HUGO**.

sym+ a variant form of **syn+** (before b, m, or p).

sym- prefix (in *chemical nomenclature*) denoting symmetric; e.g. *sym*-trinitrobenzene.

symbiont an organism that lives as a partner in a **symbiosis**.

symbiosis (*pl.* **symbioses**) a long-term association between individuals belonging to two different species. The term is often used in a restricted sense to denote associations that are beneficial to one or both partners, although strictly it refers equally to neutral or harmful associations. —**symbiotic** *adj.*

symmetry precise correspondence between two halves with respect to both size and shape. A molecule has an n -fold **axis of symmetry** if rotation around this axis through an angle of $360^\circ/n$ produces an arrangement indistinguishable from the original (n being a small integer). For example, water has a twofold ($n = 2$) axis of symmetry. A plane of symmetry is a plane drawn through a molecule such that a mirror placed in



sympathectomy

this plane forms an image of one half of the molecule that is superimposable on the other half. For example, a plane of symmetry is present in *meso-tartaric acid*. An alternating axis of symmetry or rotation-reflection axis is an axis such as is present in a molecule when rotation around the axis through an angle of $360^\circ/n$ followed by reflection in a plane perpendicular to the axis produces a molecule superimposable on the original molecule. For example, a compound with such a twofold ($n = 2$) axis is fumaric acid. A centre of symmetry is a position (point) such that if a straight line is drawn from every atom to the point, and is continued beyond the point, an equivalent atom is found at a distance equal to that between the point and the original atom. A molecule with such a centre has a twofold alternating axis of symmetry. An example of a molecule with a centre of symmetry is fumaric acid.

sympathectomy the surgical section of sympathetic nerve fibres.

sympathetic nervous system one of the two divisions of the autonomic nervous system the other being the parasympathetic nervous system. Most, but not all, sympathetic nerves release norepinephrine as a neurotransmitter.

sympatheticomimetic an alternative term for sympathomimetic.

sympathomimetic or sympatheticomimetic 1 describing a substance or action that stimulates the sympathetic nervous system; adrenergic. 2 a sympathomimetic agent, especially a sympathomimetic amine.

symport or **cotransport** the process of solute translocation in which two solutes equilibrate across an osmotic barrier, and the translocation of one solute is coupled to the translocation of the other in the same direction. Compare **antiport**.

symporter any substance or structural feature that promotes the exchange diffusion of two specific substances across a membrane by **symport**.

syn- *abbr.* for **synclinal** (def. 1); see also **conformation**.

syn+ *comb. form* signifying similarity, association, or concerted action.

Synacthen see **tetracosactide**.

synaeresis a variant spelling of **syneresis**.

synapse or (rare) **synapsis** the junction between a nerve fibre of one neuron and another neuron; the site of interneuronal communication. As the nerve fibre approaches the synapse it enlarges into a specialized structure, the **presynaptic nerve ending**, which contains mitochondria and **synaptic vesicles**. At the tip of the nerve ending is the **presynaptic membrane**; facing it, and separated from it by a minute cleft – the **synaptic cleft** – is a specialized area of membrane on the receiving cell, known as the **postsynaptic membrane**. In response to the arrival of nerve impulses, the presynaptic nerve ending secretes molecules of **neurotransmitters** into the synaptic cleft. These diffuse across the cleft and transmit the signal to the postsynaptic membrane. See also **neuromuscular junction**. —**synaptic adj.**

synapsin I a major neuron-specific protein present in virtually all synaptic vesicles. Phosphorylated at multiple sites by Ca^{2+} -calmodulin-dependent and cyclic AMP-dependent protein kinases, it coats synaptic vesicles and binds to the cytoskeleton. Example, precursor for human synapsin IA: database code SYN1_HUMAN, 705 amino acids (73.92 kDa); synapsin IB is derived by alternative splicing; synapsins II are related.

synapsis 1 the pairing of chromosomes during **meiosis**. 2 a less common word for **synapse**.

synaptic cleft the extracellular space, typically some 20 nm across, that separates the outer membrane of the presynaptic nerve ending from the postsynaptic membrane of the receiving neuron (or other target cell) in a **synapse**. Chemical mediation of nerve transmission occurs across the synaptic cleft.

synaptic vesicle a vesicle, 20–65 nm in diameter, occurring within the cytoplasm of the presynaptic nerve ending and often closely related to the presynaptic membrane (see **synapse**). Synaptic vesicles store **neurotransmitters** and release them by fusing with the presynaptic membrane.

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synaptic vesicle-associated membrane protein another name for **synaptobrevin**.

synaptobrevin or **sinaptobrevin** or **synaptic vesicle-associated membrane protein** (*abbr.*: VAMP) a SNAP receptor involved in the docking and fusion complex (see **SNARE**) of synaptic vesicles and playing a central role in neural exocytosis. Synaptobrevin isoforms 1 and 2 (VAMP-1 and VAMP-2) are known. Both are expressed in all the main rat tissues, but differential expression is found, e.g. VAMP-1 is restricted to exocrine pancreas and kidney tubular cells, VAMP-2 predominates in islets of Langerhans and kidney glomerular cells. Example from *Drosophila melanogaster* (similar to fish and mammalian proteins): database code SYB_DROME, 152 amino acids (16.69 kDa).

synaptojanin a protein found in brain in nerve terminals, that has inositol 5-phosphatase activity and binds to the SH3 domain of **amphiphysin**, colocalizing with that protein in double immunofluorescence micrographs. It appears to participate with **dynamin** in synaptic vesicle recycling. It has homology with the yeast protein Sac1, which is genetically implicated in phospholipid metabolism and the function of the actin cytoskeleton. mRNA sequence from *Rattus norvegicus*: database code RNU45479.

synaptonemal complex a structure that holds paired chromosomes together during prophase I of **meiosis** and that promotes genetic recombination.

synaptophysin an integral membrane glycoprotein that occurs in presynaptic vesicles of neurons and in similar vesicles of the adrenal medulla. A marker protein for neuroendocrine cells and neoplasms, it has the attributes of a vesicular channel protein. Example from rat: database code SYN_RAT, 265 amino acids (29.03 kDa).

synaptosome any of the discrete particles (nerve-ending particles) formed from the clublike presynaptic nerve endings that resist disruption and are snapped or torn off their attachments when brain tissue is homogenized in media isosmotic to plasma. In synaptosomes the main structural features of the nerve endings are preserved.

synaptosome-associated protein see **SNAP** (def. 2).

synaptotagmin an abundant integral membrane protein of synaptic vesicles and chromaffin granules with a proposed role in membrane interactions during trafficking of synaptic vesicles at the active zone of the synapse. The primary structure contains two copies of a repeat that is homologous to the regulatory domain of protein kinase C. Example, homotetramer, human synaptotagmin I (*other name*: p65): database code SYT1_HUMAN, 422 amino acids (47.52 kDa).

synarchic regulation see **synarchy**.

synarchy cooperation between two intracellular messengers (e.g. cyclic AMP and Ca^{2+}) that act in concert to activate intracellular pathways.

syncatalytic synchronous with the catalytic process.

synchrotron an apparatus used in nuclear physics that accelerates particles of atomic size through an electric field. The particles are initially charged by an ion source, and their path may be linear (**linear accelerator**) or of spiral form (**cyclotron**). The acceleration imparts a high kinetic energy to the particles, which can lead to structural changes in the particles. One use of a synchrotron, therefore, is the production of radioactive or other nuclides for use in biology or medicine. A synchrotron radiation source is a circular storage ring in which electrons are accelerated in a magnetic field.

synclinal 1 *abbr.*: **syn-**; describing one of the two possible conformations of the base and sugar ring in a nucleotide, the other being the **anticlinal** conformation. See **conformation**. 2 or (sometimes) **gauche** or **skew** *abbr.*: **sc-**; describing a particular staggered conformation of a molecule in which the **torsion angle** is within $\pm 60^\circ$ of 0° . See **conformation**.

syncytium (*pl.* **syncytia**) a mass of cytoplasm containing several nuclei enclosed within a single plasma membrane. Syncytia are normally derived from single cells that fuse or fail to



complete cell division. Examples are muscle fibres, formed by fusion of **myoblasts**, and residual bodies formed during **spermatogenesis**.

SYND2 *abbr.* for syndecan-2. See **syndecan**.

syndecan a type I integral membrane proteoglycan that links the cytoskeleton to the interstitial matrix. It bears heparan sulfate and chondroitin sulfate and binds to bFGF (see **fibroblast growth factor**) with high affinity, and also to components of the extracellular matrix. Its intracellular domain binds to the actin cytoskeleton. Examples (precursors) from human, syndecan-1: database code SDC1_HUMAN, 310 amino acids (32.44 kDa); syndecan-2 (*abbr.*: SYND2); (*other names*: fibroglycan; heparan sulfate proteoglycan core protein; (*abbr.*: HSPG); database code SDC2_HUMAN, 201 amino acids (22.15 kDa).

syndein *an alternative name for ankryin*.

syndesine a di(α -amino acid) isolated from hydrolysates of certain **collagens**, in which it forms covalent cross-links between adjacent polypeptide strands. It is formed by **aldol condensation** between the side chains of a residue of **allysine** and one of **hydroxyallysine**; two isomers are possible.

syndiotactic having or relating to a regular alternation of differences in stereochemical structure in the repeating units of a polymer. *Compare isotactic*.

syndiotactic polymer a regular polymer composed of molecules having a regular repetition of alternating enantiomeric configurational base units at least at one main-chain chiral or prochiral atom in the configurational base unit.

syndrome a set of concurrent symptoms characteristic of a particular pathological state.

synemin a protein, M_r 230 000, that is found in the Z disk of skeletal and cardiac muscle cells and in **intermediate filaments** of some nonmuscle cells. It binds tightly to **desmin**.

synenkephalin or **proenkephalin 1-70** a derivative of **proenkephalin** secreted as an intact molecule or as a part of precursors in the adult brain and adrenal medulla. It has the immunoreactivity of proenkephalin but does not contain the enkephalin sequence.

syneresis or **synaeresis** **1** (*in chemistry*) the separation of liquid from a gel caused by contraction of the gel structure. **2** (*in pathology*) the gradual contraction of a blood clot that occurs after its formation, leading to a more solid mass that seals the ruptured blood vessel.

synergism **1** the phenomenon whereby the effect of two or more agonists together is greater than the sum of their effects when used individually. **2** the concerted action of two muscles to enhance their combined effects. *Compare antagonism*. —**synergistic** or **synergistic adj.**

synergist **1** an agent that increases the effectiveness of another agent. **2** a muscle that acts in concert with another muscle to increase its effect. *Compare antagonist*.

synexin or **annexin VII** a cytosolic Ca^{2+} -dependent lipid-binding protein belonging to the **annexin** family. Originally isolated from adrenal medullary cells, isoforms have since been found in skeletal muscle, brain, and liver, and also in *Dictyostelium discoideum*. Adrenal medulla and liver have isoforms of 47 kDa and 51 kDa, while skeletal muscle has only the 51 kDa isoform. Synexin promotes membrane fusion, aggregates isolated chromaffin granules, and forms voltage-dependent calcium channels in artificial and natural membranes.

syngamy the union of the nuclei of two gametes to form the single nucleus of the zygote during fertilization; sexual reproduction. —**syngamic** or **syngamous adj.**

Synge, Richard Laurence Millington (1914–94), British biochemist and peptide chemist; Nobel Laureate in Chemistry (1952) jointly with A. J. P. Martin 'for their invention of partition chromatography'.

syngeneic of or pertaining to cell types in an experimental mammalian chimera that are antigenetically similar. *Compare allogeneic*.

synonym codon a **synonym triplet** in messenger RNA.

synonymous mutation *an alternative name for silent mutation*.

synonym triplet any of the two or more sequences of three purine and/or pyrimidine bases in nucleic acids that may specify the same amino acid. In messenger RNA the term is synonymous with **synonym codon**.

synovia or **synovial fluid** the fluid surrounding a joint or filling a tendon sheath.

synovial membrane or **synovium** the membrane of mesothelial cells and connective tissue that surrounds a joint; it secretes the **synovia**.

synovium *an alternative name for synovial membrane*.

syneriplanar *see conformation*.

syntaxin a SNAP receptor (**SNARE**) involved in docking of synaptic vesicles at the presynaptic zone of a **synapse**. Example, syntaxin A from rat: database code SYNA_RAT, 288 amino acids (33.03 kDa).

syntenic genes gene loci or genes that lie on the same chromosome.

synthase a generic name that may be added to the name of a substance to designate an enzyme catalysing a reaction in which the named substance is synthesized. The synthases are distributed amongst various systematic classes of enzymes – e.g. glycogen synthase is in class EC 2 and malate synthase is in class EC 4 – although until 1984 no enzyme of class EC 6 (ligases) could be so named. Now the term is used also to form the recommended names of a few enzymes where the reaction is complex and there is a common name for the product (e.g. EC 6.3.2.11, carnosine synthase). The majority of enzymes in the EC 6 class now have recommended names ending in **ligase**, e.g. EC 6.5.1.3, RNA ligase (ATP), which has also been termed polyribonucleotide synthase (ATP). In the past, the term synthase was frequently used interchangeably with **synthetase**.

synthase-phosphorylase kinase EC 2.7.1.37; *recommended name*: protein kinase; *other names*: phosphorylase *b* kinase kinase; glycogen synthase *a* kinase; hydroxyalkyl-protein kinase; serine(threonine) protein kinase. A serine/threonine **protein kinase** that catalyses phosphorylation of several proteins as indicated by the names above; it is activated by phosphorylation by **cyclic AMP-dependent protein kinase**. In addition to regulating the glycogen phosphorylase/synthase enzymes, it can activate the $\text{ATP}, \text{Mg}^{2+}$ -dependent (AMD) protein phosphatase and other proteins. Example, GSK-3 (rat), a serine protein kinase implicated in the hormonal control of several regulatory proteins including glycogen synthase, Myb protein, and the transcription factor product of *c-jun*; monomer: database code KG3B_RAT, 420 amino acids (46.69 kDa). *See also phosphorylase kinase*.

synthesis (*in chemistry*) the production of a more complex molecule from simpler reagents by one or a series of reactions. —**synthesize** or **synthesise vb.**

synthesizer or **synthesiser** an instrument that automates a chemical synthesis, especially of peptides or oligonucleotides.

synthetase **1** an old class name, still widely used but no longer recommended, for any enzyme of class EC 6; these enzymes are now classified under the systematic class name **ligase**. **2** a generic name that may be added to the name of a substance to form the common name of any ligase-class enzyme that catalyses a reaction producing the named substance with concomitant breaking of a diphosphate linkage in a nucleoside triphosphate; e.g. acetyl-CoA synthetase (EC 6.2.1.1, acetate-CoA ligase), glutamine synthetase (EC 6.3.1.2, glutamate-ammonia ligase is recommended). Until 1984 names so formed were often the recommended names of such enzymes; now, the generic names **ligase** or **synthase** are used to form the recommended names, although common names formed from synthetase are still acceptable. In the past, the term synthetase was frequently used interchangeably with **synthase**.

synthetic **1** of, or pertaining to, **synthesis**. **2** (of a substance or material) chemically synthesized as opposed to prepared from recognizable natural materials. Such a product is identical chemically to that isolated from the natural source.

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synxenic

synuclein a brain presynaptic protein that is expressed also in other tissues, but at very low levels. It is highly conserved, rodent and zebrafish α synucleins being 95% and 86% identical to human. Some mutations are linked to familial **Parkinson's disease**. Example: SYUA_HUMAN, 140 amino acids (14.46 kDa).

synxenic pertaining to the growth of organisms of a single species in the presence of one or more other species. A culture in which one other species is involved is described as **monoxenic**; with two other species it is **dixenic**; and with three it is **trixenic**. Cultures with more than three additional species are described as **polyxenic**. Compare **axenic**. —**synxenity** *n.*; **synxenize** or **synxenise** *vb.*

synxenite the principal organism being studied in a **synxenic** culture.

syrup or **sirup** **1** a thick, sticky, viscous liquid consisting of a concentrated aqueous solution of sucrose. **2** the concentrated juice of a fruit or plant, especially of sugar cane. **3** any solution that is thick, sticky, and viscous.

system A one of the transport systems responsible for the uptake of neutral amino acids by animal cells. It is most reactive with amino acids having short, polar or linear side chains, including the nonmetabolizable 2-aminoisobutyric acid. Transport is sodium ion-dependent and is reduced by lowered extracellular pH. **System ASC**, prominent in erythrocytes, is distinguished from system A on the basis of pH insensitivity and higher stereospecificity. See also **sodium/neutral amino acid cotransporter**, **system L**, **system N**.

system ASC see **system A**.

systematic name 1 (*in organic-chemical nomenclature*) a name of a molecular entity that is composed wholly of specially coined or selected syllables, with or without numerical affixes and locants; e.g. oxazole, pentane, cyclohex-2-en-1-ol. Such a name unequivocally specifies the complete chemical structure of a particular molecular entity. Compare **semisystematic name**, **trivial name**. **2** (*in enzyme nomenclature*) a name of an enzyme that describes its catalytic action as exactly as possible, thus identifying the enzyme precisely. Such names are formed in accordance with definite rules, which provide the basis for classifying enzymes according to the type of reaction catalysed and for assigning code numbers to them. One important consequence of this method of classifying is that, with certain exceptions, all proteins having the same catalytic property share the same systematic name and code number irrespective of their source and structure. To simplify classification, the direction of the reaction catalysed is chosen to be the same for all enzymes in a given class, even if only the reverse reaction has been actually demonstrated experimentally for a particular enzyme. A systematic enzyme name consists of two parts: the first contains the name of the substrate or, in the case of a bimolecular reaction, of the two substrates separated by a colon (with small and equal spaces before and after the

colon); and the second, ending in '+ase', indicates the nature of the catalysed reaction.

systematics *functioning as sing.* **1** the study of systems, especially the principles of classification. **2** (*in biology*) the discipline that deals with classification and nomenclature.

Système International *abbr.*: SI; **Système International d'Unités**; a system of units based on the metric system. See **SI units**.

systemic concerning the whole body of an animal or a whole plant rather than an individual part.

systemic circulation the blood vessels supplying all parts of the body except the lungs.

systemin an 18-residue polypeptide, isolated from tomato leaves, that acts as a potent activator of plant proteinase inhibitor genes. It induces synthesis of two proteinase inhibitors in tomato and potato leaves. It is the only plant polypeptide hormone-like signalling molecule presently known. It has the sequence

AVQSKPPSKRDPPKMQTD,

and is proteolytically processed from a 200-residue precursor, **prosystemin**. Example: database code SYST_LYCES, 200 amino acids (23.00 kDa). Plant proteinase inhibitors are induced by wounding, e.g. by insects or pathogens, and may be part of the plant defence mechanisms.

system L one of the transport systems responsible for the uptake of neutral amino acids by animal cells. It is most reactive with branched-chain and aromatic amino acids such as leucine, isoleucine, valine, phenylalanine, and the nonmetabolizable analogue 2-aminobicyclo-[2,2,1]-heptane-2-carboxylic acid (BCH). The transport mechanism is sodium-ion-independent and may be stimulated by lowered extracellular pH. **System L** has been described in a wide variety of animal cell types, including avian and mammalian. Example from *Pseudomonas aeruginosa*: database code BRAZ_PSEAE, 437 amino acids (45.27 kDa). See also **sodium/neutral amino-acid cotransporter**, **system A**, **system N**.

system N one of the transport systems responsible for the uptake of amino acids by animal cells. It is a sodium-ion-dependent system specific to asparagine, glutamine, and histidine. See also **sodium/neutral amino acid cotransporter**, **system A**, **system L**.

systole a rhythmical repeated contraction, as of the chambers of the heart or of a pulsating vacuole. Compare **diastole**. —**systolic** *adj.*

Szent-Györgyi Albert Szent-Györgyi von Nagyrápolt (1893–1986), Hungarian-born US biochemist distinguished for his work in a variety of fields, including the isolation of ascorbic acid, the role of dicarboxylic acids in oxidative metabolism, and the biochemistry of muscle contraction; identified **adenylic acid**; Nobel Laureate in Physiology or Medicine (1937) 'for his discoveries in connection with the biological combustion processes, with special reference to Vitamin C and the catalysis of fumaric acid'.

Tt

t 1 symbol for a tert+ (i.e. **tertiary**) in a chemical name (alternative to *t*). **b** **triton** (def. 2). **2** *abbr.* for terminator sequence.

t symbol for 1 time. **2** Celsius temperature (alternative to θ). **3** transference number.

t- *abbr.* for 1 tert-. **2** *trans*-.

t_{1/2} symbol for half-life. *See also* **T**_{1/2}.

T symbol for 1 tera+ (SI prefix denoting 10¹² times). **2** tritium. **3** tesla. **4** a residue of the α -amino acid L-threonine (alternative to Thr). **5** a residue of the base thymine in a nucleic-acid sequence. **6** a residue of the ribonucleoside ribosylthymine (not thymidine, i.e. deoxyribosylthymine; alternative to Thd). **7** twist number.

T symbol for 1 thermodynamic temperature. **2** transmittance. **3** period. **4** kinetic energy (alternative to E_K or K).

-T conformational descriptor designating the **twist conformation** (def. 2) of a five-membered ring form of a monosaccharide or monosaccharide derivative. Locants of ring atoms that lie on the side of the structure's reference plane from which the numbering appears clockwise are indicated by left superscripts and those that lie on the other side of the reference plane are indicated by right subscripts; e.g. 1,2-*O*-isopropylidene- β -L-idofuranose-³T₂. *See also* **conformation**.

T_{1/2} symbol for half-life of a radionuclide (alternative to $t_{1/2}$).

T₁ symbol for spin-lattice relaxation time, longitudinal relaxation time.

T₂ symbol for spin-lattice relaxation time, transverse relaxation time.

T_m or **t**_m or **θ** _m symbol for melting temperature.

T_t symbol for transition temperature.

T1 *abbr.* for bacteriophage T1, a **T-odd phage**.

T2 1 or **T**₂ symbol for L-diiodothyronine. **2** *abbr.* for bacteriophage T2, a **T-even phage**.

T3 1 or **T**₃ symbol for L-triiodothyronine. **2** *abbr.* for bacteriophage T3, a **T-odd phage**.

T4 1 or **T**₄ symbol for L-thyroxine. **2** *abbr.* for bacteriophage T4, a **T-even phage**.

T6 *abbr.* for bacteriophage T6, a **T-even phage**.

T7 *abbr.* for bacteriophage T7, a **T-odd phage**.

tA symbol for N⁶-(threonylcarbamoyl)adenosine; *see* **hypermodified base**.

Ta symbol for tantalum.

tachometer an instrument that measures the speed at which any object is rotating. —**tachometric** *adj.*; **tachometry** *n*.

tachy+ *comb. form* denoting fast or rapid.

tachykinin any of a group of closely related vasoactive peptides characterized by a rapid stimulant effect on vascular (and extravascular) smooth muscle, producing hypotension (compare **bradykinin**). They also directly effect nervous tissue and stimulate salivary and lachrymal secretion. Naturally occurring tachykinins are oligopeptide amides containing 10–12 amino-acid residues and terminating in the common sequence -Phe-Xaa-Gly-Leu-Met-NH₂ (where Xaa = Phe, Tyr, Ile, or Val). Mammalian tachykinins include neuromedin K, **neurokinin A** (formerly substance K), and **substance P**; others (e.g. **eledoisin**, **physalaemin**) occur in amphibians and octopods. Substance P and neurokinin A are generated from a prepro-tachykinin protein precursor (*see* **protachykinin β precursor**). The primary mRNA transcript is differentially processed into three mature mRNAs: α , β , and γ ; the β and γ prepro-tachykinin mRNAs are predominant in the rat, each encoding substance P and neurokinin A. Example, human (precursor): database code TKNB_HUMAN, 129 amino acids (15.00 kDa).

tachykinin receptor any of a number of membrane proteins that bind **tachykinins** and mediate their transmembrane action. They fall according to the relative potency of agonists into three classes: (1) NK₁ (407 amino acids, human), agonist po-

tenacy: substance P (SP) > neurokinin A (NKA) > neurokinin B (NKB); (2) NK₂ (398 amino acids, human), agonist potency: NKA > NKB >> SP; and (3) NK₃ (452 amino acids, rat), agonist potency: NKB > NKA > SP. All are of the seven-transmembrane-domain type and share significant sequence similarity with G-protein-coupled receptors. The effector pathway in all cases is by activation of phosphoinositide-specific phospholipase C. Example of putative receptor from human: database code TXKR_HUMAN, 440 amino acids (49.43 kDa).

tachyphylaxis a decline in the response to repeated applications of doses of agonists. *See also* **desensitization**, **fade**.

tachyplesin any of a group of arthropodous peptide antibiotics, 17 or 18 amino acids long and with a C-terminal arginine α -amide. They are abundant in acid extracts of hemocyte debris of horseshoe crabs (*Tachyplesus* spp.). They strongly inhibit the growth of Gram-positive and Gram-negative bacteria, and of fungi such as *Candida albicans*. Example from *Tachyplesus gigas*: database code TAC3_TACG1, 17 amino acids (2.24 kDa).

tac promoter a hybrid **promoter**, constructed *in vitro*, that contains the -35 sequence of the *trp* promoter and the -10 sequence of the *lacZ* promoter. *See* **minus 10 region**.

Tacrolimus *see* **FK506**.

tactic polymer a regular polymer in whose molecules there is an ordered structure with respect to the configurations around at least one main-chain site of stereoisomerism per **constitutional base unit**.

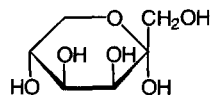
tactile capable of exciting the sense of touch; relating to the sense of touch (as a sense organ).

tactoid a **paracrystal**-like aggregate; the term may be applied to the rodlike filaments of hemoglobin that form in **sickle cells**, or to assemblies of virus particles (e.g. of tobacco mosaic virus).

tag 1 to identify by a marker; for example, a cell may be tagged by attachment of a (fluorescent or radiolabelled) antibody. **2** an identifier used in tagging.

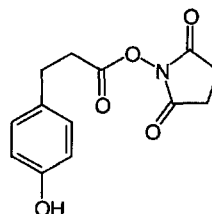
Tagamet *see* **cimetidine**.

D-tagatose a *nonsystematic name* for D-lyxo-2-hexulose, a hydrolysis product of some plant gums.



α -D-pyranose form

Tagit a *proprietary name* for N-succinimidyl 3-(4-hydroxyphenyl) propionate; a substance used in the preparation of **Bolton and Hunter reagent** for tagging peptides and proteins with radioiodine.



tail 1 an extension growing or appending from a point of origin



tailing

or attachment. For example, the fatty-acyl groups of a phospholipid and the glycerol to which they are bonded are referred to as the tail. Compare **headgroup**. **2** (in chromatography) a region of more slowly moving or decomposed material extending back from a band towards the origin. See also **poly(A) tail**, **tailing**.

tailing 1 the addition of a stretch of identical nucleotides to the ends of a restriction fragment using **terminal deoxynucleotidyl transferase**. **2** the spreading out of a band or spot in a chromatographic process.

Tal symbol for a residue, or sometimes a molecule, of **talose**.

talaric acid one of the **hexaric acids**; see also **aldaric acid**.

talc a soft mineral composed of hydrated magnesium silicate, $Mg_3Si_4O_{10}(OH)_2$, with small quantities of metal impurities. In its powdered form it is known as talcum powder, or French chalk, and in granular form as soapstone (steatite). Powdered talc is used as a lubricant, as a filler in paper and other products, and in cosmetics, etc.

talcum powder powdered talc.

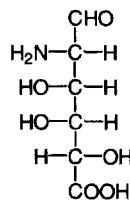
talín a protein involved in connections between the cytoskeleton and the plasma membrane. A phosphoprotein, it binds strongly to **vinculin**, and less strongly to **integrins**. It is a founder member of the **TERM** family. Example from mouse: database code TALI_MOUSE, 2541 amino acids (269.53 kDa).

talo+ or (before a vowel) **talos+** comb. form of **talose**.

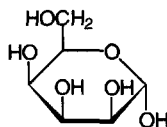
talo- prefix (in chemical nomenclature) indicating a particular configuration, a set of (usually four) contiguous CHOH groups as in the acyclic form of D- or L-**talose**. See also **mono-saccharide**.

taloepitín N-(6-deoxy- α -L-talopyranosyl-oxyphospho)-L-leucyl-L-tryptophan; an inhibitor of several metalloproteinases, including **thermolysin**. Compare **phosphoramidon**.

talosaminouronic acid a component of the **peptidoglycan** of Archaea.



talose symbol: Tal; the trivial name for the aldohexose **talo-hexose**, which differs from glucose in the configuration of the groups at C-2 and C-4; there are two enantiomers.



α -D-talose

Tamm, Igor Yevgenyevich (1885–1971), Russian-born Soviet physicist; Nobel Laureate in Physics (1958) jointly with P. A. Cherenkov and I. M. Frank 'for the discovery and the interpretation of the Cherenkov effect'.

Tamm-Horsfall glycoprotein or **uromodulin** the principal glycoprotein of human urine. It occurs in other species, e.g. the cow and rat. The molecule is filamentous with an M_r of several million and is composed of subunits of $M_r \approx 80\,000$; it contains about 25% carbohydrate and a small amount of lipid. An abnormally high M_r is observed in cystic fibrosis. Example from human (precursor): database code UROM_HUMAN.

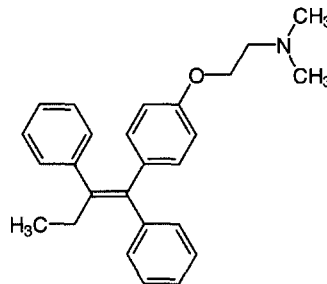
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640 amino acids (69.76 kDa). See also **proteinuria**. [After Igor Tamm (1922–71), US virologist and Frank Lappin Horsfall (1906–71).]

Tamoxifen [Z]-2-[4-(1,2-diphenyl-1-butenyl)-phenoxy]-N,N-dimethylethanamine; the proprietary name of an estrogen antagonist, used in the treatment of breast cancer and suggested for use as a prophylactic drug in women with a family history of breast cancer. The drug has complex pharmacological properties and can behave as a pure estrogen agonist, a partial agonist, or an antagonist, depending on the species of animal, the target organ examined, and the endpoint measured. It is a **genotoxic** carcinogen in the rat.



tan abbr. for tangent (def. 1).

tandem-crossed immunoelectrophoresis see **crossed immunoelectrophoresis**.

tandem repeat 1 (of DNA sequences) an arrangement in which two or more copies of a particular base sequence are situated immediately one after the other. Tandem repeats are commonly found in **telomeres**, the repeated sequence being characteristic of species and source of DNA, and of relatively short length (up to about 10 bases). They also occur in **highly repetitive DNA**. **2** (of genes) a cluster of adjacent multiple copies of the same gene. The genes for ribosomal RNA (rRNA) are characteristically contained in tandem clusters, sometimes referred to as **ribosomal DNA** (rDNA). See also **concatemer**.

Tanford-Kirkwood theory a theory of protein titration curves based on a model in which the charges are taken to be discrete unit charges located at fixed positions on the protein molecule. [After Charles Tanford (1921–) and John Gamble Kirkwood (1907–59).]

tangent 1 abbr.: tan; a function of an angle, being, in a right-angled triangle, the ratio of the side opposite the given angle (if acute) to that of the side opposite the other acute angle. The tangent of an obtuse angle is numerically equal to that of its supplement but of opposite sign. **2** a straight line that touches, but does not cut, a curve or a curved surface.

tangential-flow filtration a convenient and rapid method of concentrating cells, cell organelles, or other particulate material, or for eliminating such material from solutions. Using a special apparatus, the suspension containing the material is caused to flow over ('tangential flow') a semipermeable membrane; only material smaller than the membrane pores will pass through the membrane, forming the filtrate, leaving larger matter to be collected (retentate).

Tangier disease a rare autosomal recessive condition in humans in which there is a deficiency or complete absence of **high-density lipoprotein**. Homozygotes have very low levels of **apolipoprotein A-I**. There is accumulation of cholesterol esters in cells of the mononuclear phagocyte system. The condition is relatively benign as organ function is not affected. [After Tangier Island, Chesapeake Bay; the disease was first found in inhabitants of this island.]

tannic acid or **tannin** or **gallotannin** or **gallotannic acid** an imprecise term used for a mixture of hydrolysable **tannins** (def. 1).

tannin 1 any of a group of polyphenolic plant products that



can be used to tan animal skins to produce leather. The term was introduced in 1796. Tannins are divided into **hydrolysable tannins** (the larger group) and **condensed tannins**. The former consist of a core of a polyhydric alcohol, usually glucose, esterified with either gallic acid or hexahydroxydiphenic acid. Condensed tannins are polymers of flavans and never contain sugar residues; they are often termed **flavolans**. Tannins form insoluble protein tannates with proteins, which, when the protein is a skin protein, forms the basis of **tanning** (def. 1). The liberation of tannins during the extraction of plant enzymes can cause difficulties because they tan enzymes, i.e. form insoluble tannates. The addition of polyvinylpyrrolidone may help to avoid this problem. Good tanning agents should have an M_r of between 500 and 3000 and contain sufficient phenolic hydroxyl groups (1–2% of M_r) to form effective cross-links with protein. **2 a common name for tannic acid.**

tanning 1 a technique of considerable antiquity, being known by the Egyptians around 3000 BC, for producing leather from animal skins by the use of **tannin** (def. 1). Modern methods using chrome salts are faster, taking a few hours rather than weeks. **2** the treating of red blood cells with tannic acid. Such cells are used in a test for anti-thyroglobulin antibodies, after they have been treated with thyroglobulin. The tanning enhances sensitivity.

T antigen 1 any of the several early gene products of a genetic locus of certain transforming DNA viruses, including adenovirus, simian virus 40 (SV40), and polyoma virus. The substance originally detected in nuclei of transformed cells by immunofluorescence (hence the name antigen; the 'T' indicates tumour or transformation) led to the identification of several products, the genes of this locus encoding a number of essential early proteins that are needed for replication. Two products are formed by SV40, known as **large T antigen** (or **T antigen**) and **small t antigen** (or **t antigen**). Polyoma virus expresses three products from this locus, large T antigen, middle T antigen, and small t antigen. Large T antigen of SV40 alone is sufficient to immortalize primary rodent cells and transform established rodent cell lines. It is a multifunctional protein that is a potent transcriptional activator which promiscuously activates simple promoters containing a TATA box or initiator element and at least one upstream transcription-factor-binding site. Examples, large T antigen from SV40: database code TALA_SV40, 708 amino acids (81.58 kDa); large T antigen from polyoma virus: database code TALA_POVMK, 648 amino acids (74.71 kDa). Middle T antigen of polyoma virus is essential for polyoma-virus-mediated tumorigenesis and cell transformation. It associates with and activates cellular tyrosine kinases, including c-Src, c-Yes, and Fyn, and also binds phosphatidylinositol 3-kinase and protein phosphatase 2A. Example, middle T antigen of polyoma virus: database code TAMI_POVMA, 442 amino acids (48.62 kDa). Small t antigen of SV40 exercises a helper effect in SV40 transformation. It forms complexes with protein phosphatase 2A and regulates the phosphorylation and transcriptional transactivation of **CREB protein**. It stabilizes p53 in a manner that correlates with enhanced transformation efficiency of SV40. Example, small t antigen from polyoma virus: database code TASM_POVMA, 195 amino acids (22.81 kDa). **2** an antigen found on normal human erythrocytes that is unreactive unless the cells are first treated with neuraminidase. **3** one of the principal protein antigens in the cell wall of *Streptococcus* spp. **4 abbr. for** transplantation antigen (i.e. a class I MHC antigen); see **major histocompatibility complex**.

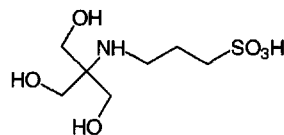
tapeworm any parasitic flatworm (phylum Platyhelminthes) belonging to the class Cestoda.

Taps or **TAPS** *abbr. for* N-tris(hydroxymethyl)methyl-3-aminopropanesulfonic acid; 3-{[2-hydroxy-1,1-bis(hydroxymethyl)ethyl]-amino}propanesulfonic acid; a **Good buffer substance**, pK_a (20 °C) = 8.4.

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Taps

Taq DNA polymerase a heat-stable DNA polymerase from *Thermus aquaticus*, a thermophilic bacterium. It is used in the **polymerase chain reaction**.

tare 1 to compensate for the weight of a container, etc. when weighing, by adjusting the balance to zero with the empty container in place before adding the material to be weighed. **2** the weight of the tared vessel or material.

target the object at which an action or process is directed. For example, a **target cell** is the cell under attack by a cytolytic or killer cell or other cytolytic agent, and a **target tissue** is the tissue that is acted on by any hormone, growth factor, or other agonist.

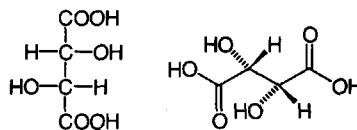
targeting subunits protein subunits that specify the location and catalytic and regulatory properties of protein phosphatases and kinases.

target repeat *see* **transposon**.

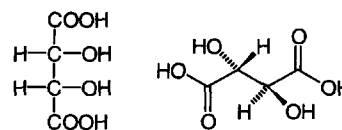
target site the site at which a **restriction endonuclease** cleaves a DNA molecule; usually the same as the **recognition site** (def. 2).

target theory (*in radiation chemistry*) a theory stating that all single hits on biologically active macromolecules by particles or quanta of ionizing radiation result in damaged and inactive molecules, that no partially active molecules are formed, and that all unhit molecules retain full activity.

tartaric acid 2,3-dihydroxybutanedioic acid; 2,3-dihydroxy-succinic acid; one of the **aldaric acids** formed by oxidation of a **tetrose** at C-1 and C-4. The $L_g(+)$ (*see* **b/L convention**), or 2*R*,3*R*, enantiomer occurs widely in plants, especially grape juice, and in fungi and bacteria; the $D_g(-)$ (or 2*S*,3*S*) enantiomer and *meso* form have limited distribution in plants. The racemic mixture, DL-tartaric acid, sometimes occurs during wine manufacture; it was originally known as **paratartaric acid**. The first determination of an absolute configuration by anomalous dispersion of X-rays was made on sodium rubidium L(+)-tartrate in 1951. The property of **optical activity** of some substances was first described by **Pasteur** in experiments with crystals of sodium ammonium tartrate.



L-form



meso-form

TATA-binding protein *abbr.:* TBP; a component of transcription factor IID (*see* **TFIID**). *See also* **RNA polymerase**.



TATA box or **Hogness box** a base-sequence element common in eukaryotic promoters, characterized by the consensus TATAAAA. It binds a general transcription factor and hence specifies the position where **transcription** is initiated.

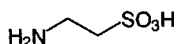
Tatum, Edward Lawrie (1909–75), US microbiologist; Nobel Laureate in Physiology or Medicine (1958) jointly with G. W. Beadle 'for their discovery that genes act by regulating definite chemical events' [prize shared with J. Lederberg].

tau 1 symbol: τ (lower case) or T (upper case); the nineteenth letter of the Greek alphabet. For uses see **Appendix A. 2** an alternative name for transcription factor TFIIIC; see **TFIIIC**.

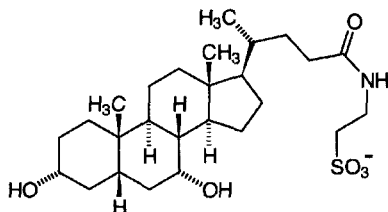
tau protein a microtubule-associated protein that promotes microtubule assembly and stabilizes microtubules. It constitutes at least a part of the paired helical filament (PHF) core found in **Alzheimer's disease**. Example from human: database code TAUI_HUMAN, 383 amino acids (39.96 kDa).

tau-protein kinase EC 2.7.1.135; an enzyme that catalyses the phosphorylation by ATP of τ -protein to O-phospho- τ -protein. Example bovine (fragment): database code S43683, 209 amino acids (23.14 kDa). See also **tau protein**.

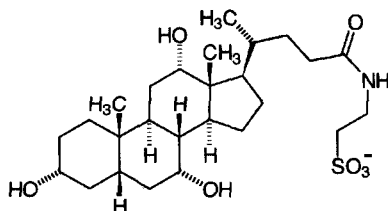
taurine 2-aminoethanesulfonic acid; a compound that is derived metabolically from cysteine by oxidation of the sulfhydryl group to sulfoxide followed by decarboxylation of the resulting cysteic acid. N-conjugated with cholic acid, it forms the bile salts **taurochenodeoxycholate**, **taurocholate**, and **taurodeoxycholate**.



taurochenodeoxycholate the anionic form of the amido conjugate of **taurine** and **chenodeoxycholate**. It is a major **bile acid**.



taurocholate the anionic form of taurocholic acid, N-cholyltaurine, a major **bile salt** of humans and other mammals. It is useful as an anionic bile-salt detergent (aggregation number 4; CMC 3–11 mM). The anions of similar taurine conjugates of other bile acids are known, e.g. **taurochenodeoxycholate** and **taurodeoxycholate**.



taurocyamine N-(aminoiminomethyl) taurine; a compound found in marine polychaete worms. The guanidine group can undergo phosphorylation, to form phosphotaurocyamine, which acts as a **phosphagen**.

taurodeoxycholate the anionic form of taurodeoxycholic acid.

acid, N-(7-deoxy)cholyltaurine. It is useful as an anionic bile-salt detergent (aggregation number 6; CMC 1–4 mM).

taurolipid any of a class of lipids consisting of a non-hydroxy fatty acid esterified with a hydroxy group of tri-, tetra-, or pentahydroxystearic acid in amide linkage with taurine. They are characteristic lipids of the protozoan *Tetrahymena*.

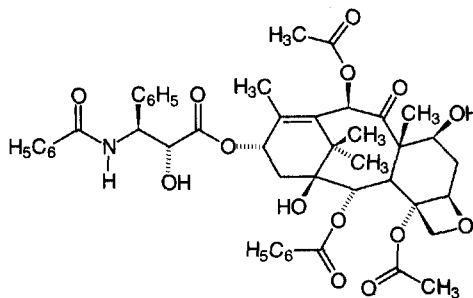
tauryl the acyl group, $\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{SO}_2-$, derived from taurine.

taut form abbr.: T form; see **tense form**, **Monod-Wyman-Changeux model for allosteric interaction**.

tautomer any one of the possible alternative structures that may exist as a result of **tautomerism**.

tautomerism a form of constitutional isomerism in which a structure may exist in two or more constitutional arrangements, particularly with respect to the position of hydrogens bonded to oxygen. The isomers, or tautomers, are readily interconvertible and exist in equilibrium. The best-known example is **keto-enol tautomerism**, in which the structures $\text{X}-\text{C}(=\text{O})-\text{CH}_2-\text{X}$ (keto) and $\text{X}-\text{C}(\text{OH})=\text{CH}-\text{X}$ (enol) exist, in proportions determined by the equilibrium position for the conditions pertaining (normally strongly favouring the keto form). Other examples include quinone-quinol and **amino-imino tautomerism**. —**tautomeric adj.**

taxol an antitumour and antileukemic agent isolated from the bark of the Pacific yew, *Taxus brevifolia*. It is used clinically to treat refractory ovarian cancer. Its anticancer activity results from its ability to inhibit mitosis and block the cell cycle in its G or M phases through enhancement of the polymerization of **tubulin** and the consequent stabilization of the **microtubules**. In view of the endangered status of *T. brevifolia*, taxol is manufactured semisynthetically from a material present in *T. baccata*. A total chemical synthesis has also been achieved. After the discovery of taxol, a compound with similar properties was isolated from the myxobacterium *Sporangium cellulosum*. This compound, **epothilone**, is more water-soluble than taxol and has the advantage that it can be produced in quantity by fermentation. Taxol and its analogues are known as **taxanes**. One proprietary name is Paclitaxel. A proprietary name of a semisynthetic analogue is Docetaxel.



taxon (pl. **taxa**) a named taxonomic group of any rank; a particular taxon comprises organisms having certain genetic characters in common. Strictly, the term should be used to denote only named groups, e.g. Chordata, Hominidae, etc., not as a generic term for the ranks themselves, e.g. phylum, family, etc.

Tay-Sachs disease or **G_{M2} gangliosidosis** an inherited neurological metabolic disorder resulting from deficiency of the lysosomal enzyme, **hexosaminidase** (form A, deficiency in α chain). This leads to the accumulation of its substrate, ganglioside **G_{M2}** (see **ganglioside**), in the brain. The disease is relatively common (1 in 2500 live births) among Ashkenazy Jews. There is motor and mental deterioration with an exaggerated startle response to sound; by 18 months of age, most patients are blind, deaf, and spastic; they die by the age of 3 years. See also **saposin**. [After Warren Tay (1843–1927), British physician and Bernard Sachs (1858–1944), US neurologist].

Tb

Tb symbol for terbium.

TBP *abbr.* for TATA-binding protein (see **TFII**, **RNA polymerase**).

tBu *abbr.* for tertiary butyl (alternative to Bu⁺).

TBW *abbr.* for total body water; see **body water**.

Tc symbol for technetium.

TCA *abbr.* for 1 tricarboxylic acid, 2 trichloroacetic acid.

TYC arm the base-paired segment of the clover-leaf model of transfer RNA to which the loop containing the base sequence TYC is attached (Y = pseudouridine).

T cell an alternative name for **T lymphocyte**.

T_H cell *abbr.* for helper cell.

T-cell growth factor *abbr.*: TCGF; an alternative name for interleukin 2.

T-cell receptor a heterodimer glycoprotein of the immunoglobulin superfamily located in the plasma membrane of T cells (**T lymphocytes**). T-cell receptors have structural similarities with class II MHC antigens (see **major histocompatibility complex**) but the α and β chains are disulfide bonded. They are responsible for recognition of target antigen in combination with MHC. After binding to antigen/MHC (class I or class II depending on the type of antigen presentation), the T-cell receptor transduces the signal into the T cell, probably through mediation by associated CD3 molecules (see **CD marker**), and brings about T-cell activation. One constant region and one variable region (see **immunoglobulin**) are located in each subunit (α and β); the subunits are related to one another. As with immunoglobulins, the variable regions are designed for specificity of recognition (but they are not products of the immunoglobulin genes). A small number of adult T-cell receptors have γ and δ chains. This class of receptor appears first in ontogeny. Examples from human leukemic cell line, constant regions; α subunit: database code TCA_HUMAN, 142 amino acids (15.91 kDa); β subunit: database code TCB_HUMAN, 177 amino acids (19.88 kDa).

T-cell-replacing factor *abbr.*: TRF; a term formerly used for some interleukins. See **interleukin 1**, **interleukin 5**.

TCGF *abbr.* for T-cell growth factor (i.e. **interleukin 2**).

T chain an alternative name for **secretory piece**.

TCP-1 *abbr.* for T-complex protein 1; a molecular chaperone that forms two stacked rings, 12–16 nm in diameter and is associated with other proteins in a 850–900 kDa complex. It is similar in function to Hsp60 (a **heat-shock protein**) in mitochondria and the **groEL** product in bacteria (see also **chaperonin**). Example from human: database code TCP1_HUMAN, 556 amino acids (60.27 kDa).

T-DNA *abbr.* for transfer DNA; that part of the *Agrobacterium* Ti plasmid that is incorporated into the genome of infected plant cells. It is approximately 23 kb long and codes for an **opine**, usually **nopaline** or **octopine**, which the transformed cells produce and the bacteria then use. At each end of the T-DNA in the Ti plasmid there is a 25 bp sequence; the right hand one of these almost identical sequences is essential for the transfer process.

TDP *abbr.* for 1 ribosylthymine diphosphate or ribothymidine diphosphate, the common names for ribosylthymine 5'-diphosphate, ribothymidine 5'-diphosphate, 5'-ribothymidylyl phosphate, thymine riboside 5'-diphosphate. 2 (sometimes) thymidine diphosphate (dTDP is recommended).

Tdr or **TdR** or **TDR** *abbr.* for thymine deoxyriboside (i.e. thymidine). Use deprecated; the symbol dThd is recommended.

TdT *abbr.* for terminal deoxynucleotidyltransferase.

Te symbol for tellurium.

TEA *abbr.* for 1 triethanolamine; N(CH₂CH₂OH)₃; a base used in the preparation of buffers, pK_a (25 °C) 7.90. 2 tetraethylammonium; used to block sympathetic and parasympathetic ganglionic transmission; normally used as the chloride.

TEAE *abbr.* for the triethylaminoethyl group, (C₂H₅)₃NCH₂CH₂-. It may be used as a substituent in hydroxylated chromatographic support matrices, e.g. cellulose, Sephadex, to form anion-exchange columns.

teichoate the anionic form of **teichoic acid**.

teichoic acid any polymer occurring in the cell wall, membrane, or capsule of Gram-positive bacteria and containing chains of glycerol phosphate or ribitol phosphate residues. Additionally, carbohydrates are linked to glycerol or ribitol, and some -OH groups are esterified with D-alanine.

tektins a family of similar proteins that are adapted to the assembly of a variety of eukaryotic cell structures. Tektins have common features that adapt them to sensitive and specific associations with molecules of their own kind; they have similar amino-acid compositions. They include actin, actin-like proteins, some microtubule proteins, mitochondrial structural protein, and erythrocyte membrane proteins.

tele a designator of the position of one of the nitrogens of the histidine imidazole ring, indicating the nitrogen furthest from the side chain. It is often symbolized by τ , as in N ^{τ} . Compare *pros*.

teleology the doctrine or study of ends or final causes, especially as related to the evidences of design or purpose in nature; the use of design, purpose, or utility as an explanation of any natural phenomenon. An example of a teleological concept in biology is that organs are made to react for the good of the organism as a whole.

teleonomy the idea that the existence of a structure or a function in an organism must have conferred an evolutionary advantage upon that organism.

teleost any member of the infraclass Teleostei, comprising fishes with rayed fins and a swim bladder; it includes most of the extant bony fishes.

telestability the transmission of three-dimensional stability from a stable part of a macromolecule to an inherently less stable part.

telokin the C-terminal sequence of smooth muscle **myosin-light-chain kinase** expressed as an independent protein. It binds calmodulin. Example from rabbit: database code TELO_RABIT, 155 amino acids (17.18 kDa).

telolysosome a **lysosome** that has accumulated so much indigestible residue that it is unable to accumulate further material to be digested. The structure still contains enzymes but these are no longer renewed. Compare **residual body**.

telomerase EC 2.7.7.-; a DNA polymerase enzyme involved in **telomere** formation, and in telomere elongation to maintain the telomere sequences during replication. The enzyme recognizes the G-rich strand of an existing telomere repeat sequence, elongating it in the 5' to 3' direction. *Tetrahymena* telomerase is a large ribonucleoprotein, a specialized example of a reverse transcriptase. The RNA component includes a template for synthesizing the G-rich repeating sequence. The **EST1** gene product is a probable example.

telomere the structure that seals the end of a chromosome. In order to counteract the tendency of the chromosome otherwise to shorten with each round of replication, the telomeric DNA consists of simple tandemly repeated sequences specific for each species. Typically one strand is G-rich and the other C-rich. The G-rich strand forms a 3'-terminal overhang about 15 residues long. These are the sequences added to the 3' terminus of chromosomal DNAs by **telomerase**. The elongation allows a primer to bind and initiate synthesis on the other strand, maintaining the length of the chromosome and preventing the loss of coding sequences. —**telomeric adj.**

telo peptide any of the peptides that protrude from the triple-helical body of **tropocollagen** and have a composition different from the triple-helical portions. They are removed by proteases, with concomitant breakage of intramolecular inter-chain bonds. They are thought to occupy near-terminal positions. The antigenic response to injected heterologous tropocollagen is directed against these telo peptide appendages external to the triple helix.

telophase the final stage of **mitosis** or **meiosis** in which the nuclei form in the daughter cells, which then revert to the resting state.

TEM or **T.E.M.** *abbr.* for 1 transmission electron microscope;

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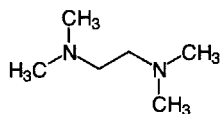
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see **electron microscope**. **2** Tris-EDTA-mercaptoethanol (buffer).

TEMED *abbr. for* *N,N,N',N'*-tetramethylethylenediamine; *N,N,N',N'*-tetramethyl-1,2-diaminoethane; a substance used as a polymerizing agent in the preparation of polyacrylamide gels. It acts to stabilize free radicals produced by ammonium persulfate.



Temin, Howard Martin (1934–94), US virologist; Nobel Laureate in Physiology or Medicine (1975) jointly with D. Baltimore and R. Dulbecco 'for their discoveries concerning the interaction between tumour viruses and the genetic material of the cell'.

temperate phage a **bacteriophage** that, following infection of its host bacterium, may either enter the lytic cycle, like a **virulent phage**, or establish a kind of symbiotic relationship with the host cell that results in its subsequent perpetuation in all the descendants of the bacterium. Bacteria carrying temperate phage are said to be **lysogenic**, while the form of the carried phage is known as a **prophage**.

temperature *abbr.*: temp.; a fundamental physical quantity of a body, region, or system that determines the direction of net heat flow between it and its surroundings. If two systems have the same temperature they are in thermal equilibrium and there is no net heat flow between them. If they have different temperatures they are not in thermal equilibrium and heat flows from the system of higher temperature to that of lower temperature. Essentially there are two different methods of measuring temperature. One is to use a scale of values set up between arbitrary fixed points based on reproducible temperature-dependent events. For example, the **Celsius temperature** scale uses a freezing point and boiling point of water as fixed points, to which are assigned the values 0 and 100, respectively; the scale between them is divided into 100 degrees. Although widely used for many practical purposes, it lacks a theoretical basis and is unsuitable in many scientific contexts. Instead, the concept of **thermodynamic temperature** is used. This is defined in terms of the second law of thermodynamics (see **thermodynamics**) and is independent of any particular substance. However, in practice thermodynamic temperatures cannot be measured directly. Consequently, a practical realization of the thermodynamic temperature scale is used, called the **International Practical Temperature Scale (IPTS)**. This is based on a number of fixed points, and is measured in **kelvins**; it defines temperature down to 13.81 kelvins, the triple point of equilibrium hydrogen.

temperature coefficient of a reaction *symbol*: Q_{10} ; the factor by which the velocity of a chemical reaction is increased by raising the temperature, t , by 10 °C. Usually it is the ratio of the rate of reaction at $(t + 10^\circ)$ to that at t , denoted by $Q_{10} = k_{t+10}/k_t$, where k_{t+10} and k_t are the rate constants at the respective temperatures. The Q_{10} value of homogeneous chemical reactions is generally in the range 2–3; many enzyme-catalysed reactions and physiological processes exhibit, over a limited range of temperatures, a Q_{10} value of ≈ 2 . See also **Arrhenius equation**.

temperature jump a technique used in **relaxation kinetics** for the study of enzyme mechanisms in which the temperature is raised rapidly (e.g. over about 1 μ s), the reaction being monitored during this and the subsequent (very short) period during which the system relaxes to a new equilibrium.

temperature-sensitive mutation a mutation that is manifest in only a limited range of temperature. The mutant carrying such a mutation behaves normally at the lower, **permissive**

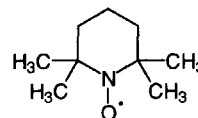
temperature, but when grown at the lower or higher, **restrictive temperature**, it shows the mutant phenotype.

template a macromolecule whose structure serves as a pattern for the synthesis of another macromolecule. The term applies particularly to a nucleic acid on which another nucleic acid of complementary base sequence is synthesized, as in genetic **transcription**, or in the case of messenger RNA, on which a polypeptide is synthesized (genetic **translation**).

template RNA an alternative name for **messenger RNA**.

template strand see **noncoding strand**.

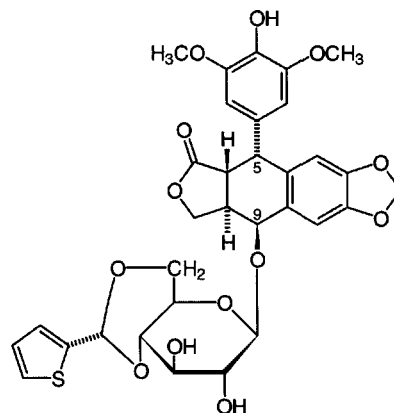
TEMPO *abbr. for* 2,2,6,6-tetramethylpiperidinoxy; 2,2,6,6-tetramethylpiperidine 1-oxyl; a stable free radical used as a spin label in electron spin-resonance spectroscopy.



temporal gene any gene that 'programs' a system in time, determining the activation of structural, architectural, and regulatory genes controlling different developmental stages in various cell types.

tenascin a substrate-adhesion molecule of the extracellular matrix that appears to inhibit cell migration and may play a role in supporting the growth of epithelial tumours. It is a hexameric protein; a homotrimer may be formed in the triple coiled-coil region and may be stabilized by disulfide rings at both ends; two such half-hexabrachions (see **hexabrachions**) may be disulfide linked within the central globule. Alternative splicing leads to four variants from a single gene in a tissue- and time-specific manner during development. It has EGF-like repeats (see **epidermal growth factor**) and **fibronectin** domains. Example from human: database code TENA_HUMAN, 2199 amino acids (240.45 kDa). *Other names*: hexabrachion; cytotoxic; neuronectin; gmem; ji; miotendinous antigen; glioma-associated extracellular matrix antigen; gp150–225.

Teniposide a semisynthetic lignan derivative prepared from **podophyllotoxin**. It is similar in action to **Etoposide**, but more potent as a cytotoxic agent *in vitro*.



ten-minute phosphate see **labile phosphate**.

tense form or taut form *abbr.*: T form; the form of an allosteric protein that has a lower affinity for a ligand than the R form (relaxed form). See **Monod-Wyman-Changeux model** (for allosteric interaction).

tenside any of the group of surface-active compounds widely



tension

known as **detergents**. The name derives from their effects on surface tension.

tension **1** the concentration of a gas expressed as the partial pressure of the gas. **2** the force exerted by stretching any material. **3** voltage, electromagnetic force, or potential difference, as in high tension, low tension.

tensor **1** (*in mathematics*) a quantity expressing the ratio in which the length of a **vector** is increased. *Compare* **scalar**. **2** (*in anatomy*) a muscle that stretches or tightens some part while not altering the direction of the part.

tentoxin cyclo[L-leucyl-N-methyl-(Z)-dehydrophenylalanyl-glycyl-N-methyl-L-alanyl]; a cyclic tetrapeptide, originally isolated as a metabolite of *Alternaria alternata*. It selectively induces chlorosis in some plant species when applied to the germinating seedling.

Ter *see* **reactancy of enzymes**.

tera+ *symbol:* T; SI prefix denoting 10^{12} times.

terato+ or (*before a vowel*) **terat+** *comb. form* denoting congenital abnormality.

teratocarcinoma a **teratoma** that includes carcinomatous cells.

teratogen a substance that produces malformations in embryos. —**teratogenic** *adj.*

teratoma a tumour composed of disorganized tissues that are foreign embryologically to the tissue normally found at the site. Teratomas most frequently occur in testis or ovary, and may be derived from remnants of multipotential embryological cells that can differentiate into many cell types.

TERM a family of F-actin binding proteins, named from **talin** and the related proteins **ezrin**, **radixin**, and **moesin**.

terminal amino acid the amino-acid residue at the end of a peptide or polypeptide chain. The **C-terminal amino acid** is the end amino acid that has a free α -carboxyl group (*see* **C terminus**); the **N-terminal amino acid** is the end amino acid that has a free α -amino group (*see* **N terminus**).

terminal deletion *see* **deletion**.

terminal deoxynucleotidyltransferase *abbr.* TdT; EC 2.7.7.31; *recommended name:* DNA nucleotidylexotransferase; *systematic name:* nucleoside-triphosphate:DNA deoxynucleotidylexotransferase; *other names:* terminal addition enzyme; terminal transferase; terminal deoxyribonucleotidyltransferase. An enzyme, commonly isolated from calf thymus, that catalyses the successive transfer of deoxynucleotide residues from the corresponding nucleoside triphosphates to the 3' end of an oligo- or polydeoxynucleotide. It is a template-independent, Mg^{2+} -dependent **DNA polymerase**. Example (bovine; enzyme used in molecular biology): database code TDT_BOVIN, 520 amino acids (59.60 kDa).

terminal transferase *see* **terminal deoxynucleotidyltransferase**.

terminal web the network of filaments forming the part of the **cytoskeleton** that lies close under the plasma membrane. It is a specialized cortex, to which actin filament bundles may be attached, especially, e.g., at the base of microvilli of intestinal epithelial cells. It contains a dense network of **spectrin** molecules that connect the inner **intermediate filaments** to the actin filaments that are connected to the plasma membrane.

termination codon or **terminator codon** any **codon** that signals the termination of a growing polypeptide chain. *See also* **amber codon**, **ochre codon**, **stop codon**.

termination factor any protein that participates in the termination of either transcription (*see* **rho protein**) or translation (*see* **release factor**).

terminator (of transcription) a DNA sequence lying just downstream of the coding segment of a gene that is recognized by RNA polymerase as a signal to stop synthesizing messenger RNA during **transcription**.

termolecular reaction *see* **molecularity**.

ternary consisting of three parts.

ternary complex any complex formed between three chemical entities, e.g. a complex formed between an enzyme and two substrate molecules.

terpene any member of a class of hydrocarbons occurring par-

ticularly in most essential oils but represented throughout the biological world. Terpenes are composed essentially of multiples of **isoprene** (C_5H_8) units, and may be acyclic, cyclic, or multicyclic, saturated or unsaturated, and can be substituted by various functional groups, e.g. OH. Terpenes containing 10, 15, 20, 30, 40, or a large number, of carbon atoms per molecule are known as **mono-**, **sesqui-**, **di-**, **tri-**, **tetra-**, or **poly-**terpenes respectively. The class is often taken to include alcohol, aldehyde, and carboxylic acid derivatives of terpene hydrocarbons. Isoprene is the only naturally occurring **hemiterpene**, i.e. having five carbons, but the phosphorylated compounds Δ^2 - and Δ^3 -isopentenyl diphosphates are key intermediates in terpene biosynthesis. *See also* **polyprenyl**.

terpenoid **1** resembling a **terpene** in chemical structure. **2** any member of a class of compounds characterized by an isoprenoid chemical structure (as in a **terpene**) and often including their oxygenated and hydrogenated derivatives.

Terramycin a proprietary name for **oxytetracycline**.

tert- *abbr.:* t-; *prefix (in chemical nomenclature)* signifying **tertiary** (def. 1).

tertiary (in chemistry) **1** a *prefix:* *tert-* (*abbr.:* t-); describing an alkyl compound (e.g. an alkanol) in which the functional group (e.g. a hydroxyl group) is attached to a carbon atom linked to three others. **b** describing the carbon atom bearing the functional group in such a compound. **2** describing an amide or an amine in which three appropriate groups are attached to a nitrogen atom. **3** describing a salt formed by replacing all three of the ionizable hydrogen atoms of a tri-basic acid by one, two, or three other cations of appropriate valency. *Compare* **primary**, **secondary**.

tertiary structure the level of protein structure at which an entire polypeptide chain has folded into a three-dimensional structure. The tertiary structure results from interactions between amino-acid residues, that may be widely separated in the primary structure, but may be brought into proximity by the folding of the polypeptide chain. The forces of interaction between residues include hydrogen bonds and electrostatic attraction. The structure is stabilized in some proteins by disulfide bonds. In multichain proteins, the term tertiary structure applies to individual chains. *Compare* **primary structure**, **secondary structure**, **quaternary structure**.

Tes or **TES** *abbr. for* N-tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid; 2-[[2-hydroxy-1,1-bis(hydroxymethyl)ethyl]amino]ethanesulfonic acid; a **Good buffer substance**; pK_a (20 °C) = 7.5.

tesla *symbol:* T; the SI derived unit of **magnetic flux density**. It is equal to one weber of magnetic flux per square metre; i.e. $1\text{ T} = 1\text{ Wb m}^{-2}$. [After Nikola Tesla (1856–1943), Croatian-born US physicist.]

testicular of, or relating to, the **testis**.

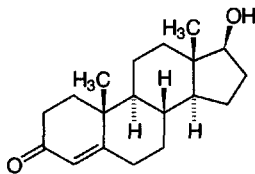
testicular feminization or **testicular feminisation** an inherited X-linked condition in which genetic males have developmental abnormalities ranging from a complete external female phenotype to partially masculinized ambiguous genitalia. The primary defect is an unresponsiveness to **dihydrotestosterone**; defects in an X-linked gene coding for a dihydrotestosterone receptor may be responsible.

testis (*pl.* **testes**) the male gonad; in vertebrates there is typically a pair of testes, producing spermatozoa and synthesizing **androgens**.

testosterone 4-androsten-17 β -ol-3-one; the principal and most potent androgen. It is formed from androstenedione by 3 α (17 β)-hydroxysteroid dehydrogenase (NAD^+) (EC 1.1.1.239), in Leydig cells of the testis. Testosterone is responsible during embryonic development for virilization, and promotes and maintains male secondary sex characteristics throughout life. It is also synthesized in the ovary from which some is secreted into the blood, but most is converted to estradiol. Its actions are in large part mediated by **dihydrotestosterone**.

tetanus

sterone, to which it is converted in target cells. See also **androgen receptor**.



tetanus 1 a disease resulting from infection with the bacterium *Clostridium tetani*. See also **tetanus toxin**. **2** or **tetany** a sustained contraction of a muscle. —**tetanal** or **tetanoid** *adj.*

tetanus toxin the protein toxin of *Clostridium tetani*, the causative agent of tetanus. It is a heterodimer of light and heavy chains. It blocks neural exocytosis and thus norepinephrine release, possibly as a result of a Zn^{2+} -dependent proteolysis of **synaptobrevin** by the light chain, but also by inhibition of a neuronal transglutaminase (protein-glutamine γ -glutamyltransferase, EC 2.3.2.13). The protein is synthesized as a precursor: database code TETX_CLOTE, 1314 amino acids (150.55 kDa); this is cleaved to yield the light chain (residues 1–456) and heavy chain (residues 457–1314).

tetanus toxoid the **toxoid** prepared from *Clostridium tetani* toxin and included in anti-tetanus immunogens.

tetany an alternative term for **tetanus** (def. 2).

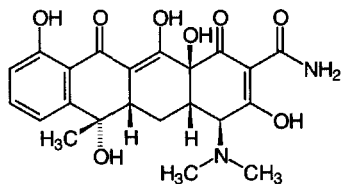
tetra+ or (before a vowel) **tetr+** *comb. form* 1 denoting four, fourfold, every fourth; see also **tetrakis+** (def. 1). **2** (in chemical nomenclature) (distinguish from **tetrakis+** (def. 2)) **a** indicating the presence in a molecular entity of four identical specified unsubstituted groups; e.g. tetrachloroethylene, tetramethylammonium chloride. **b** indicating the presence in a molecular entity of four identical inorganic oxoacid residues in linear anhydride linkage; e.g. sodium tetraborate.

tetracos+ or (before a vowel) **tetracos+** *comb. form* indicating twenty-four.

tetracosactide or **tetracosactrin** or **cosyntropin** generic names for β^{1-24} -corticotropin, a synthetic polypeptide identical to the N-terminal 24 residues of **corticotropin**, and having all of its hormonal activity. It is sometimes known as 1–24 ACTH. Proprietary names include Synacthen.

tetracosapeptide a 24-residue peptide.

tetracycline 1 any of a group of broad-spectrum antibiotics which are also effective against rickettsial organisms, mycoplasmas, and certain protozoa. The bacteriostatic activity of the tetracyclines depends on their direct inhibition of protein synthesis. They are active against protein synthesis on both bacterial 70S and eukaryotic 80S ribosomes although 70S ribosomes are rather more sensitive. The tetracyclines are much more effective against protein synthesis in intact bacteria than against that in eukaryotic cells because of selective absorption by bacterial cells. The tetracyclines inhibit the binding of aminoacyl-tRNA to the acceptor site on the ribosome. **2** any of a group of antibiotics based on the tetracycline nucleus. They include aureomycin (7-chlortetracycline); tetracycline (5-oxytetracycline); declomycin (6-demethyl-7-chlortetracycline); doxycycline (6-deoxy-5-oxytetracycline). See also **tetracycline resistant**.



tetracycline resistant *abbr.*: **TC^r**; resistant to the effects of the antibiotic **tetracycline**. The most commonly encountered mode of resistance depends on a diminished cellular accumulation of these compounds. In enteric bacteria two **transposons**, Tn1721 and Tn10, commonly carry the genes for tetracycline resistance. Tn10 encodes two proteins: a membrane-located protein (M_r 36 000) that appears to mediate resistance and a repressor protein associated with the inducibility of tetracycline resistance. In Gram-positive bacteria the induction mechanism does not appear to involve a repressor protein but depends upon a translational attenuation mechanism. The membrane protein is responsible for a pump that specifically pumps the tetracycline out of the cells against a concentration gradient. Foreign DNA may be fused to the tetracycline-resistance gene in a cloning vector thereby enabling bacterial clones containing the foreign DNA to be selected from those lacking resistance to the antibiotic. The plasmid pBR322 contains genes for resistance to both tetracycline and ampicillin. Insertion of a foreign DNA at certain sites inactivates either or both genes.

tetracycline sensitive *abbr.*: **TC^s**; sensitive to the effects of **tetracycline**.

tetradeca+ or (before a vowel) **tetradec+** *comb. form* indicating fourteen.

tetradecadienoic acid any 14-carbon straight-chain fatty acid having two double bonds. Tetradeca-3*E*,5*Z*-dienoic acid (**megatonic acid**) and the (3*Z*,5*Z*)-isomer are sex attractants for *Attagenus megatoma* and for *A. elongatulus* respectively.

tetradecapeptide a 14-residue peptide.

tetradecenoic acid any 14-carbon monounsaturated fatty acid. Tetradec-9*Z*-enoic acid (**myristoleic acid**) is the most common naturally occurring form, but the 4*Z*-isomer occurs in tohaku oil and the 5*Z*-isomer (**physeteric acid**) in sperm-whale oil.

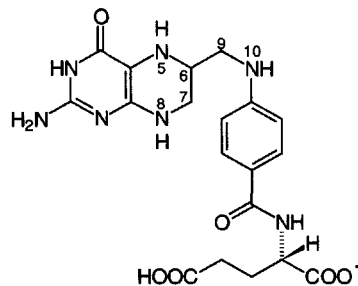
tetraenoic describing an acid having four double bonds. In long-chain fatty acids of the *n*-6 or *n*-3 families (see **fatty acid**), these will be **methylene-interrupted double bonds**.

tetragonal symmetry an arrangement, e.g. of a crystal or of ligands around a metal ion in solution, having a single four-fold axis of **symmetry**.

tetrahydro+ *comb. form* (in chemical nomenclature) indicating the presence of four hydrogen atoms at positions of specific interest (in relation to the parent compound, or as specified by a locant).

tetrahydrobiopterin see **biopterin**.

tetrahydrofolate *abbr.*: **H₄folate** or (not recommended) **THF**; *N*-[4-[(2-amino-3,4,5,6,7,8-hexahydro-4-oxo-pteridin-6-yl)-methyl]amino]benzoyl]-L-glutamate; folate bearing additional hydrogens at positions 5, 6, 7, and 8 of the pterin moiety. It is formed as the reduction product of the action of **dihydrofolate reductase** on dihydrofolate with NADPH as co-substrate.



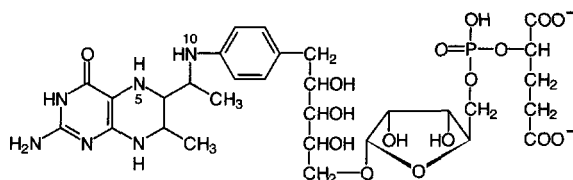
tetrahydrofuran *abbr.*: **THF**; tetramethene oxide; an organic solvent that is useful for bringing into aqueous solution com-

tetrahydromethanopterin

pounds that are otherwise sparingly soluble in water. It is a cyclic ether that is moderately soluble in water.



tetrahydromethanopterin *abbr.*: H₄MPT; a derivative of 7-methylpterin (2-amino-4-hydroxy-7-methylpteridine) that participates in the pathway of methane formation in **methanogenic** bacteria. It undergoes a series of interconversions involving N⁵ and N¹⁰, analogous to those of the **folate coenzymes**, with formation of formyl-H₄MPT, methenyl-H₄MPT, methylene-H₄MPT, and methyl-H₄MPT.



Tetrahymena a genus of ciliated protozoans, members of which are used in studies of ciliary axonemes, self-splicing RNA, and telomere reproduction.

tetrakis+ *prefix* 1 denoting four times, quadrupled; *see also* **tetra+** (def. 1). 2 (in chemical nomenclature) (distinguish from **tetra+** (def. 2)) a indicating the presence in a molecular entity of four identical specified groups each substituted in the same way; e.g. tetrakis(2-hydroxyethyl)ammonium chloride. b indicating the presence at separate positions in a molecular entity of four identical inorganic oxoacid residues; e.g. inositol 1,3,4,5-tetrakisphosphate.

tetrakisphosphate any compound containing four independent phosphoric residues in ester linkages, at positions indicated by locants; e.g. inositol 1,3,4,5-tetrakisphosphate.

tetramer a molecular complex having four components or subunits. If the subunits are identical, the term **homotetramer** is used; if any one is different from any other, the complex is called a **heterotetramer**.

tetraose tetrasaccharide; used as a stem name for any oligosaccharide consisting of four residues per molecule of the same sugar in a single type of linkage, as in cellotetraose. *Distinguish from* **tetraose** and **tetrose**.

+tetraose *suffix* used in nomenclature of glycolipids to indicate a molecule containing an oligosaccharide moiety composed of four sugar residues (which may be the same or different), as in gangliotetraose. *Distinguish from* **tetraose**.

tetrapeptide a four-residue peptide.

tetraplex DNA or **quadruplex DNA** a four-stranded DNA structure adopted by sequences rich in guanine bases. At least two major classes of such structures are known. The first involves the folding back of a repetitive sequence of guanines resulting in antiparallel strands. The other is characterized by the association of four independent parallel strands. *See also* **triplex DNA**.

tetrapyrrole a structure found in **heme** and other **porphyrins**, **chlorophylls**, **bacteriochlorophylls**, **phytochromes**, **phycobilins**, and **corrinoids** (e.g. vitamin B₁₂). It consists of four pyrroles variously substituted and linked to each other through carbons at the α position; in the case of hemes, these links are formed by one-carbon units (methylidene group). Characteristically, a metal is found coordinated to the pyrrole nitrogens, including iron (in hemes), magnesium (in chlorophylls), copper (in

turacin, the pigment of turaco bird feathers), and nickel (in **factor F₄₃₀**, a prosthetic group in methanogenic bacteria). Synthesis is via aminolevulinic acid, which is synthesized from glycine and succinyl-CoA in animals and nonphotosynthetic organisms by 5-aminolevulinic synthase. For further steps in the pathway, *see* **porphobilinogen synthase**, **hydroxymethylbilane synthase**, **uroporphyrinogen-III synthase**, and **uroporphyrinogen decarboxylase**.

tetraric acid an **aldaric acid** derived by oxidation of a **tetrose** at C-1 and C-4. There are three tetraric acids, D-, L-, and **meso-tartronic acids**.

tetrasaccharide an oligosaccharide with four **monosaccharide** units.

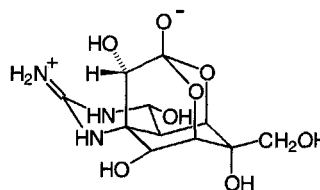
tetrasomic describing any polysomic cell, tissue, or individual in which one chromosome is represented four times in an otherwise **diploid** chromosome complement. *Compare* **monosomic**, **nullisomic**, **trisomic**. —**tetrasomy** *n*.

tetratricoseptide repeat *abbr.*: TPR; a degenerate consensus sequence of 34 amino acids found in multiple copies in several fungal and other proteins. These sequences participate in RNA synthesis regulation, protein import, and *Drosophila* development. Each sequence has two regions that are stereochemically complementary in α -helical conformation and form a hole and a knob surrounded by hydrophobic residues.

tetrazolium salt any of a group of compounds that are readily reduced to highly coloured, sparingly soluble formazan compounds, and are employed in the assay of some dehydrogenases. They include Blue Tetrazolium and Nitroblue Tetrazolium. **1H-tetrazole** is used in the chemical synthesis of oligonucleotides.

tetritol any alditol having a chain of four carbon atoms in the molecule.

tetrodotoxin *abbr.*: TTX; *also known as* **fugu toxin**; a toxin found in the liver and ovaries of the fugu, or puffer fish (order Tetraodontiformes, from which the toxin derives its name). It is a specific, reversible blocker of voltage-gated sodium channels. The LD₅₀ in mice is 8 μ g kg⁻¹. It does not block potassium or chloride channels.



tetrose any aldose having a chain of four carbon atoms in the molecule. *Distinguish from* **tetraose**.

tetrolucose any ketose having a chain of four carbon atoms in the molecule.

T-even phage any one of the **bacteriophages** T2, T4, or T6 of *Escherichia coli*. These DNA phages share a number of morphological, biochemical, and other features that distinguish them as a group from **T-odd phages** and are important historically in research on phages.

Texas Red a red-emitting fluorophore that is a sulfonyl chloride derivative of Sulforhodamine 101. It may be used with **FITC** in dual-label flow microfluorimetric and fluorescent microscopic studies because its excitation and emission spectra do not overlap with those of fluorescein.

TF *abbr.* for 1 tissue factor. 2 transcription factor.

TFII any of a group of **transcription factors** for eukaryotic RNA polymerase II promoters. **TFIIA** is needed for **TFIID** binding. Example from yeast, a heterodimer comprising large and small chains; large chain: database code TOA1_YEAST, 286 amino acids (32.17 kDa); small chain: database code TOA2_YEAST, 286 amino acids (32.17 kDa).



122 amino acids (13.46 kDa). **TFIIB** binds to RNA polymerase. Example from human: database code TF2B_HUMAN, 316 amino acids (34.79 kDa). **TFIID** has two components, a TATA-binding protein that binds to the **TATA box**, and one of a number of accessory factors; it shows weak homology with bacterial **sigma factors**. Example from human: database code TF2D_HUMAN, 339 amino acids (37.65 kDa). **TFIIE** binds to RNA polymerase; it also shows weak homology with bacterial sigma factors. Example from human, a tetramer of two alpha chains and two beta chains; alpha chain: database code T2EA_HUMAN, 439 amino acids (49.42 kDa); beta chain: database code T2EB_HUMAN, 291 amino acids (33.01 kDa).

TFIII any of a group of **transcription factors** for eukaryotic RNA polymerase III promoters. The examples are all from yeast. **TFIIIA** is a zinc-finger protein (see **zinc finger**) that binds both DNA and the 5S RNA transcript; database code S20050, 429 amino acids (49.97 kDa). **TFIIIB** is an RNA polymerase-binding protein. 70 kDa subunit: database code TF3B_YEAST, 596 amino acids (66.83 kDa). **TFIIIC** (or **tau**) is a heterotrimer, subunit database codes: TFC1_YEAST, 649 amino acids (73.47 kDa); TFC3_YEAST, 1160 amino acids (131.96 kDa); TFC4_YEAST, 1025 amino acids (120.09 kDa).

TFA *abbr.* for trifluoroacetic acid.

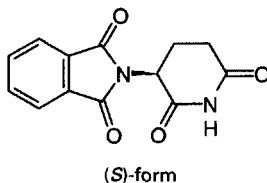
T form *abbr.* for tense form (or taut form).

TGF *abbr.* for transforming growth factor.

Th *symbol* for thorium.

thalassemia or (*esp. Brit*) **thalassaemia** any of a group of diseases resulting from inherited defects in the rate of synthesis of one of the types of polypeptide chain of **hemoglobin**. Such defects lead in turn to ineffective erythropoiesis, hemolysis, and a variable degree of anemia. In **alpha thalassemia** the defect is in the production of alpha chains and there is concomitant excessive beta- and gamma-chain production leading to the formation of various abnormal hemoglobins. In **beta thalassaemia** the formation of beta chains is defective with an excessive synthesis of alpha chains and the continued production of fetal hemoglobin ($\alpha_2\gamma_2$).

thalidomide *N*-(2,6-dioxo-3-piperidyl)phthalimide; α -phthalimidoglutarimide; a synthetic tranquilizer and sedative. It was used originally (in the 1950s) as a racemate, which causes severe fetal abnormalities such as the absence or reduction in lengths of one or more limbs. The L(*S*) enantiomer apparently has the teratogenic activity (see **teratogen**), while the D(*R*) enantiomer has the desired sedative property. The situation is unclear since thalidomide racemizes rapidly *in vivo*.

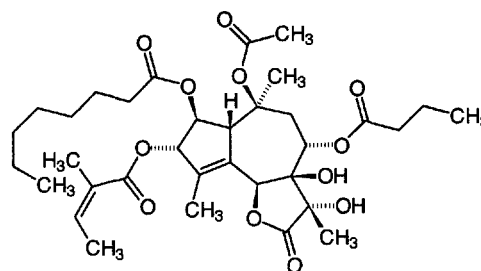


(*S*)-form

THAM *abbr.* for tris(hydroxymethyl)aminomethane; see **Tris**.

thanatogene a hypothetical gene whose expression is supposed to lead to cell death in terminally differentiated cells. [From Greek *thanatos*, death.]

thapsigargin a tumour-promoting sesquiterpene lactone. It inhibits endoplasmic reticulum **calcium-transporting ATPase** and mobilizes intracellular calcium.



thapsigargin

thaumatin a sweet-tasting, basic, 20 kDa protein isolated from the berries of the bushy African plant, *Thaumatococcus danielli*; there are five different forms. Its sweetness in aqueous solution is 2500 times greater than that of sucrose; it has sequence similarities with **monellin**.

Thd *symbol* for a residue of the ribonucleoside ribothymidine; ribosylthymine; thymine riboside (alternative to T). *Compare* dThd.

Thd5'P *symbol* for ribothymidine 5'-phosphate.

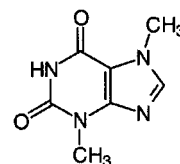
Thd5'PP *symbol* for ribothymidine 5'-diphosphate (alternative to ppT).

Thd5'PPP *symbol* for ribothymidine 5'-triphosphate (alternative to pppT).

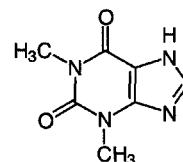
thebaine one of the alkaloids present in opium.

T-helper cell *see* **helper cell**.

theobromine 3,7-dimethylxanthine; the principal alkaloid of the cacao bean. Its actions are similar to those of **theophylline**. *See also* **caffeine**.



theophylline 1,3-dimethylxanthine; a competitive inhibitor of cyclic AMP phosphodiesterase (a 10 mM solution completely inhibits the beef heart enzyme). It has diuretic and cardiac stimulatory action, and relaxes smooth muscle; it is a bronchodilator. *See also* **caffeine**, **theobromine**.



Theorell, Axel Hugo Theodor (1903–82), Swedish biochemist; Nobel Laureate in Physiology or Medicine (1955) 'for his discoveries concerning the nature and mode of action of oxidation enzymes'. *See also* **Theorell-Chance mechanism**.

Theorell-Chance mechanism (*in enzyme kinetics*) an enzyme mechanism in which it is obligatory for substrates to bind in a certain order and in which no kinetically significant ternary complex is formed. [After A. H. T. **Theorell** and Britton Chance (1913–), US biochemist.]

theoretical plate a term used in the analysis of chromatography column performance; the higher the number of theor-

therm+

etical plates achieved, the greater the efficiency of the column. The number of theoretical plates is derived by dividing the peak width by the retention time. The term derives from the concept that the column can notionally be divided into a large number of 'slices' or 'plates' at each of which the separation process will progress. *See also* **effective theoretical plate number**.

therm+ a variant form of **thermo+** (before a vowel).

thermal analysis *see* **differential thermal analysis**, **differential scanning calorimetry**.

thermal conductance *symbol*: G ; the ability of a material to conduct heat, measured as the ratio of the heat flow rate, Φ , to the temperature difference, i.e. $G = \Phi/\Delta T$. SI units are watts per kelvin.

thermal conductivity *symbol*: λ or k ; a measure of the ability of a material to transmit heat. It is the heat flux in W m^{-2} divided by the temperature gradient; i.e. $\lambda = J_q/(dT/dl)$, where J_q is the heat flux and dT is the thermodynamic temperature difference over the length, dl . It is measured in watts per metre per kelvin.

thermal denaturation profile a curve relating some physical property of a biopolymer in solution (e.g. viscosity, optical absorbance, or optical rotation) to temperature. The shape of the curve indicates the occurrence of physical changes in the biopolymer, e.g. separation of the two strands of double-stranded nucleic acid or alteration in the tertiary structure of a protein. *See also* **denaturation**, **melting temperature**.

thermal equilibrium the condition of a system when there is no tendency for heat to flow within it, or to or from its surroundings; all parts of the system and its surroundings are at the same temperature.

thermistor a semiconductor whose resistance is sensitive to temperature. [From thermal resistor.]

thermitase EC 3.4.21.66; *other name*: thermophilic *Streptomyces* serine protease; an enzyme that catalyses the hydrolysis of proteins, including collagen. It is a Ca^{2+} -binding serine proteinase from *Thermoactinomyces vulgaris*, and is a homologue of **subtilisin**. It is inhibited by *p*-mercuribenzoate, which reacts with the single cysteine residue near the active-site histidine. Database code THET_THEVU, 279 amino acids (28.33 kDa); 3-D structure known.

thermo+ or (before a vowel) **therm+** *comb. form* indicating a relationship to heat or heat transference.

thermobarometer a control manometer and flask used in a Warburg apparatus, or other constant-volume-type respirometer, to compensate for changes in temperature and barometric pressure during the course of an experiment.

thermochromism any reversible change of colour of a substance with change of temperature. *Compare* **electrochromism**, **photochromism**. —**thermochromic** *adj.*

thermocouple a device consisting of two pieces of dissimilar metals joined together at a point, their free ends being connected to an instrument to measure electromotive force (emf). The emf produced at the metal-metal junction is a function of the junction temperature, hence a thermocouple may be used to measure temperature. The phenomenon is known as the Seebeck effect.

thermodynamicase (sometimes) an alternative name for **protein disulfide isomerase**.

thermodynamic equilibrium constant *see* **equilibrium constant**.

thermodynamics the study of the laws that govern the conversion of energy from one form to another, the direction in which heat will flow from one system to another, and the availability of energy to do work. The **zeroth law of thermodynamics** states that if two systems are each in thermal equilibrium with a third system, then they must be in thermal equilibrium with each other. The **first law of thermodynamics** or the **law of conservation of energy** specifies that in an isolated, adiabatic system the sum of the internal energies of all kinds, U , remains constant, whatever may be the changes within the system itself. If a system of constant mass gains

heat, q , and has work, w , done on it then a change in its internal energy, ΔU , occurs such that $\Delta U = q + w$. (Where the system loses heat, $q < 0$, and where work is done by it, $w < 0$.) The **second law of thermodynamics** is concerned with the restrictions in the direction of energy flow in natural systems and takes account of the irreversibility of natural processes. It specifies that the entropy of any system, S , unlike energy, is not conserved but increases with time in any spontaneous process in an isolated system. In the limiting case of a reversible process, the increase in entropy of a system, dS , is defined as the heat absorbed by the system, dq_{rev} , divided by the thermodynamic temperature, T ; i.e. $dS = dq_{\text{rev}}/T$. The **third law of thermodynamics** is concerned with the fact that the difference between the change in enthalpy, ΔH , and the change in the free energy, ΔG , of a system diminishes as the temperature of the system diminishes, and the value of $(\Delta H - \Delta G)$ tends to zero as the thermodynamic temperature tends to zero. This requires that the change in the heat capacity of the system (at constant pressure), ΔC_p , and ΔS must fall to zero at 0 K.

thermodynamic temperature or (formerly) **absolute temperature** *symbol*: T ; a fundamental physical quantity – one of the seven SI base quantities – defined in terms of the laws of thermodynamics, independently of any particular substance. The SI unit is the **kelvin**. *See also* **temperature**.

thermogenesis the generation of heat by specialized tissues in mammals and in flowers or inflorescences of plants, e.g. arum lilies (*Arum maculatum*). In mammals, this function is performed by **brown adipose tissue**. —**thermogenic** *adj.*

thermogenin another name for **brown fat uncoupling protein**.

thermolabile unstable at elevated temperatures. *Compare* **thermostable**. —**thermolability** *n.*

thermolysin EC 3.4.24.27; *other name*: *Bacillus thermoproteolyticus* neutral proteinase; a notably heat-stable, zinc-containing metalloendoproteinase produced by *B. thermoproteolyticus*; it also contains four bound calcium ions. It catalyses the preferential cleavage: Xaa-|-Leu > Xaa-|-Phe. Thermolysin is typical of several such bacterial enzymes: database code NRL_3TLN, 316 amino acids (34.30 kDa); 3-D structure is known.

thermolysis the uncatalysed cleavage of one or more covalent bonds caused by exposing a chemical compound to a raised temperature. *See also* **pyrolysis**. —**thermolytic** *adj.*

thermometer any instrument for measuring temperature. All thermometers depend on the change with temperature of some easily measured physical property such as the expansion of mercury (or other liquids) or the change in electrical resistance. *See also* **thermocouple**.

thermometry the branch of physics concerned with thermometers and the measurement of temperature.

thermoneutron activation analysis an alternative name for **radioactivation analysis**.

thermoosmosis the flow of solvent, especially water, across a membrane as a result of a temperature gradient across the membrane.

thermophile an organism, especially a microorganism, that can tolerate high temperatures and that grows optimally at temperatures above 45 °C. *Compare* **mesophile**, **psychrophile**. —**thermophilic** *adj.*

thermoregulator any device or mechanism (including physiological) that controls the temperature of an environment. —**thermoregulatory** *adj.*

thermospray or **electron spray mass spectrometer** (*abbr.*: ESMS); an instrument that directly applies a proportion of the eluent of a liquid chromatograph to a mass spectrometer for analysis of resolved components.

thermostable maintaining characteristics or function under a change (normally increased) of temperature. *Compare* **thermolabile**. —**thermostability** *adj.* *Compare* **thermolabile**.

thermostat 1 an automatic device for regulating temperature, especially maintaining a constant temperature. Such devices often work by cutting off the supply of heat when the required



thermotitration

temperature is exceeded and restoring it when the temperature falls below the required value. **2** to provide with or to control using a thermostat. —**thermostatic** *adj.*

thermotitration the determination of the endpoint of a reaction, or the point at which the reaction reaches equilibrium, by measurement of heat of reaction.

THESIT an alternative name for $C_{12}E_9$ (see **C₁₂E₉**), or Lubrol PX (see **Lubrol**).

theta *symbol*: θ (lower case) or Θ (upper case); the eighth letter of the Greek alphabet. For uses see **Appendix A**.

theta structure the structure that forms during replication of circular duplex DNA by the opening up of the two strands. Under electron microscopy it has the appearance of a Greek theta.

L-thevetose 6-deoxy-4-O-methyl-L-glucose; a component of some cardiac glycosides.

THF *abbr.* for **1** tetrahydrofuran. **2** tetrahydrofolate (often used but not encouraged; **H₄folate** preferred).

thia+ or (before a vowel) **thi+** *comb. form* containing sulfur, especially in place of carbon. See also **thio+**.

thiamine or (sometimes) **thiamin** or (formerly) **aneurin(e)** an alternative name for **vitamin B₁**.

thiamine diphosphate or (sometimes) **thiamin diphosphate** or (formerly) **thiamin pyrophosphate** *abbr.*: **TPP**; see **vitamin B₁**.

thick filament a longitudinal assembly of **myosin** molecules occurring in muscle. The thick filaments are the major constituents of the A band of the **sarcomere** in skeletal muscle; they interact with the **thin filaments** to bring about muscle contraction (see **sliding filament model**).

thin filament a molecular complex of **actin**, **troponin** (composed of the three peptides troponins I, T, and C), and **tropomyosin**, found in muscle tissue. In striated muscle the thin filaments span the I band and interdigitate between the ends of the **thick filaments** in the A band. Interaction between the thin and thick filaments brings about muscle contraction (see **sliding filament model**).

thin-film dialysis a type of **dialysis** that maintains the **diffusate** and the **retentate** in thin layers in intimate contact with the dialysis membrane in order to lessen the contribution to the overall rate of the process made by diffusion of solutes in free solution. Very rapid removal or exchange of small molecules can occur, especially if the volume ratio of diffusate to retentate is high. With appropriate membranes the technique may also be used for analytical studies of the size and conformation of peptides and proteins in solution.

thin-layer chromatography *abbr.*: **TLC**; chromatography on a thin layer of any of various powdered materials, including silicic acid, aluminium oxide, or cellulose. The layer, characteristically between 0.05 and 1 mm thick, is made to adhere to a sheet of glass, aluminium, or plastic. Thicker layers accept heavier loading, while thinner layers resolve more efficiently and rapidly. Resolving power is high, runs are rapid, and, because support materials such as silicic acid or aluminium oxide are inert to them, corrosive sprays, such as sulfuric acid, can be used to convert organic materials to carbon (a process known as charring), revealing them as black spots.

thio+ or (sometimes before a vowel) **thi+** *comb. form* **1** relating to or containing sulfur. **2** (in chemical nomenclature) denoting replacement of oxygen in a compound by (bivalent) sulfur. In trivial names of compounds it is used as a prefix (as in **thiouracil**; compare **uracil**), whereas in the systematic names of individual compounds it is used as an affix and is usually placed before the affix denoting an oxygen atom or an oxygen-containing group (as in **ethanethiol**; compare **ethanol**), although for organic sulfides, if named analogously to ethers by substitutive nomenclature, it replaces **oxy+** (as in 3-(methylthio)propionate. Compare **thia+**. See also **mercapto+**.

thioacetal see **acetal**.

thioctic acid a less common name for **lipoic acid**.

thiocyanate **1** the anion, $N\equiv C-S^-$, derived from thiocyanic acid. **2** any salt of thiocyanic acid. **3** any organic compound

containing the monovalent group $-S-C\equiv N$. Compare **isothiocyanate**.

thioester any organic compound formed, actually or formally, by the elimination of the elements of water either (1) between the carboxyl group of a carboxylic acid and a thiol, or (2) between the thiocarboxyl group of a thiocarboxylic acid and an alcohol or phenol. Thioesters formed by mechanism (1) are of general formula $R^1-CO-SR^2$, whereas those formed by mechanism (2) may be of general formula either $R^1-CO-SR^2$ or $R^1-CS-OR^2$.

thioether an old name for any organic sulfur-containing compound of general formula R^1-S-R^2 . Sulfide is now the recommended generic name.

thiogalactoside transacetylase see **galactoside acetyltransferase**.

thioglucosidase EC 3.2.3.1; *systematic name*: thioglucoside glucosylhydrolase; *other name*: sinigrase. An enzyme that hydrolyses thioglucosides to the constituent thiol and sugar, e.g. sinigrin to glucose and *N*-(sulfoxy)-3-mercapto-3-butenimide.

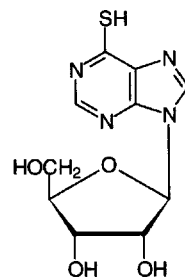
thioglycoside or **S-glycoside** any glycoside in which a glycosyl group is substituted into a thiol group in another compound. A naturally occurring example is **sinigrin**.

thiohemiacetal see **acetal**.

thiohemiketal see **acetal**.

thiohypoxanthine an alternative name for **mercaptopurine**.

thioinosine *symbol*: **sI** or **Sno**; 6-thioinosine; thiohypoxanthine riboside; mercaptopurine ribonucleoside; 9- β -D-ribofuranosyl-6-mercaptopurine; a nucleoside not occurring naturally. See also **mercaptopurine**.



thioketal see **acetal**.

thiokinase (sometimes) any of certain ATP-requiring enzymes of sub-class EC 6.2.1 (acid thiol ligases) that synthesize CoA thioesters of various fatty acids. These enzymes (recommended names given first) are: (1) **acetate-CoA ligase**, which acts on acetate, propanoate, and propenoate; (2) **butyrate-CoA ligase**; EC 6.2.1.2; *other names*: fatty acid thiokinase (medium chain); butyryl-CoA synthetase; butanoate-CoA ligase (AMP forming); this acts on C_4 to C_{11} acids and on corresponding 3-hydroxy- and 2,3- or 3,4-unsaturated acids; (3) **long-chain-fatty-acid-CoA ligase**; EC 6.2.1.3; *other names*: fatty acid thiokinase (long chain); acyl-CoA synthetase; acid:CoA ligase (AMP forming); this acts on a wide range of long-chain saturated fatty acids, the exact range of chain lengths varying somewhat according to the tissue of origin of the enzyme; and (4) **acetate-CoA ligase (ADP-forming)**; also acts on propionate and (very slowly) on butanoate.

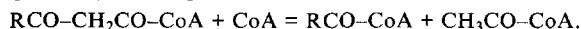
thiol **1** or (formerly) **mercaptan** any organic compound containing $-SH$ as the principal group directly attached to a carbon atom. **2** or **sulphydryl** the (covalently linked) $-SH$ group in any chemical compound.

+thiol *suffix* (in organic chemical nomenclature) indicating the presence of a **thiol** group attached to a carbon atom. Compare **mercapto+**.

thiolactomycin an antibiotic isolated from *Nocardia* spp. It selectively inhibits type II straight-chain fatty-acid synthases (i.e. the monofunctional enzyme system such as that in

Escherichia coli) but not the type I enzymes (i.e. the multifunctional enzyme system such as that in *Saccharomyces cerevisiae*).

thiolase a commonly used but misleading generic name for certain enzymes of differing chain-length specificity within subgroup EC 2.3.1 (acyltransferases) that catalyse the **thiolysis** by coenzyme A of CoA thioesters of β -oxo fatty acids according to the general equation:



Thiolases are thiol enzymes, their *S*-acyl derivatives being formed as intermediates in the reactions catalysed. Although the overall reactions are reversible, the equilibrium positions are greatly in favour of cleavage. These enzymes include: **acetyl-CoA C-acetyltransferase**; EC 2.3.1.9 and **acetyl-CoA C-acyltransferase**, EC 2.3.1.16.

thiolate 1 the anion, R-S^- , derived from a **thiol** (def. 1). 2 any salt of a thiol. 3 to introduce a thiol group into an organic compound. —**thiolation** *n*.

thiol proteinase an old name for **cysteine endopeptidase**.

thioltransacetylase **A** see **dihydrolipoamide S-acetyltransferase**.

thiolysis lysis of a covalent organic compound by a thiol; a reaction analogous to **alcoholysis** in which bond cleavage is mediated by nucleophilic attack by the $-\text{SH}$ group of an alkanethiol. The term is applied in particular to the **thiolase**-catalysed cleavage of a β -oxoacyl-CoA by coenzyme A in each cycle of chain-length reduction during fatty-acid catabolism. —**thiolytic** *adj*.

thionein the apoprotein of a **metallothionein**.

thionin any of a group of small proteins, characterized by two conserved motifs, that occur in plants and are toxic to animals. Example, leaf thionin from barley (precursor): database code THN3_HORVU, 137 amino acids (14.65 kDa); the toxic principle is formed from amino-acid residues 29–34.

thionucleoside any *N*-glycoside of any **thiopurine** or **thiopyrimidine**.

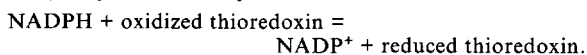
thionucleotide any phosphoric ester of any **thionucleoside**.

thiopurine any **purine** in which a thiol group has replaced a hydroxyl group. Compare **mercaptapurine**.

thiopyrimidine any **pyrimidine** in which a thiol group has replaced a hydroxyl group.

thioredoxin any member of a widely occurring class of heat-stable hydrogen-carrier proteins of ≈ 12 kDa that serve as hydrogen donors in various reduction reactions, notably of ribonucleoside diphosphates to the deoxy analogues, and in the light-dependent reductive activation of various chloroplast enzymes (including a number of those in the pentose-phosphate cycle). Chloroplast thioredoxin occurs in two forms (designated *f* and *m*), which show specificity in their enzyme-activating properties; a third form (designated *c*), of unknown function, occurs outside chloroplasts. Thioredoxins can exist in disulfide (i.e. oxidized) and dithiol (i.e. reduced) states, interconversion being catalysed by ferredoxin–thioredoxin reductases. Example, thioredoxin 1 from *Anacystis nidulans*: database code TH11_ANANI, 106 amino acids (11.52 kDa). See also **thioredoxin reductase (NADPH)**.

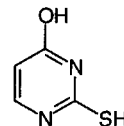
thioredoxin reductase (NADPH) EC 1.6.4.5; a flavoprotein (FAD) enzyme that catalyses the reaction:



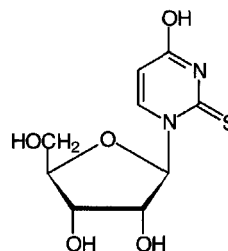
Example from *Escherichia coli*: database code TRXB_ECOLI, 320 amino acids (34.45 kDa); a homodimer.

thiotemplate mechanism or **protein thiotemplate mechanism** the mechanism of biosynthesis of *Bacillus brevis* peptide antibiotics, e.g. **gramicidin S** and **tyrocidin A**, in which the precursor amino acids, after activation, are bound as thioesters to the synthetase enzymes.

thiouracil *symbol*: Sur; 2-thiouracil; 4-hydroxy-2-mercaptopyrimidine; 4-oxy-2-thiopyrimidine; a naturally occurring thiopyrimidine, found in seeds of members of the cabbage family (e.g. *Brassica* spp.) and as a minor base in certain types of ribonucleic acid, especially transfer RNA (see also **thiouridine**). Thiouracil, 6-methylthiouracil, and 6-propylthiouracil are clinically useful for treatment of hyperthyroidism and thyrotoxicosis, their action being to inhibit biosynthesis of the thyroid hormones at the peroxidase-catalysed iodination and coupling steps.



thiouridine *symbol*: S or sU or Srd; either of two constitutional isomers occurring as minor nucleosides in certain types of ribonucleic acid, especially transfer RNA (tRNA): **2-thiouridine** (2-thiouracil riboside; 1- β -D-ribofuranosyl-4-oxy-2-thiopyrimidine; *symbol*: ^2S or $s^2\text{U}$), or its 5-methyl, 5-(methoxycarbonylmethyl), or 5-(methoxycarbonylmethyl) derivatives, which are found in some tRNA species in the first position of the anticodon; and **4-thiouridine** (4-thiouracil riboside; 1- β -D-ribofuranosyl-2-oxy-4-thiopyrimidine; *symbol*: ^4S or $s^4\text{U}$), which is found in some tRNA species elsewhere than in the anticodon.



2-thiouridine

third law of thermodynamics see **thermodynamics**.

thixotrope a gel that exhibits **thixotropy**.

thixotropy the property shown by certain fluids and gels of becoming less viscous when subjected to stress, e.g. by shaking or stirring, and of becoming more viscous again when the stress is relieved. —**thixotropic** *adj*.

Thr *symbol* for a residue of the α -amino acid L-threonine (alternative to T).

THR a gene family encoding thyroid receptors; part of a superfamily including receptors for steroids, retinoic acid, and vitamin D₃; see **erbA**.

threitol active erythritol, *threo*-1,2,3,4-tetrahydroxybutane, *threo*-1,2,3,4-butanetetrol; an enantiomeric pair of polyols, configurationally isomeric with **erythritol**. D-Threitol occurs in certain fungi, e.g. the honey fungus, *Armillaria mellea*.

threo- a configurational prefix to the systematic name of a polyhydric alcohol, especially of a monosaccharide, used to indicate the particular stereochemical configuration of a set of two contiguous $>\text{CHOH}$ groups that occurs in D-threose (see **threose** for structure) or L-threose, e.g. D-*threo*-2-pentulose for D-xylulose. Compare **erythro-**.

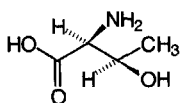
threoninate 1 threonine anion; the anion, $\text{CH}_3\text{-CH(OH)-CH(NH}_2\text{)-COO}^-$, derived from **threonine**. 2 any salt containing threonine anion. 3 any ester of threonine.

threonine the trivial name for α -amino- β -hydroxybutyric acid; (2*R**,3*S**)-2-amino-3-hydroxybutanoic acid; $\text{CH}_3\text{-CH(OH)-CH(NH}_2\text{)-COOH}$; an α -amino acid with two chiral centres.



threonine aldolase

Because molecules of threonine possess a second chiral centre, at C-3, in addition to the chiral centre at C-2 common to all α -amino acids other than glycine, the enantiomers L-threonine, (2*S*,3*R*)-2-amino-3-hydroxybutanoic acid (*symbol*: T or Thr), and D-threonine, (2*R*,3*S*)-2-amino-3-hydroxybutanoic acid (*symbol*: D-Thr or DThr), are diastereoisomeric with those of **allothreonine**, (2*R**,3*R**)-2-amino-3-hydroxybutanoic acid. [Note: The enantiomers of threonine may also be named semi-systematically as derivatives of threose: L_s-threonine in amino-acid nomenclature is synonymous with 2-amino-2,4-dideoxy-D_s-threonic acid in carbohydrate nomenclature, and D_s-threonine with 2-amino-2,4-dideoxy-L_s-threonic acid (the subscript letters d or g being added to the configurational prefixes where there might be uncertainty regarding the reference centre of chirality; see *d/l convention*).] L-Threonine is a coded amino acid found in peptide linkage in proteins; codon: ACA, ACC, ACG, or ACU. In mammals it is an essential dietary amino acid, is glucogenic, and its deficiency causes fatty liver. The fungal toxins phalloidin and phalloin contain one residue of D-threonine per molecule.



L-threonine

threonine aldolase see **glycine hydroxymethyltransferase**.

threonine dehydratase EC 4.2.1.16; *systematic name*: L-threonine hydro-lyase (deaminating); *other names*: threonine deaminase; L-serine dehydratase; serine deaminase. A pyridoxal-phosphate enzyme that catalyses a reaction between L-threonine and water to form 2-oxobutanoate, NH₃, and water; it acts also on L-serine. There are two types (examples from *Escherichia coli*): type 1 is responsible for the first step of isoleucine biosynthesis (serine is an alternative but poorer substrate); database code THD1_ECOLI, 514 amino acids (56.13 kDa); type 2 is a catabolic enzyme that acts on both serine and threonine, and is properly considered as a hydroxy-amino-acid deaminase; database code THD2_ECOLI, 329 amino acids (35.19 kDa).

threonine synthase EC 4.2.99.2; *systematic name*: O-phospho-L-homoserine phospho-lyase (adding water). A pyridoxal-phosphate enzyme that catalyses the formation of L-threonine from O-phospho-L-homoserine and water with release of orthophosphate. Example from *Escherichia coli*: database code THRC_ECOLI, 428 amino acids (47.06 kDa); it shows sequence homology with the serine dehydratase/threonine dehydratase group.

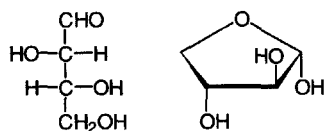
threoninium threonine cation; the cation, CH₃-CH(OH)-CH(NH₃⁺)-COOH, derived from **threonine**.

threoino the alkylamino group, CH₃-CH(OH)-CH(COOH)-NH-, derived from **threonine**.

threonin-*O*³-yl the alkoxy group, HOOC-CH(NH₂)-CH(CH₃)-O-, derived from **threonine**.

threonyl the acyl group, CH₃-CH(OH)-CH(NH₂)-CO-, derived from **threonine**.

threose the trivial name for the aldotetrose *threo*-tetrose; it has D and L enantiomers, which are respectively diastereoisomeric with those of erythrose. See also **threo**-.



α -D-threose

threshold the value that must be achieved or exceeded for an effect to become discernible. For example, the **kidney threshold of excretion**, for any substance, is the plasma concentration at which not all of the substance that appears in the filtrate can be reabsorbed, so that it appears in the urine.

THRF *abbr.* for thyrotropic hormone releasing factor; i.e. **thyrotropin-releasing hormone**.

THRH *abbr.* for thyrotropic hormone releasing hormone; i.e. **thyrotropin-releasing hormone**.

thrombin EC 3.4.21.5; *other name*: fibrinogenase; a serine proteinase, *M_r* 33 700-33 900, the natural coagulant of **fibrinogen**. Thrombin appears in the blood following activation of the coagulation system, as a result of proteolysis of **prothrombin**. It catalyses the preferential cleavage of Arg-|-Gly, activating fibrinogen to **fibrin** and releasing fibrinopeptides A and B. Its effects are mediated by the **thrombin receptor**. See also **blood coagulation**.

thrombin receptor a protein of platelets and vascular endothelial cells that binds **thrombin** and mediates its intracellular effects. It is a G-protein-coupled receptor. Example from human (precursor): database code THRR_HUMAN, 425 amino acids (47.35 kDa); this is a glycoprotein; the signal peptide comprises amino acids 1-26, and amino acids 27-41 are removed by thrombin, after which the new N terminus functions as a tethered ligand and the receptor is thereby activated.

thrombocyte an alternative name for **platelet**.

thrombogenic 1 tending to promote the formation of a blood clot (i.e. a **thrombus**). 2 having a precursor function for, or tending to promote, the formation of **thrombin**.

β -thromboglobulin a homotetramer of a peptide derived from **platelet basic protein precursor** by removal of the first four N-terminal residues from LA-PF4. It is released by activated platelets. It belongs to the *intercrine a* family (see **chemokine**) and mediates a number of actions in the inflammatory response, including chemotaxis of leukocytes.

thrombokinas an alternative name for factor Xa; see **blood coagulation**.

thrombolysis the degradation of a blood clot (i.e. **thrombus**) as a result of physiological or pharmacological action.

thrombolytic 1 tending to promote **thrombolysis**. 2 a drug or other substance promoting thrombolysis.

thrombomodulin *abbr.*: TM; *other name*: fetomodulin; a specific endothelial cell receptor that forms a 1:1 stoichiometric complex with thrombin, the complex being responsible for conversion of protein C to activated protein C. It is a type I membrane protein. Example (precursor) from human: database code TRBM_HUMAN, 575 amino acids (60.26 kDa).

thromboplastin an alternative name for factor III; see **blood coagulation**.

thrombospondin or **glycoprotein G** a thrombin-sensitive major glycoprotein located in the alpha granules of human platelets. It is secreted on thrombin activation and binds to the platelet membrane in the presence of Ca²⁺. It is an adhesion molecule that mediates cell-cell and cell-matrix interactions and can bind to fibrinogen, fibronectin, laminin, and type V collagen. It is a trimeric disulfide-linked protein and contains three types of repeating amino-acid sequence; the first is 57 amino acids long and shows homology with circumsporozoite protein from a malarial parasite; the second is 50-60 amino acids long and shows homology with **epidermal growth factor**; the third occurs as a continuous eightfold repeat of a 38-residue sequence; structural homology with **calmodulin** indicates that these repeats constitute multiple calcium-binding sites. Example, human thrombospondin 1 precursor: database code TSP1_HUMAN, 1170 amino acids (129.27 kDa). There are at least three such related proteins in humans.

thromboxane any of a class of **prostanoids** synthesized by **thromboxane-A synthase** (EC 5.3.99.5) from prostaglandin H (the product of prostaglandin-endoperoxide synthase) as thromboxane A (TXA). Depending on whether the substrate of the enzyme is dihomom-(6,9,12)-linolenate, arachidonate, or eicosa-